Physical Chemistry in Action

Carlos Bravo-Diaz
Sonia Losada-Barreiro
José G. Santos
Margarita E. Aliaga

From Experimental Kinetic Data to Reaction Mechanisms

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Physical Chemistry in Action

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Carlos Bravo-Diaz · Sonia Losada-Barreiro ·
José G. Santos · Margarita E. Aliaga

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Carlos Bravo-Diaz
Departamento de Química Física
Universidade de Vigo
Vigo, Spain

Sonia Losada-Barreiro
Departamento de Química Física
Universidade de Vigo
Vigo, Spain

José G. Santos
Facultad de Química y de Farmacia
Pontificia Universidad Católica de Chile
Santiago de Chile, Chile

Margarita E. Aliaga
Facultad de Química y de Farmacia
Pontificia Universidad Católica de Chile
Santiago de Chile, Chile

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Preface

The origin of this book can be grounded on the experience of the authors in both academia and research (physical organic chemistry). Along the past years, from the very beginning of their journey as Ph.D. students, the authors realized that very little in their background (as well as in that of other students and/or young researchers) prepared them to face the problem of investigating reaction mechanisms. They also noticed that colleagues in other universities working in the same or similar fields were facing analogous problems when attempting to prepare future researchers in designing kinetic experiments and/or to carrying out kinetic and mechanistic studies.

In spite that the authors are aware that there are many excellent textbooks on the theory of chemical kinetics published, they also recognize that information and/or guidelines on practical kinetics (hands-on) in solution is scarce, very frequently dispersed in the literature and, unfortunately, sometimes not easily available. In most cases, such information on practical kinetics is transmitted orally, being provided on the spot by instructors or Ph.D. advisors with some previous experience in chemical kinetics, information that they obtained from their advisors and/or from their own experience as scientists.

The reasons for such lack of information on practical kinetics are various, but in most cases they are closely related to the limited time given during the chemistry career for the various topics. Students usually take one or, at most, two courses on kinetics, organic and analytical chemistry, where only basic background information is provided and the acquired knowledge is neither enough nor makes practical connections between the various aspects required to undertake kinetic and/or mechanistic research.

The majority of the numerous texts that were published over the years on various aspects of reaction kinetics did not approach the topic nor made any attempt to teach young—or senior—researchers that, for whatever reason, require investigating a reaction mechanism in solution how to obtain experimental data to carry out a kinetic study, how to treat them, how to extract important kinetic information from them, and how to use variations in the composition of the reaction mixture to finally propose a mechanism for the investigated reaction. We, therefore, felt that a text covering practical aspects of investigating reaction mechanisms was necessary and

even desirable in attempting to bridge both theoretical and practical aspects of any kinetic and/or mechanistic study.

Bearing this in mind, this book is mainly addressed to researchers (either young graduate students or senior) who have some background in both analytical and physical chemistry (mainly in chemical kinetics) and that, for whatever reason, they need to undertake the study of the mechanism of a chemical reaction of interest in solution. The first aim of this book is, therefore, to integrate fundamental understanding of reaction kinetics with the practical, hands-on, design of experiments with the ultimate goal of showing the readers how to plan a mechanistic study, to make them realize the relationships between experimental results and conclusions with the ultimate goal of helping them to finally propose the most reasonable reaction mechanism. Encyclopedic completeness is sacrificed for providing an emphasis on these principles, with the aim of presenting them in a way as clear as possible, and by using several literature examples to enhance basic understanding rather than racing immediately to more sophisticated or specialized applications, which sometimes are covered in the own examples provided.

For the sake of clarity and simplicity, in this book we will limit ourselves to reactions in homogeneous solution where reactants are dissolved in a particular solvent. The main reason is because most synthetically important reactions take place in the liquid phase and because the approach can be easily extended to reactions taking place in biological systems and in, some instances, in the environment. We will thus make no attempts to cover gas phase or surface (two-phase) or solid-state kinetics or reactions that take place between two neat liquids. We will also make no attempts to cover theories of reaction kinetics and common terms as "activation energy" or "transition state" will be employed without explanation. In addition, to keep focus, clarity and to avoid in as much as possible intricate mathematics, derivations of equations will not be given in full and the interested reader should consult monographs on reaction kinetics and/or mathematics when needed. We will also limit to a minimum the information provided to characterize colloidal systems as micelles, microemulsions and emulsions, e.g., their physical properties, as excellent texts are available in the literature, but will rather focus in illustrating their effects on chemical reactivity.

In many aspects, this book may be considered to be a practical culmination of a long journey through the physical organic chemistry field that always attracted much attention to all of us, an adventure that has been sometimes tortuous but that, fortunately, was never carried out alone. We have been accompanied, from the very beginning of our training in the academia as Ph.D. students, by numerous advisors and colleagues who helped us to understand the complexities of the topic, sharing with us their wisdom and expertise to finally transmitting the acquired knowledge and expertise to new students, both undergraduate and graduate, who may continue working in the fascinating world of the chemical kinetics and, more broadly, in the physical organic chemistry field. In spite that they all deserve special mention for their guidance and endless support, we omitted on purpose their names because of the limited space in this book, as mentioning to all of them individually is an almost impossible task. We offer to all of them our gratitude and sincere apologies. We also

want to thank our Universities and funding agencies for providing financial support and resources that allowed us to acquire our expertise in the field.

We are also grateful to Springer for giving us the opportunity to publish this book. Special thanks go to Dr. Sofía Costa (Springer editor) for guiding us through the problems of preparing the first documents to highlight our ideas in the peer-review process, which indeed have been extremely helpful, and to Mr. Ravi Vengadachalam for his help throughout the whole production process.

Vigo, Spain

Carlos Bravo-Diaz
(On the behalf of all coauthors)

Competing Interests The authors have no competing interests to declare that are relevant to the content of this manuscript.

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About the Authors

Carlos Bravo-Diaz is a Full Professor of Physical Chemistry at the University of Vigo since 1995 and Honorary Professor of Marie Curie Sklodowska University since 2018. Ph.D. (1991) University of Santiago de Compostela. Associate research fellow (1991–1993, Rutgers, the State University of New Jersey, USA). Advisor of two postdoctoral and 10 Ph.D. students, and of more than 20 graduate and undergraduate students. Current research collaborations include groups at Portugal, Chile, USA, Brazil, Poland and Italy. Most (> 97%) of his research work was (and still is) focused in analyzing and interpreting chemical reactivity in bulk solution and in multiphasic colloidal systems. He has coauthored more than 150 peer-reviewed research papers in international journals, including several reviews and book chapters. Editor of one research book (Springer) and coauthor of several book chapters (Springer).He has taught, for years, several kinetic (and other physical chemistry) courses at undergraduate, graduate and Ph.D. level. He also coauthored several educational papers in the Journal of Chemical Education and published several proceedings from papers accepted in International Education Conferences (Edulearn).

Sonia Losada-Barreiro is Professor of Physical Chemistry at University of Vigo. In 2009, she completed her Ph.D. at University of Vigo focused on (1) the kinetics and mechanisms of the reactions between arenediazonium ions and antioxidants and (2) the study of the distribution of antioxidants in emulsions O/W. Nowadays, her research is aimed at understanding reactions and interactions at the interfaces of colloidal systems to control chemical processes in multiphasic systems. Her research involves the modelization of chemical reactions at interfaces of colloidal systems to understand the role of medium and concentration effects on the distribution of reactants between the different regions of the multiphasic systems. She authored or co-authored more than 52 peer-reviewed papers, 10 reviews and four book chapters.

José G. Santos received his Ph.D. in 1971 from the Complutense University of Madrid, (Spain), studying Kinetics and Mechanisms of organic reactions at the Department of Radiochemistry of the Rocasolano Physical Chemistry Institute of Madrid (doctoral advisor: Prof. Juan Manuel de la Figuera). Since 1971, he served

as a full-time research professor at the Pontifical Catholic University of Chile. Prof. Santos and has taught different Undergraduate and Postgraduate courses in Physical Chemistry. He has been head of Research and Postgraduate studies at his Faculty and a member of the Conicyt chemistry study group. He has supervised more than 100 undergraduate theses and 11 doctoral theses. He has coauthored more than 150 original research articles in peer-reviewed journals. His research work has focused on the kinetics and mechanism of organic reactions, substituent and solvent effects. In the last years, my research interests focused on reactivity studies in compartmentalized media (micelles, microemulsions, etc.), and in supramolecular chemistry.

Margarita E. Aliaga received her Ph.D. degree in 2007, studying Kinetic and Mechanisms of organic reactions at the Department of Physical Chemistry, Pontificia Universidad Católica de Chile (Ph.D. advisor: Prof. José G. Santos). She subsequently (2008–2009) joined a postdoctoral researcher's work at the Institute of Nutrition and Food Technology, University of Chile (Prof. Hernan Speisky). In 2009 she obtained a position as Assistant professor at the Pontificia Universidad Católica de Chile, where she continued working as Associate professor and recently (2024) as Full professor. Prof. M.E. Aliaga has taught different undergraduate and postgraduate courses on Physical Chemistry. Advisor of postdoctoral research (ANID), Ph.D. students, and undergraduate students. She has coauthored more than 70 peer-reviewed research papers in international journals, including original papers, reviews, and book chapter. Her research fields focus on kinetics and mechanism of organic reactions, reactivity studies in compartmentalized reaction media, fluorescent- and colorimetric probes and supramolecular chemistry.

Chapter 1
Introduction



Chemical kinetics is one of the oldest—classical—branches of physical chemistry intrinsically tied to the understanding of reaction mechanisms and to the assignation of specific rate constants to elementary mechanistic steps. Chemical kinetics is a science in itself, with its own concepts and methods of reasoning, but that makes use of knowledge from other areas of chemistry and/or science (thermodynamics, quantum chemistry, physics, biology, etc.). Its main objective is the determination, interpretation, and use of the laws which govern the rates of chemical reactions. It is concerned with the "how, why and when" of chemical reactions, being a central intellectual cornerstone to the discipline of chemistry. And yet, it is of enormous practical importance as well because chemical reactions control our environment, our life progress, and our ability to prepare new materials.

Some landmarks in the history of chemical kinetics are associated to attempts to make it quantitative in interpreting the mechanisms of the various chemical reactions. For instance, the steady-state approximation of Bodenstein, collision theory, and Lindemann mechanism involving activated molecules for unimolecular dissociation, the various approaches of the Michaelis–Menten multistep mechanisms for enzyme kinetics, and progress in theoretical kinetics with landmarks as the transition state theory is just some of the various examples that could be mentioned. For simple reactions, one can even calculate rate constants for individual reaction steps by accurate solution of the Schrödinger equation using scattering theory. Theoretical calculations often involve (a) the calculation of potential energy surfaces for electronically adiabatic processes (Born–Oppenhemier approximation) and (b) calculation of the dynamics per se. For chemical reactions, where bonds are broken and formed, the potential energy surfaces are always system-specific and therefore must be obtained from quantum mechanical calculations either by direct dynamics or by the intermediacy of fitting a potential energy function.

For truly kinetic investigations, it is far from enough to limit ourselves to measure the rate of appearance of a product or the disappearance of a reagent. It is also necessary to identify the factors that determine these rates, to quantify their effects, and to

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Table 1.1 Some branches of science where kinetic concepts are employed and to which kinetics is relevant, including some representative examples

Branch	Application of kinetics
Organic, inorganic chemistry	Reaction mechanisms, catalysis design
Biology (including biochemistry and forensic biology)	Physiological processes, bacteria growth
Chemical engineering	Reactor configuration, design, and scale-up
Physics	Diffusion, viscosity, nuclear processes
Photochemistry	Photochemical reactions (earth and planetary chemical reactions, solar conversion)
Electrochemistry	Electrode processes and sensing
Pharmacology	Drug action, stability, and release
Geology	Flow processes

try to explain them. Indeed, prediction and control of the course of a chemical reaction are major goals in many fields of science and technology (including chemistry, environment, biology, atmosphere, etc.) and are sought by scientists interested in understanding the fate of chemical reactions. Table 1.1 shows some of the branches of science to which kinetics is pertinent. In many cases, the simple knowledge of what are the reactants and the products is not enough to understand and define all chemical transformations the various molecules undergo even though in some cases experienced chemists may deduce reaction mechanisms after erudite mental processes which, at the end, need to be confirmed by experimental observations. Thus, theory alone does not allow us to elucidate mechanisms, particularly those of complex reactions, and such information needs to be acquired from experimentation.

The sort of information needed to investigate the mechanisms of chemical reactions can be briefly summarized in terms of:

- The factors that affect the rates of reaction and their dependence on concentration.
- The mathematical relationships between the rate constants and the various concentrations.
- The effects of temperature and solution composition on the rate of the reactions.
- The proposal of a reaction mechanism which summarizes all information above.

To achieve the goal and to obtain as much information as possible about the fate of reactants during a chemical transformation, we must try to identify all products, determine their yields, detect all intermediates, analyze the variations in the reaction rates as a function of concentration of reacting species, temperature, ionic strength, and a number of other parameters. Sometimes the search may be frustrating, because even a well-planned selection of variation of parameters can offer useful experimental results, but still several potential routes (mechanisms) may be consistent with the experimental observations, making impossible to assign one single mechanism to the reaction being studied.

As in practice we are unable to investigate all possible combinations of experimental conditions—simply because they are too numerous—only a restricted number of combinations are chosen to reveal the most useful information about the mechanism of the reaction. If, for whatever reasons, other combinations of experimental parameters are chosen and investigated at later stages, then the resulting information can either confirm the proposed mechanism or make necessary a revision of it. If the latter holds, then a new mechanism should be adopted so that it takes into account the new and old experimental observations.

In the process of proposing a reaction mechanism, the fundamental criterion of evaluation should be its capability to deal with existing experimental evidence and its predictive ability under new experimental circumstances. When two or more potential mechanisms can be used to describe the reaction, then the simplest form is often preferred always bearing in mind that new experimental results may make necessary to exclude it and to replace it by a more complicated one. Indeed, elucidation of reaction mechanisms in solution is not an easy task because most of the reactions consist of several elementary steps, which occur either consecutively or as parallel reactions. Despite the difficulties that arise from trying to find unique mechanisms, kinetic studies are always of pivotal importance since they provide the means of controlling and optimizing the transformation of chemical compounds.

A crucial point in any kinetic study is the measurement of time intervals. Several complications related with the determination of the exact time corresponding to a given composition are eliminated when the course of the reaction is tracked continuously and one or more concentrations are monitored directly in the reaction mixture. This is usually associated to the use of computer-controlled instruments which, in addition to control the time intervals, allows one to store analytical data for further processing. For handling and evaluation of kinetic data it is frequently necessary to know the initial or final concentrations (or the proportional physical quantity) of reacting species.

Reactions are usually started by rapid mixing of given volumes of two solutions, each of which separately contains one of the reacting components. Such a mixing must be done in such a way to achieve, after mixing, the desired experimental conditions in the shortest time interval possible. Both solutions are usually brought to the desired reaction temperature in advance either by placing them into a water bath or in two vessels with thermostating capability (usually thermostated water jackets).

A somewhat complicated situation can arise if, for whatever reason, a direct monitoring of the reaction is not possible. It then may become necessary to take samples from the reaction mixture for further processing. If the volume of the reaction mixture is large, samples can be taken with pipettes or syringes at controlled time intervals, quenching the reaction as soon as possible, for example, by cooling it down immediately, changing the acidity of the mixture, or adding a quenching reactant that reacts rapidly and quantitatively with one of the components of the reaction mixture. In the latter case, efficient mixing by agitation or stirring is highly recommended upon addition of the sample to quenching liquid.

Initial concentrations of reactants are usually known, and thus, no problems are expected when the kinetics is followed by direct determination of concentration in

individual samples. When physical methods are used to measure concentrations, the value of the measured physical quantity at time zero must be obtained either from an independent experiment in the absence of one of the reactants or by extrapolation to zero time. As these time intervals should refer to the beginning of the reaction, the estimate of the time at which the mixing can be considered as complete becomes of particular importance. For systems in which, for any reason, the value of the quantity at zero or infinite time is uncertain, the values can be obtained theoretically from the fitting of experimental data and should be consistent with the expected experimental values.

It is possible to use a combination of chemical and/or physical measurements to study concentration changes of the reaction mixture as long as the set characterizes the composition of the solution. This is why it is important to ensure the homogeneity of the solution. Indeed, the success of a kinetic study depends, to a high degree, on the choice of a suitable analytical method, that must be specific, sufficiently accurate, and rapid. In addition, the method of choice must not affect the reaction rate or otherwise interfere with the studied reaction and, to minimize the effects of components other than the reactants in the reaction mixture on the sought quantity, a calibration of the concentration response should be carried out preferably by using authentic samples of reactants and products under conditions as close as those employed in the kinetic run.

When the reaction under study is simple, it is frequently sufficient to monitor the concentration of just one component. On the contrary, if the investigated system consists of several elementary steps (consecutive or side reactions) and if the formation of one or more intermediates takes place in the course of the reaction, it is frequently necessary to monitor the concentration changes of two or more components.

Several other practical considerations may also be necessary to take into consideration depending on the particular research needs or the purpose of the kinetic study. Among them, the following ones are suggested:

1. The extent of the study. The researcher must make an estimation of the number of compounds whose reactions are to be investigated, deciding whether to investigate a reaction for a single reactant or if a group of related reactions should be studied. In addition, the researcher must also decide if it is necessary to investigate the effects of solution composition and how many repetitive measurements are required for the determination of a single rate constant.
2. The researcher needs to estimate the average time needed for one kinetic run.
3. The speed of analyses may play an important role in the timetable for systems where it is necessary to take samples and analyze them after quenching. It is usually necessary to add the reaction time and the time for handling the samples.
4. An important part in the planning of any kinetic study is the time required for the evaluation of the kinetic data. The most efficient time approach uses computers interfaced to the analytical instrument to avoid introducing the experimental data into the computer for further processing. The time needed for handling the data depends on the analytical method and on the type of reaction involved.

5. A factor that may play an important role in the planning of any kinetic study is the total consumption of the reacting species, particularly in the case of reactions involving expensive reactants, which must be of the highest purity available. In spite that the amount of a substance needed for the determination of concentrations at several times depends on its concentration and the volume of the reaction mixture, the lowest possible concentration is usually determined by the sensitivity of the analytical method used to monitor the reaction.

The practical implementation of the above considerations and suggestions will be done in next chapters. We will begin in Chap. 2 recalling the attention of the reader about the first decisions to be made when undertaking a kinetic study: what method should be employed to monitor a chemical reaction to measure concentrations, the importance of controlling temperature and how to monitor fast reactions in solution. In Chap. 3, we will assess representative examples of kinetic profiles considering the relationship with the concentration of relevant physical properties, and then we showed its data treatment in the respective rate equations in common elementary reactions. Besides, we will approach different ways of obtaining relationships between mechanisms and rate laws: empirical rate constants, determination of reaction orders, elementary steps and their combinations, calculation of rate constants, etc.

In Chap. 4, we will also introduce the kinetics of common complex reactions (reversible, parallel, consecutive), discussing the very useful approximations of pre-equilibrium and steady state in proposing reaction mechanisms, together with their advantages and limitations. More complex reactions schemes, including the role of the solvent on chemical reactions, and catalysis will be considered in Chaps. 5–7. Chapter 8 is focused on the elucidation of reaction mechanisms on the basis of the analysis of the variations of the rate constant upon changing the solution composition (acidity). Finally, Chap. 9 will approach the statistical, nonlinear regression analyses and the description of some common commercially available software for performing nonlinear regression analysis.

The organization of the book is differentiated from others that are currently available in one or more of the following aspects:

a. The field of chemical kinetics is treated in some depth just to provide the necessary background to undertake a kinetic study.
b. Special emphasis is placed in the analyses of experimental data to obtain rate constants and rate laws.
c. The text contains several representative examples taken from the literature to illustrate the various concepts and mechanisms covered throughout the text.
d. A whole chapter is devoted to analyze the effects of solution composition to identify potential mechanisms and to lay the basis of some experience to the reader to propose any other mechanisms not covered here.
e. Two chapters are devoted to cover kinetics in colloidal systems either in aqueous and non-aqueous systems because, in the real world, many reactions do not take

place in bulk solution but rather in multiphasic systems. The aim of these chapters is mainly to show how to investigate their effects on chemical reactivity and how to model chemical reactivity in those systems.

The integration of all these elements is the solely responsibility of the authors.

Chapter 2
Techniques and Methods to Monitor Chemical Reactions



Preliminary Note: A literature survey reveals that, very frequently, the concepts embodied in the words "kinetic methods", "kinetic techniques", "instrumental techniques", and "instrumental methods" are used indistinctly by the authors because they are frequently context-dependent and because there are only ambiguous differences between them. For the sake of clarity and simplicity, hereafter in this chapter we will employ the words "technique" (instrumental or kinetic) to describe the instrument(s) (spectrophotometer, fluorimeter, etc.) and/or the physical property (e.g., light absorption, emission, electric potential, etc.) employed to identify the components of a reaction mixture and/or to measure their concentrations; meanwhile the word "method" (instrumental, kinetic or experimental) will be employed in a broader sense to include not only the analytical technique employed to measure concentrations but also the way the reactants are mixed and homogenized and, eventually, perturbed or separated from equilibrium (flow, relaxation, competition, etc.).

The expressions "reaction rate constant" and "reaction rate coefficient" are also used indistinctly by authors. According to the IUPAC [1, 2], both terms are equivalent and so hereafter we will also employ them indistinctly meaning the proportionality constant between the rate of a reaction and the molar concentrations of the reactants.

2.1 Introduction: Preliminary Considerations Before Undertaking Kinetic Studies

Any kinetic experiment consists, basically, in measuring the concentrations of one or more reactants or products at different times during the course of the reaction. The experiments are repeated under different experimental conditions so that, at the end, the researcher can achieve the main goals of the kinetic study: (1) the determination of the overall rate of the reaction and/or that of their individual elementary steps,

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and (2) the analysis of the sequence of elementary steps giving rise to the overall reaction, i.e., the proposal of the *reaction mechanism*.

Prior to initiate any kinetic study once a particular chemical reaction is deemed necessary to be investigated, the researcher needs to acquire, no matter how, some preliminary information and will have to make some preliminary choices before attempting to do any experiment to obtain kinetic data. This preliminary information is necessary to help the researcher in making a judicious choice of the appropriate techniques and instrumental methods that could be employed to monitor the reaction and to, eventually, identify and quantify any and all components that could be formed in the course of the reaction, including the formation of intermediates. The technique(s) of choice must be suitable according to the reaction timescales and, obviously, should be accessible to the researcher. Other important preliminary information that may influence the choice of the experimental technique is related to the required experimental conditions under which the kinetic study will be carried out, including the range of temperatures, acidity, the nature of solvent, and/or the stability of the reactants/products under such experimental conditions. This initial information is also necessary to achieve the greatest economy in the time spent on the kinetic study, and it is essential to estimate the accuracy that is needed in the measurement of concentrations and time. Such estimates can, in turn, play an important role in the choice of the method, instrumental technique and/or experimental conditions.

Most probably, the first decision to be made to carry out a kinetic study is related the choice of the technique(s) that can be employed to monitor the reaction. The huge variety of existing chemical species and types of reaction (and those to be discovered in future) means that the timescale over which chemical reactions occur covers many orders of magnitude, from very slow reactions (~years), such as iron rusting, to extremely fast reactions, such as the electron transfer processes which may go down to a few femtoseconds (i.e., ~10^{-15} s). It is worth noting here that the concepts of "slow" or "fast" reactions are relative. Hereafter, we will employ them for qualitative purposes, bearing in mind that they are commonly referred to the time necessary to mix the reagents and start the process of collecting kinetic data, i.e., a few seconds, so that a few dozens of data points can be collected before the reaction is over. Some adjectives such as "moderately" or "very" may also be employed to make qualitative classification of reaction time scales as that in Table 2.1. The researcher, thus, must be aware of the various methods and techniques available to investigate reactions whose lifetimes vary over several orders of magnitude and, most importantly, if they can, or cannot, be easily employed under different experimental conditions.

Independently of the details of the necessary experimental arrangements needed to carry out the kinetic study, the researcher must also bear in mind that to undertake successfully any kinetic experiment, the reactants need to be mixed to initiate the reaction on a timescale that is negligible relative to that of the reaction, and then monitoring the concentration(s) of one or more reactants and/or products as a function of time. This means that, in practice, a finite time interval is needed to mix reactants, homogenize the mixture to obtain an identical distribution of all chemical species in the reaction mixture and to carry out the measurement of the concentration. This

Table 2.1 Qualitative classification of the speed of chemical reactions according to their half-lives

Type of reaction	Half-life (s)	Time for completion
Very slow	$>10^8$	Months–years
Slow	10^5–10^8	Weeks
Moderate	1–10^5	Minutes–hours
Fast	10^{-6}–1	Seconds
Very fast	10^{-12}–10^{-6}	Microseconds

Intervals are approximate

time depends on the ability of the researcher, but in the best cases is not less than, roughly speaking, 5–10 s. The researcher has the possibility to control, within certain limits, the rate of the reaction by changing the experimental conditions (concentrations, temperature, acidity, etc.) so that we can study experimentally a considerable number of reactions over convenient time scales. But, in any case, an appropriate and practical judgment needs to be done by choosing one or more potential analytical techniques whose response times are much shorter that the half-life of the reaction, taking into consideration the time needed for handling the reactants, mixing them and homogenizing the resulting mixture.

Whatever are the reaction lifetimes, the technique(s), and/or method(s) to be employed by the researcher to carry out successfully any kinetic study has to meet some basic requirements:

- The technique(s) to be employed should have the capability of performing kinetic measurements with the desired accuracy under controlled conditions (for example, at controlled temperature). At the same time, the technique should allow to perform the kinetic study under different experimental conditions (e.g., different concentrations, temperatures, acidities, etc.).
- The techniques/method used should not interfere with the course of the chemical reaction nor alter the composition of the reaction mixture. Hence the use of non-destructive techniques is highly advisable and advantageous.
- The technique should permit to monitor the temporal evolution of as many reactants or products as possible, that is, should allow to measure changes in the concentrations of as many species as possible with time under the chosen set of experimental conditions.
- Ideally, the researcher would like to automate and computerize the measurement of the physical property employed to monitor the reaction. Data acquisition techniques should be employed to finally enable fast data collection and further treatment of the experimental data. The instrumentation employed should allow to collect enough relevant kinetic data (typically, at least 20 data points) during the course of the reaction, preferably during the first three half-lives.

Another important factor that needs to be considered—for economy and environmental reasons—is the total consumption of the reacting species. The amount of a substance needed for the obtaining one concentration–time curve depends on

the concentration and the volume of the reaction mixture. The lowest concentrations needed are usually defined by the sensitivity of the instrumental technique employed, for example, for optical absorption the limit is roughly 10^{-4} M, whereas electrochemical and/or fluorescence techniques can detect concentrations as low as 10^{-6} M. While the concentration limit is frequently imposed by the analytical method employed, the imagination and expertise of the researcher to carry out a kinetic study are demonstrated by:

1. The adaption of experimental procedures to small volumes.
2. An appropriate choice of the reaction vessels and mode of measurement.
3. Adapting procedures requiring sampling (which require large amounts of material) to those where concentration changes are measured directly in the reaction mixture (requiring, therefore, comparatively less material).

2.1.1 Accuracy of Rate Measurements

The methods employed to study the kinetics of the reactions must be specific, accurate, and rapid. While it is relatively simple to monitor the kinetics of "slow" reactions, highly sophisticated techniques are required to study "fast" reactions. In the next sections we will consider some of them, providing details to help the researchers to make a judicious choice of the technique, but before that, it may be of some interest to get a preliminary feeling of the nature of the errors in kinetic data so that the researcher can get an idea of data reliability (further details are given in Chap. 9). At the end, the accuracy with which a complex quantity such as the rate constant can be determined depends on the accuracy of the measurement of all concentrations involved, of the measurement of time and on the way in which experimental data are handled to finally obtain the rate constant.

The inherent errors in any physical measurement can be classified as random and systematic. Random measurements in experimental measurements are caused by unknown and unpredictable changes in the experiment, occurring in the measuring instrument or in environmental conditions, and are relatively simple to deal with. Some examples of inherent errors include changes in the electric noise of electrical instruments or sudden changes in the environment temperature. Systematic errors in experimental measurements include those that are not random and arise from the measuring instruments either because there is something wrong with the instrument or its data handling system of because the instrument is wrongly used by the researcher. The *precision* of a measurement is limited by the random errors. Systematic errors are more difficult to detect and so also more difficult to handle. The *accuracy* of a measurement is often reduced by the systematic errors, which are difficult to detect even for experienced researchers.

The a priori, detection of systematic errors is more an art rather than a science, since it implies a foreknowledge of the real behavior of the system. Systematic errors sometimes show themselves as uniform trends. In these cases, deviations of the experimental data from mean values show a consistent trend over a range of

experimental conditions rather than being randomly distributed. For such a deviation to be significant, it must exceed the relative error in the measurement of the rate coefficient. Examples of systematic errors are, for example, those due to small amounts of secondary reactions (which are the most frequent) or those due to instrument limitations.

Most frequently, the rate constants are calculated from data providing the concentration of one reactant or product at different times. It is important to achieve the highest possible accuracy in the measurement of the sensitive parameters. For instance, it is usually possible to measure times with a higher accuracy than concentrations, and thus the greatest attention should be paid to the measurement of concentrations, since the inaccuracy associated with the measurement of one parameter may be so large that dominates the effects due to the combined errors in all other measured quantities.

Similarly, it is important to control other factors affecting the reaction rate—temperature, acidity, pressure, radiation, etc. It is usually very advantageous to keep constant (or as constant as possible) all those factors affecting the rate coefficient during the course of the reaction. Obviously, it is only of limited advantage to employ special, accurate, instrumental techniques for the determination of the concentration if, for example, temperature is not sufficiently controlled and changes during the course of the reaction.

It is also important to notice that obtaining highly accurate values of rate constants may not be always the ultimate goal of a kinetic study. The required accuracy depends on the ulterior use of the determined rate coefficients. For example, a high accuracy is usually needed in structural studies where the effects of substituents are being investigated (rate constants with accuracies lower than 10%), but much higher accuracy (3–5%) is needed when investigating the effects of solvents, ionic strength, isotope effects, etc.

In many cases, the study of the effects of various parameters on the values of the rate coefficients is of general interest and of much greater importance than an accurate measurement of the value of a single rate coefficient. It is frequently advisable, because more information is gained, carrying out 20 measurements at 10 different pH with an accuracy of $\pm 10\%$ than 10 measurements at only one pH with an accuracy of $\pm 2\%$. Similarly, it is much better to obtain rate coefficients at 3–5 temperatures with accuracies of $\pm 10\%$ than to obtain reproducible values ($<2\%$) at a single temperature.

The principal random error in the rate coefficient is determined by the precision of the analyses and the length of the time interval chosen. Analytical errors are about the same for all orders ranging 0–4% and thus mainly depend on the extent of reaction occurring between the first and last points. It should be noted that for completely random errors, the relative error in the rate coefficient may be significantly improved by taking a great number of measurements preferably during the time in which the concentration changes by two–threefold, i.e., one needs to monitor the reaction during at least 3–4 half-lives and to acquire the majority ($>70\%$) of the experimental data within the first and second half-lives. Thus, where possible, it is advisable to monitor the disappearance of reactants or the formation of products during the initial stages of the reaction and measure the concentration once the reaction is over.

In any case, the reader must bear all the time in mind that published values of rate coefficients or other kinetic parameters become part of the chemical literature, and such data can be used in future in unsuspicious/unanticipated ways by other researchers, and thus it is if primary importance to provide as may details as possible of the exact conditions employed in the measurements.

2.1.2 *Effects of Temperature on the Precision of the Rate Coefficients*

The dominant variable in a reaction is the temperature and this variable must be carefully controlled to measure rate coefficients (k) with any precision. The effect of temperature on k can be calculated from the differential form of the Arrhenius equation (Eq. 2.1), where A is the so-called pre-exponential factor and Ea the activation energy of the reaction.

$$k = Ae^{-\frac{E_a}{\mathrm{RT}}} \Rightarrow \frac{\mathrm{d}(\ln k)}{\mathrm{d}T} = \frac{E_a}{\mathrm{RT}^2} \Rightarrow \frac{\Delta k}{k} = \frac{E_a}{\mathrm{RT}}\frac{\Delta T}{T} \tag{2.1}$$

It follows that for most thermally activated reactions (E_a ~ 80 kJ/mol), the ratio $E_a/\mathrm{RT} \approx 35$ at the practical room temperature of $T = 300$ K, and therefore to obtain k with a precision of $\pm 1\%$, T must be known to ± 0.1 °C. At much higher temperatures ($T > 100$ °C), it is difficult to obtain control of temperatures with a precision better than $\pm 1\%$, which in turn means errors of $\pm 5\%$ in k.

The researcher must be aware that an additional error source may arise from strongly exothermic or endothermic reactions, where the heat transfer to the thermostat may not be as fast as the reaction rate, and therefore the internal real temperature may not match that of the thermostat. Fortunately, this frequently is not a serious problem for reactions in solution because the solvent is able to act as an effective thermostat because its concentration is usually much higher (up to several orders of magnitude) than that of the reactants.

2.1.3 *Weighting of Experimental Data Points*

When obtaining (concentration, time) pairs of data points, there should not be any difference between the procedure of obtaining a number N data points during a single run and that of making N duplicate runs obtaining one pair of data per run. Each pair of data provides a set of values from which the rate coefficient can be calculated, and if all the data are taken from a single run, then the best value of the rate coefficient is the average from all the data point taken two at a time. Thus, the first procedure is less costly of time and equipment.

The integral method of computing the rate coefficient (see Chap. 3 for further information on integrated rate equations) generally places most weight on the initial concentration, since that quantity appears in the integrated rate equation. Such practice is wise only when the initial concentration is known with better precision than any of the others. If there is some other data point for which the precision is better, then it is better to recast the equations in a form which refers the other concentrations to this point, i.e., it is advisable to shift the coordinate axis. Graphical representations of the data usually overcome these difficulties because the experimental data points are plotted independently of a knowledge of the initial concentration. Mathematical treatments for reactions of several orders have been published [3].

2.1.4 Main Components of Kinetic Studies

Figure 2.1 summarizes schematically the three main modules that integrate any kinetic study, showing the time scales of different methods currently employed to monitor chemical reactions as well as some of the most common physical properties employed. In principle, it is possible to use any chemical or physical property—or a combination of them—to monitor changes in the concentration of reactants or products with time provided that the set of properties characterizes the composition of the reaction mixture.

From the experimental point of view, the biggest challenge that must face any research is the choice of a technique/method that best adapts to the reaction that will be studied, the order of magnitude of its rate constant and the availability of the required equipment. The researcher must bear in mind, and balance, the need of controlling parameters such as temperature, solvent, nature of the reactant's intermediates and products, if the reaction goes to completion irreversibly or will exit an equilibrium between reactants and products.

2.1.5 Instrumental Techniques

Manual or classical methods to quantify the concentrations of reactants/products (e.g., gravimetric or volumetric) (sometimes also called wet chemistry) are still used in many laboratories, mainly for analytical purposes, but the need of fast measurements and of determining low concentrations of reactants/products made scientists to exploit the advantages of techniques where values of physical properties (e.g., electric potential, conductivity, light absorption/emission, mass to charge ratio, etc.) that are proportional to concentrations can be obtained for quantitative measurements.

Because classical methods are not routinely employed, we will mostly focus on the modern instrumental methods, that exploit the measurement of any physical or chemical property of the reactants and/or products that is proportional to its concentration to monitor chemical reactions. The practical use of these modern instrumental

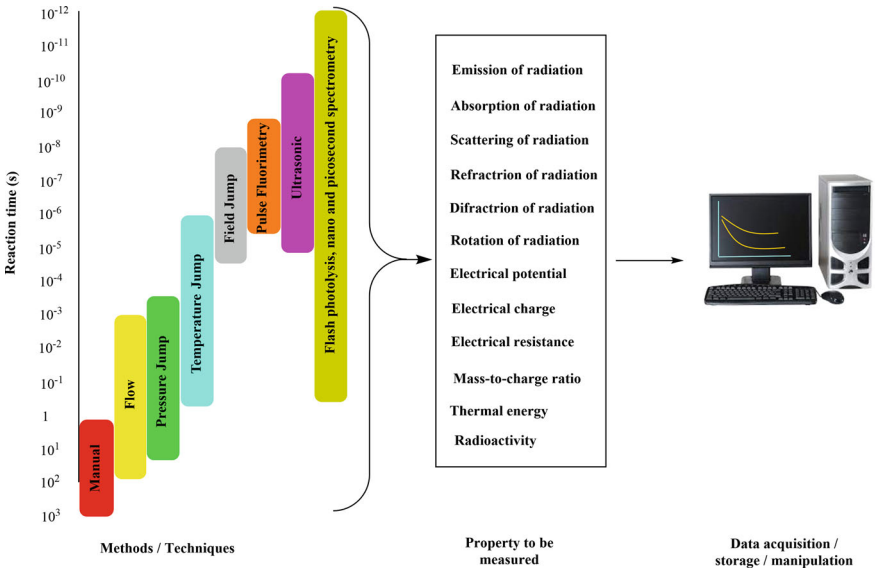


Fig. 2.1 The three main elements of a kinetic study. The first one relates to the methods/techniques employed and depends on the time range and the type of reaction to be studied. The second one relates to the property/properties to be measured to detect changes in the concentration of reactants or products during the course of the reaction and can be employed alone or in combination with other properties. The choice of the property also depends on the required time resolution. The third module relates to how the information on the measured property is acquired, stored, and further manipulated, and it is commonly done through computer-controlled instruments/detectors

methods and the measurement of physical properties was constrained, and sometimes still is, by the limitations in the development of electronics and computer industries, paralleling the lack of reliable, simple, and/or cheap instrumentation. Table 2.2 summarizes some of the most common techniques currently employed to monitor chemical reactions. The reader must be aware that this is not an exhaustive list and many other methods may be available. Very frequently, the limits are in the imagination of the researcher to set up particular combinations of methods/techniques for investigating the kinetics of particular reactions or reactions under particular experimental conditions.

Some spectroscopic techniques involve an exchange of energy between a photon and the sample. The energy can be absorbed by the sample (absorption spectroscopy) or emitted by an excited photon upon returning to the lower state energy (emission spectroscopy). Spectroscopic techniques that do not involve exchange of energy between photons and sample (diffraction, refraction, scattering, etc.) are not commonly used in chemical kinetics. Electrochemical techniques involve three basic experimental designs: (1) measure potential when current is zero, (2) measuring potential while controlling current, and (3) measure current while controlling potential. Each of these experimental designs relies on Ohm's law ($V = IR$) and uses different types of instruments.

Table 2.2 Examples of spectroscopic and electrochemical techniques employed to monitor chemical reactions

Common spectroscopic techniques		
Type of energy transfer	Region and sketchy wavelength range (m)	Example
Absorption	Radio wave (10^{0}–10^{3})	EPR, NMR
	Microwave (10^{-3}–10^{0})	Microwave
	IR (10^{-6}–10^{-3})	Infrared
	UV–Vis (10^{-8}–10^{-6})	UV–Vis, atomic
	X-ray (10^{-11}–10^{-8})	X-ray absorption
	γ-ray (10^{-16}–10^{-11})	Mossbauer
Photoluminescence	UV–Vis	Fluorescence phosphorescence
	X-ray	X-ray fluorescence
Chemiluminescence	UV–Vis	Chemiluminescence
Common electrochemical techniques		
Static ($I = 0$)	No current flow	Potentiometry
Dynamic: current flow ($I \neq 0$)	Fixed potential	Amperometry, coulometry
	Variable potential	Polarography, voltammetry (pulse, cyclic)

Physical methods generally need an authentic sample for standard calibration. It is important to realize that the accuracy of any measurement cannot be greater than the accuracy with which the composition of the standard is known. Physical quantities can be measured in quenched samples, but in most cases, it is advantageous to measure directly in reaction mixtures, which is possible with continuous recording of the property. For practical purposes, we will only describe briefly here some of the most important techniques because the topic is covered enough in various physical chemistry and analytical undergraduate courses.

2.1.5.1 Spectroscopy

Spectroscopy is a technique that involves the interaction of electromagnetic radiation with the molecules, Fig. 2.2. It can be used to identify compounds in a mixture if not all of them absorb at the same wavelength, to quantify its concentration and to study some of its physical properties. Commonly used spectroscopic techniques include microwave (MW), infrared (IR), and ultraviolet–visible (UV–Vis). UV–Vis is probably the most widely employed technique to monitor chemical reactions in solution because of its feasibility and simplicity and because the required instrumentation is relatively cheap compared to that of the other techniques. Figure 2.2 shows the detected changes in the absorption spectrum of the reacting system 4-nitrobenzenediazonium (4PNBD) and glycine in aqueous buffered solution as a

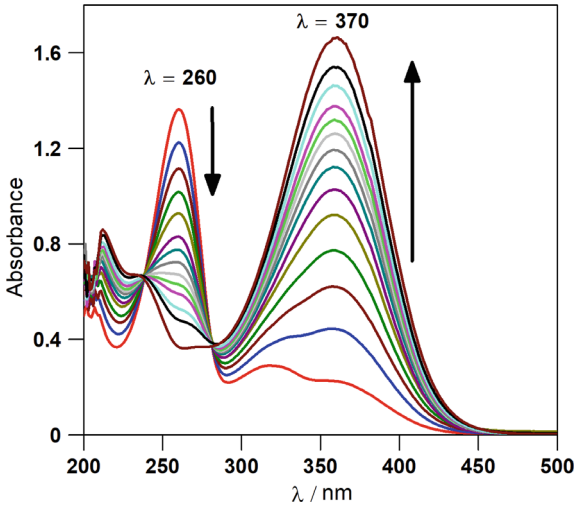


Fig. 2.2 Representative spectra obtained at different times for the reaction between 4PNBD and glycine in aqueous buffered solution. The arrows show the disappearance of the main absorption band of 4PNBD ($\lambda = 260$ nm) and the new band, associated to the formation of the corresponding triazene ($\lambda = 370$ nm). Reproduced with permission from ref. [4], Copyright © 2007 John Wiley & Sons, Ltd.

function of time. Two main bands are observed, the first one centered at 260 nm is associated to the disappearance of 4PNBD and the second one centered at 370 nm is consequence of the formation of the reaction product (a triazene).

2.1.5.2 Electrochemistry

Electrochemical analyses exploit the redox properties of reactants or products to measure the potential and/or current of the electroactive species in electrochemical cells, Fig. 2.3. Therefore, technique is appropriate to study the redox behavior of reactants and quantify the concentration of electroactive neutral species and/or of ions in the sample. Several types of electrochemical techniques can be employed including potentiometry, coulometry, amperometry, voltammetry, etc., each one with different characteristics from those of the others. A typical electrochemical cell consists of a three-electrode system that is purged with a non-electroactive gas (typically nitrogen) to minimize the presence of oxygen, Fig. 2.3. The cell is thermostated by a temperature-controlled fluid coming from an external thermostat. Further details on electrochemical cells and their set-up can be found in any electrochemical or analytical textbook [5, 6].

Figure 2.4 illustrates the use of the linear sweep voltammetry technique to monitor the reaction between 4-nitrobenzenediazoniumm ions (4PNBD) and aminoacids (glycine and serine), where up to five reduction peaks are detected, allowing simultaneous monitoring of 4NBD loss, product formation, and the formation of a transient intermediate.

The reduction peaks at $E_p = -0.5$ and -1.0 V (peaks IV and V, respectively) were not detected in the absence of aminoacids and thus are associated to the products formed in the course of the reaction. The peaks ate $E_p = -0.1$ and -0.8 V (peaks I

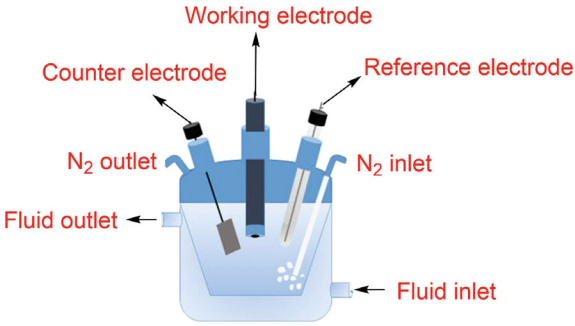


Fig. 2.3 Typical thermostated electrochemical cell for voltammetry

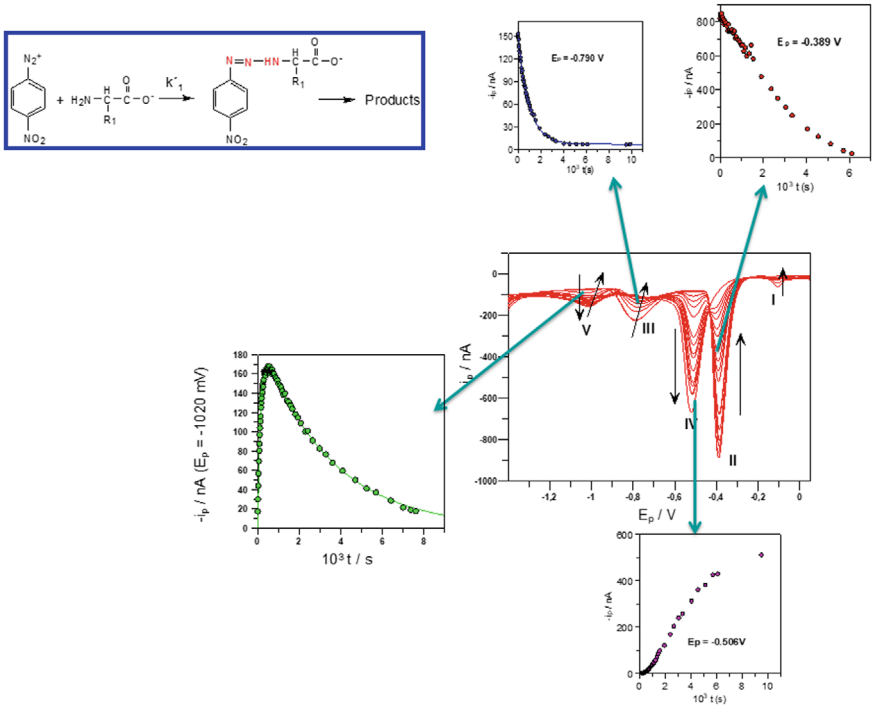


Fig. 2.4 Overlapped linear sweep voltammograms of the reaction between 4PNBD and serine in aqueous buffered (Britton-Robinson, BR, buffer, pH = 6) solution obtained at different times. Upon addition of serine to an aqueous acid solution of 4-nitrobenzenediazonium salt, up to five reduction peaks are detected. The arrows are indicating the trend in the variations of the peak currents, ip, with time. Peaks I, II, and III are associated to the reduction of the diazonium ($–N_2^+$) group (I and III) and the $–NO_2$ group of 4NBD (peak II). The new reduction peaks, not observed in the absence of the amino acid, are detected at peak potentials of $E_p \approx -0.5$ V (peak IV) and $E_p \approx -1.0$ V (peak V). Reproduced from ref. [4] with permission, Copyright © 2007 John Wiley & Sons, Ltd.

and III, respectively) are associated to the reduction of the diazonium group, $-N_2^+$, and in the presence of the aminoacids ([aminoacid] >> [4PNBD]), their peak currents decrease exponentially with time following clean first-order kinetics for more than 3 half-lives. Peak II ($E_p = -0.4$ V) is associated to the reduction of the nitro group of 4PNBD. The variation of their peak currents ($E_p = -1.0$ V) with time is biphasic increasing with time up to a maximum after which a decrease is detected, consistent with the formation and subsequent decomposition of a transient intermediate. The peak at $E_p \sim -0.5$ V (peak IV) increases with time after an induction period and is, thus, associated to product formation, but that of peak V ($E_p \approx 1.0$ V) shows a biphasic behavior, with i_p values increasing up to a maximum after which a decrease is observed until no reduction peak is detected (i.e., this peak corresponds to a transient intermediate). Note that the highest peak current achieved for peak V is reached when 85–90% of 4NBD has been consumed as indicated by the i_p value of peak III. Further details can be found elsewhere [4].

2.1.5.3 Separation Techniques

Separation techniques are an essential tool to identify the components of a reaction mixture enabling researchers not only to isolate and identify individual components but also to quantify them. However, the relative "slowness" of the chromatographic analyses, the difficulties in maintaining the reaction mixture at constant temperature and the fact that the reaction mixture will be in contact with the metallic parts of the instrument and/or with the mobile phase limits seriously their use to monitor reactions because of the extra cautions that need to be taken to ensure that the changes observed are only due to the investigated chemical reaction and not by potential side effects. Different types of chromatography include gas (GC), liquid (LC), thin-layer (TLC), electrophoresis, etc.

Figure 2.5 illustrates the use of LC to monitor the formation of dediazoniation products. For the present case, the use of a quenching agent is needed to avoid side reactions of the arenediazonium ions with the mobile phase and metallic parts of the chromatograph.

A direct injection of the reaction mixture was not advisable because may lead to side reactions with the mobile phase and the metal parts of the HPLC. Reactions were run in 15–20 vials (prepared under the same experimental conditions) and were quenched at progressively longer intervals of time by adding an aliquot of a 2-napthol-6-sulfonic acid (2N6S) aqueous solution so that the final [2N6S] >> [PNBD]. Further details on the method can be found elsewhere [9].

2.1.5.4 Mass Spectrometry

Mass spectrometry (MS) is a technique that involves measuring the mass-to-charge ratio of ions produced from a sample using electric and magnetic fields. MS can

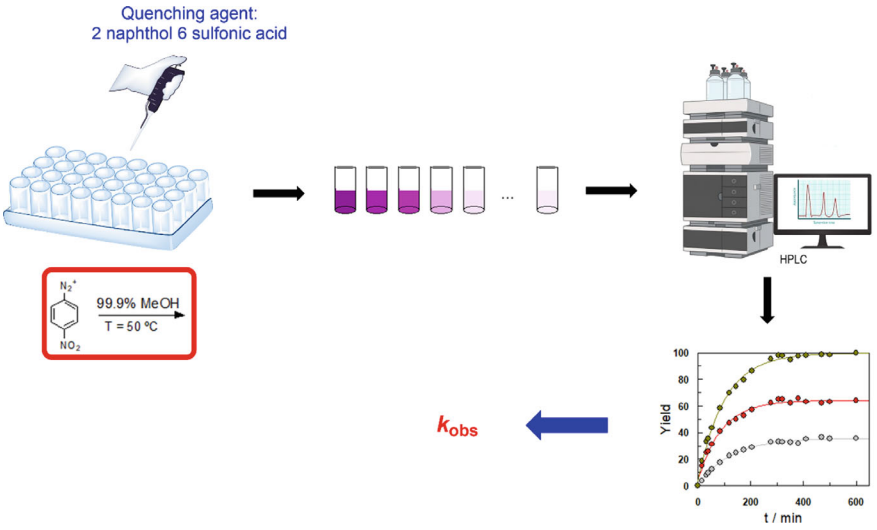


Fig. 2.5 Experimental protocol employed to determine the observed rate constants (k_{obs}) for the formation of dediazoniation products of 4-nitrobenzenediazonium (PNBD) tetrafluoroborate in MeOH by chromatography (HPLC, UV–Vis detection) [7, 8]

identify, quantify specific compounds in a reaction mixture, determine their molecular weight, and study their chemical properties. MS can be combined with other techniques, commonly separation techniques such as liquid chromatography (LC) and/or gas chromatography (GC), electrospray ionization (ESI), and matrix-assisted laser desorption/ionization (MALDI) leading to different types of mass spectrometry (LC–MS, GC–MS, ESI–MS, MALDI–MS), each one with its unique advantages and limitations. A nice example on the use of HPLC–MS to monitor the kinetics of enzymatic reactions has been provided by Hsich et al. [10].

2.2 Methods and Devices for Controlling Temperature and Medium Effects

Temperature is a key variable in many kinetic experiments because the rate constants depend on temperature. Such dependence is usually interpreted in terms of the Arrhenius Eq. 2.1, where A is the so-called pre-exponential factor and E_a is the activation energy of the reaction. From Eq. 2.1 it follows that, for a given reaction, the size of change in the rate constant k with T depends on the value of the energy, and as a rule of thumb, for thermally activated reactions with E_a values ranging 80–100 kJ mol^{-1}, a change in temperature of 10° roughly doubles the activation value of the rate constant.

For many laboratories, a benchtop circulator bath, a heater/cooler bath, or just a chiller works fine to acquire temperature control. These units save lab floor space and come in versions that can shake samples and circulate the water if uniform temperature is needed. At the end, the choice of a suitable heating/cooling thermostat to control the temperature depends on the task at hand as well as the likely ones ahead. Relatively basic units work for a wide range of kinetic studies, and some of the smaller units are even portable.

2.2.1 Thermoelectric Temperature Control—Peltier Effect

The thermoelectric temperature control relies on electricity for cooling or heating according to the Peltier effect, which is a thermoelectric effect caused when a potential difference is applied across a thermocouple causing a temperature difference between the junctions of different materials in the thermocouple, Fig. 2.6. The effect creates a temperature difference by transferring heat between two electrical junctions. A voltage is applied across joined conductors to create an electric current. When the current flows through the junctions of the two conductors, heat is removed at one junction and cooling occurs. Heat is deposited at the other junction.

Peltier devices are composed of two elements of a semiconductor which is constructed from doped Bismuth Telluride. Upon application of a direct current power source, these devices generate a cooling action, countered by a generation of heat on the opposite side of the device. Thermoelectric "modules" are placed in a sandwich-type arrangement between high-performance aluminum heat sinks and one or more axial fans. Peltier assemblies are coupled with electronic devices to "pump" the heat from the inside of an enclosure to the outside.

With this system, both high heating and/or cooling rates can be achieved. The best Peltier devices can reach temperature values below ambient temperature, typically from −5 to 60 °C (some sophisticated Peltier devices can reach lower and higher temperatures), allowing a quick and accurate (±0.01 °C) control of temperature. An advantage of the system is that the size of the Peltier devices is relatively small, and the investment and operating costs are relatively low.

2.2.2 Liquid Bath Temperature Control

This kind of temperature control often uses water or thermo-oil-controlled thermostats circulating through appropriate tubes. Temperature control can be carried out between −40 and +200 °C. They are relatively cheap but have the disadvantages of showing low heating and cooling rates, heat loss because of the tubing, and, sometimes, inconvenient handling of liquids in the laboratory. Cautions must be taken to ensure that the temperature set in the thermostat is the same as that in the reaction chamber because heat losses during circulation may be important, and thus tubing

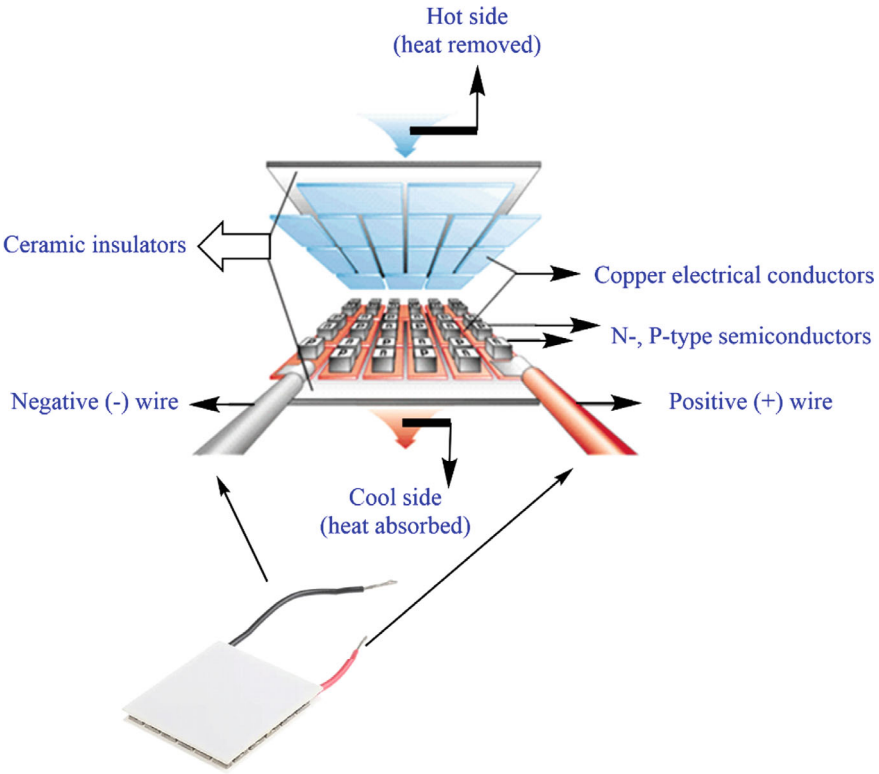


Fig. 2.6 Scheme of a Peltier device. In a typical application of the Peltier module, the object to be cooled is placed on the cold plate of the module and a heat sink is placed on the hot side. Because the hot and cool sides are close to each other, extra heat dissipaters (usually made of an Al alloy) and fans are employed

should be isolated to minimize heat losses in as much as possible during the transport of the thermostating fluid. Figure 2.7 illustrates the use of a water bath to control the temperature of a multi-cell holder attached to a diode-array spectrophotometer.

2.2.3 Medium Effects: Solvents, Acidity, and Ionic Strength

Control of the composition of reaction medium (mainly the nature and concentration of the solvent, acidity and ionic strength) is, together with temperature, of outmost importance for reactions taking place in solution. The type and composition of the solvent, and mainly its dielectric constant, are crucial because the reactants and other components need to be dissolved. Reacting species are usually present in very low molar concentrations compared to that of the solvent, and thus maintaining constant



Fig. 2.7 Detail of the thermostated 8-cell carrier of one of the diode-array spectrophotometers employed in our laboratory. Control of temperature is achieved by employing circulating water from a thermostat that warrants an accuracy in T of ± 0.1 °C. To minimize heat losses (and temperature drops), the polypropylene tubes are isolated with pipe insulation foam tubes. Note the hole on the left that allows the UV–Vis radiation (of selected wavelength) from the lamps to fall on the samples. The diode-array detector is located inside the device on the right of the cell holder

the solvent composition during the course of the reaction generally presents no problems and it is sufficient to prepare the reaction mixtures in the solvent of choice. Solvents can be classified in different ways according to their physical and chemical properties in molecular, ionic (molten salts or salt mixtures, ionic liquids) and metals. Further division based on physical and chemical properties leads to various categories including inert (unreactive, with low or zero dipole moments and low polarizability), inert-polarizable (aromatics, polyhalogenated hydrocarbons), protogenic (hydrogen-bonding proton donors), protophilic (hydrogen-bonding proton acceptors), amphioprotic (having both protogenic and protophilic properties), and dipolar-aprotic. Table 2.3 lists some examples of these classes of molecular solvents, where plain water has undergone a magnificent renaissance as solvent for organic reactions not only because it is an environmentally safe solvent but also due to its important capacity for hydrophobic hydration of apolar solutes. The customary classification of solvents according to their chemical constitution allows certain qualitative predictions that can be summarized in the rule of thumb "similar dissolves similar", and a proper choice of solvent based on its chemical reactivity helps to avoid undesirable reactions between solvent and reactants. The interested reader is referred to Chap. 5, where the effects of solvents on chemical reactivity are discussed.

Once an appropriate solvent has been chosen, cautions must be taken with its purity (to avoid side reactions), with the solubility limits of the reactants (to avoid precipitates and phase separations), and to ensure the physical/chemical stability of those solutions with time (to minimize reactant and solvent wastes and to allow repetition of experiments under the same conditions). In some instances, addition of small amounts of cosolvents, adjustments of temperature, acidity and ionic strength

Table 2.3 Some examples of solvents

Molecular solvent	Examples	Potential cosolvents
Inert	Aliphatic hydrocarbons, fluorocarbons	Benzene, diethy ether tetrachloromethane
Inert-polarizable	Benzene, tetrachloromethane, carbon disulfide,	Diethy ether, ethyl acetate, hydrocarbons
Protogenic	Trichloromethane	Ethanol, hydrocarbons
Protophilic	Tertiary amines	
Amphiprotic	Water, alcohols; ammonia	Acetone, alcohols, acetonitrine, dioxane
Dipolar-aprotic	Dimethylformamide, acetonitrile, dimethyl sulfoxide	Water, diethyl ether, benzene

may be necessary to increase the solubility of poorly solvent-soluble reactants. Thus, in general, the preparation of stock solutions of known concentration of the reactants and other components does not present serious difficulties.

Reactants, intermediates or products may show acid–base properties. Thus, it is important to know whether those substances are present in the reaction mixture as acids or bases and what is/are the reactive specie(s). To maintain the ratio of concentrations of conjugate acid and base pair during the course of the reaction, it is essential to maintain the pH of the reaction mixture constant. This is possible by continuous addition of an acidic or basic solution to the reaction mixture or, much more practical and simpler, by buffering. Continuous addition of an acid or base solution requires the use of concentrated solutions (to minimize changes in the volume of the reaction what would lead to changes in concentrations) to exactly balance the acid or base consumed. This approach is not commonly employed because of the inherent experimental difficulties that require its implementation.

Buffering is the most practical and widely employed approach to maintain a constant pH. Buffers can be defined as solutions which resist a change in pH when small amounts of acid or base are added; therefore, they should contain an acidic species to neutralize hydroxyl ions and a basic species to neutralize protons, that is, they are composed of conjugate acid–base pair. Table 2.4 shows some of the most common buffers employed in chemistry and biology. Buffers chosen to control the acidity in the course of chemical reactions need to meet some basic requirements:

- Their components must be soluble in the chosen solvent and must not react with the starting materials nor with the intermediates and products.
- The rates of exchange between the conjugated pairs (i.e., the rates of the reactions required to establish equilibrium between the acid–base conjugate pairs) is fast compared with the rate of the investigated reaction (this condition is fulfilled for the majority of buffers, except the carbonate buffers).
- The buffer solution used must show sufficient buffer capacity, meaning that their use is commonly restricted to the pH range where they show the best buffering properties (i.e., for $\mathrm{pH} = \mathrm{p}K_\mathrm{a} \pm 1$)

Table 2.4 Selected buffers and their pK_a values at 25 °C [11–13]

Trivial name	Buffer name	pK_a
Phosphate (pK_1)	–	2.15
Malate (pK_1)	–	3.40
Succinate (pK_1)	–	4.21
Citrate (pK_2)	–	4.76
Acetate	–	4.76
Pyridine	–	5.23
MES	2-(N-Morpholino)ethanesulfonic acid	6.10
ADA	N-2-Acetamidoiminodiacetic acid	6.59
EDPS (pK_1)	N,N-Bis(3-sulfopropyl) ethylenediamine	6.65
PIPES	Piperazine-N,N-bis(2-ethanesulfonic acid)	6.76
ACES	N-2-Acetamido-2-hydroxyethanesulfonic acid	6.78
BES	N,N-Bis-(2-hydroxyethyl)-2-aminoethanesulfonic acid	7.09
MOPS	3-(N-Morpholino) propanesulfonic acid	7.20
TES	2-[Tris(hydroxymethyl) methylamino]ethanesulfonic acid	7.40
HEPES	N-2-Hydroxyethylpiperazine-N-2-ethanesulfonic acid	7.48
TEA	Triethanolamine	7.76
POPSO	Piperazine-N,N-bis(2-hydroxypropanesulfonic acid)	7.85
TRIS	Tris(hydroxymethyl) aminomethane	8.06
TAPS	3-([Tris(hydroxymethyl)methyl] amino)propane sulfonic acid	8.40
PIBS	1,4-Bis(4-sulfobutyl)piperazine	8.60
CHES	Cyclohexylamino ethanesulfonic acid	9.55
CAPS	3-(Cyclohexylamino) propanesulfonic acid	10.40
Piperidine	–	11.12

- It is advisable that the concentrations of the buffer components should be at least 20-fold higher than the initial concentration of any of the reacting species to consider—in practice—that the change in the concentrations of the buffer components negligible.
- It is necessary to ensure that the presence of the buffer does not interfere with the detection mode (i.e., the analytical method) employed to measure the concentrations of the reacting species or products.

Once a suitable buffer has been chosen, it must be dissolved and titrated to the desired pH with the aid of a previously calibrated pH-meter. Stock solutions should be made with (preferably) deionized and distilled water and should be renewed frequently to prevent fungal or bacterial growth, especially in the pH 6–8 range. It may also be necessary to add chelating agents such as EDTA to ensure interferences from metals in the solution.

When analyses of the pH-dependence of the reaction rate are required, it is advantageous to keep the concentration of one of the buffer components constant and vary only the concentration of the second component. For example, in the preparation of citrate or acetate buffers for kinetic measurements, it is recommended to keep the concentration of citric or acetic acids constant and vary that of sodium citrate or acetate, followed by keeping the concentration of the salts constant and varying those of the acid.

Although the use of simple buffers (citric, acetate, phosphate) is common practice, there may be situations where the use of buffers that span their pH range over more than 3 pH units is desirable. In these cases, mixtures of buffers to cover the pH range of interest should suffice as the pH passes the p*Ka* of one buffer, and their effect is negligible and therefore has no further action, but the pH is still regulated by other buffers present in the mixture. Examples of composite buffers that have been widely employed in kinetic studies are the Britton–Robinson (BR, pH range 2–12, prepared with a mixture of boric, phosphoric and acetic acids) or McIlvaine (pH range 2–8, prepared with citric and sodium dihydrogen phosphate). A detailed description of buffer mixtures which provide a wide range of buffering capacity with constant ionic strength is available [11].

The reader must be aware that the nature of the solvent has an impact on the pKa values. For example, the pKa of aniline in plain water is pKa $= 4.7$, but in methanol and ethanol are of pKa $= 6.2$ and pKa $= 5.7$, respectively. So, extra cautions need to be taken when mixtures of solvents are employed to carry out the chemical reactions.

Another parameter that affects the rate of the reaction is the ionic strength, I, an effect usually called "*kinetic electrolyte effect*" and that mainly affects when ions or ionizable compounds are involved and their concentration changes in the course of the reaction, and to a lesser degree when the reactants are neutral. If the latter case, it is usually sufficient to add an excess of an inert electrolyte such as sodium perchlorate.

The choice of the electrolyte to control the ionic strength is not always straightforward, as its addition to the reaction mixture may induce sometimes precipitation of the reactants, and the following points need to be taken into consideration:

- The electrolyte must be soluble and inert.
- 1:1 electrolytes should be the first choice as the use of highly charged electrolytes may lead to a poorer control of the ionic strength because of weighing or volumetric errors of small amounts of electrolyte.
- Cations and anions should have poor complexing abilities and should have low charge-to-radii values to avoid creation of large local electric fields that may eventually bind cations and/or affect the ionic atmosphere or nearby ions. For example, perchlorate ions are poor complexing agents where the negative charge is dispersed over the Cl and O atoms, resulting in an extremely high solubility in both aqueous and non-aqueous solvents.
- Electrolytes with cations and anions of similar mobilities are preferred as they will reduce liquid-junction potentials in glass electrodes when measuring pH.

- Cautions must be taken to avoid precipitation of the salt bridges of glass electrodes; for example, the use of $NaClO_4$ may lead to precipitation of the $KClO_4$ employed in the salt bridge.

With these considerations in mind, experience dictates that the most appropriate electrolytes for maintenance of the ionic strength are $NaClO_4$ and in less degree NaCl and KCl. When ions or ionizable compounds are involved, as for example in buffered solutions, either a massive concentration of a neutral salt is added or reaction mixture or the ionic strength is calculated by using Eq. 2.2,

$$I = \frac{1}{2}\sum_i C_i z_i^2, \tag{2.2}$$

where C is the concentration of all ionic species i with charge z_i, and a sufficient amount of neutral salt is added to maintain I constant. Note that while the ionic strength I of solutions containing 1:1 salts equals numerically with their concentration C, I becomes larger than C as the absolute values of the ionic charges increases. For example, the ionic strength of a 1 M NaCl solution is $I = 1$ M, and the ionic strength of a 1 M $Al_2(SO_4)_3$ would be $I = 15$ M. It follows that Eq. 2.2 is only valid for low to moderate ionic strengths; meanwhile more sophisticated treatments are needed for medium or high ionic strengths.

For a bimolecular reaction of the type $A^{z_A} + B^{z_B} \rightarrow$ Products the variation in the rate constant with the ionic strength can be expressed in terms of the Debye–Hückel limiting law, Eq. 2.3. Thus, for neutral reactants, the effect of the ionic strength is negligible; meanwhile when the reactants have the same charge, increases in the reaction rate will be observed upon increasing the ionic strength. Conversely, the reaction rate decreases if the ions involved are of opposite charges.

$$\log k_{\text{obs}} = \log k_0 + z_A z_B I^{\frac{1}{2}}. \tag{2.3}$$

2.3 Chemical and Physical Methods for Monitoring Chemical Reactions

There are many ways to monitor the progress of a reaction in attempting to determine the mechanism of the reaction. For convenience, we can broadly classify them in two groups: chemical and physical methods

2.3.1 Chemical Methods

Chemical methods are applicable to homogeneous reactions in gas or liquid phase and are discontinuous, meaning that concentrations in the reaction mixture are not measured continuously but only at known time intervals. The reaction mixture is sampled at defined time intervals; the samples are then manipulated and analyzed by an off-line method which may involve titration procedures, spectroscopic or chromatographic techniques. More advanced techniques include IR, NMR, EPR, spectroscopies, mass spectrometry and reflectance measurements.

In a typical experiment, the reaction mixture is placed in one or more sealed vials (thermostated at the desired temperature) and the reaction is, somehow, quickly stopped. This can be done in different ways, for instance, by rapidly placing the vial in a cold bath to stop the reaction (sudden decrease in the reaction rate) or by quenching the reaction by employing a chemical derivatization method.

In the latter case, an aliquot of the reaction mixture is added to vials containing a solution of a suitable quenching reagent that reacts rapidly with one of the reactants to form a stable compound, and once the reaction is over, the resulting samples are analyzed by employing any suitable analytical technique. An illustrative example is shown in Fig. 2.5). Alternatively, one can add, at defined times, aliquots of the reaction mixture to previously prepared vials containing the quenching reactant. The first approach has been exploited to monitor the formation of reaction products with time by HPLC and the latter to monitor reactions in opaque systems such as emulsions (Fig. 2.8).

The main difficulty of the chemical method is, most probably, associated to its ability to stop the reaction of interest so that precise concentration–time data can be collected. Hence, the method only applies to reactions that are not too fast because of the time spent to quench the reaction or to cool down the reaction mixture, which should be as low as possible. These difficulties and limitations make that physical methods are the preferred ones because of their versatility.

2.3.2 Physical Methods

Physical methods are based on the collection of values of a physical property (related to the concentration of the species that is monitored) with time. Most of the techniques commonly employed are *batch techniques*, in which reaction is initiated at a single chosen time and concentrations are then followed as a function of time after initiation. For the sake of simplicity, c*ontinuous techniques*, in which reaction is continuously initiated and the time dependence of the reaction mixture composition is inferred from, for example, the concentrations in different regions of the reaction vessel, will not be covered here, and the reader is referred to specialized books and reviews.

The acquisition of values of the physical property can be done in systems where one component can be separated from the reaction mixture (for instance, in reactions

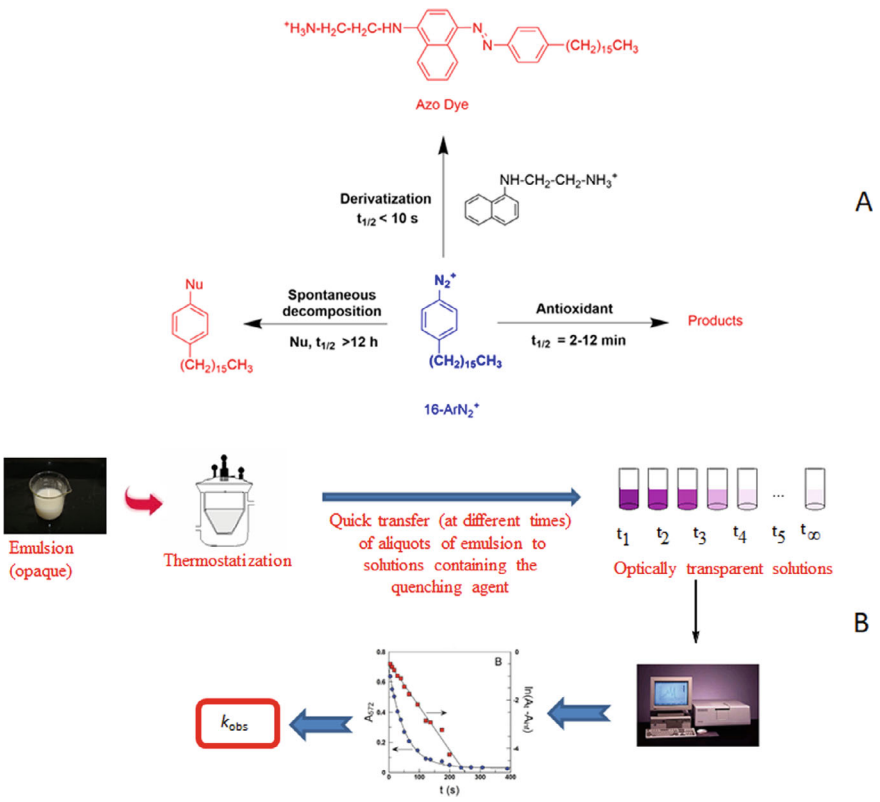


Fig. 2.8 Illustrative example of a chemical (derivatization) method employed to monitor reactions in opaque systems such as emulsions [14]. The reaction of interest is that between 16-ArN_2^+ and an antioxidant (AOs, Figure A), whose half-life ranges 2–12 min. To quench the reaction, the quenching agent N-(1-naphthyl)ethylenediamine (NED) was employed. The half-life of the quenching reaction is <10 s, so that the reaction with the AO is effectively quenched. The coupling agent was dissolved in a series of BuOH/EtOH mixtures, all containing the same concentration of NED so that the final concentration of NED is much higher than that of 16-ArN_2^+ and mixture is optically transparent. The absorbance of the azo dye (which is proportional to the amount of unreacted 16-ArN_2^+) can be measured with the aid of a spectrophotometer. The observed rate constant for the reaction between 16-ArN_2^+ and AO can be determined from the variations in the absorbance of the azo dye with time as illustrated in Figure B.

in solution where gases are produced, so that the pressure of the evolved gas can be measured (manometric method), by measuring physical properties of the reaction mixture (density, viscosity, thermal conductivity, dilatometry, etc.,) or by determining a physical property of one or more of the components of the reaction mixture (spectroscopic, electrochemical, optical rotation, etc. methods). There are also a number of other methods where one or more of the components of the reaction mixture are separately assayed, and these include chromatographic, mass spectrometry, and hybridization techniques).

In all cases, the choice of one or other method to monitor the reaction is conditioned by the lifetime of the reaction so that researchers must make preliminary choices of the required instruments to convert information of the physical or chemical characteristics of the reaction components to information that can be manipulated and interpreted.

Thus, one can distinguish at least three necessary stages to undertake any kinetic study: (1) the choice of an appropriate method to monitor de reaction, (2) the choice of the physical property characteristic of the reactants/products, to be used, and (3) a system to retrieve the desired information, that can also be employed for data storage and further data manipulation. For example, when irradiating a sample with UV–vis light, the resulting information is contained in the phenomena that results from the interaction of the electromagnetic radiation with a given compound (for instance, an absorption spectrum). The intensities of the radiation (representing a non-electrical information) are transformed, by means of transducers, to an electrical current whose value is proportional to the intensity of the radiation. Currents are measured before and after the radiation reaches the sample and are converted to voltage, whose value is proportional to the current intensity.

"Slow" or "moderately rapid" reactions are usually initiated by mixing the reactants together by hand or by employing any suitable mechanical device. The progress of the reaction is monitored by employing appropriate instrumental methods (see Sect. 2.4). Modern instrumental analytical methods make use of analysis based on chemical, physical, and physicochemical changes that are exhibited by substances undergoing chemical reactions. In principle, any suitable property of the system, proportional to the concentration of any reactant/product, can be employed to monitor the reaction. The instrument converts information about the physical/chemical characteristics of the reactants to information that can be manipulated by means of signal processors, detectors, transducers and other electronic devices that transform non-electrical information to electric one to finally show it in a non-electrical way that can be manipulated by humans as illustrated in Fig. 2.9.

The way the reactants are mixed to initiate the reaction is critical when undertaking a kinetic study. For very "slow" or moderately "slow" reactions, reactants can be mixed manually, meaning that some time is elapsed between the initiation of the reaction and when data collection starts. Flow techniques, based on rapid mixing of reagents, are commonly employed to study reactions occurring on timescales down to 10^{-3} s, for example fast enzyme and other catalytic reactions. These methods are relatively "slow" compared with those necessary for "fast" reactions. Relaxation methods are ~1000-fold faster than flow methods and are commonly employed to monitor reactions where short-term disturbances of equilibria are induced and where no turn-over can be observed, for example, ligand binding, isomerizations, induced or spontaneous conformational changes, etc. Radiation methods, including flash photolysis, pico- and femtosecond spectroscopy, employ light impulses and are even more rapid. They allow monitoring reactions occurring in photosynthesis,

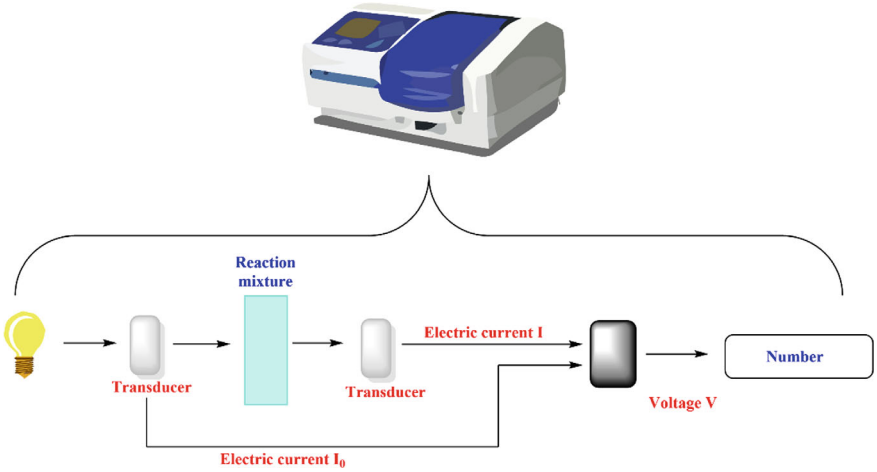


Fig. 2.9 Typical diagram showing some of the various types of information and their conversion in a UV–vis spectrophotometer. The information begins in the sample as the concentration of, for instance, a reactant. The information is teased from the sample by irradiating it with an electromagnetic radiation that interacts with the sample to produce absorption in a region of the spectrum. The radiation intensity is transformed in an electrical signal before and after reaching the sample and both values to finally arriving at a number related to the intensity. The mathematical relationship between the electrical output and the input radiant power is called the *transfer function* of the transducer

vision, binding of O_2 and CO to hemoglobin and myoglobin, etc. Flashes are generated by discharge flash lamps allowing resolutions of a few milliseconds; meanwhile pulse lasers are generated by discharging condensers making the nano- and pico-second range available [15].

Though the detection of the change depends on the method and the required time resolution, similar detection techniques can be applied, alone or in combination, with the methods mentioned above leading to particular instrumental methodologies. For example, absorption and emission photometry are routine methods widely used universally, especially because they are highly affordable by most laboratories, but their sensitivity is relatively low. For especial applications, circular dicroism (CD), nuclear magnetic and electron spin resonances (NMR and EPR, respectively), mass spectrometers, electrochemical detectors, and specific electrodes are combined with fast techniques. Nowadays all these methods and detection modes are computer controlled, and generally data acquisition, registration, recording, and subsequent data evaluation or treatment are not limiting.

2.4 Methods to Monitor "Slow" Reactions: Manual Methods

Manual methods refer to those experimental arrangements where reactants are, somehow, mixed manually or with the aid of a suitable mechanical device. Because mixing times are in the seconds range, these methods can only be employed to monitor reactions that are "slow" or very slow. Once the reaction is initiated, the progress of the reaction can be monitored by determining changes in the concentration, in the mass of reactants or products, or by measuring changes in the volume or pressure of the reaction mixture.

Modern manual methods use four major instrumental techniques: spectroscopic, chromatographic, electrochemical, and mass spectrometry, although there is a wide range of other techniques available. Microwave, IR, Raman, visible, or ultraviolet radiation is all extensively used for identification and quantification of different species because spectral transitions are fingerprints of molecules. In the gas phase, spectra show sharp lines allowing easy identification of the different species. In solution, the spectra are more complex but still allow, very frequently, identification and quantification by measuring changes in the frequency, intensity, or line widths of the spectral lines.

Chromatography allows separation of the components of the reaction mixture and various modes of detection are available, including spectroscopic, electrochemical (liquid phase), and mass-to-charge ratio (gas phase) techniques. The main virtue of chromatographic techniques is probably their versatility, accuracy, speed of analysis, and versatility to handle complex mixtures, allowing analysis times in the order of a few minutes. Only small samples are required to carry out a complete chromatographic analyses and very small amounts of reactants or products, $<10^{-6}$ mol, can be detected depending on the mode of detection. However, because of the relatively slow flow of the mobile phase, the method is unsuitable to monitor reactions with lifetimes shorter than a few minutes. In addition, when employed for kinetic studies, extra cautions need to be taken to make sure that the components of the chromatograph (mainly the mobile and stationary phases and the tubing) do not interfere in the reaction of interest. Control of the reaction temperature through the chromatographic system may also be a problem.

Complex gas reactions such as combustion, explosion, ion–ion and ion–molecule reactions, isotopic analyses, and kinetic isotope are frequently studied by employing mass spectrometers. In mass spectrometry techniques, the samples are vaporized and bombarded with electrons so that molecules are ionized. Detectors measure the mass/charge ratio from which the molecular weight is determined, and the molecule identified and, eventually, quantified. Only very small samples are required, and detection limits are below 10^{-12} M. For complex mixtures, MS are linked to chromatography that separates first the components.

2.4.1 Case Study: UV–Vis Spectroscopy as a Tool for Investigating Chemical Reactions

Spectroscopic methods are widely used in almost all modern laboratories as a routine method of quantitative analysis and, in some instances, can be employed to investigate the kinetics and mechanisms of chemical reactions. Measurements in the microwave region (NMR, EPR, etc.) have been employed to monitor fast reactions, but the cost of the instruments, the relatively high concentrations of reactants required and the need of using deuterated solvents (usually quite expensive) make them to be restricted to special situations. Measurements in the infrared region require special procedures and instrumentation because solutions often contain water, which complicates IR measurements. Fluorescence emission—though very sensitive—is restricted to a small group of molecules. Thus, the discussion below will be mainly focused on UV–Vis spectroscopic methods even though it may also be applicable to other spectroscopic methods [16].

UV–vis spectroscopy has become the most popular methodology employed to investigate the kinetics of reactions in solution. The method is unrivaled primarily because of its simplicity, accuracy, rapidity, and cost-effectiveness. Quantitative measurements covering concentration as low as 10^{-6} M can be performed, and the spectrophotometric cells can be easily thermostated for an accurate control of temperature in the course of the reaction. In addition, modern instruments containing diode-array detectors allow simultaneous measurements of absorbance at multiple wavelengths, making possible to monitor simultaneously the disappearance and/or formation of various components of the reaction mixture. These instrumental settings are particularly useful to study consecutive and competitive reactions where intermediates are formed and subsequently decomposed. In addition, the advances in the electronics of the instruments make that modern UV–vis spectrophotometers can be equipped with multi-cell holders, allowing the simultaneous monitorization of several (moderately slow) reactions.

The basic set-up of a UV–vis spectrometry system is a light source (deuterium and tungsten lamps), a sample, and a spectrometer or spectral analyzer. Light from the light source passes through the sample, and the different wavelengths are absorbed to different extents in a sample. The spectrometer then analyzes the intensity after passing through the sample light (I) and compares it to the light intensity measured with no sample present (I_0) for each wavelength λ, Fig. 2.9. The extent of absorption is given by the Lambert–Beer law: $A = \varepsilon d[c]$, where A is the absorbance ($A = -\log I/I_0$), ε is the molar absorption coefficient, d is the optical path length, and $[c]$ is the concentration of the reactant or product in the sample. Figure 2.10 illustrates the typical experimental sequence followed when UV–vis spectroscopy is employed to monitor a chemical reaction.

Before deciding for the UV–Vis spectroscopic method, several experimental aspects need be considered:

- At least one of the reactants or products must have an absorption band whose intensity changes as the reaction proceeds.

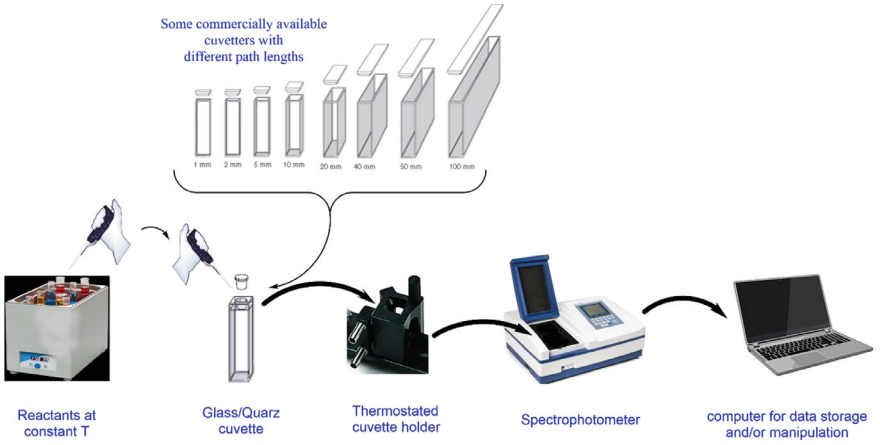


Fig. 2.10 Illustrative example of a typical kinetic experiment aimed to determine the observed rate constant of a chemical reaction. The spectrophotometer cuvette is chosen according to the desired path length (the most common path length is 10 mm) and can be either rectangular. Once the reactant solutions are thermostated at the desired temperature T, they are mixed in the spectrophotometer cuvette with the aid of pipettes to initiate the reaction. The cuvette is rapidly shaken to homogenize the mixture and placed in the cuvette holder, which is also thermostated at the temperature T with the aid of a Peltier system or from water proceeding from an auxiliary heater/cooler. The cuvette holder is located inside the spectrophotometer. The data acquisition process is initiated (this is usually computer controlled), and once the reaction is over, the pairs of data (Absorbance, time) are fitted to the desired kinetic equation to finally obtain the observed rate constant k_{obs}.

- Effective mixing of the reagents and homogeneity of the reaction mixture must be ensured.
- It is convenient (although not essential) that the spectroscopic signal (absorbance due to the monitored component) is to be linearly proportional to its concentration in the range of concentrations to be used (see Chap. 3). This implies that the signal *vs.* concentration relationship (i.e., Lambert–Beer law) must be previously investigated to ensure the linearity and to establish the range of concentrations where such lineal dependence holds. In these cases, the absorbance values can be employed directly to determine the rate constants (Chap. 3), otherwise some previous mathematical treatment is required before determining rate constants to transform the value of the absorbance into concentrations.
- The reaction medium must be sufficiently transparent to the radiation in the region of absorbance of the reactants or products. Sometimes it is possible to correct for a background absorption of the solvent by using a blank cell or dual-beam instruments, but this generally worsens the signal-to-noise ratio.
- The extent of overlap of the chosen signal with other signals from other species (including reactants, intermediates or products) should be minimal, and although this is not a strictly an essential condition, it certainly facilitates the experimental work and further interpretation of results.

For a practical use of UV–vis spectroscopy to monitor reactions, a number of instrumental settings other than the proper choice of the size of the cuvette (i.e., path length) need to be considered. Among them, the source of radiation (lamp selection), the wavelength, slit width and a proper selection of the data interval and acquisition times are of special importance. As in any detection method, the key to success in measurement is achieving high sensitivity through a high signal-to-noise (S/N) ratio. The S/N ratio of a specific measurement greatly depends on the excitation and emission characteristics of the absorbing/emitting species and on the wavelength and slit width settings used.

2.4.1.1 Lamp Selection

The lamp selection (tungsten, deuterium, or xenon) depends on the technique (absorption or emission) and on the wavelength chosen to monitor the reaction. Deuterium lamps are employed for radiations in the UV range; meanwhile tungsten lamps are employed for visible wavelengths and xenon lamps are employed in fluorescence measurements.

2.4.1.2 Some Tips to Choose an Appropriate Wavelength to Monitor the Reaction

The course of a reaction can be monitored either by recording the whole absorption spectra or by measuring changes in the absorbance at a given wavelength. For preliminary studies, recording the whole spectra (i.e., obtaining absorbance-wavelength plots) is highly advisable to identify the reacting species, intermediates and products. Recording the whole spectra facilitates to choose wavelengths at which future measurements can be carried out in subsequent systematic studies. Theoretically, it is possible to determine the concentrations of n absorbing components by measuring the absorbance and molar absorptivities at n wavelengths. However, the accuracy of the determined values is usually so small that such an analysis is frequently meaningless.

For kinetic purposes, optimal wavelengths to monitor changes in the concentration of only one component (reactant or product) of the reaction mixture are those where the largest changes in absorbance or fluorescence are achieved, that is, the largest changes between the intensity of the signal at the beginning of the reaction ($t = 0$) and when the reaction is over. For example, in Fig. 2.2, optimal wavelengths for monitoring the loss of the starting material and the formation of the reaction products are those at 260 and 370 nm, respectively.

Ideally, wavelengths should be chosen so that the signal comes from only one absorbing/emitting reactant, but this is not usually the case. When the absorption spectrum yields two absorption bands which either both increase or decrease and the ratio of the absorbances remains constant at various wavelengths, the choice of the wavelength is not essential and in general it is possible to follow the change of the absorbance with time at any wavelength where its value changes.

A somewhat more complex situation can be found if the ratios of absorbances are not constant, i.e., when several absorption maxima change with time and the ratios between absorbances also change. Maxima may increase or decrease, and even some maxima may increase while other decrease, and as long as the absorption bands do not overlap, it is possible to make use of the change in the maxima in a similar way as in the case where only one absorbing species is present. If the bands do not completely separate, i.e., if the absorbances of the individual components overlap, then frequently it is necessary to measure the changes in absorbance at two or more wavelengths, although sometimes—depending on the particular case—simplifications can be done by using, for instance, the first or second derivative of the absorbance–wavelength curve.

An analysis of the changes in the spectra with time allows the researcher to distinguish the role of the absorbing species in the reaction: the absorbance of reactants decrease meanwhile that of products increase (Fig. 2.2). The presence of intermediates can be detected from absorption bands which first increase and later decrease. Formation of intermediates can also be inferred from the shape of the absorption spectra. In reactions that are simple, one-step, processes, the absorption spectra of reactants and products may intersect in one or more wavelengths. At such wavelengths, the absorbance does not change with time and such points are called *isobestic* (Fig. 2.2). The existence of one or more isobestic points suggests that the reaction studied is neither complicated by the formation and accumulation of relatively stable intermediates nor by side reactions.

When the reaction takes place in solution, the solvent or solution components (e.g., buffers or non-reacting components) other than the reactants or products may absorb at particular wavelengths. In general, such background absorbance should be subtracted from the measured absorbance (the use of double-beam instruments can help to compensate such effect).

2.4.1.3 Slit Width

The selection of the slit with of the spectrophotometer is also an important parameter. The slits of a spectrophotometer are rectangular apertures through with light enters into, and exits from, the monochromator and its main purpose is to control the spectral resolution of the monochromator, i.e., its ability to differentiate between close wavelengths. In standard monochromators, the entrance and exit slits have equal width; the wider the slit width, the larger interval of wavelengths that passes through the monochromator. Increases in the slit width lead to increases in two independent optical effects: (1) the total slit area increases, increasing the spatial fraction of the light source intensity that enters the monochromator, and (2) increases the spectral fraction that enters the monochromator, that is, more photons of different wavelengths get through the monochromator. Depending on the instrument, this width is fixed or can be controlled by the operator.

Overall, the optimum slit width depends on the purpose of the measurement. To accurate identify compounds through their spectrum, slits should be as low as

possible to avoid polichromaticity deviations from the Lambert–Beer law. In general, the wider the slit is, the higher the noise is and the lower the resolution is. Restrictions in the slit width can be relaxed when the main purpose of the measurement is quantitative analysis of the concentration of the absorbing components, because in the vast majority of practical cases, quantitative measurements proceed from previously prepared calibration plots from commercial samples rather than depending on absolute absorbance measurements. For kinetic purposes, slit widths of 2 nm are frequently enough, but spectrophotometers bearing the possibility of changing slit widths from 0.1 to 5 nm are available.

2.4.1.4 Data Acquisition Time and Time Intervals

The selection of the time interval and total acquisition time are also important parameters. The time interval refers to the elapsed time between two consecutive measurements. This time should be short-enough to allow the operator to finally obtain at least 20–30 data points so that roughly half of them are collected within the first half-life of the reaction. The problem of selecting optimal time intervals is the establishment of a time scale that often involves a knowledge of the reaction time. Most of the data points need to be collected in the time intervals where the largest changes in concentration are detected, that is, during the first stages of the reaction.

The total acquisition time should be large enough to allow monitoring the reaction for at least 3–4 half-lives. For example, if N is the total number of (absorbance, time) pairs of data points to be collected, then a convenient distribution of the data points could be in such a way that approximately half of total data points ($N/2$) are collected during the first half-life of the reaction, one fourth ($N/4$) during the second half-life and one fifth ($N/5$) during the third one, so that approximately 95% of data are collected during the first three half-lives of the reaction.

For the handling and evaluation of the data, it is frequently necessary to know the initial and/or final concentrations (or the value of the proportional physical parameter) of the reacting species. The initial concentration of the reactants is usually known, and thus there should not be problems when the kinetics is monitored by direct determination of the concentration in individual samples. When physical methods are employed, the value of the initial measured quantity at time zero must be obtained either from an independent experiment (for example, in the absence of one of the reactants or under conditions where the reaction does not take place) or by extrapolation to zero time. In the latter case, it is necessary to ensure rapid and complete mixing of the reaction components and measure the physical quantity after several precisely known time intervals in the initial stages of the reaction.

2.5 Methods to Monitor "Fast" Reactions

2.5.1 Continuous-Flow Method

This was probably the first technique employed for monitoring fast reactions. It is based on the rapid mixing of two separate solutions to finally reproduce the desired reaction conditions. Typically, all components of the reaction mixture except one (the limiting reactant) are place in one chamber (e.g., a syringe); meanwhile the solution containing the limiting reagent is placed in a second syringe. The pistons of both syringes are pushed by a driving device at constant velocity and the solutions enter into a mixing chamber where the reaction starts. The reacting solution flows continuously through an observation tube where the detectors are located. In optical devices, the tube serves as a cuvette and must be constructed appropriately from glass, quartz glass, or other suitable material. In electrochemical devices, the tube must contain the electrochemical cell. The critical point in these instruments is the mixing chamber that needs to be designed to ensure a complete, homogeneous, mixing within fractions of milliseconds, requiring an optimal—almost perfect—construction (Fig. 2.11).

The distance between the starting point of the reaction in the mixing chamber and the various observation points in the tube is directly proportional to the time progress of the reaction, so that the age of the reaction can be defined in terms of the ratio between the flow distance and the flow rate, Eq. 2.4.

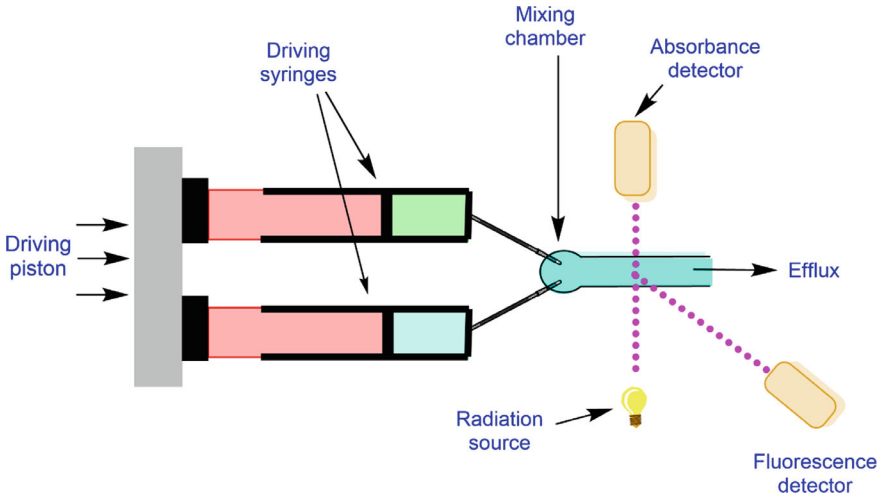


Fig. 2.11 Schematic representation of a continuous-flow instrument. For simplicity, the connections of the detectors with intermediate electronic devices and with a computer are not indicated. All components of the apparatus coming into contact with solutions must be carefully maintained at the desired temperature with the aid of heating/cooling devices (e.g., thermostat or Peltier) to avoid streaks in the flow system

$$\text{Age(s)} = \frac{\text{Flow distance(cm)}}{\text{Flow}\left(\text{cm s}^{-1}\right)}. \tag{2.4}$$

At fixed flow distances and flow rates, the age of the reaction is always constant and the reaction can be monitored. To characterize the time progress of the reaction, several points at different distances from the mixing chamber are obtained, and fast registration of the data points is not required. One might think that increasing the flow rate would make flow methods faster. This is not true as homogeneous mixing within milliseconds requires optimal design and construction of mixing chambers, otherwise nozzles may produce strong turbulences, insufficient mixing, friction effects, and heating of the reactant solution.

An important advantage of flow methods is the reduction of the time between the starting of the reaction and the start of detection (dead time). The dead time determines the time resolution, and it is the most critical factor for instruments measuring fast reactions and values as small as 0.2 ms can be reached nowadays. The main drawback of the experimental set-up is that it requires very large amounts of reaction mixtures and thus large amounts of chemical wastes. For example, for a typical flow rate of 10 m s^{-1}, observing a distinct point for 1 s in a 5 mm diameter tube requires ~200 ml generating such amount of waste (volume in 1 s $= \pi r^2 h = \pi * 0.25^2 * 1000$ mm s^{-1}). Incorporation of computers and improvement in the sensitivity of detectors and reduction of tube diameters allowed to decrease substantially the waste volumes down to 10 ml for one measurement but still produce large amounts of waste.

2.5.2 Stopped-Flow Methods

A similar arrangement than that employed in the continuous-flow methods is employed in stopped-flow apparatus, Fig. 2.12. In this case, the pistons to empty the syringes containing the reactants are pushed rapidly by one shot with a driven block either hydraulically or by gas pressure. Frequently, the two reaction syringes have identical volumes so that the reactants in each syringe are dissolved in a 1:1 ratio after mixing. Some stopped-flow apparatuses are also built-up with two lateral reservoirs to allow rapid refilling of the reaction syringes for repeated measurements and to minimize the presence of air bubbles in the system. The observation cell is placed behind the mixing chamber followed by a stop syringe. Absorption (UV–Vis) detectors are placed perpendicular to the flow direction to allow free passage of the optical beam meanwhile in emission detectors (fluorescence) the emitted beam is detected perpendicular to the excitation source in the third space dimension. The UV–Vis detectors should be more sensitive than regular bench spectrophotometers to register small changes in absorbance within a short period of time. The radiation source should provide high light intensity and larger slits than in conventional spectrophotometers which are frequently employed to increase sensitivity. Double-beam devices, where the reactant solutions pass through reference cuvettes, each half of

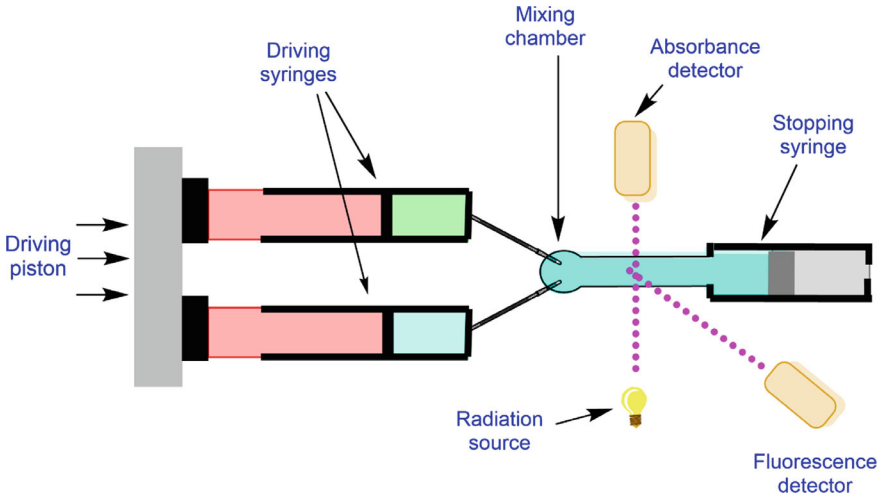


Fig. 2.12 General design of a stopped-flow instrument with absorption and fluorescence detection. For simplicity, the connections of the detectors with intermediate electronic devices and with a computer are not indicated. All components of the apparatus coming into contact with solutions must be carefully maintained at the desired temperature with the aid of heating/cooling devices (e.g., thermostat or Peltier) to avoid streaks in the flow system [17, 18]

the thickness of the sample cuvette, before entering the mixing chamber, so that the difference signal between the reference (before mixing) and the reaction mixture is recorded.

The basic stopped-flow arrangement described above can be combined with various other settings for specific applications. For example, NMR-stopped-flow units are frequently employed to monitor conformational changes but possess poorer time resolution and require higher reactant concentrations. Light-scattering stopped-flow apparatus is employed using laser radiation and X-ray radiation (from a synchrotron) allowing a rapid mix of reactants to monitor fast enzymatic reactions. The stopped-flow arrangements can be even combined with calorimeters with thermo-elements or thermistors serving as sensors to monitor enzyme reactions and with selective glass electrodes (e.g., pH stopped-flow apparatus) to monitor changes in proton concentrations.

2.5.3 Relaxation Methods

When a chemical system comes to equilibrium such as that in Eq. 2.5, we can easily determine the ratio of the rate constants for the forward and reverse reactions, which equals numerically to the equilibrium constant, but not the individual values. To measure them, the equilibrium should be perturbed in some way and the rates of relaxation to a new equilibrium position provide information about the forward and

reverse rates of the reaction. These methods are, therefore, in principle applicable to chemical systems where an overall net change in the concentration does not occur. It is worth noting that, since systems at equilibrium are already well-mixed, relaxation methods overcome the mixing problems associated with the flow methods (Sect. 2.5.2).

$$A \underset{k_r}{\overset{k_f}{\rightleftharpoons}} B. \tag{2.5}$$

There are two main ways of perturbing the equilibrium, by suddenly changing the temperature (temperature jump, T-jump) or by changing the pressure (pressure jump, *P*-jump). Other concentration-jump methods can be imagined, for example, methods where sudden changes in acidity—pH-jump—have also been described, but, in general, the equilibrium constants are sufficiently sensitive to temperature or pressure changes that make them unnecessary. We will describe briefly the T- and P-jump methods as they are the most common. Mathematical treatment of the corresponding kinetic equations will be described in Chap. 4.

2.5.3.1 Temperature Jump

This is, probably, the best-known relaxation method, which has been available for several decades. The perturbation consists of a sudden increase in the temperature of the reaction mixture by Joule heating resulting from the resistance to an electrical current flowing through it, Fig. 2.13. Fast temperature changes are usually achieved after discharging high-voltage capacitators, typically producing jumps of several degrees in as little as 1–2 μs. Modern laser devices can be employed to produce large increases in temperature in much shorter times (several *ns*), thus expanding the range of reactions that can be studied with these methods.

Once perturbed, the temperature of the heated volume decays to ambient temperature and the rates of relaxation are measured by means of any suitable detector.

2.5.3.2 Pressure Jump

This method, which has also been available for several decades, the equilibrium is altered by applying a rapid change in pressure. Pressure increases of up to 200 atmospheres can be achieved in small samples (<50 μL) in a few microseconds. Such changes are usually obtained by employing piezoelectric crystals. The method has several distinct advantages over the *T*-jump method:

- Reactions can be followed on long time scales when required because the pressure remains after the jump.

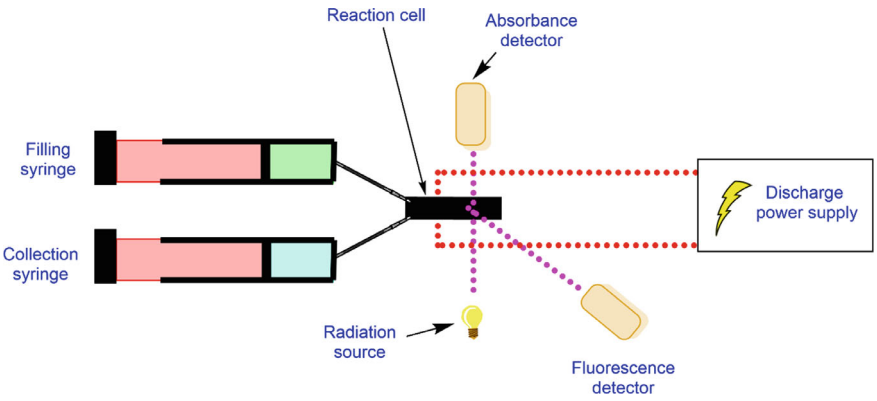


Fig. 2.13 General layout of a T-jump apparatus. Reactants and products are in equilibrium in the reaction cell, which is quickly perturbed by a rapid change in temperature. There is a new equilibrium constant at the higher temperature, but the initial concentrations are balanced for the lower temperature. The system therefore relaxes and the reaction proceeds until the concentrations have reached their new equilibrium values. The reaction is followed spectrophotometrically by UV/Vis (absorbance and/or fluorescence) using fast response detectors and signal processing so that very fast processes can be characterized by changes in the optical signal

- Samples do not require electrolytes for efficient pressure jump (as required in *T*-jump experiments for efficient Joule heating).
- Jumps can be recorded in both directions.
- Collection of data with very good signal-to-noise ratio is possible because jumps can be repeated several times (even though the pressure-induced perturbation is smaller than those produced in *T*-jumps.

2.5.3.3 pH-Jump

Acid–base equilibria can be studied by rapidly changing the acidity of the reaction medium. To induce pH changes, chemicals that rapidly transfer protons to the medium or capture them are employed. Typically, chromophores with different ionization constants in the ground and excited states are added to perturb the acid–base equilibrium. The chromophores are perturbed with the aid of pulsed lasers, so that light absorption will rapidly excite the chromophore which then transfers a proton to the solvent, increasing rapidly the acidity of the medium. A typical chromophore widely employed in aqueous solutions is 2-naphthol-6-sulfonate, whose pK_a in the ground state is 9.1 meanwhile in the excited state is 2, and whose pseudo-first-order rate constant for transferring the proton to water is ca 10^9 s^{-1}. Changes in acidity from pH = 7 to 4 have been achieved in <10 ns [19].

2.5.4 *Flash Photolysis*

Flash photolysis is a technique that exploits the use of short pulses of light to generate temporary (transient) species in a sample, e.g., a radical intermediate or a short-lived excited state. As the sample relaxes back to the ground state or the final reaction products, the changes occurring are recorded as a function of time using a spectroscopic technique, Fig. 2.14. It is the technique of choice to study reaction kinetics, the lifetimes of non-emissive triplet states, and to detect elusive intermediates in light-triggered processes. Therefore it is very useful in fundamental photochemistry, materials science, and photobiology [20].

Transient absorption can be performed in a wide range of timescales, from picoseconds to seconds. Experiments in the nanosecond to seconds range have many applications involving detection of free radicals, excited states, and charge transfer reaction intermediates. A flash photolysis spectrometer typically monitors the change in absorption as a function of time, utilizing a second light source (probe) besides the excitation (pump) pulse going through the sample. This technique is also known as transient absorption (TA).

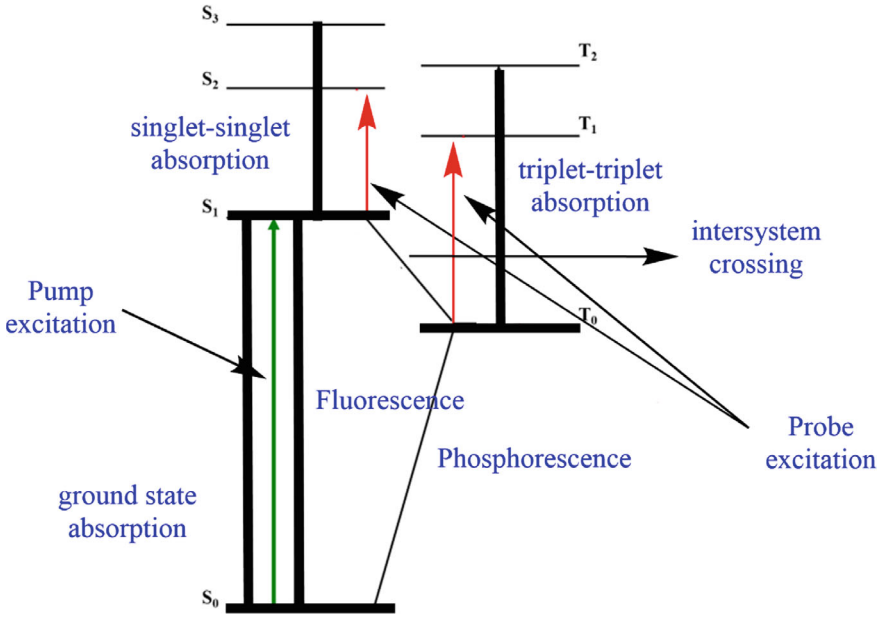


Fig. 2.14 Energy diagram (schematic) for transient absorption (TA) of a triplet state with pump laser excitation to the state S1, which then populates the triplet T1 by intersystem crossing. The change in T1 absorbance as a function of time is recorded by monitoring the probe beam. As the excited state population decays from the T1 to the ground state S_0, the absorbance will decrease. The lifetime of the triplet state may be determined from its phosphorescence lifetime

2.6 External Probes to Monitor Chemical Reactions: Sensors

The discussion in Sect. 2.5 was based on the assumption that there is a suitable change in some physical property of the reacting system. This is, of course, the ideal case, where the property employed to monitor the reaction is intrinsic to the system. There are, however, situations where, no matter why, there is no suitable change in an intrinsic property (or otherwise cannot be employed to measure concentrations), requiring some kind of manipulation of the reaction mixture.

This is the case, for instance, where external probes are needed and employed. External probes are typically molecules introduced in the reacting system by covalent attachment of a suitable chromophoric probe to one of the reaction partners. This methodology is widely employed to monitor biochemical reactions where a variety of fluorescent probes that react either with thiol or amine groups in oligonucleotides and proteins are frequently employed. For these procedure, auxiliary experiments should be performed to determine the binding constants of the ligand to the protein to ensure an efficient attachment [21, 22]. Alternative approaches, where genetic engineered techniques are employed to create proteins containing residues that can be specifically labeled, however, it makes necessary to fully characterize the new synthetized protein and that the modification does not affect the biological activity of the protein.

A number of other reactions may be monitored by linking it to secondary processes that provide suitable physical properties to monitor the reaction. The essential requirement is that the linked process is much faster than the reaction being studied, so that it does not affect the rate of the process being investigated. Examples are reactions where protons or other ions are either taken up or released; in these cases the use of pH indicators or ion-selective electrodes can be employed to monitor the reaction.

The use of sensors has also been exploited to monitor reactions [23]. A chemical sensor can be defined as a device that transforms chemical information, ranging from the concentration of a specific sample component to total composition analysis, into an analytically useful signal. The obtained chemical information mentioned above may originate from a chemical reaction of the reactant of interest or from a physical property of the system investigated. Biosensors are a particular sensor device which is sensitive to biomolecules, reacting rapidly with them and yielding a new component with, for example, suitable spectroscopic properties (typically emission).

In general, the chemical sensor devices are composed of two main functional units: a receptor and a transducer. Other necessary units are electronic devices, that process the signal from the transducers, and the display that shows the output signal in the form of numeric, graphic tabular or even an image, depending on the requirements of the user. The receptors are the module where the chemical information is transformed into a form of energy which is measured by the transducer. The transducer is, therefore, the part capable of transforming the energy carrying the chemical information about the sample into a useful analytical signal. Transducers can be

optical, based on measurements of radiation (absorbance, reflectance, luminescence, fluorescence, etc.), electrochemical, based on measurements of electric current or potential (voltammetric, potentiometric), mass sensitive devices, magnetic, thermometric, etc. In the last decades, there has been a huge increase in the development of biosensors to detect molecules of interest with medical purposes.

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Chapter 3
Basic Kinetic Concepts. Reaction Rates and Rate Laws for Some Common Elementary Reactions. Relationships Between Concentrations and Physical Properties

3.1 Basic Kinetic Concepts. Reaction Rate, and Rate Laws for Some Common Elementary Reactions

For a general and irreversible chemical reaction:

$$a\mathrm{A} + b\mathrm{B} \rightarrow c\mathrm{C} + d\mathrm{D}.$$

The **reaction rate** (denoted indistinctly by the symbols r or v) is related to the rate of disappearance of reactants or formation of products, according to Eq. 3.1 [1].

$$r = -\frac{1}{a}\frac{\mathrm{d}[A]}{\mathrm{d}t} = -\frac{1}{b}\frac{\mathrm{d}[B]}{\mathrm{d}t} = \frac{1}{c}\frac{\mathrm{d}[C]}{\mathrm{d}t} = \frac{1}{d}\frac{\mathrm{d}[D]}{\mathrm{d}t}, \tag{3.1}$$

where a, b, c and d are the stoichiometric coefficients and the magnitude between brackets [ ] denotes molar concentrations. The rates of consumption of reactants and of formation of products are not the same, in general, for all substances involved in a chemical reaction because their stoichiometric factors may be different. For example, for the reaction $A + 3B \rightarrow 2C$, the rates of change if their concentrations are $r_\mathrm{A} = 1/3r_\mathrm{B} = 1/2r_\mathrm{C}$. For the sake of convenience, it is useful to define a rate that is the same for all the species in terms of the extent of the reaction ξ, defined as

$$\xi = \frac{\Delta n_C}{c} = \frac{\Delta n_D}{d} = -\frac{\Delta n_A}{a} = -\frac{\Delta n_B}{b}, \tag{3.2}$$

where Δn is the variation in number of moles of the different species in a given time. The rate of the reaction is

$$r = -\frac{\mathrm{d}n_A}{aV\mathrm{d}t} = -\frac{\mathrm{d}n_B}{bV\mathrm{d}t} = \frac{\mathrm{d}n_C}{cV\mathrm{d}t} = \frac{\mathrm{d}n_D}{\mathrm{d}V\mathrm{d}t} = \frac{1}{V}\frac{\mathrm{d}\xi}{\mathrm{d}t} \tag{3.3}$$

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Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_3

Because $[X] = n_x/V$, it follows that

$$\frac{dn_x}{dt} = \frac{V d[X]}{dt} + \frac{X dV}{dt} \Rightarrow r = \frac{1}{x}\frac{d[X]}{dt} + \frac{1}{x}\frac{[X]dV}{V dt}, \tag{3.4}$$

where x is the stoichiometric coefficient of the reactant or product X and V is the volume of the reaction mixture. Equation (3.4) is the general definition of the rate of reaction. As shown, it depends on both the change of the concentrations and on the change of volume during the reaction. Note that even if no reactions were occurring, an increase in volume would bring about a change in the concentration of the species present, whose rate is $-[X]dV/Vdt$ and therefore the second term in Eq. (3.4) makes the rate of reaction equal to zero in the case of no reaction.

For the sake of simplicity, reactions are usually carried out in reactors of constant volume, so that $dV/dt = 0$ and, under these circumstances, the rate of the reaction simplifies to commonly known expression for the rate of reaction (3.5), where $[X]$ is the molar concentration of a reactant or product and x is its stoichiometric factor (positive for products, negative for reactants).

$$r = \frac{1}{x}\frac{d[X]}{dt} \tag{3.5}$$

For most reactions in solution, the empirical **rate of reaction** (or, simply, the **rate law**) is defined by an equation that relates the rate of reaction and the of concentrations of chemical species as indicated by Eq. (3.6):

$$v = k[A]^{\alpha}[B]^{\beta}, \tag{3.6}$$

where k is the **rate constant** (or rate coefficient) and the exponents α and β are the (partial) orders of the reaction with respect to the species A and B, respectively. The sum of α and β ($n = \alpha + \beta$) in the rate law is the total order of reaction. Their values can be a whole number (positive or negative) or a fraction. The values α and β need to be determined experimentally, they are independent of concentration and time and their values are not necessarily the same as the stoichiometric coefficients of the general reaction. The values of the rate constant k depend on temperature and ionic strength, and its units depend on the order of the reaction. The rate of consumption of, for example, a reactant is given by $r_A = k_A[A]^{\alpha}[B]^{\beta}$, and since the exponents must be equal to those in Eq. (3.6), the rate of the reaction is, in general, different from the rate of consumption or formation. For the reaction $A + 3B \rightarrow 2C$, $k = k_A = 1/3k_B = 1/2\ k_C$.

Not all reactions behave as described by Eq. (3.6) and the term "order" is meaningless for reactions that do not follow such an equation. For example, reactions catalyzed by enzymes (Chap. 6) do not have a defined order and necessarily occur by composite mechanisms, that is, they take place through several steps in opposite to some others that take place in one single step. A distinction is made between

"elementary reactions", which are reactions that occur in a single step with no experimentally detectable reaction intermediates, and "composite" or "complex" reactions, which involve more than one elementary reaction. These reactions will be analyzed in Chap. 4. Elementary reactions involving one, two, or three molecules are called unimolecular, bimolecular, and termolecular (or "trimolecular"), respectively. There are no known elementary reactions involving four or more molecules. The molecularity of an elementary reaction is, thus, the number of reactants involved in the individual chemical event.

3.1.1 Some Elementary Reactions

3.1.1.1 Zero-Order Reaction with 1 Reactant

If the rate of reaction $A \rightarrow P$ does not depend on the concentration of A, the reaction is of "zero-order" and follows a zero-order kinetics. The rate law is expressed as

$$v = -\frac{\mathrm{d}[A]}{\mathrm{d}t} = k[A]^0 = k. \tag{3.7}$$

Upon integration, Eq. (3.8) holds

$$[A] = [A]_0 - kt. \tag{3.8}$$

Equation (3.8) indicates that if a reaction is of zero-order, a plot of the variation of the concentration of the reagent $[A]$ with time should be linear with a slope equal to the rate constant.

From Eq. (3.8) it follows that the half-life $t_{1/2}$ of the reaction (time required for $[A]$ to decrease to half of its initial value) is given by Eq. (3.9), which predicts that, for zero-order reactions, the half-lives of the reaction depend on the concentration of the reactant, being the second half-life shorter than the first one, and the third-one shorter than the second one.

$$t_{1/2} = \frac{[A]_0}{2k} \tag{3.9}$$

3.1.1.2 First-Order Reaction with 1 Reactant

First-order reactions are those whose rate is proportional to the first power of concentration, so that if the elementary reaction $A \rightarrow P$ is irreversible and of first order with respect to A, then the rate of the reaction is given by Eq. (3.10).

$$v = -\frac{\mathrm{d}[A]}{\mathrm{d}t} = k[A]^1 = k[A], \tag{3.10}$$

which upon integration, yields Eq. (3.11).

$$[A] = [A_0]\mathrm{e}^{-kt}, \tag{3.11}$$

which can be linearized by taking logarithms to Eq. (3.12).

$$\ln[A] = \ln[A_0] - kt \tag{3.12}$$

Similar mathematical relationships can be derived for product P formation, Eq. (3.13), bearing in mind that $[A] = [A]_0 - [P]$, $[\mathrm{A}]_0 = [P]_\infty$, where $[P]_\infty$ is the concentration of P at "infinite" time, i.e., at the end of the reaction.

$$P = P_\infty(1 - \mathrm{e}^{-kt}) \tag{3.13}$$

Figure 3.1 illustrates the variations of the concentration of reactants and products with time for a first-order reaction.

From Eq. 311 it follows that the half-life $\mathrm{t}_{1/2}$ of the reaction is given by Eq. (3.14).

$$t_{1/2} = \frac{\ln 2}{k} \tag{3.14}$$

Equation (3.14) suggests a simple way to check if the reaction is first order: the reaction is carried out at different initial concentrations of A and the measured half-lives should be the same independently of the initial concentration (Fig. 3.1).

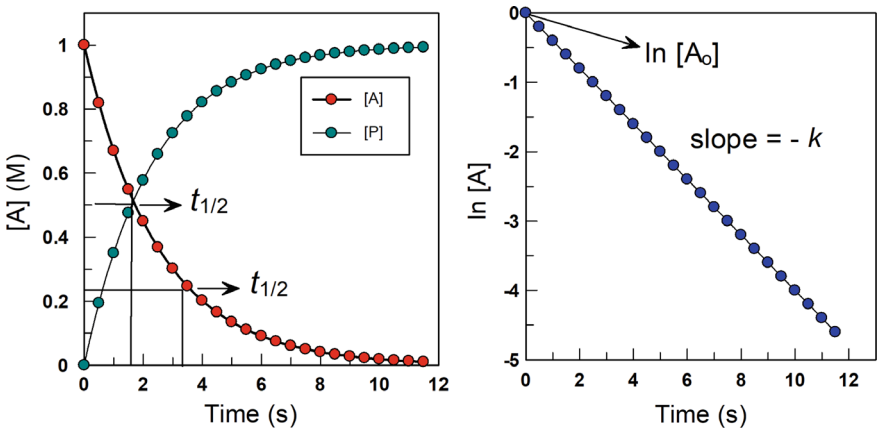


Fig. 3.1 Simulated exponential and linear plots for a model first-order reaction according to Eqs. (3.11)–(3.13). $k = 0.4\ \mathrm{s}^{-1}$

3.1.2 Second-Order Reactions

3.1.2.1 Second-Order Reaction with 1 Reactant

Second-order reactions are those whose rate is proportional to the square of a reactant concentration or to the product of two reactant concentrations. If the reaction $A \rightarrow P$, is of second order in A, the resulting kinetic equation is:

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = -k[A]^2, \tag{3.15}$$

which after integration, leads to Eq. (3.16)

$$\frac{1}{[A]} = \frac{1}{[A_0]} + kt, \tag{3.16}$$

which predicts that, for second-order reactions, plots of $1/[A]$ versus t should be straight lines with a slope equal to the rate constant value and an intercept equal to $1/[A_0]$.

3.1.2.2 Second-Order Reaction with 2 Reactants

For reactions of the type $A + B \rightarrow P$ which are first order in A and B (and therefore overall second order) the rate law is given by Eq. (3.17):

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = -\frac{\mathrm{d}[B]}{\mathrm{d}t} = \frac{\mathrm{d}[P]}{\mathrm{d}t} = k[A][B]. \tag{3.17}$$

The concentrations of A and B at any time are given by Eqs. (3.18) and (3.19), where x is the concentration of the product at any time and $[A]_0$ and $[B]_0$ are the initial concentrations of A and B, respectively.

$$[A] = [A]_0 - x \tag{3.18}$$

$$[B] = [B]_0 - x \tag{3.19}$$

Substituting in (3.17), Eqs. (3.20) and (3.21) can be derived

$$-\frac{\mathrm{dA}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}, \tag{3.20}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t} = k(A_0 - x)(B_0 - x), \tag{3.21}$$

which after integration, yields Eq. (3.22)

$$\frac{1}{A_0 - B_0}\left[\ln\frac{B_0(A_0 - x)}{A_0(B_0 - x)}\right] = kt. \tag{3.22}$$

According to this equation, a graph of the left-hand side of Eq. (3.22) versus t should be straight with slope k.

3.1.3 *Reactions of nth-Order*

In the general case of a reaction of order n, where only one reactant participates or, if more, are all the participating reactants with the same concentrations, the rate law equation is:

$$v = \frac{\mathrm{d}[A]}{\mathrm{d}t} = -k[A]^n \Rightarrow \frac{\mathrm{d}[A]}{[A]^n} = -k\mathrm{d}t. \tag{3.23}$$

After integration, Eq. (3.24) and (3.25) can be derived

$$\int_{[A]_\circ}^{[A]} \frac{\mathrm{d}[A]}{[A]^n} = -k\int_0^t \mathrm{d}t, \tag{3.24}$$

$$kt = \frac{1}{n-1}\left\{\frac{1}{[A]^{n-1}} - \frac{1}{[A]_0^{n-1}}\right\} \tag{3.25}$$

This derivatization can be generalized to show the different rate laws for the different orders of reaction. Table 3.1 summarizes some of these equations, together with their half-lives.

Table 3.1 Equations related to zero, first, second and $n^{\text{-th}}$ reaction, orders

Model	Rate law	Equations	$t_{1/2}$
Zero order	$v = k$	$[A] = [A]_0 - kt$	$t_{1/2} = [A]_0/2k$
First order	$v = k[A]$	$[A] = [A]_0 e^{-kt}$ $\ln[A] = \ln[A]_0 - kt$	$t_{1/2} = \ln 2/k$
Second order if $[A]_0 = [B]_0$	$v = k[A]^2$	$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$	$t_{1/2} = 1/[A]_0 k$
Second order if $[A]_0 \neq [B]_0$	$v = k[A][B]$	$\frac{1}{[B]_0 - [A]_0}\left[\ln\frac{[A]_0}{[B]_0}\frac{([B]_0 - x)}{([A]_0 - x)}\right] = kt$	–
n^{th} order	$v = k[A]^n$	$\frac{1}{n-1}\left(\frac{1}{([A]_0 - x)^{n-1}} - \frac{1}{[A]_0^{n-1}}\right) = kt$	$t_{1/2} = \frac{2^{n-1}-1}{k(n-1)[A]_0^{n-1}}$

A commonly employed approximation to simplify the determination of rate laws is the **isolation method** (see detail in Sect. 3.3). The researcher sets, experimentally, in large excess the concentrations of all the reactants except one (see Sect. 3.3.1) so that the concentrations of those reagents in excess can be considered, in practice, constant. The "apparent" order of the reaction will be that of the "isolated" reactant and can be easily determined by any of the methods described above. The orders of the other reactants are determined in a similar way.

3.2 Relationship Between Physical Properties *P* of Reactants/Products and Their Concentrations

As indicated in Chap. 2, there are two main methods to quantify the concentrations of reactants and/or products: the manual or classical methods (e.g., gravimetric or volumetric) and physical methods (that measure physical properties of the solution components, e.g., electric potential, conductivity, light absorption/emission, mass to charge ratio, etc.). Classical methods are not routinely employed in kinetic measurements because they only allow to monitor "slow" reactions, and thus we will mostly focus in the modern instrumental methods, that exploit the measurement of any physical or chemical property of the reactants and/or products that is proportional to its concentration to monitor chemical reactions.

Physical properties (P) suitable for monitoring chemical reactions are those that present an appreciable variation in their values within the concentration range of reactants required by the experimental conditions to be employed. The mathematical treatment is considerably simplified if the physical property is additive; that is, if the value of the physical property is the sum of the values corresponding to individual components (including inert components, e.g., the solvent). Once the values of the properties are collected, they can be introduced in any commercial computer program for further manipulation, for example, to fit the data to any equation by means of linear and nonlinear least-squares regression.

It is also very convenient, in order to simplify the manipulation of experimental data, to choose a property so that there is a linear relationship between the physical property and the concentration. In practice there are quite a few physical properties that meet these requirements, at least under non-extreme conditions; for example, radiation absorbance, electrical conductivity, emission fluorescence, electrochemical, and polarized light deflection (Sect. 2.3). However, it should be kept in mind that significant deviations can often occur due to the non-ideality of the system and, therefore, linear relationships between physical property and concentration will only be valid for relatively small concentration ranges.

To date, a number of kinetic methods have been reported in which a physical property of the reactant system is measured at different time intervals during the course of the reaction. In this way the reaction mixture is not disturbed by sample collection and the reaction is allowed to proceed to completion. Some techniques used

as continuous methods in solution have been described in the previous chapter and their usefulness for the treatment of kinetic data is presented below. The concentration of the reactant/product selected to monitor the chemical reaction can be determined by measuring the value of any physical property (P) of the reactant/product, which must be proportional to its concentration. For simplicity, let us consider the reaction $A \rightarrow S$.

In general, the value of any measured property P is the sum of various contributions, including that of the solvent and that of the reactor: P (reaction mixture) $= P_{\mathrm{A}} + P_{\mathrm{S}} + P_{\mathrm{solvent}} + P_{\mathrm{reactor}}$. Conditions are usually chosen to make the contributions of the solvent and of the reactor as a constant (for example, by choosing conditions where the value of the property is negligible or by recalibrating the instrument by making a "zero" (blank), so that the only contribution to the measured value of the property is due to the reactant $P = b[A]$, where $[A]$ stands for the concentration of a reactant and b is a proportionality constant.

Representative examples of physical properties, which can be found a proportional relationship with the concentration of a reactant A are shown in Table 3.2 and explained in the next section.

The resulting equations which relate the measured physical property P and time for zero-, first- and second-order reactions are summarized in Table 3.3. Those equations were derived by assuming that the relationship between P and concentration of the analyte is of type P (reaction mixture) $= P_A + P_{\mathrm{Solvent}} = b[A]$ where $[A]$ stands for the concentration of a reactant. Conditions are usually chosen so that $P_{\mathrm{solvent}} = 0$, for example, by making "blank" with a solution containing all components except the control species or by choosing a property whose value is zero (for example, by

Table 3.2 Representative relationships used in kinetic studies

Physical property P	Relationship between physical property P and concentration of a reactant A
Pressure	$P_A = (\mathrm{RT})[A]$
UV–Vis	$A_A = (\varepsilon l)[A]$
Fluorescence	$F_I = I_0\ (2.303\ \varepsilon\ [A] l)\phi$
Conductometry	$\kappa_A = \lambda_A[A]$
Polarimetry	$\alpha_A = (\alpha_0{}^{\upsilon})\ \delta[A]$
Surface-enhanced Raman	$I_T = EF_T I_{T0} \Theta_T S_T N_T$

P_A = pressure of reactant A (g); T = temperature (K); R = universal gas constant; A_A = absorbance of reactant A; ε = molar extinction coefficient or molar absorptivity; l = length of the light path; F_{I} = intensity of fluorescence emission; I_0 = intensity of incident light; ϕ = quantum yield of the fluorophore; κ_A = conductivity of A; λ_A = molar conductivity of A; α_A = angle of rotation caused by a solution of A; $\alpha_0{}^{\upsilon}$ = specific rotation caused by a solution of A for radiation of a fixed frequency υ and temperature T; δ = optical path length; I_{T} = SERS intensity; $\mathrm{EF_T}$ = average SERS enhancement factor of the reactant A; I_{T0} = ordinary Raman intensity of the reactant A; Θ_{T} = surface coverage; S_{T} = active SERS surface area; N_{T} = reactant surface density at the highest molecular packing

choosing a suitable wavelength so that any of the solution components (except the reactant) absorbs or emits radiation.

Assuming the initial concentration of A at $t = 0$ is $[A_0]$ and $[S_0] = 0$. At a given time, t, the concentration has changed by an amount x so that the concentration of A decreases to $([A]_0 - [x])$ and the concentration of product S increases to $[x]$. The changes of physical property for the reactant A at time $t = 0$, $t =$ t and $t = \infty$ are given by Eq. (3.26).

$$\begin{array}{lccl} & A \longrightarrow & S & \\ t = 0 & [A]_0 & 0 & P_0 = b_A[A]_0 \\ t = \infty & 0 & [S]_\infty & P_\infty = b_S[A]_0 \\ t = t & ([A_0] - [x]) & [x] & P_t = b_A([A_0] - [x]) + b_S[x] \end{array} \tag{3.26}$$

Taking account, the differences indicated in equations in (3.26), mathematical equations can be derived for different reaction orders in terms of the value of the measured physical property P. Derivatization of rate equations in terms of any physical property P, is straightforward bearing in mind the relationships given in (3.27). For the most common reaction orders, the corresponding Eqs. (3.28)–(3.30) are given below.

$$\begin{aligned} P_0 - P_\infty &= [A]_0(b_A - b_P) \\ P_0 - P_t &= [x](b_A - b_P) \\ P_t - P_\infty &= (b_A - b_P)([A]_0 - [x]) \end{aligned} \tag{3.27}$$

Zero-order:

$$\begin{aligned} [A]_t = [A]_0 - kt &\Rightarrow ([A]_0 - [x]) - [A]_0 = -kt \\ &\Rightarrow \left(\frac{P_t - P_\infty}{b_A - b_P}\right) - \left(\frac{P_0 - P_\infty}{b_A - b_P}\right) = -kt \\ &\Rightarrow (P_t - P_\infty) = (P_0 - P_\infty) - k(b_A - b_P)t \end{aligned} \tag{3.28}$$

First-order:

Table 3.3 Rate equations for zero-, first- and second-order reactions as a function of the values of a measured property proportional to the concentration of a reactant or product

Order of reaction	Rate equations
0	$(P_t - P_\infty) = (P_0 - P_\infty) - k(b_A - b_P)t$
1	$\ln(P_t - P_\infty) = \ln(P_0 - P_\infty) - kt$
2	$\frac{1}{P_t - P_\infty} = \frac{1}{P_0 - P_\infty} + \frac{k}{(b_A - b_P)}t$

$$\ln[A]_t = \ln[A]_0 - kt \Rightarrow \ln\frac{[A]_0 - [x]}{[A]_0} = -kt$$
$$\Rightarrow \ln\frac{\frac{P_t - P_\infty}{b_A - b_P}}{\frac{P_0 - P_\infty}{b_A - b_P}} = \ln\frac{P_t - P_\infty}{P_0 - P_\infty} = -kt \tag{3.29}$$

Second-order:

$$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt \Rightarrow \frac{1}{[A]_0 - [x]} = \frac{1}{[A]_0} + kt$$
$$\Rightarrow \frac{1}{P_t - P_\infty} = \frac{1}{P_0 - P_\infty} + \frac{k}{(b_A - b_P)}t \tag{3.30}$$

Spectrophotometry

Spectroscopic methods exploit the effects of irradiating the vessel where the reaction takes place with a monochromatic electromagnetic radiation of suitable frequency (generally in the infrared, visible or ultraviolet zone) and measuring the changes in their intensity during the course of the reaction. The Lambert–Beer law provides, for dilute solutions, the relationship between the value of the property (absorption) and the concentration of the absorbing species, Eq. (3.31).

$$A = \log\frac{I_0}{I_{\text{tr}}} = \varepsilon Cl, \tag{3.31}$$

where A is the absorbance, I_0 is the initial intensity of the radiation (before passing through the sample), I_{tr} is the transmitted intensity (i.e., after passing through the sample), l is the optical path length within the sample and ε is the molar absorption coefficient or molar absorptivity.

A representative example of the method is the monitorization of the bleaching of pyrogallol red (PGR), an *ortho*-diphenolic dye, (Fig. 3.2). In aqueous solutions, at physiological pH 7.0, PGR shows a strong visible band which is highly sensitive to the presence of oxidants like AAPH (2,2′-azo-bis-(2-amidinopropane) dihydrochloride) showing a maximum in the UV–Vis spectra at 540 nm [2]. The kinetics of the reaction was studied by monitoring the disappearance of the absorption band at 540 nm (pH = 7.4), Fig. 3.2. The existence of a well-defined isosbestic point at 435 nm indicates that PGR was oxidized to stable final products, without significant intermediate accumulation.

Electric conductivity

Reactions between charged chemical species can be followed by monitoring the variation of the electrical conductivity of the solution over time [3, 4]. It is an inexpensive and simple technique, but it requires the absence of additional ions in the solvent that could affect the result of the measured property. The conductance G presented by a given volume of the solution, of a given geometry, to the passage of the electric

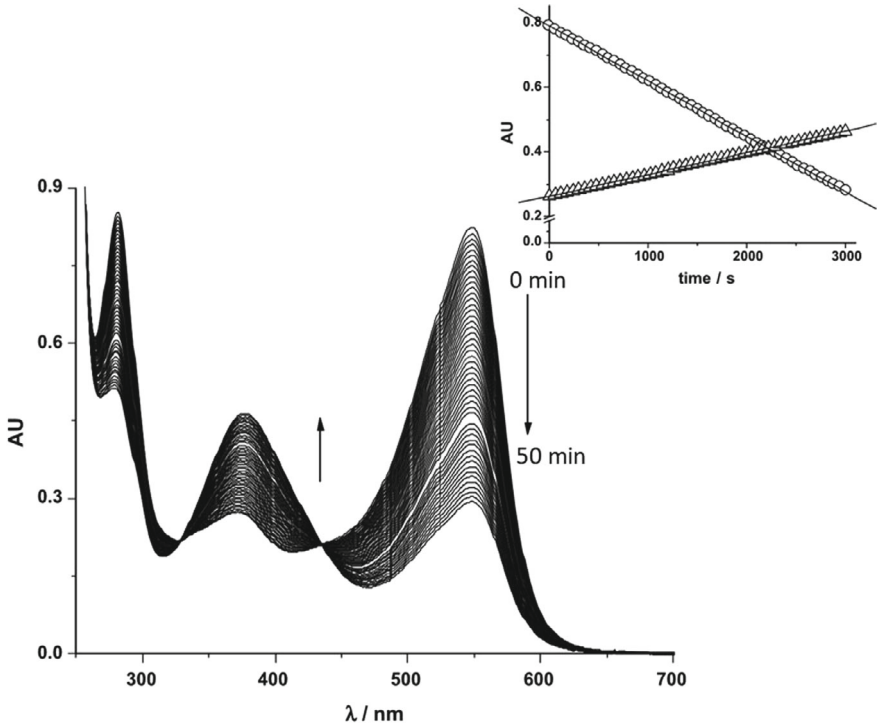


Fig. 3.2 Bleaching of PGR (30 μM) by 10 mM AAPH, in phosphate buffer (10 mM)/ethanol, 70/30 (v/v), pH 7.0 at 37 °C, during 50 min of reaction. Insert: Time course of absorption units (AU) at (○) 540 nm and (△) 380 nm. Adapted with permission from Ref. [2] Copyright © 2009, Taylor & Francis

current is measured experimentally (the conductance G is the inverse of the resistance (R) offered by the solution to the passage of the current, i.e., $G = 1/R$). The electrical conductivity (κ) of the solution can be obtained from the resistance (R) and the geometry of the given volume (l—total length- and A—cross-sectional area-), Eq. (3.32).

$$\kappa = \frac{1}{R}\frac{l}{A} = G \times K_{\text{cell}} \tag{3.32}$$

It should be noted that the cell constant K_{cell} is obtained from the dimensions of the cell used to measure the conductance G. In practice, measured K_{cell} is entered into the meter (directly or by user calibration) whereby the conversion from conductance to conductivity is calculated or directly presented.

To a first approximation, and considering dilute solutions, the conductivity κ_i of an ionic species is proportional to its concentration:

$$\kappa_i = \lambda_i[i]. \tag{3.33}$$

[$_i$] is the molar concentration of the ionic species i and the proportionality constant λ_i is called the molar conductivity of the chemical species *i*. In dilute solutions, the total conductivity of the solution is the sum of the conductivities of the ions present in the solution [5], represented by the Eq. (3.34).

$$\kappa = \sum n_i q_i \mu_i, \tag{3.34}$$

where κ is conductivity, n_i is the number moles of ions i, q_i is charge of the ion *i*, and μ_i is its mobility.

Figure 3.3 shows an example of kinetics of a nucleophilic substitution (SN1), the solvolysis reaction of 2-chloro-2-methylpropane ($(CH_3)_3CCl$) to produce 2-methyl-2-propanol and hydrochloric acid, at 25 °C [6]. The reaction is described in (3.35):

$$(CH_3)_3CCl + H_2O \rightarrow (CH_3)_3COH + H^+ + Cl^- \tag{3.35}$$

Assuming that the reaction is first-order in $(CH_3)_3CCl$ and that the solvent concentration is much greater than $(CH_3)_3CCl$, we can write down the rate law and integrate to get a conveniently plotted form: ln $[(CH_3)_3CCl]$ = ln $[(CH_3)_3CCl]_0$-*kt*. In order to obtain *k*, the first-order rate constant, from the above equation, it is necessary to determine $[(CH_3)_3CCl]$, or some quantity proportional to it, as a function of time. The conductivity (κ) was measured due to the formation of hydrochloric acid as shown in Fig. 3.3.

The conductivity, κ_i, at any time is proportional to the amount of $(CH_3)_3CCl$ used and the conductivity at infinite (κ_∞) time, is proportional to the initial $(CH_3)_3CCl$ concentration. Therefore, the integrated rate law associated with the product (considering Eq. 3.11) allows to determine the reaction rate constant, *k* (see Eq. 3.36).

Therefore, the integrated rate law associated with the product (considering Eq. 3.11) allows to measure the constant reaction rate, *k* (see Eq. 3.36).

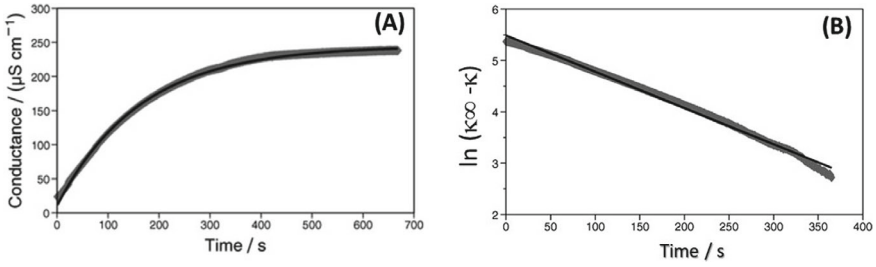


Fig. 3.3 **a** Experimental data reported for the solvolysis of 2-chloro-2-methylpropane in 15% ethanol/water mixture at 290 K and nonlinear regression analysis with Eq. (3.48) with fitted parameters ($\kappa_\infty - \kappa_0$) = 244.7 ± 0.2 μScm^{-1}, and k = 0.00615 ± 0.00002 s^{-1} and **b** linear regression fitted to Eq. (3.47) that gives k = 0.00695 ± 0.00002 s^{-1}. Adapted with permission from Ref. [6] Copyright © 2011, American Chemical Society

$$\ln(\kappa_\infty - \kappa) = \ln(\kappa_\infty - \kappa_0) - kt, \tag{3.36}$$

where κ_0 is the initial conductivity and zero time. Equation (3.36) can be rewritten for nonlinear regression analysis Eq. (3.37).

$$\kappa = \kappa_\infty - (\kappa_\infty - \kappa_0)e^{-kt} \tag{3.37}$$

Other example of the method is that for the solvolysis reaction using 2-adamantyl tosylate as the SN1 model and methyl tosylate, as the SN2 model with the aim to obtain experimental data on kinetic solvent isotope effects (KSIE) to help to identify the mechanistic change.

Fluorescence Spectroscopy

The intensity of fluorescence emission of weakly absorbing dilute samples is related to the molecular extinction coefficient by the relation showed in Eq. (3.38):

$$F_I = I_0(2.303\varepsilon Cl)\phi, \tag{3.38}$$

where I_0 = intensity of incident light; ε = molecular extinction coefficient; C = concentration in mol/L and l = pathlength of sample; and ϕ = the quantum yield of the fluorophore [7].

Therefore, the direct relationship with concentration allows working with fluorescence intensity measurements as a time-dependent variable in the corresponding graph. An example of the use of this property for monitoring chemical reactions is shown in Fig. 3.4 by developing the first ratiometric fluorescent probe for specific detection of cysteine (Cys) over homocysteine and glutathione based on the drastic distinction in the kinetic profiles [7]. The reaction is described in Scheme 3.1:

The reaction of the probe with Cys in pH 7.4 PBS (phosphate buffer containing 10% CH_3CN as cosolvent) was monitored using the emission band at 557 nm associated with the probe. In the presence of Cys, a decrease of such emission intensity was observed and an enhancement of the emission intensity at 487 nm, was reported (Fig. 3.4a). The kinetic parameters for the reaction were determined by fitting the fluorescence intensities of the samples to the pseudo first-order equation (see Sect. 3.3) [8].

The kinetic profiles were carried out under *pseudo* first-order conditions with a large excess of Cys (100–250 equiv.) over probe (5 μM), thus, the reaction was determined by fitting the fluorescence intensities of the samples to the pseudo first-order equation:

$$\ln[(F_{\max} - F_t)/F_{\max}] = -k't, \tag{3.39}$$

where F_t and $F_{\max}$ are the fluorescence intensities (for example at 487 nm) at time t and the maximum value obtained after the reaction was complete. k' is the pseudo first-order rate constant and it is a relevant parameter to characterize the probe for the specific detection of Cys over Hcy and GSH. The later can be obtained from the

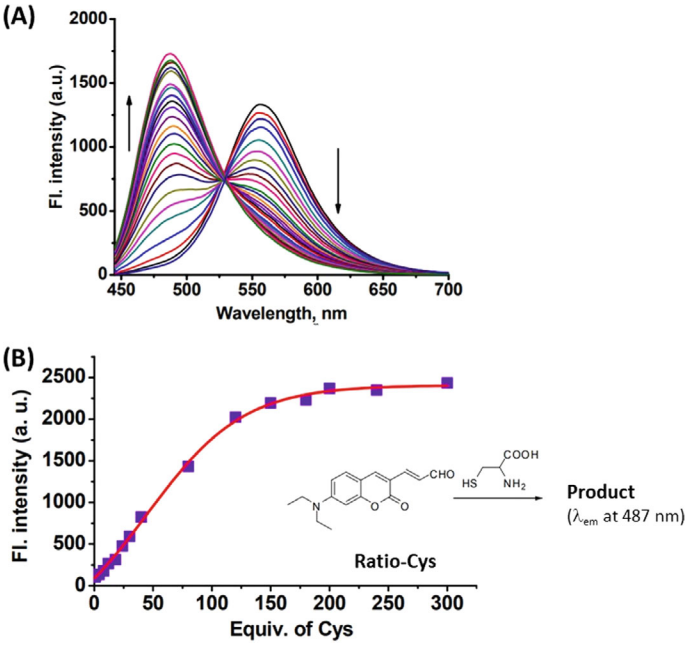
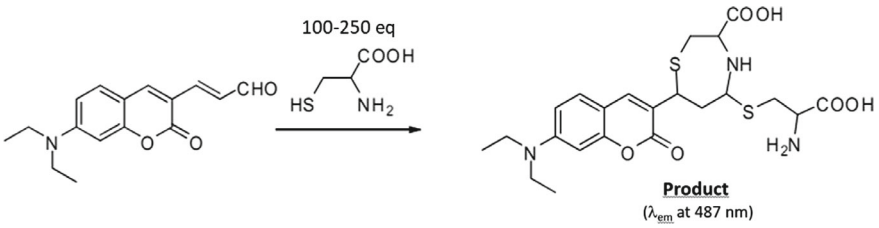


Fig. 3.4 **a** Fluorescence intensity of compound Ratio-Cys (5 μM) incubated with Cys (150 equiv.) for 0–50 min. **b** Fluorescence intensity of Ratio-Cys (5 μM) at 487 nm upon addition of Cys (0–300 equiv.). The spectra were recorded after incubation of the probe with Cys for 30 min. Excitation at 435 nm [7]. The insert shows the representative scheme reaction between the probe "Ratio-Cys" and Cys. Reprinted (adapted) with permission from Ref. [7]



Scheme 3.1 Proposed interaction between the fluorescent probe and cysteine [7]

fluorescence intensity, which is proportional to the concentration (in the low range) and that can be used to measure the kinetic profile of fluorophore species.

Electrochemistry

Electrochemistry is another tool for kinetically to study reactions involving electron transfers. A potentiometric method for measuring redox potential of the system (E) is unambiguously related to the concentrations of the oxidized and reduced forms by the Nernst equation (Eq. 3.40).

$$E^{'n}_{\mathrm{Ox/Red}} = E^{0}_{\mathrm{Ox/Red}} + \left(\frac{RT}{F}\right)\ln\left(\frac{C^{'n}_{\mathrm{Ox}}}{C_{\mathrm{Red}} - C^{'n}_{\mathrm{Ox}}}\right), \tag{3.40}$$

where $E^{'n}_{\mathrm{Ox}/Red}$ is the potential at time t_n, C_{Red} is the initial concentration of the reactant, and $C^{'n}{}_{\mathrm{Ox}}$ is the concentration of the formed oxidized form of the reactant at time t_n.

This relationship showed in Eq. (3.40) allows for the kinetics of changing E to determine a variation in the ratio of concentrations of the oxidized and reduced forms.

Moreover, kinetic potentiometric methods are potent tools for analysis, since they permit sensitive and selective determination of many samples within a few minutes with no sample pretreatment in many cases.

Cyclic voltammetry (CV) is a powerful and popular electrochemical technique commonly employed to investigate the reduction and oxidation processes of molecular species. CV is able to determine reaction kinetics rate constants of following chemical steps can be determined through measuring changes in peak currents and/or peak potentials. In particular, it is an invaluable tool for studying electron transfer-initiated chemical reactions, which includes catalysis. Additionally, experimentally determined rate constants can be compared with simulations of CV responses to confirm reaction mechanisms. According to the Randles–Sevcik Eq. (3.41) at room temperature (25 °C), it is possible to relate three key parameters: surface area, diffusion coefficient, and redox concentration [3].

$$i_{\mathrm{p}} = \left(2.69 \times 10^{5}\right) n^{3/2} A D^{1/2} C v^{1/2}, \tag{3.41}$$

where i_{p} is the peak current in A, n is the number of electrons transferred in the redox event, A is the surface area of the electrode in cm^2, D is the diffusion coefficient in $\mathrm{cm}^2\ \mathrm{s}^{-1}$, C is the bulk concentration (mol cm^{-3}) of the redox specie, and v is the scan rate in V s^{-1}.

In line with the Eq. (3.41), the peak currents (i_{p}) in an electrochemically reversible CV response are directly proportional to the bulk concentration of the species (C) undergoing electron transfer.

An example is the kinetic study of the reaction between the 3-methylbenzenediazonium ions (3MBD) with methyl gallate (MG) in aqueous solution by employing linear-sweep voltammetry, LSV [4]. The voltammograms of the diazonium salt mMBD showed two well-defined polarographic peaks, Fig. 3.5. Peak I_A was the one selected to monitor the reaction to minimize interferences since very few organic species are electroactive in this range of potentials (+ 0.02 V to − 0.1 V vs. Ag/AgCl).

The use of the electrochemical technique required optimizing instrumental and experimental parameters to obtain good sensitivity and selectivity and it is usually advisable to carry out auxiliary experiments to investigate the effects of experimental variables (acidity of the medium, temperature, and nature of the supporting electrolyte) and of instrumental parameters (initial potential, potential jump, and

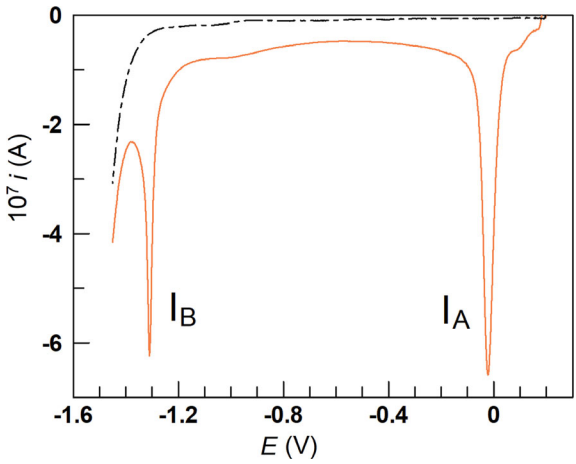


Fig. 3.5 Representative voltammogram of 3-methylbenzenediazonium (mMBD) showing the first and second reduction peaks. The first one (I_A), chosen to monitor the reaction, is associated with the transfer of one electron to yield the corresponding aryldiazenyl radical ($ArN_2^+ + e^- \rightarrow ArN_2^{\cdot}$)

scanning speed) to select the most suitable measurement conditions. Once the parameters of interest have been analyzed and optimized, the proportionality between the intensity of the reduction peak and the concentration of the species that is selected to follow the reaction (16-ArN_2^+ ions) has been verified (up to 10^{-4} M mMBD for the first reduction peak I_A) (Fig. 3.6).

Figure 3.7 shows the variations in the peak currents with time for the loss of reactant and for the formation and disappearance of a reaction intermediate. This demonstrates that LSV method has been proved to be an optimal methodology to allow simultaneous monitoring of 3MBD consumption and product formation (Fig. 3.7), the observed rate constant k_{obs} was obtained by fitting the i_p/t data for 3MBD ($E_p = -0.05$ V) to the integrated first-order equation, as shown in Fig. 3.7b.

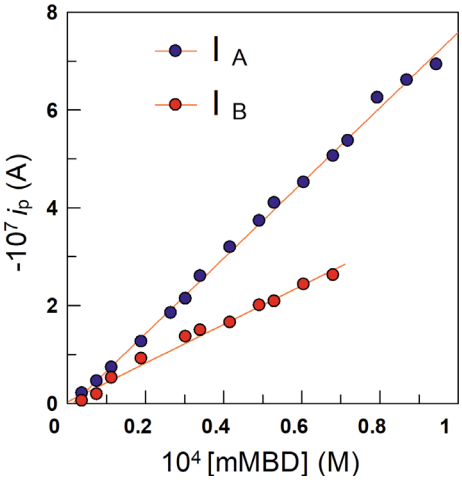


Fig. 3.6 Illustrative example of calibration plots showing the linear relationship between the measured property (in this case, the peak current) and the concentration of the reactant mMBD (*m*-methylbenzenediazonium)

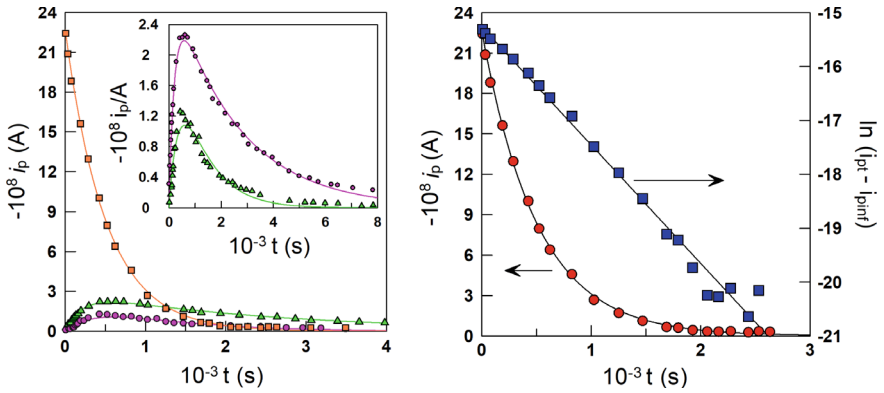


Fig. 3.7 Left. Variation with time of the peak currents of the reduction peaks at $E_p = -0.05$ V (■), $E_p = -0.5$ V (●), and $E_p = -0.64$ V (▲). Inset: Amplification of the variation of the ip of the reduction peaks associated with the formation and disappearance of the transient intermediate. The solid lines were drawn by fitting the data to a first-order equation. Right. Variation of i_p of the peak associated to mMBD loss ($E_p = -0.05$ V) and linear plot according to the first-order equation. Adapted with permission from Ref. [4]. Copyright © 2007 Verlag Helvetica Chimica Acta AG, Zürich, Switzerland

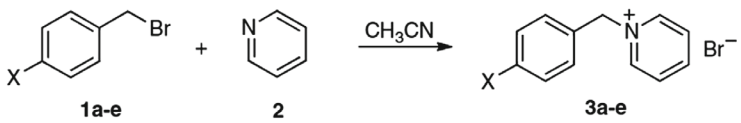
NMR spectroscopy

The ability of Nuclear Magnetic Resonance (NMR) to resolve and monitor signals of individual nuclei in different chemical environments makes it a useful method for kinetic analysis. As a quantitative analysis tool, NMR methods allow precise determination of the concentration of the analyte of interest, even in complex matrices. Generally, the analysis is based on quantification from the integration of signals from the ^{1}H spectra, but heteronuclear magnetic resonance spectroscopy also has been considered because it has as principal advantage the reduction in the number of signals in a spectrum. This has been exploited for following the kinetics of various processes including ^{15}N, ^{29}Si, ^{13}C, ^{31}P, ^{19}F and ^{79}Br spectra, among others.

Its application, however, is limited by the low sensitivity and relatively long time of data acquisition. These restrictions are particularly severe for monitoring non-equilibrium processes where reaction course is followed after system perturbation or mixing, making these experiments generally suitable only for reaction times longer than seconds. Faster rates can be obtained by the NMR analysis of dynamic equilibrium where no net changes occur, although the reaction is proceeding at a detectable rate [5].

This methodology has not been so widely disseminated, but there are interesting examples in the literature that can be studied to understand its application in chemical kinetics. ^{79}Br NMR spectroscopy was used to monitor a series of reactions in which the bromide ion is produced, including the Menschutkin reaction of pyridine with a range of substituted benzyl bromides and a Heck coupling process [5] (Scheme 3.2).

In particular, kinetic analyses of the Menschutkin reactions between the appropriate benzyl bromide **(1)** and pyridine **(2)**, at 298 K (Scheme 3.2) were carried out



X = OMe, Me, H, Cl, NO_2.

Scheme 3.2 The Menschutkin reaction between the benzyl bromides 1a–e and pyridine 2 [5]

using both ^{1}H and ^{79}Br NMR spectroscopies. Spectra were taken at regular intervals during the reaction and the extent of reaction calculated through integration of the signals corresponding to benzyl protons in the starting materials **1a–e** (ca δH4.5) and the products **3a–e** (ca δH5.8), and the bromide ion (ca δBr75). Further, these results demonstrate that sufficiently accurate ^{79}Br NMR spectra to relation with the concentration of the product and to determine rate constants of the reactions [5] as shown in Fig. 3.8. The value obtained for **1b** using this methodology was $(6.2 \pm 0.6) \times 10^{-4}$ s^{-1}, very similar result to that obtained by ^{1}H NMR $(6.9 \pm 0.4) \times 10^{-4}$ s^{-1}.

Results reported demonstrate that ^{79}Br NMR spectroscopy can be readily used to follow processes in which a bromide ion is generated. These can range from simple first-order processes to more complicated systems and in the case of the former rate constants calculated using this and other spectroscopic techniques are in good agreement.

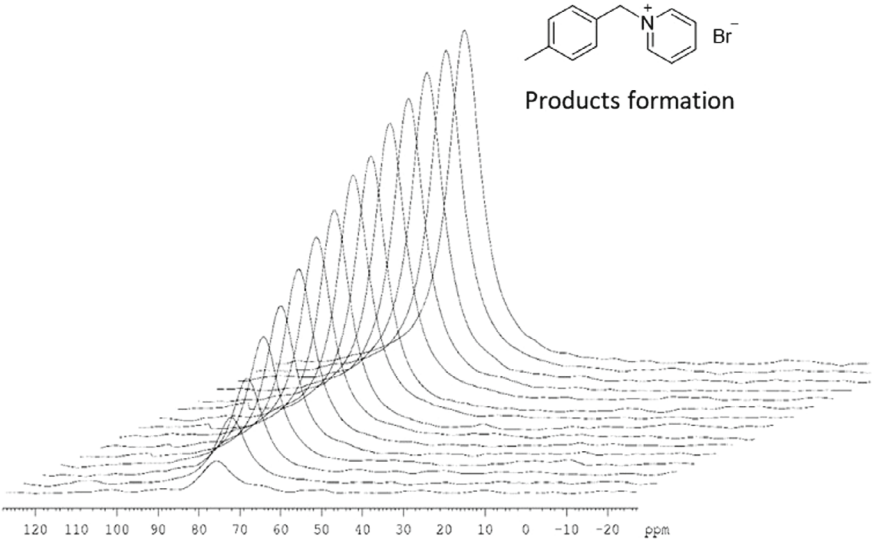
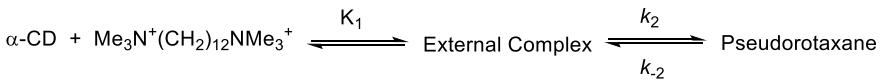


Fig. 3.8 Variations of ^{79}Br NMR spectra induced for the product formation for a Menschutkin reaction between bromide **1b** and pyridine at 298 K. The first spectrum was that acquired after 145 s and each subsequent spectrum occurs in increments of 217 s. The signal corresponds to bromide anion. Adapted with permission from Ref. [5]. Copyright © 2009 John Wiley & Sons, Ltd.



Scheme 3.3 Mechanism of [2]pseudorotaxane formation with a symmetric axle

Other interesting example, was carried out by Garcia-Rio et al. [9] who reported kinetic and thermodynamic studies on the formation of the cyclodextrin (CD)-based [2]pseudorotaxane which have been carried out by a combination of NMR and other techniques, Scheme 3.3.

In the case of NMR, the data obtained through the integration of the proton signals allow the construction of kinetic profiles such as those presented in the following Fig. 3.9a. In particular, ^{1}H NMR spectra were recorded every 197 s for a mixture of [α-CD] = 2.25 mM and [$Me_3N^+(CH_2)_{12}NMe_3^+$] = 2.5 mM, in this case the time-evolution concentration for [2]pseudorotaxane product (blue profile), uncomplexed α-CD (black profile), and uncomplexed bolaform (red profile) were depicted by integrating the N^+Me_3 signals of the free bolaform (Fig. 3.9b) and by integrating the ^{1}H signals of the cyclodextrin (Fig. 3.9c).

Isothermal titration calorimetry measurements

Isothermal titration calorimetry (ITC) is a non-destructive technique that uses differential electrical power to measure the rate of heat released or absorbed during a reaction as the concentration of substrate varies with time, positioning it as a standard kinetic technique in biology and chemistry [10]. Two general methods have been reported for monitoring enzyme kinetics: (i) single-injection method and (ii) multiple-injection method [11].

When the single-injection method is employed to collect Michaelis–Menten kinetic data on an enzyme-catalyzed reaction (see Chap. 6 on catalysis for further information), the amount of heat (Q) involved is directly related by the enthalpy change ($\Delta_r H$) for the studied reaction to the moles of product formed (n), Eq. (3.42)

$$Q = n\Delta_r H = [P]_0 V_0 \Delta_r H, \tag{3.42}$$

where V_0 is the volume of the reaction vessel, and [P] stands for the molar concentration of the product. Equation (3.42) can be rearranged and when the first derivative is applied, Eq. (3.43) can be derived.

$$r = \frac{\mathrm{d}[P]}{\mathrm{d}t} = \frac{1}{V_0 \Delta_r H} \frac{\mathrm{d}Q}{\mathrm{d}t} \tag{3.43}$$

The "signal" provided by the instrument of dQ/dt is used to determine the reaction rate, and the area under the dQ/dt signal is used to determine $\Delta_r H$. The dQ/dt signal measures the rate of conversion of substrate to product and returns to the baseline when all the substrate is exhausted. From the derived Michaelis–Menten plots, the affinity for the substrate (K_M), turnover rate (k_{cat}), and catalytic efficiency (k_{cat}/K_M =

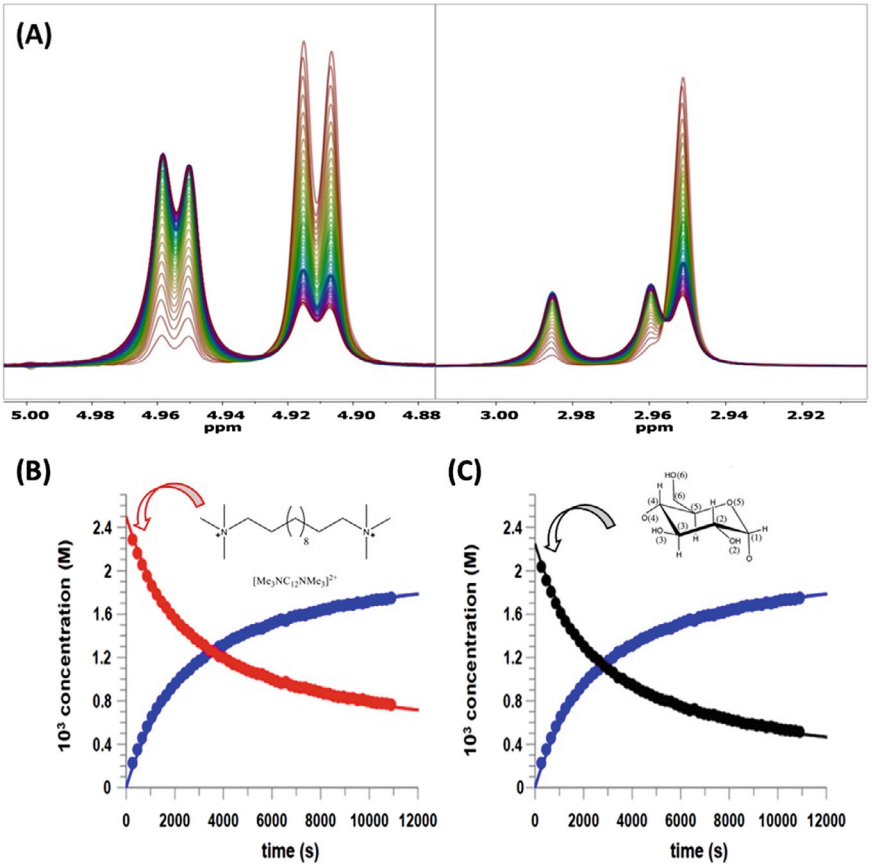
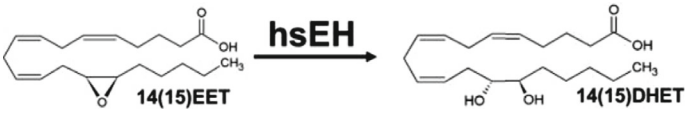


Fig. 3.9 **a** Time evolution of the ^{1}H-NMR spectra for a mixture of [α-CD] = 2.25 mM and $[Me_3N^+(CH_2)_{12}NMe_3^+]$ = 2.5 mM at 25 °C. Spectra were taken every 197 s showing the region for the anomeric hydrogen atom of α-CD (left) or the trimethylammonium groups of bolaform (right). Time-evolution concentration for [2]pseudorotaxane (blue), uncomplexed α-CD (black), and uncomplexed bolaform (red) for the mixture of [α-CD] = 2.25 mM $[Me_3N^+ (CH_2)_{12}NMe_3 +]$ = 2.5 mM in D_2O at 25 °C, **b** data obtained by integrating the N^+Me_3 signals of the free bolaform, **c** data obtained by integrating the H1 signals of the cyclodextrin. Adapted with permission from Ref. [9] Copyright © 2016, American Chemical Society

K_{sp}) values can be obtained. Scheme 3.4 and Fig. 3.10 show the general reaction and an example of a kinetic study for the epoxy-fatty acids (EpFA) hydrolysis, catalyzed by human soluble epoxide hydrolase (hsEH), employing an ITC single-injection method (SIM) [8].

The substrate 14(15)EET is one of the predominant regioisomers produced by cytochrome P450 (CYP) epoxygenases from arachidonic acid. Experimentally, the substrate solution in the syringe of the calorimeter is injected in a single step into the sample cell containing the enzyme solution, producing a heat response which



Scheme 3.4 General reaction for the hydrolysis of 14(15)EET catalyzed by human soluble epoxide hydrolase (hsEH)

endured for as long as the reaction proceeds, returning to the baseline when the reaction reached completion and all the substrate has been transformed. The ΔH_{app} was calculated by integrating this area under the peak.

$$\Delta H_{app} = \frac{1}{[S]_0 V} \int_{t=0}^{t=\infty} \frac{dQ}{dt} dt \tag{3.44}$$

The Michaelis–Menten kinetics curve fitting was obtained by manually selecting the window between the end of the injection (maximum substrate concentration, corresponding to saturating enzyme conditions) and the end of the decaying portion of the injection (minimum substrate concentration, corresponding to the beginning of the kinetics curve); see Fig. 3.10. The software fitting calculates $([S]_i;v_i)$ data points through Eqs. (3.42)–(3.44) building a Michaelis–Menten curve (Fig. 3.10c), thereby providing k_{cat} (the catalytic rate constant) and K_M (the Michaelis constant) values.

When employing the multiple-injection method, the steady-state heat rate is measured after each injection of substrate into an enzyme solution. The data produced is a plot of heat rate versus concentration of substrate, typically with (20–40) data points. Since each data point takes 3–5 min, a single experiment takes 1.5–3.5 h. For experiments with enzymes, this method works with concentrations of enzyme and substrate must be adjusted so that heat rates change significantly with each injection of substrate but are also constant after each injection. The inverse titration is not practical because of leakage of enzyme from the burette after the first injection [12].

FTIR spectroscopy

FTIR spectroscopy has a wide use and applicability mainly in the analysis of important molecules in the pharmaceutical, chemical, and polymer industries. FTIR analysis is used to better understand the molecular structure of materials, as well as the kinetics, mechanism and pathways of chemical reactions and catalytic cycles.

The ability of in situ FTIR to provide real-time tracking and monitoring of reaction progress is ideal for developing information on reaction mechanism and means to optimize reactions. Reaction information using this methodology allows for identification and profiling of variations in reactant and product concentrations as the reaction progresses. The important events of the reactions are revealed, thus the reaction rates and the rest of the kinetic parameters are determined from this real-time kinetic profiling information, as well as the compatibility with the proposed mechanisms.

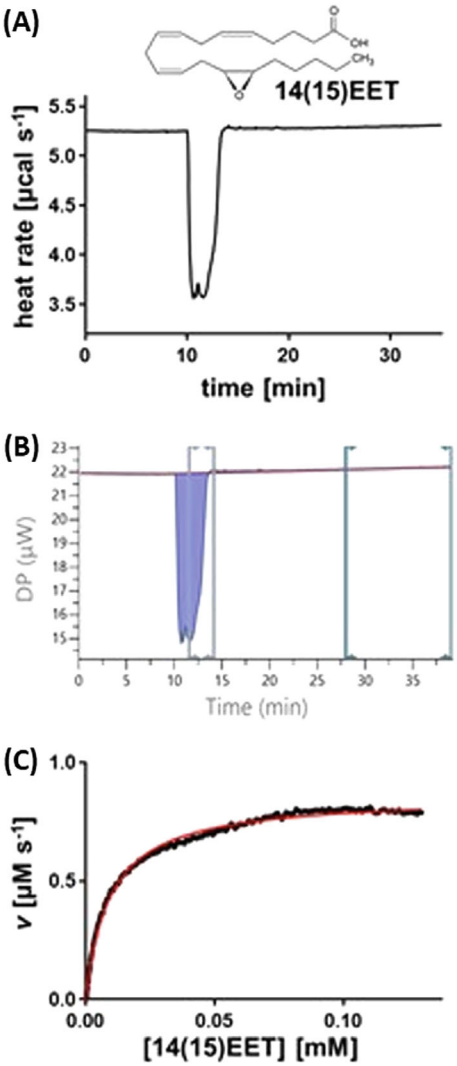


Fig. 3.10 Quantitative characterization of hsEH-mediated hydrolysis of 14(15)EET. **a** Representative thermal profile of a single-injection of 14(15)EET into hsEH CTD. **b** Screenshot from the MicroCal PEAQ-ITC analysis software. The ΔH_{app} was calculated by integrating the area under the peak (violet), upon definition of the baseline through manual adjustments of the left markers workspace (green). The data points used for the Michaelis–Menten kinetics curve fitting were obtained by manual adjustments of the right markers workspace (gray), selecting the window between the end of the injection (maximum substrate concentration—saturating enzyme conditions) and the end of the decaying portion of the peak (minimum substrate concentration—beginning of the kinetics curve). **c** Representative Michaelis–Menten fit of the $[[S]_i;v_i]$ data point extrapolated from B using the MicroCal PEAQ-ITC analysis software analysis [8]. Copyright © 2019 American Chemical Society. This publication is licensed under CC-BY

This is due to the advantages of this type of spectroscopy as one of the most powerful analytical methods for monitoring various processes such as UV-initiated curing processes which proceed within a fraction of a second.

Some representative examples involve the study of heterogeneous catalysis of the reaction of sucrose hydrolysis over Amberlite [13]. More recently, it has been reported that the transformation of ester group in ethylene–vinyl acetate copolymer (EVA) using methanol as nucleophilic agent will produce equivalent methyl acetate (MA) at the same time [14]. This is how, theoretically, FTIR is capable of capturing the difference of carbonyl group in EVA and MA, and can be applied to evaluate the progress of reaction (Fig. 3.11).

FTIR spectra of EVA changed clearly as reaction progressed. The highest absorption peak at 1743 cm^{-1} decreased quickly, while peak at 1734 cm^{-1} increased simultaneously, indicating the transformation of the ester group in EVA and the accumulation of MA in the system. The FTIR spectra associated with EVA and MA in methanol are presented in Fig. 3.12.

SERS spectroscopy

Surface-enhanced Raman spectroscopy (SERS) is a powerful tool with high potential for multiplexed detection of dilute analytes and for the investigation of a series of surface chemical reactions [15]. SERS advantages include large signal enhancement,

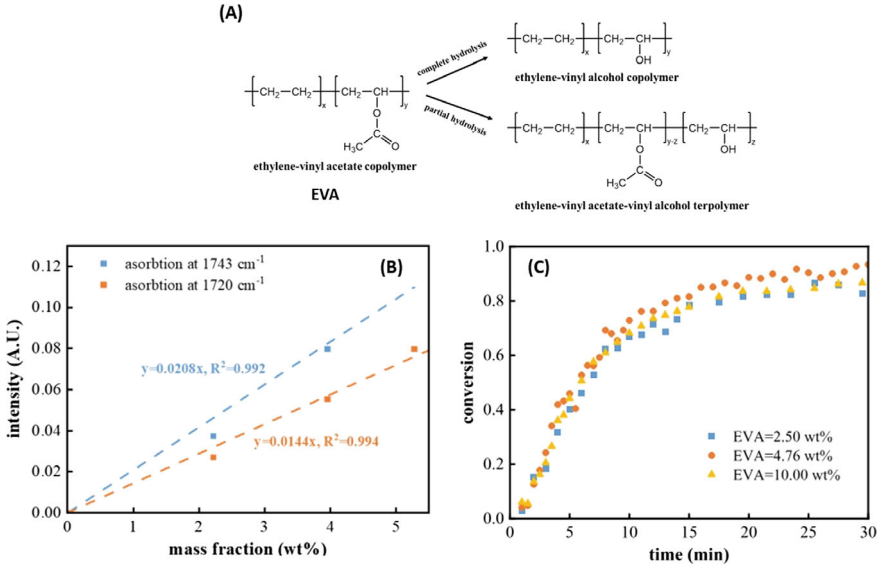


Fig. 3.11 **a** General reaction for the transformation by hydrolysis of ester groups in EVA copolymer; **b** correlation of peak height at 1743 and 1720 cm^{-1} with EVA mass fraction; **c** time profiles of conversion of ester groups at different initial EVA concentrations [14]. © 2023 Elsevier B.V. All rights reserved Copyright CCC

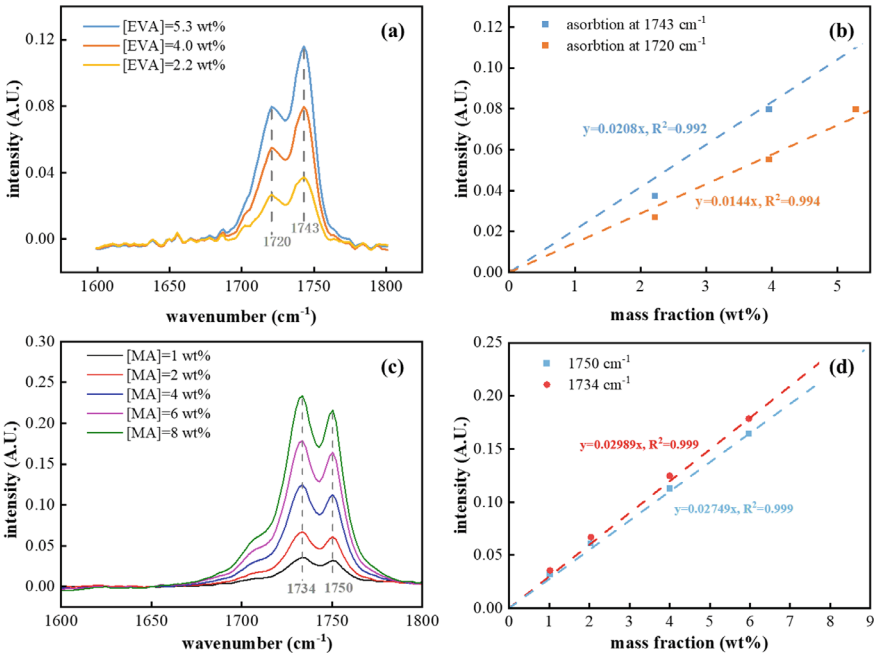


Fig. 3.12 **a** FTIR spectra of EVA in methanol; **b** correlation of peak height at 1743 and 1720 cm^{-1} with EVA mass fraction; **c** FTIR spectra of MA in methanol; **d** correlation of peak height at 1750 and 1734 cm^{-1} with MA mass fraction [14]. © 2023 Elsevier B.V. All rights reserved Copyright CCC

fingerprint chemical identification, and inherent suitability for aqueous measurements, which make SERS an effective tool for qualitative detection of dilute analytes in complex media. However, quantitative SERS remains challenging because the signal strength can be confounded by variable signal enhancements [16]. In spite of it, many studies introduced quantitative applications of SERS in various fields, and several SERS methods have been implemented for diverse applications, it may be appropriate to follow the progress of a reaction.

The relation between SERS intensity and analyte concentration (i.e., surface coverage) has to be analyzed with various adsorption isotherm models [17]. For example, according to the Langmuir adsorption model, the analyte surface coverage follows the relation (3.45a)

$$\Theta_T = \frac{K_T C}{1 + K_T C}, \tag{3.45a}$$

where Θ_T represents the surface coverage, K_T represents the adsorption equilibrium constant, and C represents the concentration of molecules in solution [10]. The SERS intensity (I_T) of the adsorbed analyte can be represented as in (3.45b):

$$I_T = EF_T I_{T0} \Theta_T S_T N_T, \tag{3.45b}$$

where EF_T corresponds to the average SERS enhancement factor of the analyte, I_{T0} corresponds to the ordinary Raman intensity of the analyte, S_T represents the active SERS surface area, and N_T represents the analyte surface density at the highest molecular packing. It is a common practice in quantitative analysis to compare the signals collected from a sample with those of standard solutions of the same molecule (known as calibration curve).

In fact, the kinetic study of the aldol condensation reaction of acetone and 4-(methylthio)benzaldehyde (MTBH), using 4-nitrothiophenol (NTP) as the internal standard, reported and shown in Fig. 3.13, serves to illustrate the application of this technique.

Figure 3.14 shows the intensity ratio of the peaks at 1700 cm^{-1} (aldehyde C = O stretch, MTBH) and 1340 cm^{-1} (NO_2 symmetric stretch, NTP) as a function of the

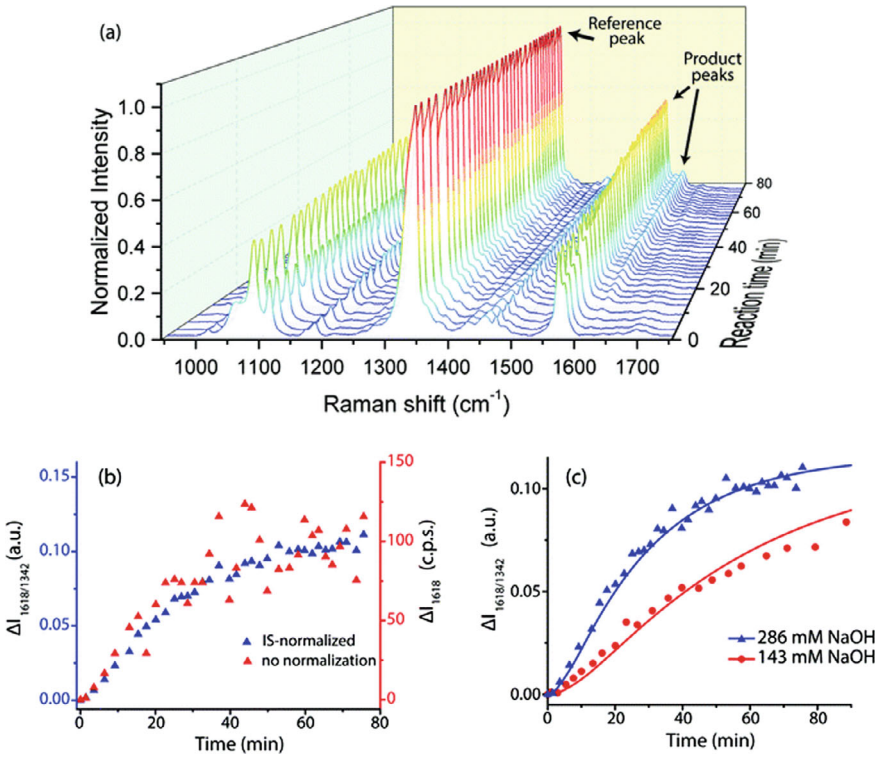


Fig. 3.13 Results of the aldol condensation reaction between free acetone and adsorbed MTBH at room temperature, with adsorbed NTP present as the internal standard (IS). Initial acetone concentration was 1.95 M. **a** 3D SERS spectrum showing time evolution during the reaction for the spectral range 950–1750 cm^{-1}. The initial NaOH concentration was 286 mM. **b** Intensity change over time of the 1618 cm^{-1} peak, comparing is-normalized data (blue) to non-normalized data (red) at the same experimental conditions as (**a**). **c** The fitted kinetic model for 1618 cm^{-1} peak, using the parameters reported in Ref. [19]

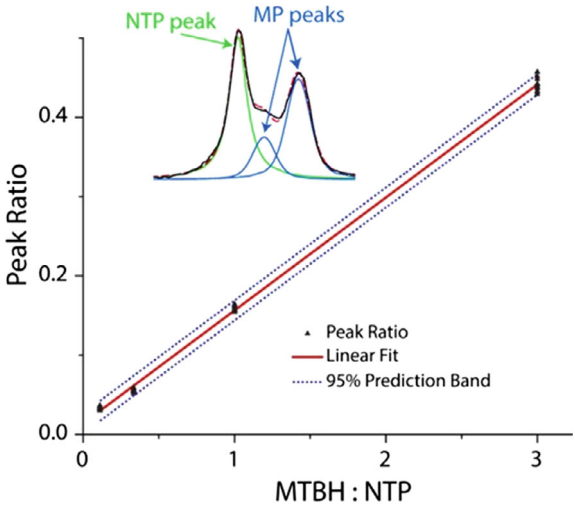


Fig. 3.14 Linear regression of MTBH/NTP coadsorption. The plot shows the intensity ratio of the peaks at 1700 cm^{-1} (aldehyde C=O stretch, MTBH) and 1340 cm^{-1} (NO_2 symmetric stretch, NTP) as a function of the volume ratio of MTBH and NTP solutions used in the adsorbing solution. Reproduced from Ref. [19] (CC BY-NC 3.0)

volume ratio of MTBH and NTP solutions used in the adsorbing solution. Reproduced from Reference Analyst 2016,141,6051–6060 (CC BY-NC 3.0).

Optical rotation

It is possible to carry out a kinetic study by monitoring the optical rotation, when reactants and/or products have different optical rotation properties. When more than one optically active substance is present in the reaction, their combined optical rotation effect upon the plane of polarized light is observed [11].

The angle of rotation (α) caused by a solution of a single pure compound is given by Eq. (3.46),

$$\alpha = [\alpha]_{\lambda}^{T} lC, \tag{3.46}$$

where $[\alpha]_{\lambda}^{T}$ is the specific rotation of the compound for radiation of a fixed wavelength λ at temperature T, l is the optical path length, and C is the concentration. Because the effect depends strongly on the temperature, the polarimeter cell must be thermostated and have a temperature-measuring device; optical rotation also depends on the solvent and if it is aqueous could depend on the pH of the solution if the compound is acidic or basic and the different forms have different specific rotation values.

The first technique used for monitoring a chemical reaction by measuring a physical property when the mutarotation of sucrose in acidic solution was studied and established the proportionality between the rate of reaction and the amount of remaining reactant.

Regarding the classic experiment of the invert sugar reaction, in an aqueous solution the sucrose molecule can be split into fructose and glucose by heat, or more effectively by the addition of protons, as shown in (3.47).

$$\underset{(\text{sucrose},\alpha_D=+66.3^\circ)}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{(\text{glucose},\alpha_D=+52.5^\circ)}{C_6H_{12}O_6} + \underset{(\text{fructose},\alpha_D=-89^\circ)}{C_6H_{12}O_6} \tag{3.47}$$

The term invert is used to indicate that upon addition of the catalyst over time the optical rotation of the solution will change sign (+ 66.3° to – 36.5°). This is because the optical rotation of the fructose molecule is opposite to and greater than that of glucose, as illustrated in the reaction (3.47).

3.3 Determining the Rate Law from Experimental Data

Considering that experimental data related to the concentration at different times have been obtained at the same temperature, the question next arises of determining a rate expression that can account for the data. The rate equation can be considered a key clue to understand the mechanism of the reaction.

Rate depends in some way on the concentration of reactant remaining as it was introduced previously, i.e., the rate is proportional to $[\text{reactant}]^n$, where n is a number well-known as (partial) order of the reaction and it indicates exactly how the rate depends on the concentration of reactant. In general, if the rate of reaction depends on the concentrations of different reactants, the rate expression is given by Eq. (3.6), where the total order of reaction is the sum of particular orders $\alpha + \beta +$, etc., and k stands for the rate constant at a given temperature. Here the kinetic analysis aim is to determine the order and k values. Several methods can be employed which are detailed in this and the next subsections.

The main methods that allow us to use the experimental data to determine the reaction orders with respect to each kinetically relevant species and therefore the rate law are:

(a) *the integration method*
(b) *the isolation method*
(c) *differential method*
(d) *initial rates method*
(e) *fractional lifetime method.*

(a) The integration method

Considering that rate laws are differential equations, we must integrate them if we want to find the concentrations as a function of time. Even the most complex rate laws may be integrated numerically. However, in a number of simple cases analytical solutions, known as integrated rate laws, are easily obtained, and prove to be very useful. We presented some of these simple cases in Table 3.1, in which the reaction rate constant is calculated by analytic or graphic approaches. If the calculated values of k exhibit good agreement with experimental values, the assumed order is correct; if not, it is necessary to assume another reaction order. The experimental data are commonly in the form of a property that is related to the concentration of reactants and/or products (for example, absorbance) versus time. If we have data the reactant concentrations as a function of the time since the equations referred to Table 3.1 all relate to reactant concentration, the data should fit an expression of the type indicated

in the Table 3.1 if the reaction order is some one of the described. For example, if we think that the reaction may be first reaction order, a plot of ln [*A*] versus time should be linear of gradient – k; if we think that the reaction may be second order, a plot of 1/[*A*] vs time should be linear of gradient k and so on. Note that if the reaction is zero-order, the rate should be independent of the concentration of the reactant.

Even if the most common values of reaction order are 1 and 2, note that this parameter cannot be an integer number and for this reason, this method is not suitable when n is not an integer or a simple fraction due to basic trial-and-error nature. If the order is complex this method could be a "hit or miss" procedure, making guesses as which orders might fit the experimental data. It is important that, such as indicated in Chap. 2, kinetic data are not limited to the very initial stages and that the reaction should be at least followed to three half-lives, then the method is much more discriminatory.

(b) **The isolation method**

As previously mentioned in this chapter, the determination of a rate law is simplified by the isolation method in which the concentrations of all the reactants except one are in large excess. Thus, for example, if for the reaction:

$$aA + bB + cC \rightarrow dD, \tag{3.48}$$

which has a rate law of the form:

$$v = k[A]^{\alpha}[B]^{\beta}[C]^{\gamma}, \tag{3.49}$$

we let B and C to be in large excess with concentrations $[B_o]$ and $[C_o]$, then it may be assumed that their concentration is constant throughout the reaction. Thus we may write:

$$v = k'[A]^{\alpha}, \tag{3.50}$$

where

$$k' = k[B]^{\beta}[C]^{\gamma}. \tag{3.51}$$

Under these conditions, we may say that the reaction has a pseudo-order rather Pseudo-order than the true-order. The process can be repeated by letting A/C and then A/B be in excess to determine α, β, γ, and k.

(c) **The differential method**

This means that at this point also it is possible to apply the **differential method**, where the kinetic expression is not integrated, but it is used in its differential form (Eq. 3.52) and transformed into a linear equation by means of logarithms.

This method allows directly and gives the values of α and k' from plots of ln(v*) versus ln[A], with a gradient* $s = \alpha$ *and an intercept* $I = ln(k')$.

$$\ln v = \ln k' + \alpha \ln[A] \tag{3.52}$$

Note that:

Under these conditions, we may say that the reaction has a *pseudo*-order rather than the true-order. The process can be repeated by letting A/C and then A/B be in excess to determine α, β, γ, and k.

(d) **Initial rate method**

The initial rates method considers only the initial portion of the reaction profile for determining the rate constant. One of the chief practical advantages of this method, was early reported by Casado et al. [18] who presented a quantitative argument showing that initial rate method is not only perfectly compatible with normal standards of experimental accuracy, but in certain ubiquitous circumstances, offers decided advantages over the integral method. The authors shall show here, by means of a numerical example that the validity of so reducing the duration of the experiment from 100% of the reaction to 80% should by no means be taken for granted.

This method is often used in conjunction with the isolation method, the rate is measured at the beginning of the reaction for several different initial concentrations of reactants. We shall suppose that the rate law for a reaction with A isolated is $v = k'[A]^{\alpha}$; then its initial rate, v_0, is given by the initial values of the concentration of A, and we write $v_0 = k'[A]_0^{\alpha}$ (3.53). Taking logarithms gives Eq. (3.54):

$$\log v_0 = \log k' + \alpha \log[A]_0. \tag{3.54}$$

For a series of initial concentrations, a plot of the logarithms of the initial rates against the logarithms of the initial concentrations of A should be a straight line with slope the order α and intercept the *pseudo*-order constant (k').

(e) **The fractional lifetime method**

As previously mentioned, the half-life time ($t_{1/2}$) is the time necessary for the initial concentration of some reagent to reduce to half of its value. To apply this method, a series of data of $t_{1/2}$ versus initial reagent concentration are required. The procedure to determine the reaction order and the reaction rate coefficient is similar to that of the integration method. The only difference is that, when integrating the reaction rate equation, the limits of the integral are from $t = 0$ and $[A]_0$, *to* $t = t_{1/2}$ and $[A] = [A]_0/2$. In this line, the method of half-life time can be extended to any fractional life time of the limiting reactant. In general, $t_{1/m}$ is the time necessary for the initial concentration to reduce to $1/m$ of its value. The integrated equations for any fractional life time are similar to those for half-life time:

For order n ($n \neq 1$):

$$k = \frac{(m^{n-1} - 1)|A|_0^{1-n}}{(n-1)t_{1/m}}. \tag{3.55}$$

For a given reaction for example, if the $[A] = 6 \times 10^{-3}$ M at the start of the reaction, then the first half-life will be the time taken for $[A]$ to drop from 3×10^{-3} M, the second half-life will be the time taken for $[A]$ to drop from 3×10^{-3} M to 1.5×10^{-3} M, the third half-life will be the time taken for $[A]$ to drop from 1.5×10^{-3} M to 0.75×10^{-3} M, and so on.

The dependence of value of the half-life on the initial concentration of reactant is determined by the order of the reaction. It can be found by substituting [reactant] = ([reactant]$_{\text{time=0}}$)/2 in the integrated expressions described in Table 3.1. The resulting expression of a particular integral order and for a general order n are indicated in Table 3.4.

As shown in Table 3.4, the half-life is independent of the initial concentration of reactant for a first-order reaction. When the reaction order is different to 1, $t_{1/2}$ is proportional to the concentration of reactant at time 0 (e.g., $[A]_0$) to the power (n-1) as shown in the general expression of a reaction of order n.

The expressions for half-life indicated in Table 3.4 may thus be employed to evaluate both the order of a reaction and its rate constant. Here two methodologies can be employed: (1) to follow the reaction during several half-lives and evaluate the dependence of three sequence of half-lives on the concentration at the start of that half-life period; (2) to address different experiments with a different initial concentration and to determine the first half-life for each experiment. By taking logarithms of both side of expression of a general order n indicated in Table 3.4, Eq. (3.56) is obtained, and the order n can be directly obtained from a plot ln $t_{1/2}$ against ln $[A]_0$. The fact that

Table 3.4 Differential equations and half-life expressions for different reaction orders

Order	Differential equation	Half-life $t_{1/2}$
Single reactant A ($A \rightarrow$ P)		
0	$-\frac{d[A]}{dt} = k[A]^0 = k$	$\frac{[A]_0}{2k}$
1	$-\frac{d[A]}{dt} = k[A]^1$	$\frac{\ln 2}{k}$
2	$-\frac{d[A]}{dt} = k[A]^2$	$\frac{1}{k[A]_0}$
3	$-\frac{d[A]}{dt} = k[A]^3$	$\frac{3}{2k([A]_0)^2}$
1/2	$-\frac{d[A]}{dt} = k[A]^{1/2}$	$\frac{\left(2-\sqrt{2}\right)([A]_0)^{1/2}}{k}$
3/2	$-\frac{d[A]}{dt} = k[A]^{3/2}$	$\frac{2\left(\sqrt{2}-1\right)}{k([A]_0)^{1/2}}$
n	$-\frac{d[A]}{dt} = k[A]^n$	$\frac{2^{n-1}-1}{k_n(n-1)([A]_0)^{n-1}}$
Reactants in their stoichiometric ratios ($A + B + \cdots \rightarrow P$)		
0	$-\frac{d[A]}{dt} = k$	$\frac{[A]_0}{2k_A}$
1	$-\frac{d[A]}{dt} = k[A]^1$	$\frac{\ln 2}{k_A}$
2	$-\frac{d[A]}{dt} = k[A]^1[B]^1$	$\frac{1}{k_A[A]_0} = \frac{1}{k_B[B]_0}$
3	$-\frac{d[A]}{dt} = k[A]^1[B]^1[C]^1$	$\frac{3}{2k_A[B]_0[C]_0}$

lineal plot on a $\ln t_{1/2}$-$\ln[A]_0$ scale provides a significant advantage, as it allows the reaction order to be determined directly, even for fractional order reactions.

$$\ln t_{1/2} = \ln \frac{\left(2^{n-1}-1\right)}{k_n(n-1)} + (1-n)\ln[A]_0 \tag{3.56}$$

It is important to note, that for an nth-order reaction ($n \neq 1$), a relevant relation can be proposed between $t_{3/4}$ and $t_{1/2}$, where $t_{3/4}$ and $t_{1/2}$ are times required for completion of 3/4 decay and 1/2 decay, respectively. Thus, it is possible to obtain a function of "n" alone, independent of the initial concentration of reagent.

$$t_{1/2} \Rightarrow [A] = \frac{1}{2}[A]_0 \Rightarrow kt_{1/2} = \frac{1}{n-1}\left\{\frac{2^{n-1}}{[A]_0^{n-1}} - \frac{1}{[A]_0^{n-1}}\right\} \tag{3.57}$$

$$t_{1/2} = \frac{2^{n-1}-1}{k(n-1)[A]_0^{n-1}} \tag{3.58}$$

$$t_{3/4} = \frac{4^{n-1}-1}{k(n-1)[A]_0^{n-1}} \tag{3.59}$$

$$\frac{t_{1/2}}{t_{3/4}} = \frac{\frac{2^{n-1}-1}{k(n-1)[A]_0^{n-1}}}{\frac{4^{n-1}-1}{k(n-1)[A]_0^{n-1}}} \Rightarrow \frac{t_{1/2}}{t_{3/4}} = \frac{2^{n-1}-1}{4^{n-1}-1} \tag{3.60}$$

Example 3.1 The spontaneous decomposition of 3-methylbenzenediazonium ions (3MBD) in acidic solution follows a $D_N + A_N$ mechanism via rate-limiting formation of a highly unstable aryl cation that further reacts with any nucleophile in the solvation shell. The absorbance of 3MBD was monitored at different times ($\lambda = 310$ nm) as indicated in the following Table E3.1. Demonstrate that decomposition of 3MBD is the first-order and the rate constant.

$$ArN_2^+ \underset{\text{slow}}{\overset{k_1}{\rightarrow}} [Ar^+N_2] \underset{\text{fast}}{\overset{k_2}{\rightarrow}} ArOH + H_3O^+ \tag{3.61}$$

Solution. To find the reaction order, let us test if reaction is the zero-, first- or second-order by the use of the integrated rate equation included in Table 3.1. Here it is important to realize that we have absorbance data and equations from Table E3.1 were expressed in terms of concentrations.

Taking account that the Lambert–Beer law is obeyed, i.e., the measured absorbance of 3MBD at time t is proportional to the concentration of 3MBD (see Eq. 3.31) where ε_i and l stand for the molar absorptivity and path length, respectively) and the mass balance at different times (zero, infinite, and time t), the absorbance value at each time can be derived after integrating corresponding equation in Table 3.1. If we considered that the absorbance of products at $\lambda = 310$ nm is equal to zero, equations in (3.62) can be simplified.

Table E3.1 Decomposition of 3MBD in acidic solution

Time (s)	A ($\lambda = 310$ nm)
1.6	0.5444
270.9	0.4960
570.9	0.4448
871	0.3984
1170.9	0.3567
1470.9	0.3190
1771.1	0.2852
2071	0.2545
2371	0.2271
2671	0.2022
2971	0.1801
3271	0.1604
3571	0.1427
3871.1	0.1269
4171	0.1129
4471	0.1002
4771.1	0.0890
5071	0.0789
5371	0.0701
5671.1	0.0621
5971	0.0549
6271	0.0487
6571	0.0431
6871	0.0382
7171.1	0.0338
7471.1	0.0298
7771.1	0.0264
8071.1	0.0234
8371.1	0.0206
8671.2	0.0182
8971.1	0.0161
9271.1	0.0143
9571.1	0.0127
9871	0.0113
10171.2	0.0099
10471.1	0.0089
10771.2	0.0079
10801	0.0078

(continued)

Table E3.1 (continued)

Time (s)	A ($\lambda = 310$ nm)
10831.1	0.0079
10861.1	0.0077

Experimental conditions: $[3MBD]_0 = 3.1 \times 10^{-4}$ M, pH = 3.7 and $T = 30$ °C

$$\begin{array}{lccl} & ArN_2^+ & \rightarrow \text{Products} & \\ t=0 & [ArN_2^+]_0 & 0 & A_0 = (\varepsilon_A l)[ArN_2^+]_0 \\ t=\infty & 0 & [P]_0 & A_\infty = (\varepsilon_P l)[ArN_2^+]_0 \\ t=1 & [ArN_2^+]_0 - x & x & A_t = (\varepsilon_A l)\left([ArN_2^+]_0 - x\right) + (\varepsilon_P l)x \end{array} \tag{3.62}$$

Taking account of Eqs. (3.26)–(3.30), mathematical relationships in terms of absorbance values can be derived for each kinetic order, Eq. (3.64) (zero-order), Eq. (3.65) (first-order) and Eq. (3.66) (second-order).

$$\begin{aligned} A_0 - A_\infty &= [ArN_2^+]_0 l(\varepsilon_A - \varepsilon_P) \\ A_0 - A_t &= xl(\varepsilon_A - \varepsilon_P) \\ A_t - A_\infty &= l(\varepsilon_A - \varepsilon_P)\left([ArN_2^+]_0 - x\right) \end{aligned} \tag{3.63}$$

$$\begin{aligned} [ArN_2^+]_t = [ArN_2^+]_0 - kt \Rightarrow \left([ArN_2^+]_0 - x\right) - [ArN_2^+]_0 = -kt \\ \Rightarrow \left(\frac{A_t - A_\infty}{\varepsilon_A l - \varepsilon_P l}\right) - \left(\frac{A_0 - A_\infty}{\varepsilon_A l - \varepsilon_P l}\right) = -kt \\ \Rightarrow (A_t - A_\infty) = (A_0 - A_\infty) - k(\varepsilon_A l - \varepsilon_P l)t \end{aligned} \tag{3.64}$$

$$\begin{aligned} \ln[ArN_2^+]_t = \ln[ArN_2^+]_0 - kt \Rightarrow \ln\frac{[ArN_2^+]_0 - x}{[ArN_2^+]_0} = -kt \Rightarrow \\ \ln\frac{\frac{A_t - A_\infty}{\varepsilon_A l - \varepsilon_P l}}{\frac{A_0 - A_\infty}{\varepsilon_A l - \varepsilon_P l}} = \ln\frac{A_t - A_\infty}{A_0 - A_\infty} = -kt \end{aligned} \tag{3.65}$$

$$\begin{aligned} \frac{1}{[ArN_2^+]_t} = \frac{1}{[ArN_2^+]_0} + kt \Rightarrow \frac{1}{[ArN_2^+]_0 - x} = \frac{1}{[ArN_2^+]_0} + kt \\ \Rightarrow \frac{1}{A_t - A_\infty} = \frac{1}{A_0 - A_\infty} + \frac{k}{l(\varepsilon_A - \varepsilon_P)}t \end{aligned} \tag{3.66}$$

By employing the derived Eqs. (3.63)–(3.66), fitting the data from Table E3.1 to derived equations indicated in Table 3.3 as shown in Fig. E3.1, clearly, the variation of $(A_t - A_\infty)$ versus time is curved so the reaction does not follow zero-order kinetics. The same thing happens when the variation of $1/(A_t - A_\infty)$ versus time is plotted. Since the plot of $\ln((A_t - A_\infty)/(A_0 - A_\infty))$ against time is linear, we can say that

kinetically the reaction is first order, with a rate constant, as given by the slope of the graph, of $k = 3.6 \times 10^{-4}\ s^{-1}$.

Example 3.2 To asses a method to determine the half-life time for the decomposition of a reactant as is the study of the spontaneous decomposition of 3-methylbenzenediazonium ions (3MBD) in acidic solution. Figure E3.2 shows a plot of the variation of the concentration of 3MBD with the time and three half-lives of the reaction were indicated.

The following data showed in Table E3.2 can be obtained on three half-lives of the reaction:

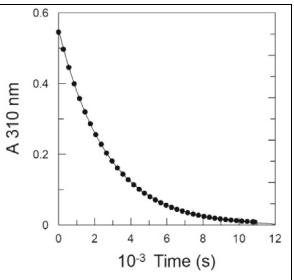
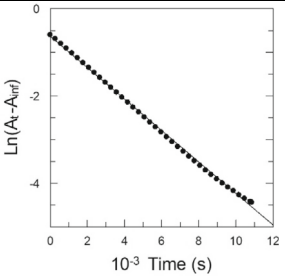
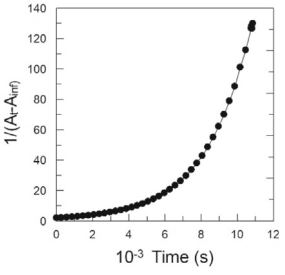
Zero-order	$(A_t - A_\infty) = (A_0 - A_\infty) - k(\varepsilon_A l - \varepsilon_P l)t$	
First-order	$\ln \frac{A_t - A_\infty}{A_0 - A_\infty} = -k$	
Second-order	$\frac{1}{A_t - A_\infty} = \frac{1}{A_0 - A_\infty} + \frac{k}{l(\varepsilon_A - \varepsilon_P)}t$	

Fig. E3.1 Expressions derived for each kinetic order in terms of absorbance and their graphical representations

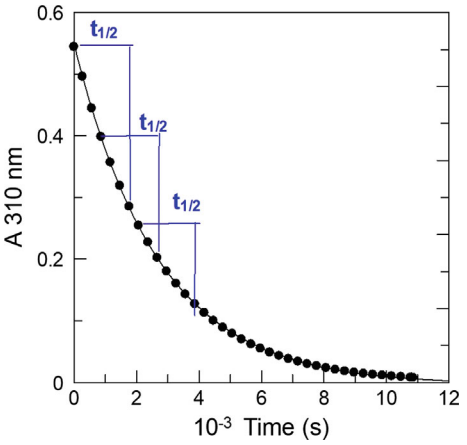


Fig. E3.2 Plot of the variation of the concentration of A (using absorbance at 310 nm) with the time and three half-lives of the reaction were indicated

Table E3.2 Data of $t_{1/2}$ at different initial concentrations of 3-methylbenzenediazonium ions (3MBD) in acidic solution

Initial concentration (10^{-3} M)	0.5444	0.3984	0.2545
Half-life (s)	1921	1950	1900

Solution. According Fig. E3.2, a significant curvature is observed showing that the reaction is not zero order with respect to [3MBD]. Three half-lives for the reaction are shown. For a first-order decay $t_{1/2}$ should be independent of initial concentration on the decay curve.

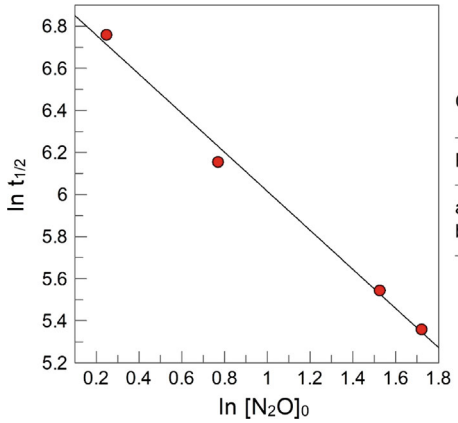
Data show the half-lives are clearly independent of the concentration at the start of the relevant time period, as is required for a first-order reaction (see equation Table 3.4). The mean half-life ($t_{1/2} = 1923$ s) may be employed to determine the rate constant by employing the equation derived for half-life time for first order, obtaining a rate constant value of $k = 3.60 \times 10^{-4}$ M.

Example 3.3 Table E3.3 shows some $t_{1/2}$ values obtained for the thermal decomposition of nitrous oxide (N_2O) when employing different initial concentrations of N_2O. Determine the order of the reaction and the rate constant value.

Solution. To find the reaction order, we can employ the following equation (derived from Eq. 3.58), so that a plot of ln $t_{1/2}$ versus ln $[N_2O]_0$ should be a straight line

Table E3.3 Data of $t_{1/2}$ for thermal decomposition of nitrous oxide (N_2O) at different initial concentrations of N_2O (1030K)

$[N_2O]_0$ (mM)	$t_{1/2}$ (s)
1.284	860
2.164	470
4.608	255
5.605	212



Correlation coefficient: -0.9967

Parameter	Value	Std. Erro
a (intercept)	6.9431	0.0653
b (gradient)	-0.9278	0.0535

Fig. E3.3 Plot of the ln $t_{1/2}$ versus ln $[N_2O]_0$, with data from Table 3.6

with a slope $s = (1 - n)$, where n is the order of the reaction. The k value can be determined by the combination of the slope and intercept (Fig. E3.3).

$$\ln t_{1/2} = \ln \frac{\left(2^{n-1} - 1\right)}{k_n(n-1)} + (1-n)\ln[N_2O]_0$$

The obtained values for gradient is $(1 - n) = -0.9278$ and for intercept 6.9431. Thus, the reaction order is 2 and the rate constant is 0.0184 s^{-1} M^{-1}.

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Chapter 4
Rates of Some Complex Reactions and Simplification of Reaction Schemes: Steady-State and Pre-equilibrium Approximations. Advantages, Limitations, and Accuracy



4.1 Introduction

The theoretical basis for the interpretation of rates of chemical reactions that was presented in Chap. 3 applies only to reactions in which a single compound is converted directly to one or more products by a unimolecular or bimolecular mechanism, or to those reactions in which two compounds react together with a bimolecular mechanism to give directly products. Most of those simple reactions have always either first or second-order rate laws. However, if the conversion of reactants into products takes place through more than one elementary step, the *overall* reaction is considered *complex* and frequently they do have complicated rate laws.

An example of a complex reaction is that between 3-methylbenzenediazonium ions and methyl gallate, Fig. 4.1, which comprises a fast ionization process (not shown for the sake of clarity), one competitive irreversible step and two consecutive steps, the first one reversible and the second one irreversible.

In considering complex reactions, the researcher should bear in mind the mechanistic consequence of the quantized nature of matter: individual molecules can undergo only simple reactions, so even the most complicated chemical transformation must be the outcome of competing and interacting simple elementary reactions. The objective of the kinetic investigation of complex reaction mechanisms is, therefore, to determine the number and nature of the elementary steps which together constitute the overall reaction.

The mechanisms of these complex reactions can be conveniently grouped according to the ways in which reactants can be converted into products, and the following classification may be convenient because many of these complex reactions involve a combination of one or more of the following type of reactions: (1) parallel, (2) reversible, (3) irreversible consecutive, (4) consecutive with an element of reversibility. Parallel reactions are those in which the same reactive species undergoes two or more competitive steps, each one with its own rate constant. Reversible reactions are those in which the products of the initial (direct) reaction can recombine

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Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_4

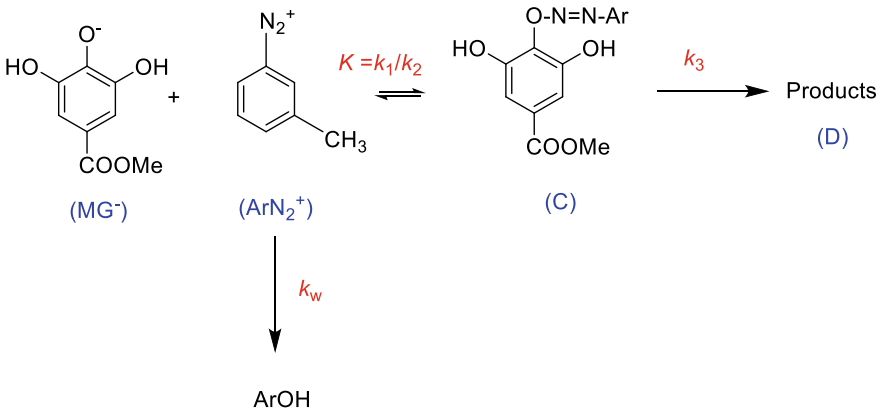


Fig. 4.1 Proposed reaction mechanism for the reaction between methyl gallate and 3-methylbenzenediazonium (ArN_2^+) that comprises two competing reactions, the spontaneous dediazoniation pathway with a rate constant k_w and a second competitive reaction leading to the reversible formation of an unstable transient intermediate C (with rate constants k_1 and k_2) that further decomposes through an irreversible step with a rate constant k_3. The very fast ionization equilibria of methyl gallate to yield the monoanionic form (which is the reactive species) is not shown for the sake of clarity

to regenerate the reactants. Irreversible consecutive reactions are those characterized by the product of the first irreversible reaction being a reactant in a subsequent irreversible step. A special case that is very common results when the first step of consecutive reactions is reversible. This situation is very relevant as it will allow us to introduce several approaches to simplify the kinetic schemes.

However, complex reactions follow reaction mechanisms involving various elementary steps, each one with their own transition state, Fig. 4.2, and the determination of the corresponding rate law is not straightforward but implies the solution of a system of differential equations, and the full analytical solution of these differential equations is only possible in the simplest cases. Sometimes, in more complicated cases, the researcher needs to take advantage of mathematical methods such as Laplace transforms, matrix or Runge–Kutta methods. In the absence of information on the orders of magnitude of the rate constants of the processes involved, the treatment of these kinetic systems is usually made by employing numerical methods (with the aid of commercially available computer programs) or even by employing stochastic methods such as the Monte Carlo one.

4.2 Parallel Reactions

Parallel (or competitive) reactions take place when a reactant **A** undergoes chemical transformation through different routes.

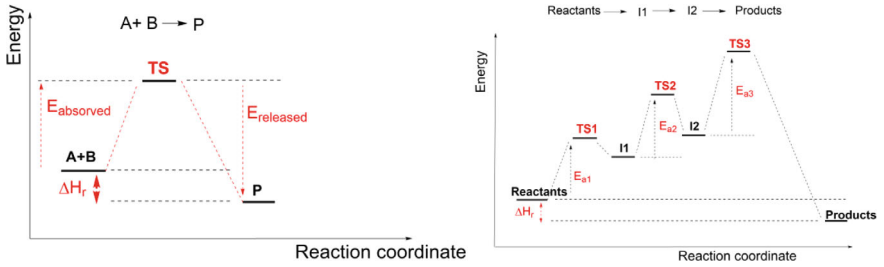


Fig. 4.2 Typical energy diagram for an elementary (left) and an irreversible consecutive complex reaction (right) showing the formation of intermediates and the corresponding transition states (defined, according to the IUPAC [1], as the state of a molecular *system from which there are equal probabilities of evolving toward states of lower energy* generally considered as reactants and products

The simplest parallel reactions scheme involves two competing first-order processes, Eq. (4.1) leading to the products **B** and **C**, each step having its own rate constant k_B and k_C, respectively.

$$\begin{aligned} A &\xrightarrow{k_B} B \\ A &\xrightarrow{k_C} C \end{aligned} \tag{4.1}$$

A somewhat more complicated case of parallel reactions is that where one or more of the reactions is bimolecular. The procedure to obtain the rate law is very similar to that for Eq. (4.1) and, for simplicity, we will not consider them but rather refer the reader to specialized books [2].

The rate for the disappearance of A and for the formation of the products B and C can be described by the differential equations (4.2)–(4.4). Note that the terms at the right-hand side are negative if the compound is consumed (e.g., A) and positive if the compound is produced (e.g., B and C).

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_B[A] + k_C[A] = (k_B + k_C)[A] \tag{4.2}$$

$$\frac{\mathrm{d}[B]}{\mathrm{d}t} = k_B[A] \tag{4.3}$$

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_B[C] \tag{4.4}$$

After integration, Eqs. (4.5) and (4.6) can be obtained, (Eq. 4.6 is the logarithmic form of Eq. 4.5), and where $[A]_0$ is the initial concentrations of the reactant.

$$[A]_t = [A]_O \mathrm{e}^{-(k_B + k_C)t} \tag{4.5}$$

$$\ln[A]_t = \ln[A]_O - (k_B + k_C)t \tag{4.6}$$

Note that a plot of $\ln[A]_t$ *versus* t yields the value of the sum of the rate constant, not the individual values of them. The symmetry of the sum makes that we cannot distinguish the cases where, for example, $k_B = 10k_C$ or $k_C = 10k_B$. Thus, the researcher should avoid the temptation of assigning individual values from fitting the time dependence of a property proportional to the concentration of the reactant or product.

Substitution of the Eq. (4.5) for **A** into Eqs. (4.3) and (4.4), the concentration changes for the formation of the products **B** and **C** can be obtained, Eqs. (4.7) and (4.8).

$$\int_{[B]_0}^{[B]_t} \mathrm{d}[B] = k_B[A]_0 \int_0^t \mathrm{e}^{-(k_B+k_C)t}\mathrm{d}t \Rightarrow [B]_t = \frac{[A]_0 k_B}{k_B + k_C}\left(1 - \mathrm{e}^{-(k_B+k_C)t}\right) \tag{4.7}$$

$$\int_{[C]_0}^{[C]_t} \mathrm{d}[C] = k_C[A]_0 \int_0^t \mathrm{e}^{-(k_B+k_C)t}\mathrm{d}t \Rightarrow [C]_t = \frac{[A]_0 k_C}{k_B + k_C}\left(1 - \mathrm{e}^{-(k_B+k_C)t}\right) \tag{4.8}$$

Equations (4.7) and (4.8) can be rewritten in terms of the final and initial values of **B** and **C**, Eqs. (4.9) and (4.10), respectively

$$\ln([B]_\infty - [B]_t) = \ln \frac{[A]_0 k_B}{k_B + k_C} - (k_B + k_C)t \tag{4.9}$$

$$\ln([C]_\infty - C]_t) = \ln \frac{[A]_0 k_C}{k_B + k_C} - (k_B + k_C)t \tag{4.10}$$

Eqs. (4.6), (4.9) and (4.10) show that only the sum of the rate constants $k_\mathrm{B} + k_\mathrm{C}$ can be obtained independently of the species that are monitored, and so to determine the individual values of k_B and k_C a second equation is needed. This second equation can be obtained from the ratio of Eqs. (4.7) and (4.8), which provides a useful relationship between the concentration of the products and the values of the rate constant, Eq. (4.11). The ratio of yields of the products is known as the *branching ratio* of the reaction and it is time independent, that is, the ratio (4.11) holds at any time of the reaction.

$$\frac{[B]_t}{[C]_t} = \frac{k_B}{k_C} \tag{4.11}$$

The reader must bear in mind that, in general, any change in the experimental conditions (temperature, solvent, addition of catalysts, etc.) may lead to a change in the ratio of product formation as the effects of the change may be different for the reactions involved. For example, one should not expect to obtain the same product

ratio upon changing the temperature because the reactions have different activation energies.

The above analysis can easily be extended to mechanisms involving more than two parallel first-order pathways, bearing in mind that the value of the observed or measured rate constant will be the sum of the values of all the rate constants of the processes involved. If the reaction mechanism involves **n** first-order steps, denoting **A** as a general reactant and $\mathbf{P_1, P_2, \ldots, P_n}$ as the different formed products, similar rate equations to (4.5)–(4.7) can be derived and solutions for the system of differential equations can be obtained, Eqs. (4.12)–(4.14).

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = (k_1 + k_2 + \cdots + k_n)[A] \Rightarrow [A] = [A]_0 \mathrm{e}^{-(k_1+k_2+\cdots+k_n)t} \tag{4.12}$$

$$\frac{\mathrm{d}[P_1]}{\mathrm{d}t} = k_1[A] \Rightarrow [P_1] = [A]_0 \frac{k_1}{(k_1 + k_2 + \cdots + k_n)} \left(1 - e^{-(k_1+k_2+\cdots+k_n)t}\right) \tag{4.13}$$

$$\frac{\mathrm{d}[P_n]}{\mathrm{d}t} = k_n[A] \Rightarrow [P_n] = [A]_0 \frac{k_n}{(k_1 + k_2 + \cdots + k_n)} \left(1 - e^{-(k_1+k_2+\cdots+k_n)t}\right) \tag{4.14}$$

It becomes clear that in parallel reactions, the yields of the products of the competitive reactions are formed according to the size of the values of the rate constants of the processes involved. Thus, optimization of the yield of a desired product with respect to that of the co-products leads to the finding of reaction conditions that increase the rate constant of the formation process with respect to that of the others.

4.3 Reversible Reactions

Reversible reactions involve two elementary reactions in which the products of the direct reaction can recombine to regenerate the original reactant (reverse reaction). The simplest reversible reactions are of the type described in Eq. (4.15), and where the values of the rates of the forward (direct) and backward (reverse) do not need to be the same but are comparable. In the case that one of them is several orders of magnitude higher than the other one, the reactions can be considered irreversible, and the reaction scheme simplifies to that in previous section.

$$A \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} B \tag{4.15}$$

The differential equations for A and B will be given by Eqs. (4.16) and (4.17).

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = -k_1[A] + k_{-1}[B] \tag{4.16}$$

$$\frac{\mathrm{d}[B]}{\mathrm{d}t} = k_1[A] - k_{-1}[B] \tag{4.17}$$

If the initial concentrations at time $t = 0$ of A and B are $[A]_0$ and 0, respectively, we can write the law of conservation of mass as $[A] + [B] = [A]_0$, so that the concentration of B at any time will be $[B] = [A]_0 - [A]$ and Eq. (4.17) can be rewritten to Eq. (4.18).

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = -k_1[A] + k_{-1}([A]_0 - [A]) = -(k_1 + k_{-1})[A] + k_{-1}[A]_0 \tag{4.18}$$

Equation (4.18) can be rearranged to Eq. (4.19) after multiplying and dividing simultaneously the second term on the right-hand side by $(k_1 + k_{-1})$

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = -(k_1 + k_{-1})\left([A] - \frac{k_{-1}}{k_1 + k_{-1}}[A]_0\right) \tag{4.19}$$

Integration of Eq. (4.19) leads to Eq. (4.20), where **I** is an integration constant.

$$\ln\left([A] - \frac{k_{-1}}{k_1 + k_{-1}}[A]_0\right) = (k_1 + k_{-1})t + I \tag{4.20}$$

The integration constant **I** can be obtained after substituting in Eq. (4.20) the initial conditions $[A] = [A]_0$ and $t = 0$, and Eq. (4.20) becomes Eq. (4.21)

$$I = \ln\left([A]_0 - \frac{k_{-1}}{k_1 + k_{-1}}[A]_0\right). \tag{4.21}$$

Replacing **I** in Eq. (4.20), leads to Eq. (4.22)

$$\ln \frac{\left([A] - \frac{k_{-1}}{k_1+k_{-1}}[A]_0\right)}{\left([A]_0 - \frac{k_{-1}}{k_1+k_{-1}}[A]_0\right)} = -(k_1 + k_{-1})t. \tag{4.22}$$

After some mathematical manipulation, Eq. (4.23) can be derived from Eq. (4.22)

$$[A] = \frac{[A]_0}{k_1 + k_{-1}}\left(k_{-1} + k_1 e^{-(k_1+k_{-1})t}\right). \tag{4.23}$$

Bearing in mind the corresponding mass balance, the concentration of B at any time is $[B] = [A]_0 - [A]$, and Eq. (4.24) can be easily obtained.

$$[B] = [A]_0 \frac{k_1}{k_1 + k_{-1}}\left(1 - \mathrm{e}^{-(k_1+k_{-1})t}\right) \tag{4.24}$$

Equations (4.23) and (4.24) provide mathematical equations describing the variations of the concentration of the species A and B with time, respectively. Note that

a decrease in the concentration of A is concomitantly accompanied by the same increase in the concentration of B and vice versa.

Equations (4.23) and (4.24) were derived under the assumption that only A was present at the beginning of the reaction. In the case that the initial concentration of B is not zero, the mass balance indicates that, at any time, $[A] + [B] = [A]_0 + [B]_0$, and thus $[B] = [A]_0 + [B]_0 - [A]$. The differential equation for A is given by Eq. (4.25)

$$\frac{d[\mathrm{A}]}{dt} = -k_1[\mathrm{A}] + k_{-1}([\mathrm{A}]_0 + [\mathrm{B}]_0 - [\mathrm{A}]). \tag{4.25}$$

Following a similar procedure as before, the changes in the concentrations of A and B are given by Eqs. (4.26)–(4.28)

$$[A] = ([A]_0 + [B]_0)\frac{k_{-1}}{k_1 + k_{-1}} + \frac{[A]_0 k_1 - [B]_0 k_{-1}}{k_1 + k_{-1}} \mathrm{e}^{-(k_1 + k_{-1})t} \tag{4.26}$$

$$[B] = \frac{k_1}{k_1 + k_{-1}}([A]_0 + [B]_0) - \frac{k_1[A]_0 - k_{-1}[B]_0}{k_1 + k_{-1}} \mathrm{e}^{-(k_1 + k_{-1})t} \tag{4.27}$$

$$\ln \frac{(k_1[A] - k_{-1}[B])}{(k_1[A]_0 - k_{-1}[B]_0)} = -(k_1 + k_{-1})t. \tag{4.28}$$

Once the equilibrium is reached, the rates of both the direct and reverse reactions equal each other (which corresponds to a formulation of the principle of microscopic reversibility), so that at equilibrium the concentrations of reactants and products do not change with time.

$$k_1[A]_e = k_{-1}[B]_e \Rightarrow \frac{[B]_e}{[A]_e} = \frac{k_1}{k_{-1}} = K, \tag{4.29}$$

where $[A]_e$ and $[B]_e$ are the equilibrium concentrations, and K becomes the thermodynamic equilibrium constant.

Reversible reactions are of fundamental importance in relaxation methods (Sect. 2.5.3) where reactions are perturbed from their equilibrium and allowed to reach it again. The evolution of the concentration of species from their equilibrium may be experimentally accessible and can be used to obtain their relaxation times ($\tau = 1/(k_1 + k_{-1})$ by combining Eqs. (4.28) and (4.29).

Figure 4.3 illustrates the variation of the concentration of species A and B for a reversible reaction at different k_1/k_{-1} ratios.

An illustrative example of a reversible, first-order, reaction is the cyclization of γ-hydroxybutyric acid (A) to yield a lactone (B) in aqueous solution. The example also provides a practical example of a back-of-the-envelope estimation of the rate constants. Figure 4.4 shows the variation in the concentration of the species A and B with time. A preliminary observation of the experimental data allows us to predict that the reaction is reversible since the concentration of reactant A does not go down to zero and its concentration remains constant with time after the reaction is completed.

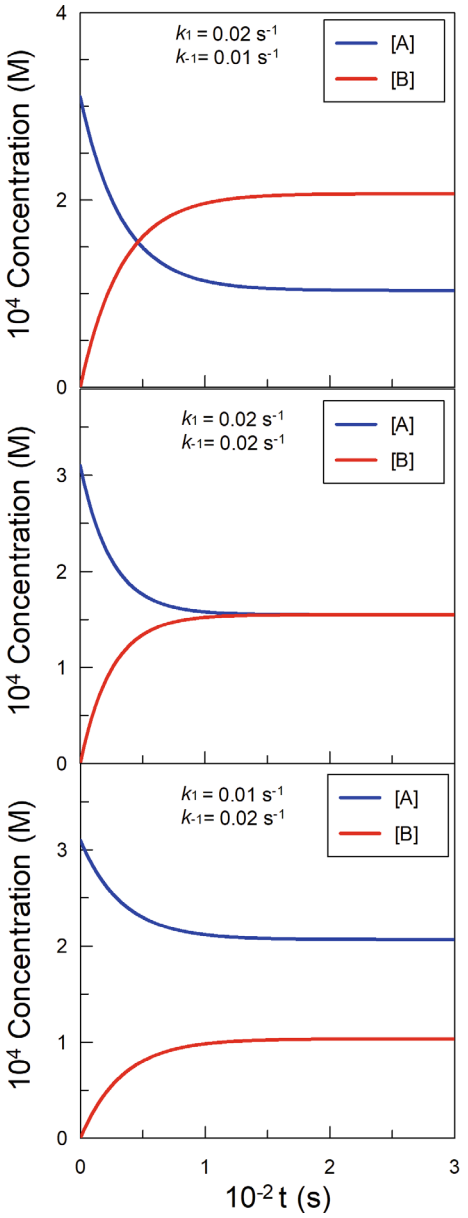


Fig. 4.3 Simulated variation of the concentration of species *A* and *B* species for a reversible reaction A $\rightleftharpoons$ B until equilibrium is reached or reactions with k_1 (forward reaction) higher than, equal to, and lower than k_{-1} (backward reaction)

From the variation of the concentration of *A*, we can estimate a value for the half-life time for the disappearance of *A*, which is the same as that for the formation of *B*. The concentration of *A* decreases from 18×10^{-3} M to ~ 4.5×10^{-3}M (dashed line in Fig. 4.4), and therefore the variation in the concentration of *A* is $\Delta[A] = 13.5$ M. By definition, the half-life is the time required to decrease the concentration by half, so that $\Delta[A]/2 = 6.7$, and the time required for a decrease in [A] from the starting concentration (18×10^{-2} M) to ($18–6.7$) $\times 10^{-2}$ M is approximately $t_{1/2} \approx 4.5 \times 10^3$ s, a value that is independent of the initial concentration of *A* as can be envisaged from the data in Fig. 4.4. According to Eq. (4.23), the value of observed (measured) rate constant is the sum of the forward and backward rate constants ($k_{obs} = k_1 + k_{-1}$) and such a value can be obtained from the half-life value ($t_{1/2} = \ln 2/k_{obs} \Rightarrow k_{obs} = 1.56 \times 10^{-4}$ s^{-1}). On the other hand, the ratio of the concentrations at equilibrium $K = k_1/k_{-1} = [B]_e/[A]_e) \approx 13.1 \times 10^{-2}/4.5 \times 10^{-2} \approx 2.9$. Hence, solving the system of two equations with two unknowns ($k_{obs} = k_1 + k_{-1}$) and $K = k_1/k_{-1}$) we can estimate the individual values of the forward and backward reactions, $k_1 \approx 1.16 \times 10^{-4}$ s^{-1} and $k_{-1} \approx 4.2 \times 10^{-5}$ s^{-1}.

The example above provides a quick, relatively easy and straightforward, way of estimating the value of the observed rate constant k_{obs} and of the equilibrium constant *K*. Nowadays, with the aid of commercial computer programs with visualization and analysis of experimental data capabilities, the k_{obs} values are most frequently determined by fitting the experimentally determined pairs of data points (([*A*], *t*) or ([*B*], *t*)) to the corresponding kinetic Eqs. (4.23) and (4.24). The equilibrium constant providing a value for the ratio of rate constants is obtained from the ratio

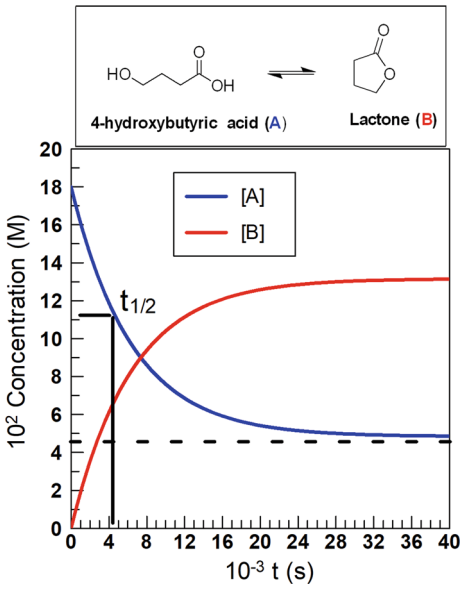


Fig. 4.4 Simulation of the variation of the concentrations of 4-hydroxybutyric acid (A) and of the lactone (B) with time until the equilibrium is reached. The reported values of the rate constants are 1.4×10^{-4} M and 4.2×10^{-5} M

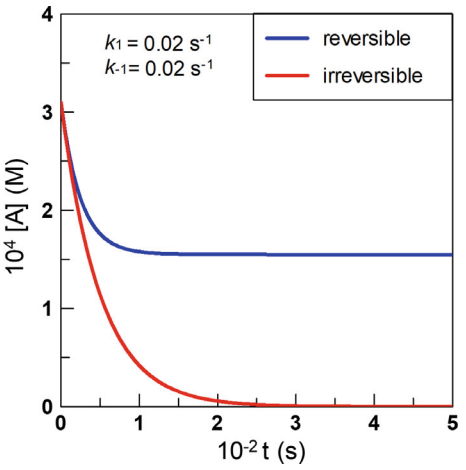


Fig. 4.5 Simulated data for concentration change of reactant A in a reversible reaction $A \Leftrightarrow B$ ($k_1 = k_{-1} = 0.02\ s^{-1}$) and an irreversible ($k_1 = 0.02\ s^{-1}$) first-order reaction

of the concentrations at "infinite time", that is, when the equilibrium is reached, Eq. (4.24).

It may be illustrative, for comparison purposes, to highlight the differences between the variation in the concentration of a reactant A with time in reversible and irreversible reactions with the same forward rate constants, Fig. 4.5. In the irreversible reaction, the concentration of the reactant decreases essentially down to zero at "infinite" time (i.e., once the reaction can be considered to be completed), which is not the case if the reaction is reversible, as substantial amounts of reactant remains in solution. In addition, the time required to reach the equilibrium condition is smaller than the time required for the irreversible reaction to be completed in spite that the values of the rate constants are the same for both cases.

4.4 Kinetic Control Versus Thermodynamic Control: Product Optimization

A combination of the mechanisms depicted in Sects. 4.3 and 4.4 is found in a number of organic reactions involving a reactant A that proceeds competitively and reversibly toward the products B and C, Eq. (4.30)

$$\begin{aligned} A &\underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} B \\ A &\underset{k_{-2}}{\overset{k_2}{\leftrightarrows}} C \end{aligned} \tag{4.30}$$

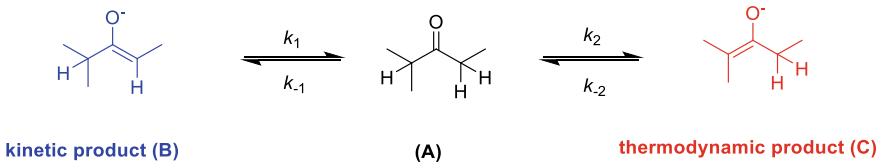


Fig. 4.6 Representative thermodynamic and kinetic products obtained for hydrogen abstraction from a carbonyl compound (*A*) in alkylation reactions

Examples of these reactions are Diels–Alder reaction, the kinetically and thermodynamically controlled synthesis of diasteroisomers which has extensively been studied or the formation of enolates from ketones (Fig. 4.6). In the latter case, the time evolution of product ratio is determined by two parameters, the rate constants and the thermodynamics.

A representative energy diagram for the reaction (4.30) is given in Fig. 4.7, where *B* is the thermodynamically stable product and *C* is the kinetically controlled one. At low temperatures, product *C* is favored because its relatively low activation energy speeds up the change from *A* to *B*. In contrast, for long reaction times, the thermodynamic product *B* tends to dominate because it is energetically more stable than the kinetic product *C*.

In the early reaction stages, the reverse routes in (4.30) may be neglected because the products *B* and *C* are not significantly accumulated to compete with the forward reaction, and in these cases, the amounts of *B* and *C* are simply associated with the forward reaction rate constants and the reaction is dominated by the kinetic control. For long-term reactions, the equilibrium concentrations of the products appear constant, $\mathrm{d}[B]/\mathrm{d}t = \mathrm{d}[C]/\mathrm{d}t = 0$ and the system reaches equilibrium. In this case, the product ratio depends only on the equilibrium constants K_1 ($= k_1/k_{-1}$) and K_2 ($= k_2/k_{-2}$), Eq. (4.31)

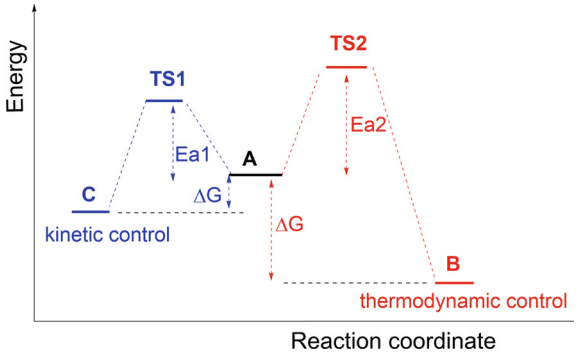


Fig. 4.7 Representative energy diagram for the competitively of conversion *A* in products *B* and *C*, in a competitive parallel first-order reactions. The reaction goes through the lower energy transition state at low temperatures to obtain the kinetic product whereas it goes through the higher energy transition state to obtain the thermodynamic product

$$\frac{[B]_{eq}}{[C]_{eq}} = \frac{[B]_{eq}}{[C]_{eq}} \frac{[A]_{eq}}{[A]_{eq}} = \frac{k_1 k_{-2}}{k_{-1} k_2} = \frac{K_1}{K_2} = K \tag{4.31}$$

Modification of experimental conditions (temperature, pH, ionic strength, molar ratio of reactants employed, pressure, solvent, etc.) can change the selectivity of the reaction and thus the final composition of the products in a reaction mixture. Different possibilities can be identified: (1) $k_1 \gg k_{-1}$ and $k_2 \gg k_{-2}$, reactions are practically irreversible parallel (Sect. 4.2), and the ratio $[B]/[C]$ is proportional to the ratio of k_1/k_2; (2) $k_1 \gg k_2$, product (B) ("kinetically favored" product) is obtained faster than product C due to its activation energy is the lower of the two; (3) $K \ll 1$ i.e., $k_{-1} \gg k_1$ and/or $k_2 \gg k_{-2}$, product (C) is preferred, is more stable, more energetically favorable product ("thermodynamically favored" product).

The reaction of O,O-diethyl 2,4-dinitrophenyl phosphate (**1**) with L-cysteine (cys) also provides a good example about kinetic and thermodynamic control[3] though in this case the reaction steps are irreversible. The following reaction mechanism was proposed for the reaction, comprising the formation of 1-(S-cysteine)2,4-dinitrobenzene in a first step, followed by a slow, second one to finally yield 1-(N-cysteine)2,4-dinitrobenzene (**3**).

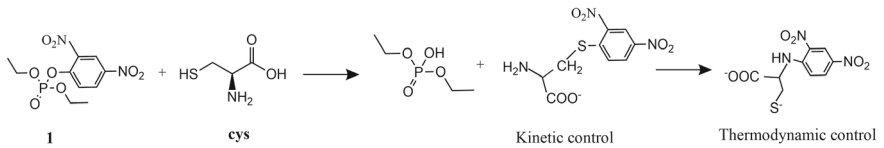


Figure 4.8 shows the sequence of ^{1}H NMR in the aromatic region of 1 upon reacting with Cys. It can be observed, at short reaction times (< 10 min), a decrease of the signal of **1** and a parallel increase of the signals 8.25, 8.80, and 9.31 ppm, attributed to the formation of 1-(S-cysteil)2,4-dinitrobenzene (**2**).

At longer reaction times, the NMR spectrum shows that the signals of the first product formed (kinetic control) disappear to form in a consecutive reaction, another more stable product 1-(N-cysteil)2,4-dinitrobenzene (thermodynamic control), with signals at 7.40, 8.63, and 9.40 ppm. This result is in accordance with those described in other S_NAr reactions of Cys, where the formed S-cysteil derivative (the kinetic control product), by an intramolecular or intermolecular second reaction, leads to the corresponding N-cysteil derivative (the thermodynamic control product) [4, 5].

A spectrophotometric kinetic study of the same reaction at three pH values (7.0, 7.5, and 8.0) by the increasing absorbance at 350 nm corresponding to product **2** was done. Considering the two pKa of cysteine ($pKa^S = 8.1$ and $pKa^N = 10.1$, therefore, at the studied pH, the form is that of thiolate ion. The nucleophilic rate coefficient for the thiolysis reaction ($k^S_N = 2.18\ s^{-1}\ M^{-1}$) was determined from the slope of linear plot of k_{obs} versus [thiolate].

The rapid increase in the absorbance was followed by a slow increase and a short displacement of the band. This reaction was zero order in amine and pH independent, in accordance with a non-catalyzed intramolecular rearrangement.

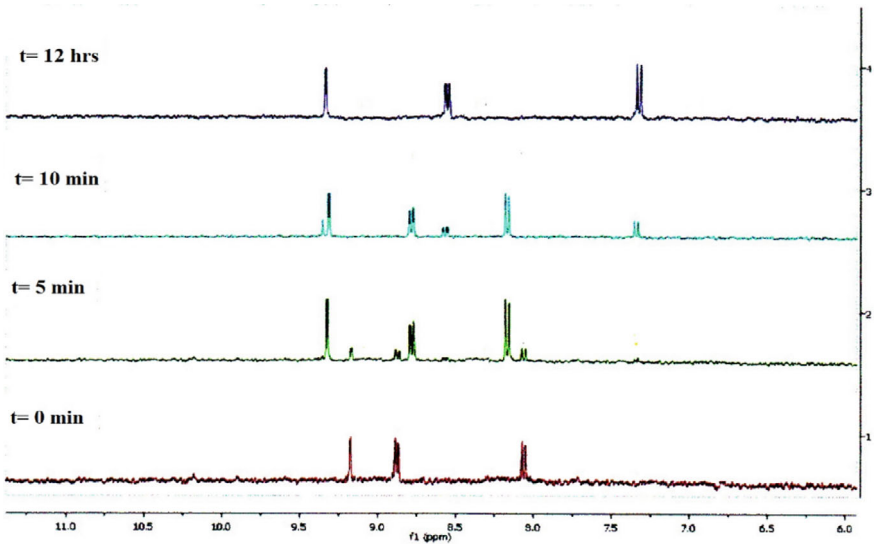


Fig. 4.8 Sequence of ^{1}H NMR spectra for the reaction of 1 with Cys in D2O–acetonitrile 60: 40% v/v. Reproduced from Ref. [3]

4.5 Consecutive Reactions

Some common reactions involve a sequence of consecutive first-order reactions such as that in (4.32). Such a mechanism assumes the formation of an intermediate species B which subsequently reacts to form the product C. The objective is to deduce a mathematical relationship for the overall rate of the reaction—the rate at which C is produced—in terms of the initial concentrations of A or final concentration of C.

$$A \xrightarrow{k_B} B \xrightarrow{k_C} C \tag{4.32}$$

For this system, the differential equations providing the changes in the concentration of all species are given by Eqs. (4.33)–(4.35).

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_B[A] \tag{4.33}$$

$$\frac{\mathrm{d}[B]}{\mathrm{d}t} = k_B[A] - k_C[B] \tag{4.34}$$

$$\frac{\mathrm{d}[C]}{\mathrm{d}t} = k_C[B] \tag{4.35}$$

If the initial concentration of A is denoted by $[A]_0$, then the integration of (4.33) is straightforward, yielding (4.36).

$$[A]_t = [A]_O e^{-k_B t} \tag{4.36}$$

Assuming that at $t = 0$, $[B]_0 = [C]_0 = 0$, substitution of Eq. (4.36) into (4.34), leads to Eq. (4.37)

$$\frac{d[B]}{dt} + k_C[B] = k_B[A]_0 e^{-k_B t}. \tag{4.37}$$

Equation (4.37) can be integrated after some mathematical manipulation by multiplying an exponential factor on both sides of equation as shown in Eqs. (4.38)–(4.40) to finally yield, after integration (assuming that $[B] = 0$ at $t = 0$), Eq. (4.41)

$$\frac{d[B]}{dt} e^{k_C t} + k_C[B] e^{k_C t} = k_B[A]_0 e^{(k_C - k_B)t} \tag{4.38}$$

$$\frac{d\left([B] e^{k_C t}\right)}{dt} = \frac{d[B]}{dt} e^{k_C t} + k_C[B] e^{k_C t} \tag{4.39}$$

$$d\left([B] e^{k_C t}\right) = k_B[A]_0 e^{(k_C - k_B)t} dt \tag{4.40}$$

$$[B] = k_B[A]_0 \frac{e^{-k_B t} - e^{-k_C t}}{k_C - k_B}. \tag{4.41}$$

The time-dependent change in the concentration of the C (Eq. 4.42) can be calculated by substituting the values of $[A]$ (Eq. 4.36) and $[B]$ (Eq. 4.41) into the mass balance equation $[A]_0 = [A] + [B] + [C]$, bearing in mind that at completion (i.e., infinite time), $[A]_{\text{inf}} = [B]_{\text{inf}} = 0$ and $[C]_{\text{inf}} = [A]_0$,

$$[C]_t = [C]_\infty \left(1 - e^{-k_B t} - \frac{k_B}{k_C - k_B}(e^{-k_B t} - e^{-k_C t}\right). \tag{4.42}$$

The concentration–time relationships for A, B, and C are illustrated in Figs. 4.9a–c for three representative reactions with $k_B > k_C$, $k_B \sim k_C$, and $k_B < k_C$. The variation of the concentration of the reactant A with time follows a first-order decay, while that of intermediate B increases, goes through a maximum and then decreases to zero. The position of the maximum concentration in $[B]$ with respect to the cross-over between the curves of A and C depends upon the relative values of the rates of formation and consumption. Obviously, if $k_B \gg k_C$, B is a relatively long-lived intermediate and an appreciable concentration may develop, meanwhile if $k_B \ll k_C$ the intermediate B is short-lived and may not even be detectable. The shape of the curve for $[C]$ is particularly interesting and is characteristic of this type of complex reaction. During a relatively short period of time (called *induction period*) the rate of formation of C is low; it then increases as the curve turns upward and the rate of formation of C is approximately linear, but then it slows down again as the curve levels off and approaches its asymptotic value as the reaction is completed (i.e., at "infinite" time).

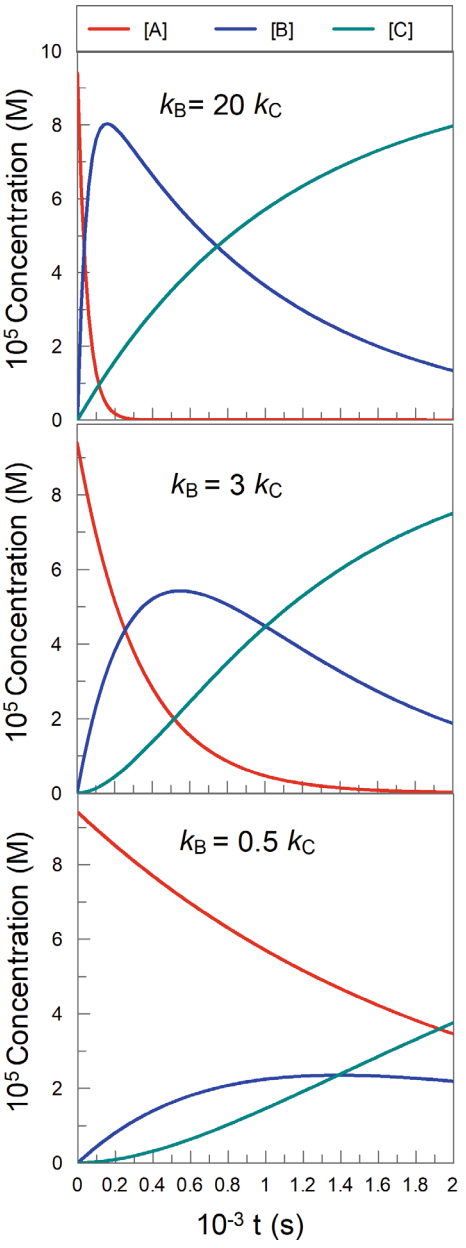


Fig. 4.9 Concentration changes of species involved in the consecutive reaction mechanism $A \xrightarrow{k_B} B \xrightarrow{k_C} C$, with $[A]_t = [A]_0$ and $[B]_t = [C]_t = 0$ at $t = 0$. Dependence of concentration of the reagent (A), intermediate (B) and product (C) on time for a first-order consecutive reaction where the ratio of the rate constants k_B and k_C is varied

The rate constant k_B can be determined from the linear variation of ln[A] with time, Eq. (4.36) as shown previously. Strictly, this type of reaction does not have an apparently simple overall rate law, and consequently we cannot determine a single overall experimental rate constant. Sometimes, the rate constants need to be obtained by fitting the concentration–time data with the aid of computer programs to find the best k_B and k_C values that best fit the experimental data. However, under some circumstances, simplifications can be made. Under some circumstances, the rate constant k_C can be obtained in a similar way from the logarithmic variation of [B] or [C] once the concentration of the reactant A can be considered negligible, (i.e., at times $t > 5\ t_{1/2}$ for the loss of A).

Similar mathematical treatments can be carried out when second-order reactions are involved [by employing, for example, the isolation method (Chap. 3)], and to reactions where an equilibrium is involved.

4.6 Simplification of Kinetic Schemes

As complex reactions become more complicated, their mathematical analysis becomes increasingly difficult and frequently can only be achieved by making simplifying assumptions. Three types of approximations are frequently used, (1) the method of isolation, (2) the pre-equilibrium and (3) the steady state. In addition, in many mechanisms it is possible to identify the steps that exert a significant effect on the overall rate of reaction. This step is usually termed as the rate-determining or rate-controlling step of the reaction, and when exists and can be identified, it allows a substantial simplification of the mechanisms and to obtain simple relationships for the kinetics of the overall reaction. It may be worth noting here that sometimes a distinction is made between rate-determining and rate-controlling steps. The former is generally used when the overall rate is equal to the rate of the first step in a consecutive mechanism, meanwhile the latter is employed when a later reaction is involved, the overall rate involving equilibrium constants as those considered in the next section. However, such a distinction is not recognized universally and hereafter we will use them indistinctly.

The isolation method was already described in Chap. 3. Here we will deal with the other ones. In many complex mechanisms, analytical solutions of the differential rate equations are only possible by using numerical methods leading to approximate solutions to the formulated reaction mechanism. Alternatively, we can make use of our chemical knowledge of the system to make approximations that approximately describe the kinetic scheme (approximations 2 and 3 mentioned before). The difficulties may arise in recognizing the chemical situation when the simplifying approximation needs to be used, and in appreciating the limitations of the method and under which conditions can, or cannot, be used. We will try to illustrate these approximations in the next sections, highlight the conditions under which they can be used and the conditions under which the approximate and exact solutions are in good agreement.

4.6.1 Rate-Determining Step

In a reaction mechanism just considered before (Eq. 4.32), if the intermediate B is converted very rapidly into products, the rate of the overall reaction is the rate of the formation of the intermediate, that is, $v = k_B[A]$ since as soon as B is formed it is transformed into C. The initial step is, therefore, the *rate-determining* step and the exact condition required for this two-step mechanism is $k_B \ll k_C$. Alternatively, if we assume that the rate of the second step is very small compared to that of the first step, $k_B \gg k_C$, the overall rate is $v = k_C[B]$ and the second step is now rate-determining.

In a similar way, if the mechanism involves several sequential first-order steps, Eq. (4.43), it becomes obvious that one of the reaction steps has the smallest rate constant, so that the rate of the overall reaction cannot be higher than that of this step, that is, the rate of product formation P is determined by the rate constant of this rate-determining step. The mechanism of the reaction simplifies to that of Eq. (4.44) and the differential equation describing the formation of the product P given by Eq. (4.45)

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \xrightarrow{k_3} D \xrightarrow{k_4} E \xrightarrow{k_5} P \tag{4.43}$$

$$B \xrightarrow{k_2} P \text{ (when } k_2 \ll k_1, k_3, k_4, k_5) \tag{4.44}$$

$$\frac{\mathrm{d[P]}}{\mathrm{d}t} \approx k_2[B]. \tag{4.45}$$

Identification of the rate-determining step of a complex mechanism helps, therefore, to simplify considerably the reaction scheme as only the previous steps need to be considered to obtain the rate law.

4.6.2 Pre-equilibrium Approximation

The complex reaction scheme given by Eq. (4.46) is representative: one which gives rise to an overall rate law, expressed as a differential equation, with too many variables to allow an exact solution easily.

$$A \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} B \xrightarrow{k_2} P \tag{4.46}$$

Similar mechanisms are those shown in Eqs. (4.47) and (4.48), where a second-order step is involved, in all cases, the reactant A reacts via a reactive intermediate B that reacts with C to finally yield the product P.

$$A + B \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} C \xrightarrow{k_2} P \tag{4.47}$$

$$A \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} B + C \xrightarrow{k_2} P \tag{4.48}$$

Let´s consider the scheme (4.46) for simplicity. The differential equations for the various species are given by Eqs. (4.49)–(4.51)

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1[A] - k_{-1}[B] \tag{4.49}$$

$$-\frac{\mathrm{d[B]}}{\mathrm{d}t} = k_1[A] - k_{-1}[B] - k_2[B] \tag{4.50}$$

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[B]. \tag{4.51}$$

These equations cannot be solved analytically and, in general, one should use numerical methods to solve them. However, the mathematical treatment can be considerably simplified if we make some assumptions when comparing the rates of the backward reaction $B \rightarrow A$ and the second forward reaction $B \rightarrow P$.

The pre-equilibrium (or fast equilibrium) approximation assumes that the reactive intermediate returns to reactants much faster than it proceeds to give product, that is, when $k_{-1}\,[B] \gg k_2[B]$. In this case, Eq. (4.52) holds

$$k_1[A] \approx k_{-1}[B] \tag{4.52}$$

and therefore Eq. (4.51) becomes (4.53)

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[B] = k_2K[A] = k[A]. \tag{4.53}$$

That is, the formation of P is first order with an experimentally obtained first-order rate constant $k = (k_1k_2)/k_{-1} = k_2K$, where K is the equilibrium constant for the pre-equilibrium.

In the case of the reaction (4.47), the differential rate equation becomes Eq. (4.54)

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[C] = k_2K[A][B] = k[A][B]. \tag{4.54}$$

So that the formation of C is second order with an experimentally obtained second-order rate constant $k = (k_1k_2)/k_{-1} = k_2K$, where K is the equilibrium constant for the pre-equilibrium.

Examples of reactions where this approximation holds are the reaction that takes place between arenediazonium ions (ArN_2^+) and methyl gallate (MG), Fig. 4.1,

the nucleophilic substitution reactions of some alkyl halides, Eq. (4.55), where a rapid equilibrium is established in a first reversible step followed by a second, rate-determining, bimolecular step where the nucleophile attacks the carbocation to yield the final substitution product R-Nuc.

$$\begin{aligned} RX &\rightleftharpoons R^+ + \mathrm{X}^-(k_1,\ k_{-1}) \\ R^+ + \mathrm{Nuc}:^- &\xrightarrow{k_2} R\text{ - Nuc} \end{aligned} \tag{4.55}$$

A third example is found when considering the reaction of acetone with chlorine in aqueous acid solution, whose reaction scheme is depicted in (4.56)

$$\begin{aligned} \mathrm{CH_3-CO-CH_3 + H_3O^+} &\underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{CH_3-COH=CH_2} \\ \mathrm{CH_3-COH=CH_2 + Cl_2} &\xrightarrow[\mathrm{H_2O}]{k_2} \mathrm{CH_3-CO-CH_2Cl.} \end{aligned} \tag{4.56}$$

4.6.3 *Quasi-Steady-State Approximation (QSSA)*

The quasi-steady-state approximation (QSSA) has been first proposed by the chemist Max Ernst August Bodenstein in 1913. The technique simplifies the analysis of complex reaction mechanisms by assuming that the concentration of the intermediate is very low and constant during a large fraction of the total reaction time. Without employing this approach, many kinetic problems would be intractable.

We will illustrate the application of the approach to the reaction (4.46). The technique assumes that once the intermediate B is formed, it reacts almost instantaneously into the product P, so that its concentration remains low and does not change much during most part of the course of the reaction time compared to the concentration changes of the other species, Hence, the steady-state approximation presumes that $k_{-1} \ll k_2$. In this case, the production of the intermediate is "slow" compared with its decomposition in the second step and therefore the overall rate of the reaction is limited by the rate of this rate-determining step.

Under this assumption, the differential equation for the change in the concentration of B is given by Eq. (4.57), from where we can obtain an approximate mathematical relationship for $[B]$ in terms of the initial concentration of the reactant, Eq. (4.58).

$$\frac{\mathrm{d}[B]}{\mathrm{d}t} = k_1[A] - k_{-1}[B] + k_2[B] \approx 0 \tag{4.57}$$

$$[B] = \frac{k_1[A]}{k_{-1} + k_2} \tag{4.58}$$

The rate of production of P is thus given by Eq. (4.59), and the overall rate constant is given by Eq. (4.60).

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[B] = \frac{k_2 k_1[A]}{k_{-1} + k_2} \tag{4.59}$$

$$k = \frac{k_2 k_1}{k_{-1} + k_2} \tag{4.60}$$

Two limiting situations can be depicted. When $k_{-1} \gg k_2$, then the overall rate constant becomes $k = k_2 k_1 / k_{-1} = k_2 K$, which is the same equation as that obtained when considering the pre-equilibrium condition. When $k_{-1} \ll k_2$, the overall rate constant is $k = k_1$, that is, the reaction is of first order.

If the complex reaction given by (4.48) is the representative one, application of QSSA leads to Eq. (4.61) for the intermediate B

$$-\frac{\mathrm{d[B]}}{\mathrm{d}t} = k_{-1}[B] - k_1[A] + k_2[B][C] = 0 \Rightarrow [B] = \frac{k_1[A]}{k_{-1} + k_2[C]} \tag{4.61}$$

The rate of disappearance of reactants A and C and that for the formation of P is given by Eq. (4.62), and the overall rate constant is given by Eq. (4.63).

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = -\frac{\mathrm{d}[C]}{\mathrm{d}t} = \frac{\mathrm{d}[P]}{\mathrm{d}t} = \frac{k_1 k_2[A][C]}{k_{-1} + k_2[C]} \tag{4.62}$$

$$k = \frac{k_1 k_2[C]}{k_{-1} + k_2[C]} \tag{4.63}$$

Note that, in contrast with rate laws obtained in Sects. 4.3 and 4.4, Eq. (4.63) is not simple because the concentration of one of the reactants (C) appears in both numerator and denominator of the overall rate equation. This reaction type provides us with a further example of a very common technique in kinetics. If we make experimentally the concentration of one of the reactants in large excess compared to that of the other, we can consider its concentration effectively constant throughout the reaction and the second-order reaction may be converted into a *pseudo* first-order reaction. For instance, if in the present case $[C]$ is in large excess with respect to $[A]$ so that it can be considered constant, we can monitor the decrease in $[A]$ or the increase in $[P]$ with time to obtain the *pseudo* first-order rate constant given by Eq. (4.63).

We can go further because Eq. (4.64) can be derived after rearrangement of Eq. (4.63)

$$1/k = (k_{-1}/k_1 k_2)(1/C) + 1/k_1. \tag{4.64}$$

Thus, if the reaction is studied at several constant known concentrations of C, and we can obtain the overall *pseudo* first-order rate constant k, in each case, a plot of 1/

k versus $1/[C]$ should be linear with a gradient $g = k_{-1}/k_1k_2$ and an intercept IN $= 1/k_1$. We are able to estimate the value for the forward rate constant but still unable to determine the individual values of k_{-1} and k_2.

The following tips can be useful for obtaining rate expressions of complex reaction schemes by employing QSSA. It is important to bear in mind that QSSA only works for reactive intermediates:

1. Identify all reactants, intermediates, and products.
2. Formulate a steady-state equation for each reactive intermediate
3. Solve the set of equations to obtain the expression of the concentration of each intermediate in terms of concentrations of reactants and/or products. If the equations depend on more than one reactive intermediate and share some common terms, look for combinations (sums or differences) of the equations that simplify the set of mathematical equations. The search of these combinations is often a tricky part of the mathematical treatment and frequently needs some practice.
4. Write down the overall rate equation in terms of a reactant or product whose concentration is experimentally accessible. The concentrations of reactive intermediates must not appear in the final expression for the rate law.
5. When very complex rate expressions are obtained, one should look for simplifications or approximations on the basis of the experimental conditions (for example, using the isolation method).

4.6.4 *Testing the Accuracy of the Quasi-Steady-State Approximation*

Let us consider, for simplicity, the easiest complex reaction comprising consecutive steps (4.32), and the exact mathematical relationships for the variation of $[A]$, $[B]$, and $[C]$ with time are given by Eqs. (4.36), (4.41) and (4.42), respectively. For the sake of convenience, the reaction scheme and the equations are reproduced below (Eqs. 4.65–4.68).

$$A \xrightarrow{k_B} B \xrightarrow{k_C} C \tag{4.65}$$

$$[A]_t = [A]_0 \mathrm{e}^{-k_B t} \tag{4.66}$$

$$[B] = k_B [A]_0 \frac{\mathrm{e}^{-k_B t} - \mathrm{e}^{-k_C t}}{k_C - k_B} \tag{4.67}$$

$$[C]_t = [C]_\infty \left(1 - \mathrm{e}^{-k_B t} - \frac{k_B}{k_C - k_B} (\mathrm{e}^{-k_B t} - \mathrm{e}^{-k_C t} \right) \tag{4.68}$$

The QSSA assumes that, after an initial brief period, the concentration of B is very low and achieves a steady state with its rate of formation and disappearance just

balance, so that we can consider the variation of the concentration of B with time is approximately constant, i.e., $d[B]/dt = 0$. Under this assumption, the corresponding rate equations are given by Eqs. (4.69) and (4.70)

$$[B] = \frac{k_B}{k_C}[A]_0 e^{-k_1 t} \tag{4.69}$$

$$[C]_t = [C]_\infty \left(1 - e^{-k_B t}\right). \tag{4.70}$$

Equation (4.69) expresses the variation of the concentration of C with time, and so it describes the rate of the overall reaction. We can now compare Eqs. (4.69) and (4.70), which have been obtained under the assumption of the QSSA with Eqs. (4.67) and (4.68), respectively, obtained after integration of the corresponding differential equations without making any assumption, i.e., they are exact.

We find that when $k_C \gg k_B$ Eq. (4.67) gives Eq. (4.69) and when $k_C \gg k_B$ and $k_B t \gg 1$, Eq. 4.68 gives Eq. (4.70). This, then, are the conditions under which QSSA introduce no serious errors. The first of these conditions could have been predicted intuitively by consideration of the reaction scheme (4.65) when B is a very reactive, short-lived intermediate with $k_C \gg k_B$. The second condition ($k_B t \gg 1$) just indicates that the approximation is not valid in the very early part of the reaction (induction period) while the concentration of B is being built up from zero to its steady-state concentration.

The question that immediately arises is how much higher the value of the rate constant k_C has to be with respect to that of k_B to fulfill the condition $k_C \gg k_B$. Figure 4.10 compares the concentration of intermediate determined by employing QSSA (Eq. 4.69) with the equation derived without employing QSSA (Eq. 4.67) for different values of the rate constants. The exact and approximate curves cross each other exactly at the maximum.

QSSA provides a good description of the variation of the concentration of B when $k_C > 25k_B$ but significant differences between the variations of the concentration of the intermediate determined by QSSA and by employing the exact mathematical expression are detected when $k_C \leq 10k_B$.

In complex reactions comprising coupled first-order reactions as those in Eq. (4.67) there is generally no need to use QSSA because such reaction schemes can be solved exactly and the approximation involves the disadvantage that not all the rate constants can be determined (for example, k_C in the mentioned reaction scheme, Eq. 4.67). But if there are strong reasons to believe that the intermediates are short-lived in a reaction that is complex enough to prevent exact analysis, the QSSA is invaluable.

The reader must bear in mind that the assumptions of the steady-state approximation may not be valid in a particular case, providing a poor description of the reaction in spite of simplifying the reaction scheme on the paper. Failure of the rate equations derived from the QSSA provides enough evidence to presume that the steady-state approximation is not justified or that the hypothetical reaction scheme is, in some

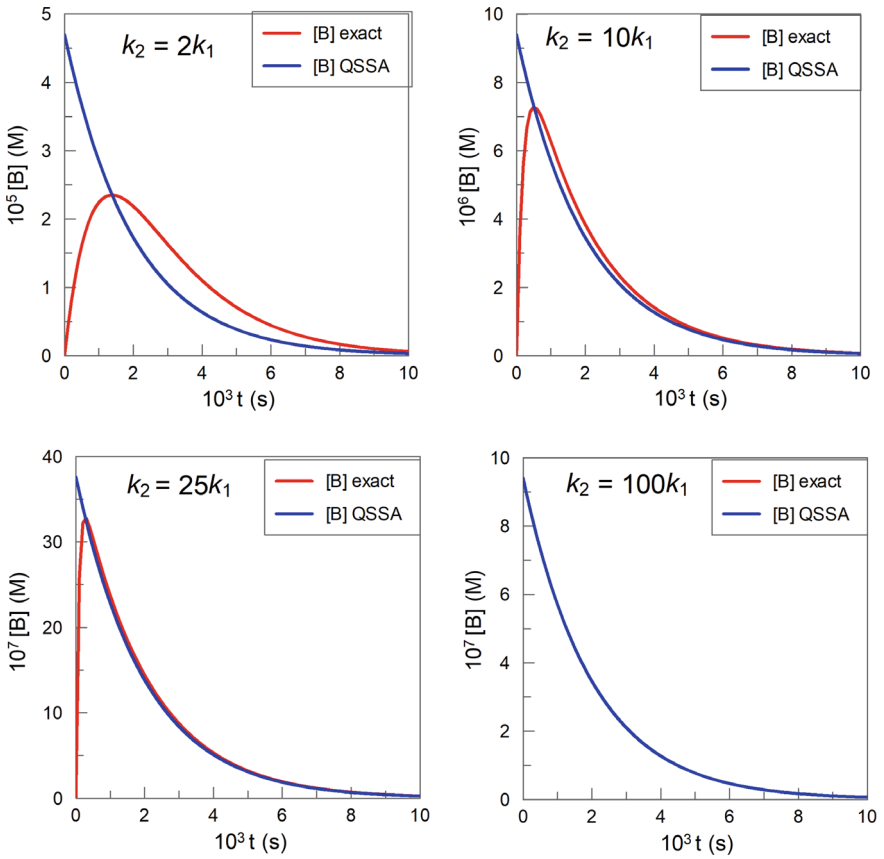


Fig. 4.10 Simulation of the variations of the concentration of the intermediate B with time determined by employing the steady-state approximation (Eq. 4.69) and by employing the exact equation (Eq. 4.43) as a function of the relative k_2/k_1 values for a first-order consecutive reaction $A \rightarrow B \rightarrow C$. Simulation was done by employing the following data: $[A]_0 = 9.4 \times 10^{-5}$M, k_1 (A $\rightarrow$ B) $= 5 \times 10^{-4}$ s^{-1})

other respect, inadequate to describe the reaction. Conversely, a good kinetic description of the experimental results can be taken as evidence of their validity. It should be, however, desirable to have independent evidence of the formation of reactive intermediates (for example, by employing spectroscopic or electrochemical techniques) or by using chemical traps.

4.7 Case Study

The case study presented to illustrate the application of the approaches discussed above to simplify complex reaction mechanisms is related to the spontaneous decomposition of arenediazonium (ArN_2^+) ions in binary alcohol-water mixtures, Eq. (4.71). In these reactions the solvent molecules act as nucleophiles (Nu = H_2O, ROH) leading to the formation of the corresponding phenols (ArOH) and ethers (ArOR). The overall reaction is given by Eq. (4.71)

$$\begin{aligned} &Ar\text{-}N_2^+ + H_2O \rightarrow \ Ar\text{-}OH + H^+ \\ &Ar\text{-}N_2^+ + ROH \rightarrow \ Ar\text{-}OR + H^+. \end{aligned} \tag{4.71}$$

The main experimental observations are summarized below:

(A) The loss of $ArN_2^+X^-$ was monitored spectrophotometrically at a suitable wavelength and the corresponding observed rate constant k_{obs} was determined.

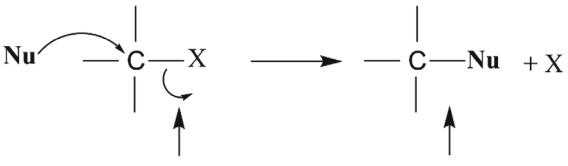
B) Reaction products were identified by HPLC by employing authentic samples according to their retention times. The formation of dediazoniation products (coming from reaction with nucleophiles present or added at concentrations much higher than those of the reactant) were monitored by and the corresponding rate constants were determined. It turned out that these rate constants were equal, within experimental errors, to that determined for the loss of ArN_2^+. So, the rate constant for product formation and reactive loss were equal to each other.

(C) The effects of nucleophile concentration (present in the solvent or added at concentrations much higher than those of the reactant) were determined by monitoring the loss of ArN_2^+ and product formation. No significant changes in the measured rate constants were detected, concluding that the nucleophile concentration only affects the product distribution and not the rate of the reaction.

Kinetic analysis of the reaction: potential mechanisms at a glance.

A simple analysis of the reaction allows one to propose three alternative pathways for the reaction:

(1) *One-step mechanism*, in which an existing bond is broken at the same time that a new bond is formed. This mechanism can be illustrated schematically as follows:



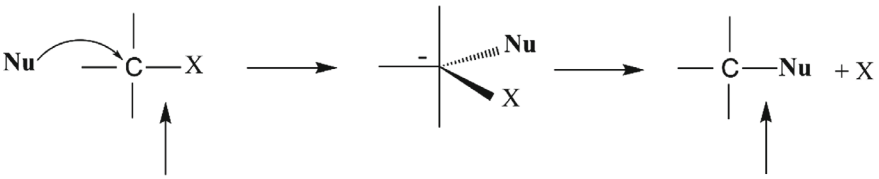
this bond is broken... **as** ... this bond is formed

This potential mechanism assumes, therefore, a bimolecular process and the rate of the reaction is given by Eq. (4.72):

$$r = k\left[ArN_2^+\right][Nu]. \tag{4.72}$$

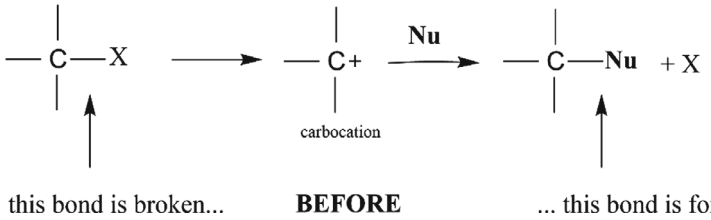
Given that the nucleophile concentration (solvent molecules or added on purpose) is always much higher than that of ArN_2^+ (isolation method), its concentration can be assumed to remain essentially constant and so the measured (or observed) rate constant $k_{obs} = k[Nu]$, that is, its value should depend on the concentration of the nucleophile, which is not observed experimentally (experimental observation C) and therefore the mechanism should be discarded.

(2) *Two-step mechanism*, in which the existing bond is broken after the new bond is formed. This mechanism can be illustrated schematically as follows:



This potential mechanism must be discarded immediately because it assumes that there will be 10 electrons around the C atom violating the octet rule.

(3) **Two-step mechanism**. The existing bond is broken before the new bond is formed. This mechanism implies, therefore, the existence of a transient ionic intermediate that further reactants with the nucleophiles present in solution. This mechanism can be illustrated schematically as follows:



For the particular case of arenediazonium ions in solution, the mechanism can be illustrated as in Eq. (4.73). The reader must note that all compounds are solvated and that N_2 gas is formed being, therefore, removed from solution.

$$Ar\text{-}N_2^+ \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} Ar^+ + N_2(g, \uparrow) \xrightarrow[Nu]{k_3} Ar\text{-}Nu \tag{4.73}$$

The above mechanism provides an opportunity to obtain the corresponding rate law and analyze its kinetic implications on the basis of the pre-equilibrium and steady-state approximations.

Pre-equilibrium. This approach implies that as soon as the intermediate is formed, it turns back to the parent reactants, so that the reverse step in the equilibrium is much faster than the second irreversible step, Eqs. (4.74)–(4.75).

$$\begin{aligned} &\text{Ar - N}_2^+ \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \text{Ar}^+ + \text{N}_2(g) \\ &\text{Ar}^+ + \text{Nu} \rightarrow^{k_2} \text{Ar - Nu} \end{aligned} \tag{4.74}$$

$$K = \frac{k_1}{k_{-1}} = \frac{[\text{Ar}^+]}{[\text{Ar - X}]} \Rightarrow [\text{Ar}^+] = K[\text{Ar - X}] \tag{4.75}$$

$$r = \frac{\text{d[Ar - Nu]}}{\text{d}t} = k_2[\text{Ar}^+][\text{Nu}] = k_2\text{K[Ar - X][Nu]} \tag{4.76}$$

And the observed rate constant is given by Eq. (4.76)

$$k_{obs} = k_2 K[\text{Nu}] \tag{4.77}$$

By employing this approach, we came to an equation for the rate law that predicts a dependence of the observed rate constant with the nucleophile concentration, which is not consistent with the experimental observations, and therefore must be discarded.

Steady state. This approach assumes that the concentration of the intermediate is very low and constant throughout the reaction (except at very early and final stages). This implies that there is no accumulation of the intermediate so that the second irreversible step is much faster than the backward reaction toward the reactants.

Under these conditions, the steady-state condition is given by Eq. (4.78)

$$\frac{\text{d}\left[\text{Ar}^+\right]}{\text{d}t} = k_1\left[\text{Ar - N}_2^+\right] - k_{-1}[\text{Ar}^+] - k_2[\text{Ar}^+][\text{Nu}] \approx 0. \tag{4.78}$$

And the rate of product formation is given by

$$\frac{\text{d[Ar - Nu]}}{\text{d}t} = \frac{k_1 k_2\left[\text{Ar - N}_2^+\right][\text{Nu}]}{k_{-1} + k_2[\text{Nu}]} \tag{4.79}$$

$$k_{\text{obs}} = \frac{k_1 k_2[\text{Nu}]}{k_{-1} + k_2[\text{Nu}]}. \tag{4.80}$$

This is a complex rate equation without a simple reaction order, but can be simplified considering limiting situations:

(a) If $k_{-1} \gg k_2[\text{Nu}]$, then the observed rate constant simplifies to Eq. (4.81)

$$k_{obs} = \frac{k_1 k_2 [Nu]}{k_{-1}} = k_2 K [Nu], \tag{4.81}$$

which predicts a second-order reaction, order 1 in ArN_2^+ and order 1 in [Nu]. Note that eq. is exactly the same as that obtained by employing the pre-equilibrium approach (Eq. 4.62), which predicts that the rate constant should increase upon increasing the concentration of nucleophiles, which is not observed experimentally, and therefore this approach is inconsistent with experimental results and must be discarded.

(b) If $k_{-1} \ll k_2[Nu]$, then the observed rate constant simplifies to Eq. (4.82)

$$k_{obs} = k_1 \tag{4.82}$$

which indicates that the reaction is order 1 and predicts that the observed rate constant does not depend on nucleophile concentration, being consistent, therefore, with the experimental observations.

So, the most probable mechanism, according the experimental evidence provided, is that in which the arenediazonium ion decomposes, in a slow step, to yield a reactive carbocation that further reacts with the nucleophiles available (S_N1 mechanism). Since the ions are solvated, nucleophiles must be those present in their solvation sphere [6].

Further evidence supporting such a mechanism was found from analysis of the variations of the volumes of activation and from the effects of temperature on the reaction, which provided the activation parameters for the reaction. Enthalpies of activation were high and entropies of activation were low, suggesting a transition state that has undergone substantial breakdown with little compensating bond formation as indicated in the proposed mechanism [7].

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Chapter 5
Reactions in Solution: Diffusion-Controlled and Activation-Controlled Reactions



5.1 Introduction

It has long been known that UV/Vis absorption spectra may be influenced by the phase (gas or liquid) and that the solvent can bring about a change in the position, intensity, and shape of the absorption band, a phenomena coined as solvatochromism. It has also long been known that solvents exert considerable influence on the rate of the reactions, making them to play central role in mediating and modulating the rates of many chemical reactions, being a central point in many physical-organic chemistry studies. After the first observation that solvents may change the reaction rates, the study of the solvent effects was for some years the work of physical-organic chemists, and remarkable research work was carried out by Hammett, Menschutkin, Hughes and Ingold, Winstein, Grunwald, and Reichardt, among others [1–4]. Research on solvents and solution chemistry has become a topic of interest in recent years because many of the solvents commonly used in laboratories and in the chemical industry are considered as unsafe for reasons of environmental protection. On the list of damaging chemicals, some organic solvents rank highly because they are often used in huge amounts and because they are volatile liquids that are difficult to contain.

Unfortunately, there is no suitable formal method for choosing a solvent to carry out particular chemical reactions. A judicious choice must be guided by the information available and the purpose of the work, i.e., if the solvent is employed as a reaction medium, either for synthesis or for a mechanistic study, or simply as a medium for recrystallization. In general, the chosen solvent must have a reasonable liquid range in which its properties remain basically constant, must be easy to purify, must be relatively easy separable from the reaction products (evaporation, low solubility, etc.), and must be inert to reactants and products, that is, cannot lead to side reactions destroying either the reactants or products. Frequently, some other considerations related to the workplace, environmental hazards and costs also need to be taken into consideration.

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Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_5

Solvents should not be considered a macroscopic continuum characterized only by physical constants such as density, dielectric constant, refractive index, but as a discontinue media which consists of individual, mutually interacting solvent molecules. The extent of these interactions is the balance of intermolecular forces, making some solvents to be with a pronounced internal structure (e.g. water) but others with a weak internal structure (e.g. hydrocarbons) in which the interaction between the solvent molecules is small. The interactions between species in solvents (and in solutions) are, for one side, strong enough to be analyzed with "simple" assumptions of the kinetic theory of gases, yet they are weak enough to be treated by the laws of solid-state physics.

Thus, solvents cannot be considered either as indifferent medium, where solutes are dissolved and where they diffuse to distribute themselves evenly and randomly, nor to possess an ordered structure resembling a crystal lattice. Most probably, and as a consequence of the complexity of the intermolecular interactions, the structure of liquids—in contrast to that of gases and solids—is the least-known of all aggregation states. Therefore, the experimental and theoretical examination of the structure of liquids is among the most difficult tasks of physical chemistry. For this reason, we will make no attempt to analyze the structure of solvents but will limit ourselves to briefly discuss their role in solution chemistry.

The word "solvent" implies that the component of the solution so described is in large excess with respect to other components. Of all solvents, water is of course the cheapest and closest to hand and so will be the solvent of choice for many applications, but many other are coming into use in the laboratory and industry, including organic solvents (acetone, alcohol, hydrocarbons), molten salts (ionic liquids), and molecular and metal liquids. The list of solvents available for preparative and analytical purposes is now long and growing and they can be classified according to their physicochemical properties at different levels. The most striking differences among liquid solvents are observed between molecular liquids, ionic liquids, and atomic (metal) liquids. They can be considered as extreme types and they are usually represented in the three vertices of a triangle. Intermediate types of solvents, including their mixtures, can then be located along the edges or within the triangle, Fig. 5.1

The assumption of intermolecular forces between solvent and solute led, on the other hand, to old principle that "similar dissolves similar" (*similia similibus solvuntur*). In many cases, the presence of similar functional groups in the molecules is enough to be "similar". In general, a compound dissolves better in a solvent with similar functional groups that in one of a completely different nature. Thus, a proper choice of a solvent, based on the knowledge of its chemical reactivity helps to avoid undesired reactions between solute and solvent. For example, one would not expect to be able to prepare a strictly anhydrous salt using water as solvent, but solvents such as sulfur dioxide or dry diethyl ether should rather need to be employed.

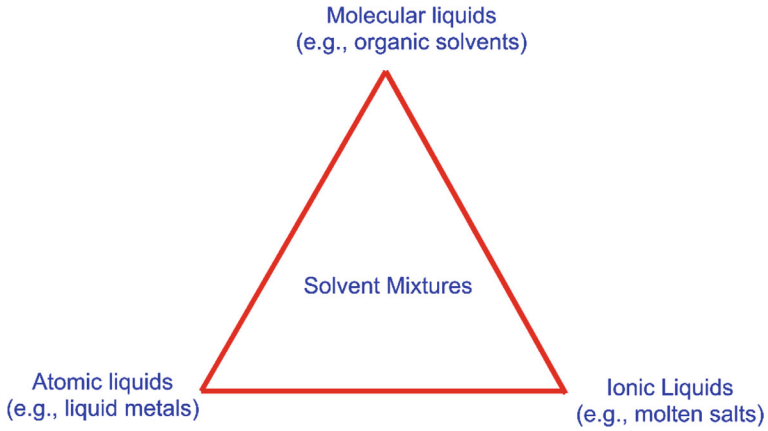


Fig. 5.1 Ternary diagram for classification of solvents according to their characteristic chemical bonds. Adapted from Reichardt [1]

5.2 Classification of Solvents

Based on physical and chemical properties of the molecular solvents, they can be classified as (Table 5.1):

I. Inert (unreactive, with low (or zero) dipole moment and polarizability.
II. Inert-polarizable, aromatics, polyhalogenated hydrocarbons).
III. Protogenic (hydrogen-bonding proton acceptors, HBA).
IV. Protophylic (hydrogen-bonding proton donators, HBD).
V. Amphiprotic (having both HBA and HBD properties).
VI. Polar-aprotic (with substantial dipole moment).

This classification may be complicated by considering the ability of solvent molecules to act as acceptors of donors of electrons (i.e., Lewis acids).

Solvents can also be classified according to their melting and boiling points, vapor pressure, refractive index, density, viscosity, surface tension, etc. The interested

Table 5.1 Examples of some molecular solvents

Solvent	Examples
Inert	Aliphatic hydrocarbons, fluorocarbons
Inert-polarizable	Benzene, tetrachloromethane, carbon disulfide
Protogenic	Trichloromethane
Protophilic	Tertiary amines
Amphiprotic	Water, alcohols, ammonia
Dipolar-aprotic	Acetonitrile, dimethyl sulfoxide, dimethylformamide

reader can find a compilation of physical data for common organic solvents in any handbook of chemistry and specialized books [1, 4, 5].

5.3 Solvent Effects on the Rates of Homogeneous Chemical Reactions

A change of solvent can considerably change both the rate and order of a homogeneous chemical reaction. Solvents can affect the reaction rates of homogeneous chemical reactions in two main ways: 1) through dynamic solvent effects, where very fast reactions are involved, and 2) through static, or equilibrium, solvent effects. In the first case, solvent reorientation may become rate-limiting and the rate constants will depend on the solvent dynamics, varying with friction (density, viscosity, internal pressure).

A judicious choice of solvent can be, thus, decisive in accelerating or decelerating a chemical reaction, and this can be of great practical importance for researchers when investigating reaction mechanisms. In some extreme cases, rate accelerations by a factor of up to ca. 10^9 can be achieved solely by a solvent change [1].

The dependence of the reaction rate on the medium can, in principle, be approached from two points of view:

1. Comparison of the rates of reaction in the gas phase and in solution.
2. Comparison of the rates of reaction in different solvents.

Because of the rather limited number of reactions that have been studied in both gas phase and solution, comparison of solvent effects has been limited to compare reaction rates in different solvents (case 2) and the reactions that have been mostly studied are those that follow ionic mechanisms, where considerable charge separation or dispersion takes place during the activation process.

5.4 Dynamic or Frictional Effects: Diffusion-Controlled and Activation-Controlled Reactions

In solution the molecules are not moving freely, as in the gas phase. The motion consists of a series of discontinuous displacements (Brownian motion), each which is followed by a residence time in the solvent cage where the molecule rapidly collides with the molecules that make up the solvent cage.

From a phenomenological point of view, we can consider the process of a bimolecular reaction where two solvated reactants are brought together. When two solvated molecules approach one another closely, the two cavities may merge, forming a single larger cavity (solvent cage) and liberating some of the solvent. When in the solvent cage, the two reacting molecules held in close proximity for some time by

surrounding solvent molecules before they can separate back to the solvated reactants or undergo further reaction if the possess enough energy to overcome the energy barrier of the reaction. The reactants in the solvent cage tend to collide a large number of times before they separate back. Such collision events are called "encounters", and for non-interacting molecules, the number of encounters is determined by the random Brownian movements of the molecules through the viscous medium, so that their diffusion may be treated employing Fick's laws of diffusion.

Suppose a reaction between two reactants A and B in solution, which approach by diffusion to form an encounter pair (AB) as indicated Eq. 5.1 and as shown schematically in Fig. 5.2. The encounter pair can break up without reaction or it can go on to form products P. Consider the case that the rate of formation of an encounter pair AB is first-order in each of the reactants A and B.

$$A_{\text{solv}} + B_{\text{solv}} \underset{k_{-d}}{\overset{k_d}{\rightleftharpoons}} (AB)_{\text{solv}} \xrightarrow{k_r} P, \tag{5.1}$$

where $(AB)_{\text{solv}}$ stands for the encounter pair, and P stands for the products. Applying the steady-state approximation to the encounter pair

$$\frac{\mathrm{d}[(AB)]}{\mathrm{d}t} = k_d[A][B] - (k_{-d} + k_r)[(AB)] = 0 \tag{5.2}$$

the rate of the reaction is given by Eq. 5.3.

$$\text{rate} = k_r[(AB)] = \frac{k_d k_r}{k_{-d} + k_r}[A][B]. \tag{5.3}$$

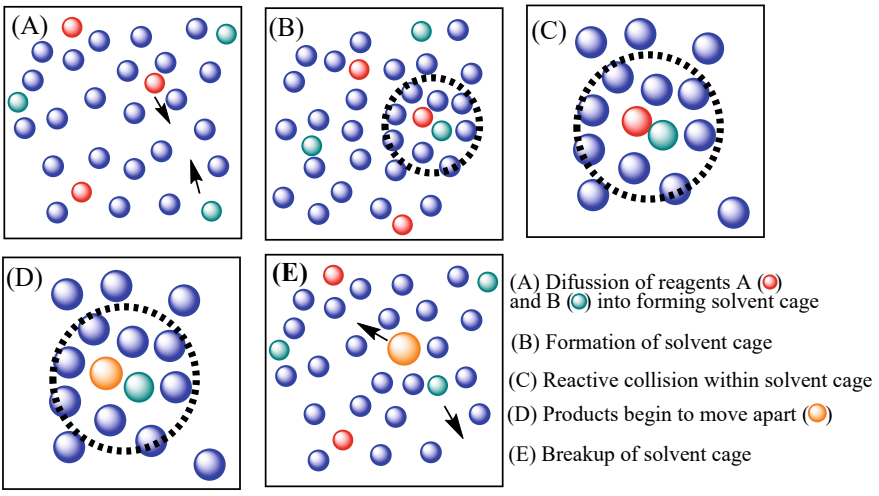


Fig. 5.2 Schematic representation of some of the processes involved in reaction in solution

The second-order rate constant (k_2) is given by Eq. 5.4

$$k_2 = \frac{k_d k_r}{k_{-d} + k_r}. \tag{5.4}$$

On the basis of the relative contributions of the rate constants in the denominator in Eq. 5.4, two limiting conditions can be described:

1. If $k_r >> k_{-d}$, the separation of A and B is not easy (for example, the reaction has a low activation energy or in a viscous solvent). This implies that the activation energy for the chemical reaction is much lower than that for the diffusion. In this case, the second-order rate constant becomes equal to the rate constant of diffusion, $k_2 = k_d$, so that the rate-determining step of the reaction is the diffusion process, and the reaction is said to be diffusion-controlled.
2. If $k_r << k_{-d}$, the rate of the reaction is much lower than the rate of the decomposition of the encounter pair back to the parent reagents, so that the second-order rate constant is now equal to $k_2 = Kk_r$, so that slow step of the reaction is the chemical reaction of the encounter pairs and the rate of the reaction is controlled by the activation process, that is, the reaction is activation-controlled.

Examples of diffusion-controlled reactions are those involving the recombination of atoms and radicals, acid–base and photochemical reactions. When the chemical reaction is diffusion controlled, this yield of the products depends on the solvent viscosity of the solvent and, as we will see in the next section, the diffusion constants present an inverse relation with the viscosity. This cage effect becomes more significant when the solvents have a high viscosity, a phenomena known as the Franck–Rabinowitch effect [6].

5.4.1 Diffusion Constants of Reactants: Diffusion-Controlled Versus Activation-Controlled Reactions

A simple mathematical model, grounded on the transition state theory (TST) and diffusion laws, permits to analyze the conditions required to distinguish between kinetically controlled and diffusion-controlled reactive systems based on molecular and solvent properties.

We will start by employing the Smoluchowski [7], Wagner [8] approximation, who applied it to the rate of aggregation of colloids. He formulated the theory of diffusion-controlled reactions by considering two uncharged, spherical, reactive molecules $A + B$ with radii r_A and r_B, respectively, Fig. 5.3. The reactants A and B diffuse together and, when in contact, the reaction only takes place if the energy of the reactants is higher than, or equal to, the activation energy, E_a, i.e., the reaction rate will be determined by the rate of passage over the reaction energy barrier. Two critical conditions need to be considered, (1) when the rate of the chemical reaction is much larger than the rate of diffusion, and (2) when the rate of the chemical reaction is

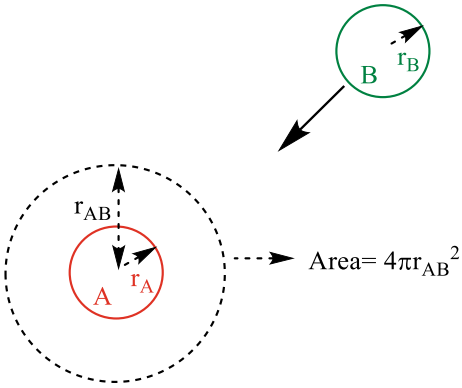


Fig. 5.3 Diffusion of molecule *B* toward molecule *A* which is assumed to be stationary and surrounded by a symmetrical (statistically) distribution of *B* molecules. The reaction takes place when the distance r_{AB} between *A* and *B* is equal to, or lower than, the sum of the radii $r_A + r_B$

less than the rate of diffusion and the system is under partial diffusion control. The derivations describing these two cases have been derived in detail [9] and will only be summarized here to illustrate the important points.

The model assumes that the observed rate does not depend on the rate at which the reactants are mixed (i.e., mixing effects are unimportant), that reactants are uniformly distributed, and that the orientation and steric effects are unimportant. Therefore, the estimated rate constants will be an upper limit because if any of the assumptions do not apply, then the true rate constant will be a fraction of that estimated.

Let's start by assuming that the molecule *A* is static at a given point of the solution and the molecule *B* is diffusing toward the stationary molecule *A*. The flux, J_B, toward the central *A* molecule can be determined form Fick's first law of diffusion given by Eq. 5.5. The flux is the amount of material passing through a spherical shell, centered on *A*, in unit time.

$$J_B = 4\pi r^2 D_B \frac{\mathrm{d}[B]_r}{\mathrm{d}r}, \tag{5.5}$$

where D_B is the coefficient of relative diffusion of *B* toward *A* and $\mathrm{d}[B]_r/\mathrm{d}r$ is the concentration gradient. Equation 5.5 describes the gradient of concentration of molecule *B* crossing the boundary of molecule *A*, Fig. 5.3.

This is the molecular description of molecules **A** and **B** colliding at boundary r_{AB}. When there is full microscopic diffusion control, any B molecule reacts immediately when located at a distance equal to the collision diameter, $r_{AB} = r_A + r_B$, such that the concentration of *B* at $r_{AB} = 0$. However, *A* molecules also diffuse, and the rate of the reaction involving all molecules *A* becomes Eq. 5.6

$$v = 4\pi (D_A + D_B) r_{AB}[A][B] \tag{5.6}$$

and the diffusion-controlled rate is given by Eq. 5.7

$$k_d = 4\pi (D_A + D_B) r_{AB}, \tag{5.7}$$

where ($D_{AB} = D_A + D_B$) is the sum of the diffusion coefficients of A and B and $r_{AB} = r_A + r_B$ is the collision radii. The units of ($D_A + D_B$) are $m^2\ s^{-1}$ and that of k_d is $m^3\ s^{-1}$. Thus, to express the rate constant in the common molar quantities and liters ($dm^3\ mol^{-1}\ s^{-1}$), the above expression needs to be multiplied by the Avogadro number N_A ($dm^3\ m^{-3}\ mol^{-1}$) and by 10^3 ($dm^3\ m^{-3}$) to finally give Eq. 5.8.

$$k = 1000 \times N_A \times 4\pi D_{AB} r_{AB}. \tag{5.8}$$

The Stoke's law for the diffusivity of spherical objects of radii r, Eq. 5.9, can be used to determine the diffusion coefficients, where k_B is the Botzmann constant and η is the viscosity of the solvent.

$$D = \frac{k_B T}{6\pi \eta r}. \tag{5.9}$$

The sum of the diffusion coefficients is

$$D_A + D_B = \frac{k_B T}{6\pi \eta}\left(\frac{1}{r_A} + \frac{1}{r_B}\right). \tag{5.10}$$

And hence the diffusion rate constant is given by Eq. 5.11, which is the basic equation for full microscopic diffusion control.

$$k_d = \frac{2k_B T}{3\eta}\frac{(r_A + r_B)^2}{r_A r_B}. \tag{5.11}$$

The viscosity of water at $T = 25\ °C$ is $\eta \sim 0.9$ cP s, and for molecules of similar size ($r_A \approx r_B$), a value of $k_d \approx 7 \times 10^9\ dm^3\ mol^{-1}\ s^{-1}$ can be estimated for the bimolecular reaction. This k_d value represents an upper limit value for diffusion-controlled reactions of neutral molecules. That is, neutral molecules cannot react faster than the rate at which they diffuse in the solvent. If reactive ions are involved, the value is expected to be somewhat higher because the electrostatic interactions increase their diffusivity. For example, values of k_d ranging (3–15) $\times\ 10^{10}\ dm^3\ mol^{-1}\ s^{-1}$ have been estimated for the reaction between hydrogen ions and various anions [9].

Equation 5.11 shows that the diffusion rate constants decrease upon increasing the viscosity of the solvent. Table 5.2 lists some typical viscosities of food fluids, and they can be 1–4 orders of magnitude greater than the viscosity of water. Thus, the corresponding k_d values can be 10^4 times lower than in water, Eq. 5.11.

Equation 5.11 also shows that the molecular size has a small influence in k_d. For molecules of the similar size ($r_A = r_B$), the ratio $(r_A + r_B)^2/(r_A r_B) = 4$, but if the size of one of the molecules is double than that of the other (e.g., $r_A = 2r_B$), the $(r_A + r_B)^2/(r_A r_B) = 4.5$, that is, an increase in the size of the molecule has little effect on the rate constant. The reason for the insensitivity of the rate to the molecular size is that smaller molecules move faster than larger ones but present smaller areas for encounters, and both opposite effects approximately cancel each other.

Table 5.2 Chart of some common solvents and their approximate viscosities at $T = 20$ °C in cP s

Fluid	η (cP s)	Fluid	η (cP s)
Acetic acid	1.12	Cyclohexane	0.89
Acetone	0.30	Diethyleneglycol	30.2
Acetonitrile	0.34	Dimethylformamide	0.80
Benzene	0.60	Dioxane	1.18
1-butanol	2.59	Ethyl acetate	0.43
Carbon disulfide	0.36	Glycerin	934
Carbon tetrachloride	0.90	Methanol	0.54
Chloroform	0.54	Toluene	0.55
Chlorobenzene	0.5	Water	1

Common solvents used in chemistry have viscosities ranging 10^{-1}–10^{2} times that of water. Data compiled from internet sources (*Properties of Solvents Used in Organic Chemistry* (http://murov.info/orglab.htm), Common Solvents Used in Organic Chemistry: Table of Properties (organicchemistrydata.org, ACS) accessed on 2024/08/28

Full diffusion control is only achieved when the rate of the chemical reaction is much greater than the rate of diffusion. If this is not the case, then the concentration of B molecules at a distance equal to the collision radii is not equal to zero because the chemical process is no longer able to remove the B molecules completely as they reach the critical distance r_{AB}. If k_{chem} is the rate constant form the chemical reaction between A and B molecules, the rate of removal of B molecules by the A molecules is $k_{\text{chem}}[B]$ and is equal to the diffusive flux J_B. Considering all A molecules, Eq. 5.12 holds

$$v = k_{\text{chem}}[AB] = \frac{k_{\text{chem}}[A][B]}{1 + \frac{k_{\text{chem}}}{4\pi D_{AB} r_{AB}}}. \tag{5.12}$$

And the rate constant is given by Eq. 5.13

In this case, the observed rate constant, k_{obs}, depends on the relative values of k_d and the rate constant of the chemical process, k_{chem}, Eq. 5.13.

$$k_{\text{obs}} = \frac{k_{\text{chem}}}{1 + \frac{k_{\text{chem}}}{4\pi D_{AB} r_{AB}}} = \frac{k_{\text{chem}}}{1 + \frac{k_{\text{chem}}}{k_d}}. \tag{5.13}$$

Note that when $k_{\text{chem}} >> k_d$, $k_{\text{obs}} = k_d$, i.e., Eq. 5.13 reduces to Eq. 5.7, meanwhile when $k_{\text{chem}} << k_d$, $k_{\text{obs}} = k_{\text{chem}}$ and there is no diffusion control.

Equation 5.13 is useful to show how to differentiate between diffusion-controlled and activation-controlled reactions. After bulk mixing is complete, the rate of reaction of **A** and **B** depends on their diffusion rates if they are near their diffusion-controlled limit, i.e., if the measured (or observed) rate constant, k_{obs} is about the same as the rate constant for the diffusion of the reactants k_d, then, the reaction is said to be under

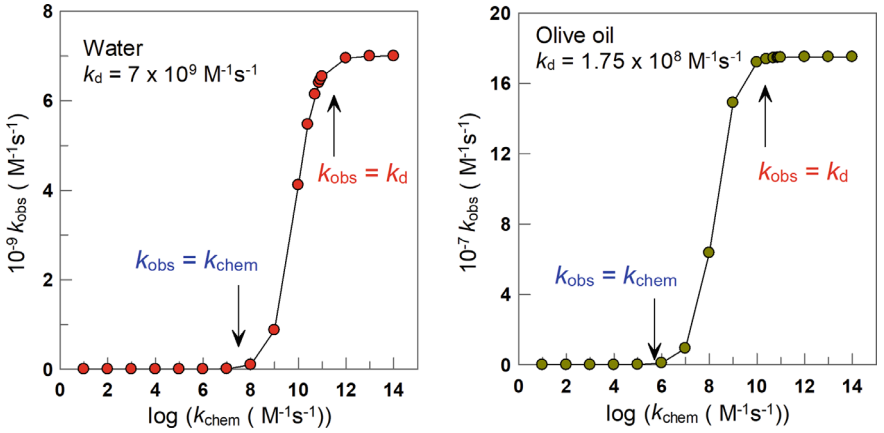


Fig. 5.4 Changes in k_{obs} as a function of log(k_{chem}) for a bimolecular reaction in aqueous solution and in olive oil, according to Eq. 5.13. Figures extracted from ref. [10]

microscopic diffusion control and the rate of the reaction is limited by the encounter frequency of **A** and **B**. Two limiting situations are possible. (i) when k_{obs} values are similar those of k_d (for instance, in the combination reaction of H^+ and OH^- ions in aqueous solution) the reaction is under full microscopic diffusion control and (ii) when k_{obs} is equal to that of the rate constant of the chemical reaction, k_{chem}, the reaction is under *microscopic activation control*. For intermediate values, between k_d and k_{chem} the reaction is under *partial microscopic diffusion control* [9].

Proton transfer reactions in water, $H_3O^+ + H_2O \rightarrow$, and proton diffusivities are about the fastest known in solution, $k \approx 1 \times 10^{10}$ M^{-1} s^{-1} and $D \approx 9.3 \times 10^{-9}$ $m^2 s^{-1}$. Assuming a limiting value of k_d for the bimolecular reaction equal to that estimated for the proton (i.e., when there is no energy barrier for the chemical reaction), the variation of k_{obs} with k_{chem} for bimolecular reactions in aqueous solution [with $k_d = 7 \times 10^9$ dm^3 mol^{-1} s^{-1} ($T = 25$ °C)] is illustrated in Fig. 5.4.

As shown, diffusion-controlled reactions ($k_{obs} \approx k_d$) are those whose rate of the chemical reaction is $k_{chem} \geq 10^{12}$ M^{-1} s^{-1}. Therefore, for kinetically controlled reactions, $k_{obs} \approx k_{chem}$, and k_{chem} should be $<10^9$ dm^3 mol^{-1} s^{-1}. For intermediate k_{chem} values (10^9–10^{11} dm^3 mol^{-1} s^{-1}), the reaction is controlled by both diffusion and activation in a different extent, depending on the particular k_{chem} value. For example, when $k_{chem} \sim k_d$ ($= 7 \times 10^9$ dm^3 mol^{-1} s^{-1}), k_{obs} ($\approx 3 \times 10^9$ dm^3 mol^{-1} s^{-1}), i.e., ~50% of diffusion control. If the same reaction takes place in a more viscous fluid such as olive oil, the upper value for k_d will be $\approx 2 \times 10^8$ dm^3 mol^{-1} s^{-1}, and only reactions with $k_{chem} < 10^6$ dm^3 mol^{-1} s^{-1} will be kinetically controlled.

Table 5.3 shows typical diffusion constants for molecules in aqueous solution at 298 K. The largest molecules in Table 5.3, hemoglobin and collagen, have, respectively, D values that are 135–1350 times smaller than that of the proton.

Table 5.3 Diffusion coefficients (D) for common molecules in water [11]

Molecule	D (10^{-9} M^{-1} s^{-1})
H_3O^+	9.3
Na^+	4.1
Sucrose	0.52
H_2O	2.3
Lysozyme	0.1
Hemoglobin	0.069
Collagen	0.0069

Table 5.4 Typical values of the rate constants obtained for bimolecular reactions in aqueous solution at $T =$ 25 °C as a function of their E_a values

E_a (kJ mol^{-1})	k_{chem} (dm^3 mol^{-1} s^{-1})
~0	$\sim 10^{10}$
<25	10^9–10^{10}
50	10^5
100	10^{-4}
>200	$<10^{-22}$

Data from van Boekel [12]

The model also considers the role of the activation energy of the chemical reaction that, according to the TST theory, largely affects the chemical reaction rate. Comparison of the k_{chem} values in aqueous solution, Table 5.4, with the diffusion rate values estimated in Eq. 5.11 indicates that for any chemical reaction with E_a higher than ~25 kJ mol^{-1}, $k_{obs} \approx k_{chem}$, i.e., for most thermally activated reactions and the rates of diffusion of the reactants are much higher than those of the chemical reactions and the system will be under dynamic equilibrium.

5.4.2 Solvent and Kinetic Isotope Effects

A routine procedure to elucidate the mechanism of a reaction exploits the substitution of an atom in a molecule by one of its isotopes, e.g., the substitution of H atoms by D atoms. This experimental praxis represents an elegant probe in the search of strong experimental evidence to finally propose the mechanism of chemical reactions. If the substitution is made on an atom of the solvent molecule, e.g., the substitution of the solvent (e.g., H_2O) by heavy water (D_2O), the procedure constitutes a powerful technique for investigating the role of the solvent isotope effects (SIE).

Substitution of an element in a compound by one of its isotopes leads to changes in the rate of reaction, the size of the change depending on the closeness of bond to the site of reaction; the biggest effects are found when the bond involved in the isotope substitution is the one cleaved in the course of the reaction. The extent of

the rate change depends on the extent to which the bond strength is affected by the isotopic substitution. The replacement of an atom by its isotope leads to mass changes affecting molecular properties (e.g., vibrational frequencies of bonds) that are sensitive to mass changes, and this in turn will depend on the relative differences in the masses of the atoms involved.

Reactions carried out in isotopically substituted solvents are used to elucidate whether the solvent participates directly or indirectly in the reaction. Three main effects can contribute to the observed effects:

1. The solvent itself is a reactant. A primary isotope effect will be observed if an O–H or O–D bond of the solvent is broken in the rate-determining step.
2. If a fast H/D exchange reaction takes place between the reactant molecules and the isotopically labeled solvent, the reactant is then labeled with the isotope (e.g. deuterium) and then the newly labeled molecule can cleave in the rate-determining step.
3. Substitution of an atom in the solvent by an isotope can lead to variations in the differential solvation of the activated complex and hence in the Gibbs energy of activation of the reaction. In this case the isotopic substitution may affect solute/solvent interactions, i.e., affect the solute solvation. This contribution is known as secondary isotope effect.

In many cases, the three contributions mentioned above are simultaneous and it is difficult to separate them. In addition to these non-specific effects, specific interactions such as hydrogen bonding between reactants and the solvent may also have an effect. Very frequently, water (as a reactant) participates in the reaction, main in those involving acid or base catalysis. Because protons which are exchangeable with solvent protons or deuterons give rise to primary and secondary isotope effects; the observed solvent isotope effects are very frequently a combination of both primary and secondary isotope effects. The solvent isotope effects are commonly employed in distinguishing between whether proton transfer occurs prior to or in the rate-determining step. When proton transfer is the rate-determining, the rate in D_2O will be less than in H_2O because of the higher dissociation energy of the D–O bond in D_2O. This is a normal primary kinetic isotope effect. However, if there is a fast reversible proton transfer prior to the rate-determining step, the reaction is faster in D_2O than in H_2O. For example, let us consider the following reaction, where S is the substrate and P the product.

$$\mathrm{S} + \mathrm{H_3O^+} \overset{\mathrm{K}}{\leftrightarrows} \mathrm{SH^+} + \mathrm{H_2O}$$

$$\mathrm{SH^+} \xrightarrow{k} \mathrm{P}.$$

The rate of the reaction is given by Eq. 5.14

$$r = \frac{\mathrm{d}[P]}{\mathrm{d}t} = k[\mathrm{SH^+}] = kK[\mathrm{S}][\mathrm{H_3O^+}]. \tag{5.14}$$

When H_2O is replaced by D_2O, the concentration of SD^+ in D_2O will be greater than that of SH^+ in H_2O because D_3O^+ is a stronger acid than H_3O^+, and, therefore, the reaction will be faster in D_2O. Similarly, in base catalyzed reactions, an increase in the rate in D_2O compared to that in H_2O is expected because DO^- is a stronger base than HO^-.

Summarizing, kinetic or equilibrium isotope effect is ultimately a consequence of the different masses of the isotopes. Effects of heavier atoms can also be primary or secondary depending on whether the bonds to the heavy atom are broken and if the rupture takes place in the rate-determining step. Because isotopic mass ratios are much closer to unity than for H–D substitutions, much smaller effects are expected when replacement of heavy atoms is done (note that the effect depends on ratio of masses, not on its difference). In practical terms, isotopic substitution of atoms other than H or D is frequently very expensive (as the isotopically substituted, and so the isotopically labeled compounds frequently are not pure materials but isotopically mixed compounds).

5.5 Static Influence of Solvents: Solvation Effects

These studies are mostly related to the static influence of solvents on rate constants, which can be understood in terms of transition state theory. According to this theory, solvents can modify the Gibbs energy of activation (as well as the corresponding activation enthalpies, activation entropies, and activation volumes) by differential solvation of the reactants and the activated complex. Since reaction rates are very sensitive to barrier heights, according to the Arrhenius equation, any change in the stability of reactants, transition state or products may have a significant effect on the reaction rate. The well-known Hughes–Ingold rules [1, 5] of solvent effects on nucleophilic substitution reaction rates, Table 5.5 are based on the equilibrium solvation of the activated complex and were proposed under the assumption that the required reorientational relaxation of solvent molecules during the activation process is sufficiently fast and that the activated complex is in thermal equilibrium with the solvent due to the frequent collisions of the reacting system with the surrounding solvent molecule.

Let us consider the reaction between the solvated reactants A and B, leading to the formation of an activated complex AB, which can return to the starting reactants or further decompose to the end products P,

$$A + B \rightleftharpoons (AB)^{\pm} \rightleftharpoons P.$$

According to the theory of absolute rates, the activated complex is envisaged being in quasi-equilibrium with the reactants, so that the equilibrium constant is defined by Eq. 5.15, where a are the activities, magnitudes between bracketss, [ ], are stoichiometric (molar) concentrations, and γ are the activity coefficients.

Table 5.5 Qualitative prediction of the solvent effects on rates of nucleophilic substitution reactions according to the Hughes–Ingold rules, who used a simple qualitative solvation model considering pure electrostatic interactions between ions or dipolar molecules and solvent molecules in initial and transition states [1, 5]

Reaction type	Initial reactants	Activated complex	Charge alteration during activation	Effect of increased solvent polarity on rate
S_N1	$R-X$	$R^{\delta+}\ldots X^{\delta-}$	Separation of unlike charges	Large increase
S_N1	$R-X^+$	$R^{\delta+}\ldots X^{\delta+}$	Dispersal of charge	Small increase
S_N2	$Y+R-X$	$Y^{\delta+}\ldots R\ldots X^{\delta-}$	Separation of unlike charges	Large increase
S_N2	Y^-+R-X	$Y^{\delta-}\ldots R\ldots X^{\delta-}$	Dispersal of charge	Small increase
S_N2	$Y+R-X^+$	$Y^{\delta+}\ldots R\ldots X^{\delta+}$	Dispersal of charge	Small increase
S_N2	Y^-+R-X^+	$Y^{\delta-}\ldots R\ldots X^{\delta+}$	Destruction of charge	Large increase

The terms "small" or "large" have only relative significance

$$K^{\ddagger}=\frac{a_{AB}}{a_A a_B}=\frac{[(AB)^{\ddagger}]}{[A][B]}\frac{\gamma^{\ddagger}}{\gamma_A\gamma_B}. \tag{5.15}$$

The formation of the product P is not significantly affected by the equilibrium between reactants and activated complex, which "exists" at the top of the energy barrier (transition state) necessary to go from reactants to products. The net reaction rate is determined by the rate at which the activated complex decomposes to products, so that the reaction rate constant is assumed to be proportional to the concentration of activated complex, Eq. 5.16.

$$k=\frac{k_B T}{h}K^{\ddagger}=\frac{\mathrm{RT}}{N_A h}K^{\ddagger}, \tag{5.16}$$

where k is the rate constant with the dimensions $(\mathrm{mol\ L^{-1}})^{1-n}(\mathrm{time})^{-1}$, k_B is the Boltzmann constant, h is the Planck constant, T the absolute temperature, R the gas constant, N_A the Avogadro number.

Since $K^{\ddagger}$ represents a quasi-thermodynamic equilibrium constant, it is possible to define thermodynamic quantities such as the Gibbs free energy, enthalpy, and entropy of activation, $\Delta G^{\ddagger}$, $\Delta H^{\ddagger}$ and $\Delta S^{\ddagger}$, respectively, so that Eq. 5.17 holds

$$k=\frac{\mathrm{RT}}{N_\mathrm{A}h}e^{\frac{\Delta G^{\ddagger}}{\mathrm{RT}}}=\frac{\mathrm{RT}}{N_\mathrm{A}h}e^{-\frac{\Delta H^{\ddagger}}{\mathrm{RT}}}e^{\frac{\Delta S^{\ddagger}}{\mathrm{R}}}. \tag{5.17}$$

According to this equation, the smaller the value of $\Delta G^{\ddagger}$, i.e., the smaller the difference between the Gibbs energy of reactants and activated complex, the greater the reaction rate is. The fastest reaction (which corresponds to $\Delta G^{\ddagger} = 0$) is given by $k_{\max} = \mathrm{RT}/N_A h$, which corresponds to the passage of the activated complex through the transition state. At normal temperature $T = 25$ °C, this value is $k_{\max} = 6 \times 10^{12}$ s^{-1}, a typical value for the frequency of molecular vibrations.

The effect of the solvent on the position of equilibrium can be considered by comparing the equilibrium constants for different solvents. For this purpose, let us consider the following unimolecular reaction in solvents $S1$ and $S2$:

$$A_{\mathrm{solv}} \rightleftharpoons B_{\mathrm{solv}},$$

where the subscript "solv" represents the molecule surrounded by solvent molecules. The Gibbs energy diagram for this reaction is as that shown in Fig. 5.5, from where it immediately follows that the variation in the standard Gibbs energy is given by Eq. 5.18

$$\Delta G^0(S2) - \Delta G^0(S1) = \Delta\Delta G_t^0(B, S1 \rightarrow S2) - \Delta\Delta G_t^0(A, S1 \rightarrow S2), \quad (5.18)$$

where ${\Delta G_t}^0$ stands for the change in the free energy of transfer from $S1$ to $S2$. Since for the equilibria, the logarithm of the equilibrium constant is proportional to the standard molar Gibbs energy change, Eq. 5.19

$$K = \frac{a_B}{a_A}, \quad \Delta G^0 = -\mathrm{RT} \ln K = \Delta H^0 - T\Delta S^0. \quad (5.19)$$

If both transfer energies are available, it is possible to achieve the following types of reaction control:

1. Cooperative effects, where the enthalpic and entropic terms have opposite signs, so that these two terms are additive.
2. Enthalpy-controlled effects, in which the two terms are opposed, but the enthalpy term is larger than the entropic one.
3. Entropy-controlled effects, in which the two terms are opposed, but the entropy term is larger than the enthalpy one.
4. Compensating effects, in which the two terms are opposed but of similar size.

A quantitative description of the influence of the solvent on the position of chemical equilibria by means of physical or empirical parameters of solvent polarity is only possible in favorable and simple cases due to the complexity of intermolecular solute/solvent interactions. Many researchers have tried to establish either theoretically or empirically, correlations between rate constants or Gibbs energies of activation and characteristic solvent parameters such as the dipole moment (solvent polarity), refractive index, solubility parameter, relative permittivity, etc. Such relationships have, unfortunately, only a very limited application; however, much progress has

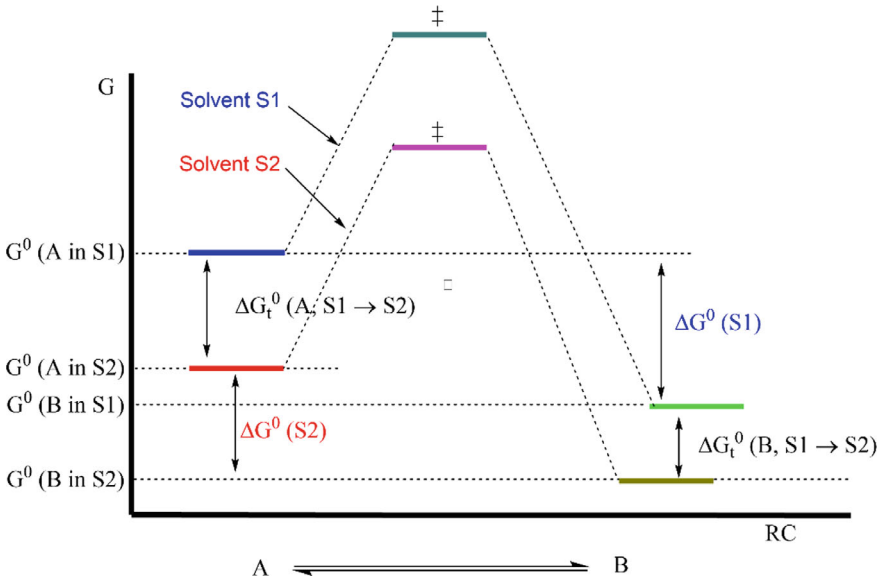


Fig. 5.5 One-dimensional Gibbs energy diagram for an equilibrium reaction $A \rightleftarrows B$ in the solvents $S1$ and $S2$

recently been made in theoretical calculations of solvation enthalpies of solutes that can participate as reaction partners in chemical equilibria.

5.6 Quantitative Measurement of Solvent Polarity: Empirical Parameters

Researchers in the physical-organic field have usually attempted to correlate the observed effects of solvents on rate constants with the polarity of the solvent, a parameter which needs to be understood in terms of the ability of a solvent for solvating dissolved charged, neutral, polar, or a polar species [1, 13]. Although the concept is grasped easily in qualitative terms, it is difficult to define precisely and even much more difficult to express quantitatively. Attempts have been made to employ physical solvent parameters such as the dipole moment, relative permittivity or the refractive index, but the approaches are unsatisfactory as they do not consider different specific solvent-solvent interactions (i.e., hydrogen bonding). Likewise, charge distributions within solvent molecules may be consequence of dipole or higher multipole moments. Up to date, there is no general mathematical relationship capable of calculating reaction rates or equilibrium constants of reactions in different solvents mostly as a consequence of the very complex solute–solvent, solute-solute, and solvent-solvent interactions, all of them taking place simultaneously.

The above mentioned reasons and others made researchers to seek for different indices of solvent polarity. Common approaches are grounded on the assumption that some particular reactions, spectral absorption or equilibrium are suitable models to infer about a relatively broad class of other solvent-dependent processes. The chosen probes or processes are generally those for which the investigated process is best understood at the molecular level.

These approaches provided quantitative indexes to solvent polarity, allowing calculations of rates and/or equilibrium constants for many reactions, as well as predictions in the spectral behavior in different solvents. As the probes or processes chosen reflect the overall balance of all intermolecular forces acting in solution, these empirical parameters are usually considered to provide a more inclusive measure of solvent polarity than any other single physical constant. However, the determined empirical solvent scales are not universal and may not be useful for all kinds of reactions and absorption systems because the contributions of intermolecular forces in the interaction between the solvent and the reference substrate may not be the same as those present in the interaction between the solvent and the substrate of interest.

Estimation of solvent polarity with the aid of chemical or physical reference processes is a very common approach in chemistry and the mathematical relationships are usually expressed in terms linear Gibbs free energy relationships, commonly known as linear free energy relationships (LFERs). LFERs are extra thermodynamic relationships which combine concepts of thermodynamics with a detailed model, thus lacking the rigor of classical thermodynamics but providing useful information which otherwise cannot be easily obtained. The advantages of these relationships are twofold. For one side, they can be used to study reaction mechanisms, as the correlation of data for a new reaction series by means of a linear Gibbs energy relationship establishes similarities between the new and the reference series. On the other hand, the use of linear Gibbs energy equations can be employed in predicting the effects of substituent or solvent changes on the reaction rates and/or equilibrium constants.

Let's consider, for instance, the following reaction:

$$A + B \rightleftharpoons (AB)^{\pm} \rightarrow P.$$

To introduce small changes to establish a reaction series we can, for example, change the reagent A by introducing different substituents. This is the case of the well-known Hammet [14, 15] Eq. 5.20, which is based on the use of *m*- and *p*-substituted benzene derivatives

$$\log \frac{k_x}{k_0} = \sigma\rho, \tag{5.20}$$

where k_X is a rate or equilibrium constant for *m*- or *p*-substituted substrates, k_0 is the rate or equilibrium constant of the standard reaction, σ is the substituent constant, and ρ is the reaction constant (by definition $\rho = 1$ for the standard reaction).

For solvent-sensitive reactions, we can also leave the reaction parameters unchanged but change the surrounding medium to modify the solvation of the reacting species.

For example, Grunwald and Winstein [16] exploited the S_N1 solvolytic reaction of t-butyl chloride (t-BuCl, 2-chloro-2-methylpropane) to define a solvent "ionizing power" parameter Y using Eq. 5.21.

$$Y = \log k^{\text{t - BuCl}} - \log k_0^{\text{t - BuCl}}, \tag{5.21}$$

where ${k_0}^{\text{t-BuCl}}$ is the first-order rate constant for the solvolysis of t-butyl chloride at 25 °C in a reference solvent, which was chosen as a binary mixture of ethanol:water (80:20) v:v and for which $Y = 0$, and $k^{\text{t-BuCl}}$ is the corresponding rate constant in another solvent. This reaction was selected as the model process because it was believed to occur by an essentially pure S_N1 mechanism, with the formation of the corresponding cation (i.e., ionization of the C–Cl bond) as the rate-determining step. Once a standard reaction and a reference solvent is chosen, a linear Gibbs energy relationship is then written as the Winstein–Grunwald Eq. 5.22.

$$\log\left(\frac{k}{k_0}\right)_{RX} = mY + c, \tag{5.22}$$

where k and k_0 are the specific rate constants for the S_N1 solvolysis of RX (in this case t-BuCl) in a given solvent and in the standard solvent, respectively, m is the sensitivity of the specific rate of solvolysis of RX to changes in the solvent "ionizing power" (Y), Y is a parameter characteristic of the given solvent, and c is the intercept (zero for an ideally behaved solvolysis). This Winstein–Grunwald equation is, thus, expected to be applicable to reactions similar to the standard one, that is, to any S_N1 reaction (Fig. 5.6).

Empirical parameters of solvent polarity can be also obtained by stablishing relationships from equilibrium measurements of solutes between two immiscible fluids. The Hansch–Leo 1-octanol/water partition constant (P_W^O) [18, 19], Eq. 5.23, is one of the best known empirical hydrophobicity parameters that are typically obtained from measurements of the distribution of solutes between two immiscible liquids. The ability of chemical compounds to be transferred to hydrophobic or lipophilic environments plays an important role in their ability to be transferred between the various regions of phases of relevant biochemical systems (e.g., cell walls, liposomes, vesicles, etc.) and in the fields of toxicology and drug delivery as the interrelation between the narcotic potencies of various substances and their partition coefficient have long been known.

$$P_W^O \equiv K_W^{\text{OCT}} = \frac{[A_{\text{OCT}}]}{[A_W]}, \tag{5.23}$$

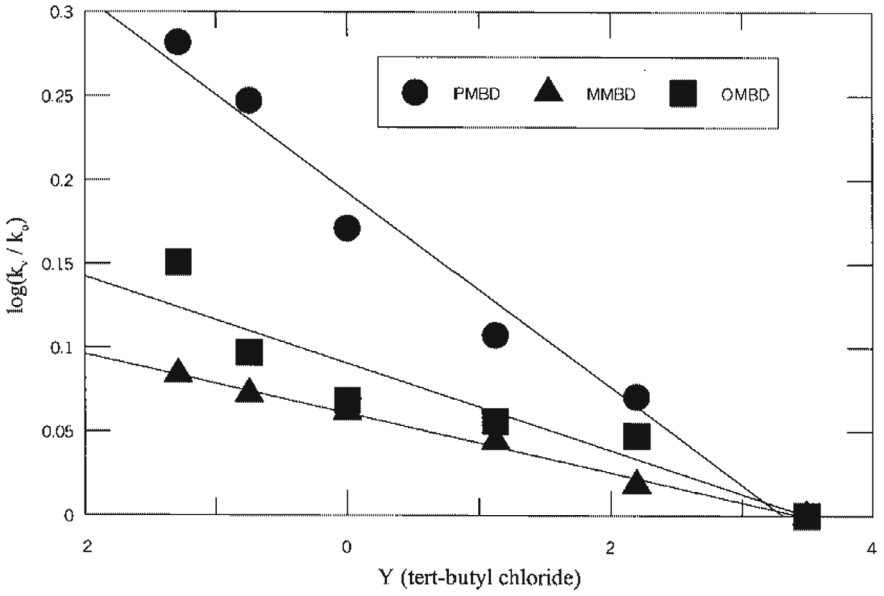


Fig. 5.6 Typical Winstein–Grunwald plots for methy substituted benzenediaznium ions (2-Methyl (OMBD), 3-methy (MMBD), and 4-methyl (PMBD)) prepared to analyze the sensitivity of dediazoniations to changes in solvent polarity. The results reflect the extraordinary insensitivity of dediazoniations to medium effects, attributed to the similarities in structure and charge distribution between the parent arenediazonium ions and the corresponding aryl cations [17]

where $[A_{\text{oct}}]$ and $[A_w]$ are the molar equilibrium concentrations of the solute A in the two immiscible phases 1-octanol and water, respectively. The 1-octanol/water partition coefficients are often used as hydrophobicity parameters in the Gibbs energy-based form of $\log P_W^O$, which is a good measure of the ease with which drugs penetrate membranes and bind to hydrophobic surfaces.

Menger and Venkataram [20] proposed another microscopic hydrophobicity substituent parameter (MH) by different ammonium salts R-NMe_3^+ X^- (R = alkyl, aryl; X^- = halide ions) to 4-nitrophenyl laurate solutions, enhancing rate constants for its basic hydrolysis. The larger rate enhancements were found for ammonium salts bearing hydrophobic R groups. Another quantitative measure of the solvent solvophobic effect has been introduced by Abraham et al. [21] who showed that ΔG^0 values of transfer of nonpolar, hydrophobic solutes from water to other solvents can be linearly correlated, providing simple quantitative values of the solvophobic power of solvents relative to two reference solvents (hexadecane and water).

Values of solvent polarity have been derived from solvent-sensitive compounds with distinctive spectroscopic properties in the various regions of the electromagnetic spectrum [22]. For instance, Spange and Weib [22], Kosower [23,

24] took advantage of intramolecular charge-transfer transition of 1-ethyl-4-(methoxycarbonyl)pyridinium iodide, which shows a marked negative solvatochromism, defining a polarity parameter, Z, Eq. 5.24, as the molar transition energy, E_T, expressed in kcal/mol. Changes in the polarity of the solvent cause hypsochromic/bathocromic shifts of the longest-wavelength charge-transfer band due to the stabilization of the electronic ground state, which is an ion-pair, relative to the electronically excited state.

$$Z \equiv E_T = \frac{hcN_A}{\lambda} = \frac{2.86 \times 10^3}{\lambda} \text{ cm}^{-1}, \tag{5.24}$$

where h is the Planck's constant, c is the speed of light, λ is the wavelength of the photon which produces the electronic excitation, and N_A is Avogadro's number.

Reichardt and Welton [1], Reichardt [3, 25] proposed the use of pyridinium N-phenolate betaine dyes (PPBD) as the standard probe molecule to overcome some of the practical limitations in the Z-value approach. The major advantage in using these dyes grounds is that the solvatochromic absorption band is at longer wavelengths than Kosower's dye, generating an extraordinarily large range for the solvatochromic shifts from ~350 to ~810 nm. The betaine dyes exhibit negative solvatochromic $\pi \rightarrow \pi^*$ absorption band with intramolecular charge-transfer. The solvent polarity parameter, $E_T(30)$, is based on the transition energy for the longest-wavelength solvatochromic absorption band of the PPBD dye. By employing betaines, it is even possible to make visual estimations of solvent polarity because most of solvatochromic shifts are within the visible region of the spectrum. $E_T(30)$ values have been determined for a large number of pure organic solvents and of binary solvent mixtures. Comprehensive empirical solvent polarity scales are available in specialized books [1].

There are many other LFERs available in the literature, many of them of wide application. The application of single solvent parameters such as the $E_T(30)$ values in correlating other solvent-dependent processes has proved to be surprisingly successful. This means that the blend of different solute/solvent interaction forces, as probed by PPBD, is nearly the same as in numerous other solvent-dependent chemical reactions and spectral absorptions. It seems, therefore, that the blend of solute/solvent interactions as measured by the $E_T(30)$ values represents a kind of "mean solvent polarity", representative of many solvent sensitive processes. For the sake of clarity, and for space reasons, the interested reader is referred to specialized books on the topic for further information [1, 4, 25].

In spite that the single empirical parameters mentioned above (and others reported in the literature) may serve as good approximation of solvent polarity, there are many examples known where the solvent-sensitive processes cannot be correlated to one single parameter and hence the above mentioned polarity scales only work well for a limited number of reactions. The solvation capability or solvation power of a solvent, which very frequently is divided into non-specific and specific solvent-solute interactions, is the result of many different kinds of interaction between the molecules of the solute and the solvent, and in attempting to take into consideration

two or more aspects of solvation, multiparameter approaches of the general form of Eq. 5.25 have been proposed.

$$A = A_0 + bB + cC + dD + \ldots. \tag{5.25}$$

In Eq. 5.25, A is the value of the solvent-dependent physicochemical property (log K, log k, $h\nu$, etc.), A_0 is a statistical quantity corresponding to the value in the gas phase or an inert solvent, B, C, D represent independent but complementary solvent parameters which are assumed to contribute to the different solute–solvent interactions, and a, b, c, etc. are regression coefficients describing the sensitivity of the property A to different solute–solvent interactions [1]. Very frequently, they higher the number of parameters employed, the better fitting to experimental results, but one takes the risk of using parameters with little or unknown physical meaning.

For example, the Gruwnwald–Winstein equation is fairly successful for the solvolytic reactions of tertiary haloalkanes and secondary alkyl sulfonates but is less satisfactory for reactions involving borderline mechanisms such as some S_N2 reactions because the m parameter seems to be solvent dependent: the reaction rate constants depend not only on the ionizing power of the solvent but also on its nucleophilicity. To account for nucleophilically solvent-assisted processes, the Grunwald–Winstein equation has been extended to a two-parameter equation of the type

$$\log(k/k_0)_{\mathrm{RX}} = l\mathbf{NT} + m\mathbf{Y}_x + c, \tag{5.26}$$

where a substrate RX is undergoing solvolysis. The l value represents the sensitivity to change in solvent nucleophilicity (NT), the m value represents the sensitivity to changes in solvent ionizing power (Y_X for a leaving group X), and c is a constant (residual) term.

Reichardt identified three points of criticism to the use of multiparametric equations for the treatment of solvent effects: (1) The separation of solvent effects into various additive contributions is somewhat arbitrary, since different solute/solvent interaction mechanisms can cooperate in a non-independent way. (2) The choice of the best parameter for every type of solute/solvent interaction is critical because of the complexity of the corresponding empirical solvent parameters, and because of their susceptibility to more than one of the multiple facets of solvent polarity. (3) To establish a multiparameter regression equation in a statistically perfect way, so many experimental data points are usually necessary that there is often no room left for the prediction of solvent effects by extrapolation or interpolation.

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Chapter 6
Catalysis



…Without catalysis, life is not possible, and in that sense, catalysis is as old as life.
Moulijn and van Santen [1]

6.1 Introduction

Catalysis is a phenomenon that increases substantially the rate of a chemical reaction by means of a small quantity of material, known as catalyst, that offers a much more energetically favorable path for the reaction compared to that without the catalyst. Catalysts accelerate reactions by orders of magnitude, enabling them to be carried under much favorable thermodynamic regimes, at much lower temperatures and pressures, compared to those of the non-catalytic process. The catalytic process is then combined with an optimized reaction and, eventually, plant design, reducing significantly both investment and operation costs of the chemical processes. To keep the reaction as near to room temperature as is practical, the chemical industry has always used catalysts to save both energy usage and costs. This makes catalysis to be, by far, the most important technology in environmental protection and the chemical industry, allowing industrially important reactions to be carried out under almost achievable conditions. All industrial catalysts have a limited lifetime, which makes the search for longer lasting catalysts high on industry's list of priorities. Nowadays, the main focus of industries is to design and develop ideal catalysts to make all chemical processes cleaner and greener. Therefore, researchers try to find catalysts that will enable industrial processes to be less polluting, reduce toxicity, produce purer products, and last longer [1–5].

In practice, many reactions that are common in the chemical industry would not be possible without catalysts, and many other would not be economical. In economic terms, the rates of chemical reactions represent an important cost factor; higher reaction rates allow higher space–time yields (kg of product per time and reactor volume) and hence smaller reaction vessels. Higher rates and higher overall catalyst yields (i.e., mass of product per unit mass of catalyst) reduce the incremental contribution

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C. Bravo-Diaz et al., *From Experimental Kinetic Data to Reaction Mechanisms*,
Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_6

of catalyst costs per unit mass of product generated. Rate constant enhancements of many orders of magnitude can be obtained in catalysis, often by very subtle changes. It should not be surprising, therefore, that catalysis has found widespread applications in industrial chemical and biochemical processes and constitute one of the most important research areas in applied science. For example, enzymes as biocatalysts have an important role in the fermentation of sugar to ethanol and the conversion of ethanol to acetic acid.

Catalysts are found in various forms: atoms, molecules, enzymes, or zeolites, among others. In addition, catalysts can be applied in different phases or environments, such as liquids or gases or at the surface of solids. Nowadays, a variety of known catalysts can be classified with respect to criteria such as structure, area of application, or composition. Table 6.1 shows some of the most important industrial catalytic processes [6]. Nowadays, the chemical industry is based on catalysis and about 90% of all the products produced by industry are made by employing catalytic processes. These catalytic routes are usually designed in such a way that all the reactants are used efficiently, and the production of waste materials or unwanted by-products is minimized.

Catalysts work because they provide an alternative reaction pathway with much lower activation energy than the uncatalyzed reaction. According to the IUPAC Goldbook [7], a catalyst is defined as "A substance that increases the rate of a reaction without modifying the overall standard Gibbs energy change in the reaction". In other words, the catalyst equally accelerates both the forward and the backward reaction: if a catalyst increases the formation rate of the products from the reactants, it will also accelerate the decomposition of the products into the reactants passing through the transition states [1, 5]. The catalyzed reaction may involve several intermediate and transition-state complexes, which are completely different from those in the mechanism for the reaction that occurs without any catalyst, see for example, Fig. 6.1.

The catalyst offers an alternative path for the reaction, which is obviously more complex, but energetically much more favorable. The diagrams above illustrate several important features on the catalysis action:

- The activation energy of the catalytic reaction is significantly smaller than that of the uncatalyzed reaction; hence, the rate of the catalytic reaction is much larger. The overall change in free energy for the catalytic reaction equals that of the uncatalyzed reaction. Hence, the catalyst does not affect the equilibrium constant for the overall reaction of $A + B$ to P. Thus, if a reaction is thermodynamically

Table 6.1 Some large-scale industrial processes where catalysts are involved

Reaction	Catalytic system
Reduction of nitrogen gases in exhaust systems	Rh, vanadium oxide
Oxidation of CO and hydrocarbons in exhaust systems	Pt, Pd
Catalytic cracking of oil	Zeolites
Preparation of HNO_3 from ammonia	Pt–Rh
Formation of vinyl chloride (ethylene + Cl_2)	Cu (as chloride)

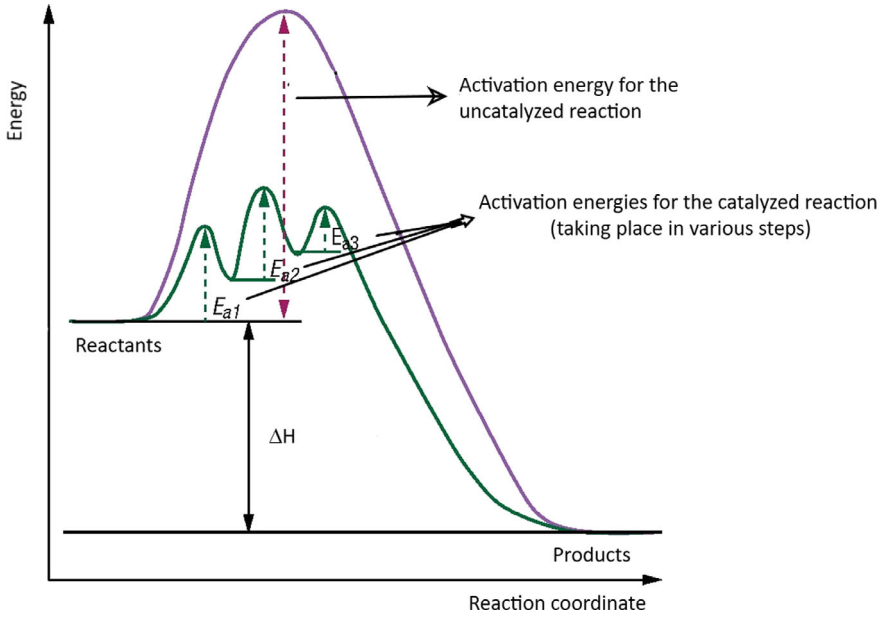


Fig. 6.1 Illustrative examples of the energy profiles of an uncatalyzed (purple line) and a catalyzed (green line) reaction. The catalyst opens a new reaction pathway making the activation energy of the catalyzed reaction to be always smaller than that of the uncatalyzed reaction

unfavorable, a catalyst cannot change this situation. C*atalysts change the kinetics but not the thermodynamics of chemical reactions.*

- The catalyst accelerates both the forward and the reverse reactions to the same extent. In other words, if a catalyst accelerates the formation of product P from A and B, it will do the same for the decomposition of *P* into *A* and *B*. Thus far it is immediately evident that there are also cases in which the combination of catalyst with reactants or products will not be successful: If the bonding between reactants and catalyst is too weak, there will be hardly any conversion of *A* and *B* into products. Conversely if the bond between the catalyst and one of the reactants, say A, is too strong, the catalyst will be mostly occupied with species *A*, and *B* is not available to form the product. If *A* and *B* both form strong bonds with the catalyst, the intermediate situation with *A* or *B* on the catalyst may be so stable that reaction becomes unlikely. Hence, a successful combination of catalyst and reaction is that in which the phenomenon of the interaction between the catalysts and reacting species is not too weak but also not too strong.

Additional mechanistic considerations are:

(a) From the mechanistic point of view, the catalyst can act in two ways: (i) forming a more reactive species with one of the reagents or (ii) favoring the conditions to increase the number of encounters between reagents.

(b) The catalyst is not consumed during the reaction but, somehow, participates in the mechanism of the reaction. There must exist a reaction step in the overall catalytic mechanism that accounts for its recovery at the end of the reaction.
(c) The action of a catalyst must be in such a way that makes the same effects in both the forward and the reverse reactions because catalysts do not change the equilibrium constants (K_{eq}) of existing equilibria.
(d) Any catalytic reaction is accompanied by a direct, non-catalytic transformation from reactants to products. The non-catalytic pathway could be negligible, but it always exists so that the rate of the reaction is the sum of the rates of the catalyzed and non-catalyzed reactions, $r = r$ (non-catalyzed) $+ r$ (catalyzed).
(e) Catalysts never make possible reactions that are not feasible from the thermodynamic point of view.

Very frequently, several other characteristics are required to consider a substance a good catalyst [5, 6]. Among them, catalysts are required to be selective (must be specific for a given reaction so that it only catalyzes the reaction of interest and prevents the formation of unwanted products), and with a low practical cost (easy to prepare and employed in small quantities). The characteristic times on which catalytic events occur vary from the time required to activate and breaking a bond in a reactant (which usually takes place in the picosecond time scale) to the completion of an entire catalytic cycle from complexation between catalyst and reactants through their separation from the product, which may take place in the microsecond to minutes to hours for complicated reactions on surfaces. On the mesoscopic level, diffusion in and outside catalytic sites may take between seconds and minutes, and the residence time of molecules in the catalytic sites may be from seconds to, effectively, infinite time if the products adsorb strongly on the catalyst.

The reader must be aware that the rate constant of the non-catalyzed (k_{nc}) and catalyzed (k_{cat}) reactions cannot be compared to describe the effectiveness of a catalyst as this is an impossibility. Let's assume that we have a bimolecular reaction of species A and B with a rate of product P formation:

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_{nc}[A][B]. \tag{6.1}$$

We don't know what the rate equation for the catalyzed reaction might look like, but it is reasonable that at least the catalyst concentration will also occur in it, e.g.:

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_{cat}[\mathrm{Cat}][A][B] \left(\text{or } \frac{\mathrm{d}[P]}{\mathrm{d}t} = k_{cat}[Cat][A], \text{ etc.} \right). \tag{6.2}$$

In practice the rate equation may take a much more complicated form than the ones shown above. Hence the order of the reaction is different even in the simplest cases, and hence a comparison of the two rate constants has little meaning. Comparisons of rates constants are meaningful only if the catalysts follow the same mechanism and if the product formation can be expressed by the same rate equation. For example, in enzyme catalysis and enzyme mimics, one often compares the rate constant of the

uncatalyzed reaction with that of the catalyzed reaction only if the mechanisms and the reaction volumes of the two reactions are the same. Only under these circumstances, comparison of rate constants may be useful. However, if an uncatalyzed reaction and the catalyzed one occur simultaneously in a system, one can generally make estimations of what part of the product is made via the catalytic route and what part isn't.

Depending on the phase (the word "phase" here refers to solid, liquid, gas, or immiscible liquids like oil and water) in which the substrate and the catalyst are, three broad types of catalysis can be considered: heterogeneous (e.g., gas–solid, liquid–liquid), homogeneous (e.g., when the substrate for a reaction and the catalysts are brought together in one phase, most often the liquid phase), and biocatalysis, which is a special case of homogeneous catalysis (some authors considered biocatalysis as heterogeneous given the much higher size of the catalysts compared to that of reactants) where the catalysts are biological compounds such as enzymes. The emphasis of this chapter will be on the fundamental kinetic and mechanistic aspects of homogeneous and microheterogeneous (where colloidal or macromolecular structures are involved) catalysis, with only brief mentions to the heterogeneous one and to biocatalysis, as their full mechanistic coverage is out of the scope of this book and they can be easily found in specialized monographs and books currently available in the literature [8–11].

6.2 Types of Catalysis

Catalysts can be applied in different phases or environments, such as liquids or gases or at the surface of solids. Nowadays, a variety of known catalysts can be classified with respect to criteria such as structure, area of application, or composition, leading to three broad types of catalysis: heterogeneous (solid-state catalysts), homogeneous catalysis, and enzymatic catalysis (biocatalysis). Microheterogeneous catalysis (enzymatic, micellar and supramolecular) is a special catalysis between heterogeneous and homogeneous sharing some but not all their main features [5, 12–14].

6.2.1 Heterogeneous Catalysis

Heterogeneous catalysis involves those systems where the chemical reaction takes place in different phases. Generally, the catalyst is in the solid phase and the reactants are gases or liquids. Some common heterogeneous catalysts are metals (Pt, Pd, Rh, Ni, Co, Ir, etc.), metal oxides, metal salts and organic material such as hydroperoxides and ion exchangers. In the solid catalyst not all the surface is active to catalyze, there are same active sites where the reaction occurs. The greater the surface is, the greater the number of active sites and the greater the reaction rate is.

There are two types of gas–solid heterogeneous catalysts: i) mass catalysts, they are those in which the active phase constitutes the whole of the catalyst and ii) supported catalysts, these correspond to those in which small amounts of catalytically active material are deposited on the surface of another material, normally an inert porous solid, called a support. The supports are porous solids such as zeolites, ceramics, active carbon, silica gel, aluminosilicates, and metal oxides. The liquid–liquid system that has gained increasing importance in organic synthesis because it is considered a green system. This system is called phase-transfer catalysis (PTC).

A typical example of a heterogeneous catalysis is the Ostwald process for making nitric acid (HNO_3) or the Haber–Bosch process for preparing ammonia from nitrogen and hydrogen.

Heterogeneous catalysis has been extensively used in the petrochemical industry, medicine production, and environmental protection, but owing to high costs and low abundance of such noble metals in nature, they cannot fulfill the increasing demand. The recovery process of heterogeneous catalysts is costless and is performed easily. These catalysts have good thermal stability and although in some cases their selectivity is a little low, they have multiple active sites, which can enhance the selectivity factor.

A special case of heterogeneous catalysis is the so-called microheterogeneous catalysis, where the reaction takes place at an interface of colloidal systems (micelles, microemulsions, liposomes, emulsions, etc.). Microheterogeneous catalysis is frequently considered a misnomer, as the reason of rate enhancement is an increase in the local concentration of reactants in the interfacial region of colloidal systems, but the term is still employed for historical reasons and because these colloidal systems usually employed to model reactions taking place in more complex biological systems.

6.2.1.1 Homogeneous Catalysis

In homogeneous catalysis both the catalyst and the reactants are in the same phase, gas or more commonly liquid, but there also are some reactions that may even take place in the solid phase. A typical example of gas phase catalysis is the decomposition of ozone in the atmosphere by chlorine atoms, $O_3 + O \rightarrow 2O_2$. Ozone decomposes spontaneously and also when irradiated (e.g., sun light) but Cl atoms accelerate the reaction tremendously. Another example of homogeneous catalyses is found in the esterification reaction, where sulfuric acid dissolves in ethanol/ethanoic acid.

Homogeneous catalysis is mostly aimed to produce specific chemical compounds (pharmaceuticals). There are a large variety of homogeneous catalysts, including Brönsted and Lewis acids, organic molecules and organometallic or coordination complexes. These catalysts have some advantages such as excellent selectivity, single active sites and provide a good contact with reactants, but they also suffer from disadvantages such as poor thermal stability and high catalyst recovery costs because being reactants and catalyst in the same phase makes difficult the recovery of the catalyst.

6.2.1.2 Biocatalysis

Biological catalysts (e.g., enzymes) are complex molecular arrangements (proteins, etc.) whose structures result in optimal, very shape-specific, active sites where reactant molecules (usually referred as substrates) guide and locate reactants in an optimum configuration for reaction. Enzymes constitute, therefore, a kind of very specific and capable catalysts, allowing important biological reactions to occur at rates necessary for the, for instance, synthesis of proteins and DNA, and the breakdown of molecules and the storage of energy in sugars. A typical example is the catalytic decomposition of hydrogen peroxide by the enzyme catalase, which increases the rate of conversion of up to 10^7 H_2O_2 molecules per second (Table 6.2).

6.3 Mechanistic Aspects of Catalysis. Modeling Chemical Reactivity

Understanding catalytic reactions at the elementary level of rupture of bonds in reactants and the formation of new bonds in products and controlling their rates is at the heart of the matter, requiring techniques for catalyst preparation, characterization, and a thorough understanding of their kinetics and mechanisms on the level of elementary reaction steps.

We will focus of the latter, and in the next sections, we will describe some of the most common kinetic and mechanistic approaches that allow us to model the reactivity of catalytic reactions.

6.3.1 Heterogeneous Gas–Solid Catalysis

We can consider a gas–solid heterogeneous catalytic process of a reaction in which a gaseous stream containing the reactants is passed through a catalytic reactor (containing a solid catalyst), and the following series of basic steps can be described (Fig. 6.2):

Table 6.2 Some examples of enzymatic reactions showing the reactants, the products, and the enzyme involved

Substrate	Products	Enzyme
Starch	Simpler sugars	Amylase
Lactose	Glucose and galactose	Lactase
Hydrogen peroxide (H_2O_2)	Water (H_2O) and oxygen (O_2)	Catalase
Bicarbonate (HCO_3^-)	Water (H_2O) and carbon dioxide (CO_2)	Carbonic anhydrase

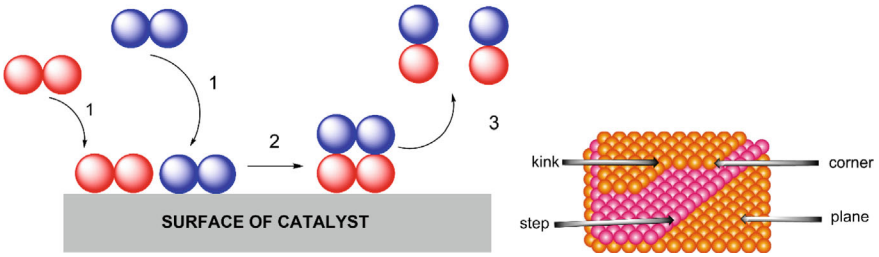


Fig. 6.2 Incoming gas species (moving usually at diffusion rates, 1) land on active sites at the surface of the catalyst forming weak bonds with the catalyst, e.g., they are adsorbed at the surface. When properly oriented, they may react (slow step 2) rearranging their electrons and the products are formed, which can be desorbed (step 3) from the active sites and diffuse away from the catalytic surface

(1) Diffusion of reactants from the bulk the phase to the catalyst surface and to the active site.
(2) Adsorption of the reagent(s) on the surface of the catalyst.
(3) Chemical reaction on the surface. This is usually the slow step of the process.
(4) Desorption of the product(s).
(5) Diffusion of the products out of the active site into the phase.

The active sites at the surface of the catalyst are defects in its crystalline network and can be kinks, corners, steps, and planes, Fig. 6.2. In most cases, the diffusion steps are fast, and the rates for these steps are being determined by porosity and diffusion coefficients, while other steps (adsorption, chemical reaction, and desorption) are determined by energetic interactions. The adsorbed species can react when properly oriented constituting the rate-determining step. An energy diagram for an uncatalyzed reaction and the same reaction, catalyzed by a heterogeneous catalyst, is shown in Fig. 6.3; illustrating the energy changes involved in the adsorption, chemical reaction, and desorption steps.

The overall process was modeled by Langmuir and Hinshelwood as illustrated in Fig. 6.4

For bimolecular reactions, the Langmuir-Hinshelwood mechanism comprising the steps indicated in Fig. 6.4 applies, where k_{aA} and k_{aB} are the rate constants of the adsorption processes of the reactants A and B, respectively, k_{dA} and k_{dB} stand for those of the desorption processes, and k_{het} is the rate constant of the rate-determining step (rds).

$$A_g + S_{\text{surface}} \underset{k_{dA}}{\overset{k_{aA}}{\rightleftharpoons}} A_{\text{ads}},$$

$$B_g + S_{\text{surface}} \underset{k_{dB}}{\overset{k_{aB}}{\rightleftharpoons}} B_{\text{ads}},$$

$$A_{\text{ads}} + B_{\text{ads}} \xrightarrow[\text{rds}]{k_{\text{het}}} P_{\text{ads}},$$

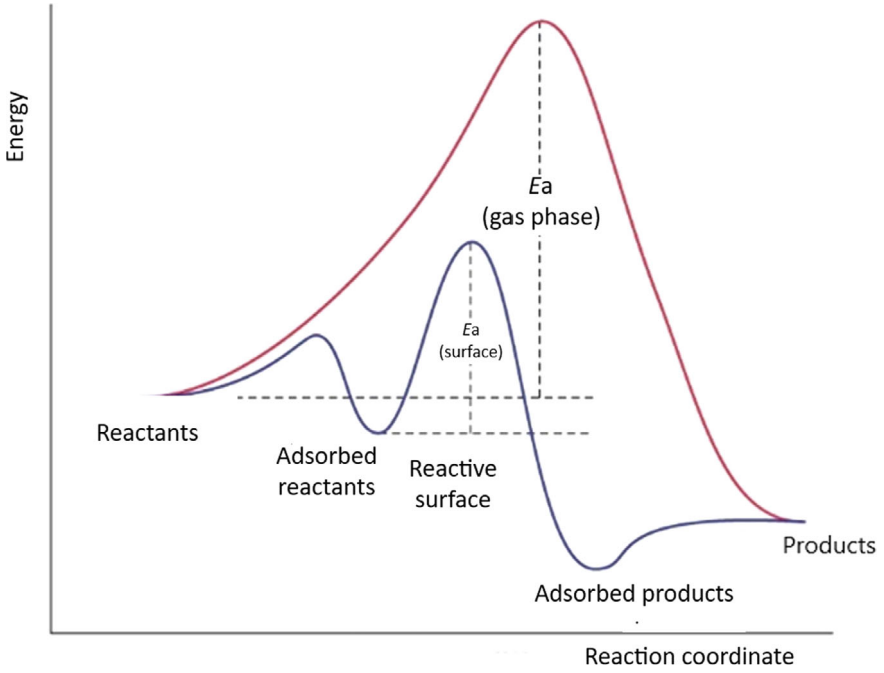


Fig. 6.3 Energy diagram for an uncatalyzed reaction (red curve) and one catalyzed by a heterogeneous catalyst (black curve)

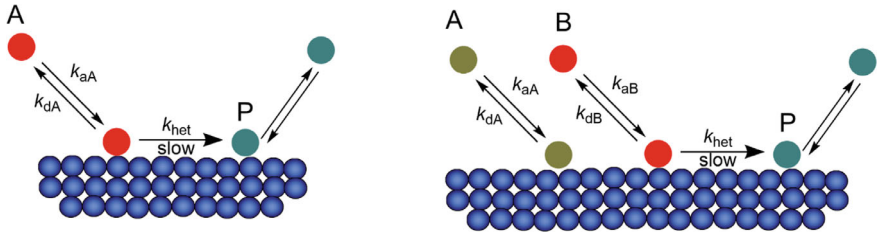


Fig. 6.4 Mechanism for unimolecular and bimolecular reactions proposed by Langmuir and Hinshelwood where the reactants A and B are adsorbed onto a surface where they react to yield products P that are finally desorbed to regenerate the catalytic surface

$$P_{\text{ads}} \xrightarrow{\text{fast}} P. \tag{6.3}$$

Denoting as θ_A and θ_B the fraction of sites occupied by molecules A and B, respectively, and θ_S the fraction of vacant sites (i.e., unoccupied sites), the total number of active sites is equal to those unoccupied plus those occupied by the molecules A and B, Eq. (6.4)

$$\theta_S + \theta_A + \theta_B = 1. \tag{6.4}$$

And the rate expressions (6.5)–(6.8) for the adsorption and desorption processes hold.

$$\frac{\mathrm{d}A_{\mathrm{ads}}}{\mathrm{d}t} = k_{aA} P_A \theta_S, \tag{6.5}$$

$$\frac{\mathrm{d}B_{\mathrm{ads}}}{\mathrm{d}t} = k_{aB} P_B \theta_S, \tag{6.6}$$

$$\frac{\mathrm{d}\theta_A}{\mathrm{d}t} = k_{dA} \theta_A, \tag{6.7}$$

$$\frac{\mathrm{d}\theta_B}{\mathrm{d}t} = k_{dB} \theta_B. \tag{6.8}$$

At the equilibrium, the rates of adsorption and desorption of A and B equal each other, so that the fraction of sites occupied by molecules A and B is given by Eqs. (6.9) and (6.10), respectively

$$k_{aA} P_A \theta_S = k_{dA} \theta_A \Rightarrow \theta_A = \frac{k_{aA} P_A \theta_S}{k_{dA}} = K_A P_A \theta_S, \tag{6.9}$$

$$k_{aB} P_B \theta_S = k_{dB} \theta_B \Rightarrow \theta_B = \frac{k_{aB} P_B \theta_S}{k_{dB}} = K_B P_B \theta_S. \tag{6.10}$$

And after substitution in Eq. (6.4), Eq. (6.11) can be derived.

$$\theta_S = \frac{1}{1 + K_A P_A + K_B P_B}. \tag{6.11}$$

And mathematical relationship given by Eq. (6.12) can be derived for the rate of reaction in terms of the adsorption equilibrium constants and the pressure of the gases.

$$r = \frac{k_{\mathrm{het}} K_A K_B P_A P_B}{(1 + K_A P_A + K_B P_B)^2}. \tag{6.12}$$

Similar mathematical derivations can be done for the unimolecular mechanism (Fig. 6.4), Eq. (6.13), and for those cases where the adsorbed reactant A reacts with a non-adsorbed reactant B (Eley–Rideal mechanism), Eq. (6.14). The reader is referred to specialized textbooks or reviews for further information.

$$r = \frac{k_{\mathrm{het}} K_A P_A}{1 + K_A P_A}, \tag{6.13}$$

$$r = \frac{k_{\text{het}} K P_A P_B}{1 + K P_A}. \tag{6.14}$$

6.3.1.1 Heterogeneous Liquid–Liquid Phase-Transfer Catalysis

The liquid–liquid heterogeneous catalysis (phase-transfer catalysis (PTC))[15, 16] has been an important development due to its great versatility in inorganic and organic synthesis and that it has been recognized as a green chemistry method since it reduces waste generation[17]. However, PTC may not be as efficient as homogeneous catalysis because of mass-transfer limitations. A phase-transfer catalysis system requires two liquid phases, each of which contains one of the substrate species and one phase contains a soluble catalyst species. One of the phases acts a reservoir of reacting anions or base for generation of organic anions, with organic reactants and catalysts (source of lipophilic cations) located in the second phase. The reacting anions are continuously introduced into the organic phase in the form of lipophilic ion pairs with lipophilic cations supplied by the catalyst.

PTC is useful in the case of reactions in which the reactants are mutually insoluble. For example, if one of the reactants is solubilized only in water and the other reactant is only soluble in an organic solvent immiscible with water. Stirring and changes in temperature can help to get them into contact to react, but their effects are generally not sufficient to achieve reasonable reaction rates. However, the addition of a complexing agent, which may also be an active catalyst itself, essentially binds with one of the substrates and facilitates transport to the other phase. The reaction takes place either at the interface or in one of the bulk phases. The complexing agent is released and returns to the first phase to "collect" more substrate for transport.

Thus, the basic requirements of an effective phase-transfer catalyst include: (1) The ability of the catalyst to form a relatively stable ion pair with the reactant. (2) A favorable partitioning in the organic phase. (3) A return to the interface so the process can be repeated. Typical agents for achieving phase-transfer catalysis include organic quaternary ammonium halides (e.g., tetrabutyl ammonium bromide, and methyl trioctyl ammonium chloride) and phosphonium halides (e.g., hexadecyl tributyl phosphonium bromide).

The PTC reaction mechanism was first proposed by Starks [18] for the reaction of aqueous cyanide (X^-) and organic 1-chloro-octane (RX).

$$\text{NaCN(aq)} + \text{1-C}_8\text{H}_{17}\text{Cl(org)} \xrightarrow{\text{QACl}} \text{1-C}_8\text{H}_{17}\text{CN(org)} + \text{NaCl(aq)}$$

The reaction does not proceed after more than 25 h in the absence of catalyst; however, the cyanide displacement takes place rapidly with only 1% of the quaternary ammonium salt $(C_6H_{13})_4N^+Cl^-$ $(= Q^+Y^-)$ added, and yields of ~ 100% of the 1-cyano-octane product (RY) are achieved in 2–3 h. The overall reaction and the proposed reaction mechanism is shown in Fig. 6.5.

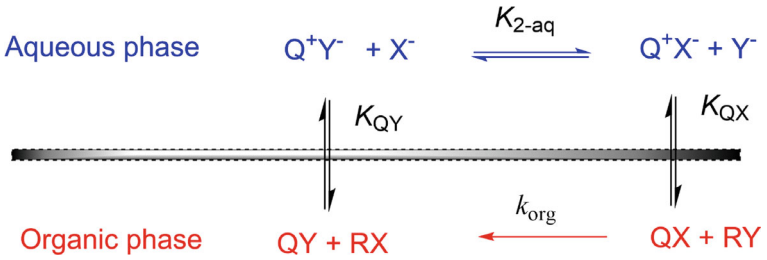


Fig. 6.5 General mechanism for phase-transfer catalysis [19]

The active anion required for the reaction (CN^-) is depicted as X^-. The ion does not dissolve to any significant extent in the organic phase that contains the substrate shown here as RY. The phase-transfer catalyst Q^+Y^- should first react with the cyanide ion (X^-) to form the active intermediate Q^+X^-, which is then transferred into the organic phase to react with the organic reactant RY to yield the displacement product RX and QY, which is then transferred to the aqueous phase to start a new catalytic cycle.

Since the reaction is carried out between immiscible phases, the nature of the interface and the physical properties of the reacting compounds at the interface become very important in promoting the desired reaction to a satisfactory level. The characteristic of the interface includes its interfacial tension, the presence of surfactants and the degree of agitation, and all of them determine the thickness of the interface, the droplet size, and the interfacial area available to transfer the reacting anion. The interfacial behavior of the reacting anion is affected by its distribution, the ease of penetration into the other phase, and the mass transfer across the interface.

According to the mechanism in Fig. 6.5, some equilibria are established in the process. The individual ion-pair distribution constants are defined by Eq. (6.15)

$$K_{QX} = \frac{[QX]_{\text{org}}}{[Q^+X^-]_{\text{aq}}} \quad K_{QY} = \frac{[QY]_{\text{org}}}{[Q^+Y^-]_{\text{aq}}}. \tag{6.15}$$

The ion-exchange equilibria across the interface can be expressed in terms of an overall exchange reaction equilibrium constant for exchange of $[X^-]_{\text{aq}}$ by $[Y^-]_{\text{aq}}$ and its transfer to the organic phase, Eq. (6.16), which represents the equilibrium constant for the ion-pair exchange in the aqueous phase.

$$K_e = \frac{[QY]_{\text{org}}[X^-]_{\text{aq}}}{[QX]_{\text{org}}[Y^-]_{\text{aq}}} = \frac{K_{QX}}{K_{QY}} K_{2-\text{aq}}, \tag{6.16}$$

where $K_{2\text{-aq}}$ is the equilibrium constant for anion exchange in the aqueous phase. Assuming that the overall reaction is controlled by the reaction in the organic phase, then the rate-determining step is given by Eq. (6.17)

$$\frac{\mathrm{d}[RX]_{\mathrm{org}}}{\mathrm{d}t} = k_{\mathrm{org}}[QX]_{\mathrm{org}}[RY]_{\mathrm{org}}. \tag{6.17}$$

The concentration of the active catalyst in the organic phase is given by Eq. (6.18)

$$[QY]_{\mathrm{org}} = \frac{K_e[QX]_{\mathrm{org}}[Y^-]_{\mathrm{aq}}}{[X^-]_{\mathrm{aq}}}. \tag{6.18}$$

The catalyst is distributed as two ion pairs in each of the organic and aqueous phases. Denoting as N_Q the total amount of catalyst (mol) added to the reaction mixture initially and Φ is the fraction of the catalyst cation Q^+ distributed in the organic phase, then the relationships given in Eqs. (6.19) and (6.20) hold:

$$[QY]_{\mathrm{org}} = \frac{N_Q\Phi}{V_{\mathrm{org}}} - [QX], \tag{6.19}$$

$$N_Q(1-\Phi) = [Q^+X^-]_{\mathrm{aq}}V_{\mathrm{aq}} + [Q^+Y^-]_{\mathrm{aq}}V_{\mathrm{aq}}. \tag{6.20}$$

Defining $[N_Q]_{\mathrm{org}} = [N_Q]/V_{\mathrm{org}}$ = total catalyst mol per unit of organic phase volume, Eq. (6.19) is converted to Eq. (6.21).

$$[QY]_{\mathrm{org}} = \frac{\Phi[N_Q]_{\mathrm{org}}}{1 + \frac{[X^-]_{\mathrm{aq}}}{K_e[Y^-]_{\mathrm{aq}}}}. \tag{6.21}$$

And the equation for the rate constant (Eq. 6.17) can be rewritten as

$$\frac{d[\mathrm{RX}]_{\mathrm{org}}}{d\mathrm{t}} = k_{\mathrm{org}}[\mathrm{RX}]_{\mathrm{org}}\frac{\Phi[N_Q]_{\mathrm{org}}}{1 + \frac{[\mathrm{X}^-]_{\mathrm{aq}}}{K_e[\mathrm{Y}^-]_{\mathrm{aq}}}}. \tag{6.22}$$

The concentrations of $[RX]_{\mathrm{org}}$, $[X^-]_{\mathrm{org}}$, and $[Y^-]_{\mathrm{org}}$ can be substituted in terms of the fractional conversion X_{RX} of RX.

$$\frac{\mathrm{d}[X_{RX}]}{\mathrm{d}t} = \frac{k_{\mathrm{org}}\Phi[N_Q]_{\mathrm{org}}(1-X_{RX})}{1 + \frac{[X^-]_{\mathrm{aq}}}{K_e[Y^-]_{\mathrm{aq}}}}. \tag{6.23}$$

After separation of variables and integration, Eq. (6.24) can be obtained

$$\frac{[X_{RX}]}{(1-X_{RX})\ln(1-X_{RX})} = \frac{K_e k_{\mathrm{org}}\Phi[N_Q]_{\mathrm{org}}t}{\ln(1-X_{RX})} + (K_e - 1). \tag{6.24}$$

Equation (6.24) predicts that a plot of the term in the left versus $t/\ln(1-X_{RX})$ should be linear and the values for k_{org} and K_e can be obtained from the slope and intercept, respectively.

6.4 Homogeneous Catalysis

Homogeneous catalysis refers to catalytic systems in which the reactants and the catalyst components are brought together in the same phase, most often a liquid phase [4]. Examples of such systems or catalysts are:

- General acid and base catalysis (e.g., ester hydrolysis).
- Lewis acids as catalysts (e.g., Diels–Alder reactions).
- Organic catalysts (e.g., thiazolium ions in Cannizzaro reactions).
- Porphyrin complexes (e.g., epoxidations, hydroxylations).
- Enzymatic processes.
- Coordination complexes (e.g., polyester condensations).

6.4.1 General Mechanism of Catalysis

It may be worthwhile to remind here some of the common recommendations made in previous chapters to derive the rate laws of catalytic reactions. When the researcher has little or no prior information on the catalytic reaction, it is highly convenient to measure the initial reaction rate as a good approach for determining the rate equation. The reason is that when the time t is close to $t_{0,}$ the experimental conditions are well defined, and the initial reaction rate, v_0, reflects these conditions. For elementary reactions, the reaction order is derived directly from the stoichiometry, and one can easily calculate the rate constant. Conversely, for non-elementary reactions (i.e., when the partial reaction orders for each component may differ from the stoichiometry of that component), the reaction order must first be found by experiment. Once the reaction order is determined, we can derive the rate equation and calculate the rate constant. A simple solution is to run several reactions and quantifying the concentration effects of each component. This gives the partial reaction order for each component.

This is a common approach in catalysis (mainly in enzymatic) and so the derivatization of kinetic equations for the general catalytic mechanism will be given in terms of the rate of the reaction rather than providing the mathematical expressions for the rate constants, which—otherwise—the reader can easily deduce from the rate equations.

The researcher has also to bear in mind that by using a large excess of a reagent one can effectively remove that reagent from the reaction rate equation. The reagent does not disappear, of course, but because its concentration is high compared with the change of concentration caused by the reaction, it remains effectively constant. This lowers the overall reaction order, creating a pseudo-order that gives information on the role of the remaining reagents.

Designating the catalyst as C and the substrate as S (that is the reactant subject to the catalytic reaction), the general mechanism of catalysis is given by the following Eq. (6.25):

Table 6.3 Nature of the reacting species in different types of catalysis. Adapted from Arnaut et al. [20]

Catalysis	C	Y	D
Heterogeneous	Solid surface	–	–
Enzymatic	Enzyme	–	–
Acid (general)	HA	A^-	H_2O
Acid (specific)	H_2O	OH^-	B
Base (general)	B	BH^+	H_2O
Base (specific)	H_2O	H_3O^+	HA

$$S + C \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} X + Y$$
$$X + D \xrightarrow{k_2} P + C, \tag{6.25}$$

where X represents an intermediate that further reacts with another substance (D) to yield the final product (P). The substance (Y) may subsequently react but is not involved in kinetically significant reactions. The species Y and D do not participate in heterogeneous and enzymatic catalysis. Table 6.3 shows some examples of the nature of reactants involved in some homogeneous catalytic processes.

For simplicity, we will consider that only the first step is reversible. Depending on the relative size of the rates of the reverse process ($v_r = k_{-1}[X][Y]$) and the second direct process ($v_{2d} = k_2[X][D]$), the pre-equilibrium or steady-state approximations hold (see Chap. 4). When $v_r \gg v_{2d}$, the pre-equilibrium approximation holds as the intermediate X is readily going back to the parent reactants, and the intermediate is called "Arrhenius-type" meanwhile when $v_r \ll v_{2d}$, the concentration of the intermediate can be considered low and constant throughout the course of the reaction, and the steady-state approximation holds. In this case, the intermediates are called "van't Hoff type".

6.4.1.1 Arrhenius Intermediates: Pre-equilibrium Approximation

This approximation applies when the rate of the reaction making the intermediate to go back to the starting reactants (backward reaction of the first step) is much faster than that leading to products. The energy profile for the catalytic reaction is shown in Fig. 6.6

The corresponding rate equation will be

$$\frac{d[P]}{dt} = k_2[X][D]. \tag{6.26}$$

According to the pre-equilibrium approximation

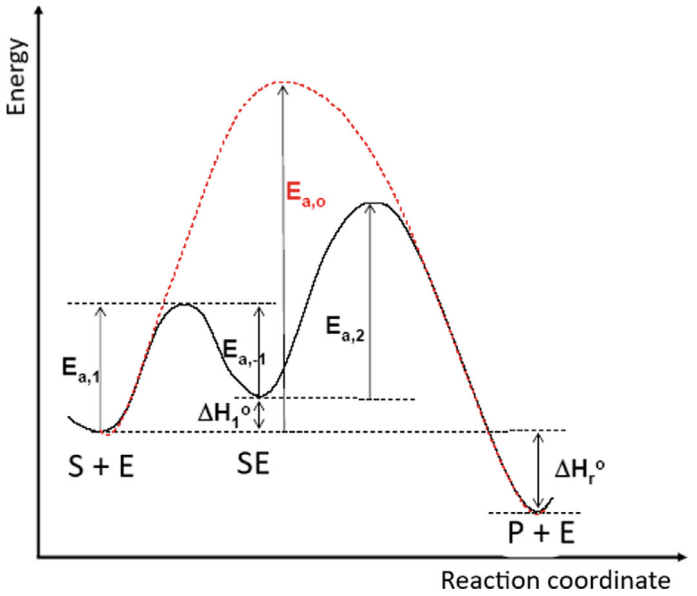


Fig. 6.6 Potential-energy profile for enzymatic catalysis between a substrate S and an enzyme E (note: here species Y and D does not participate in the reaction) and where $SC = X =$ substrate–catalyst complex. The rate-determining step is the second step, occurring after the formation of an Arrhenius intermediate ($Ea_2 > Ea_1$). Assuming that a very small quantity of substrate is present ($[S_o] \approx 0$), the activation energy of the overall enzymatic process is given by $E = E_{a1} - E_{a-1} + E_{a2}$)

$$K = \frac{k_1}{k_{-1}} = \frac{[X][Y]}{[C_{eq}][S_{eq}]}. \tag{6.27}$$

Part of C and S is transformed to X and P. A common approach is to determine initial rates, so that the amounts of products P formed are negligible. Thus, taking into consideration the corresponding mass balances

$$\begin{aligned} [S_{eq}] &= [S]_0 - [X] - [P] \approx [S]_0 - [X] \\ [C_{eq}] &= [C]_0 - [X], \end{aligned} \tag{6.28}$$

$$K = \frac{k_1}{k_{-1}} = \frac{[X][Y]}{([S]_0 - [X])([C]_0 - [X])}. \tag{6.29}$$

We can find the following limiting conditions. Typically, $[S] \gg [C]$, so that $[S_{eq}] \approx [S_o]$, the equilibrium constant simplifies to Eq. (6.30) and the rate of product formation is given by Eq. (6.31).

$$K = \frac{[X][Y]}{[S]_0([C]_0 - [X])} \Rightarrow [X] = \frac{K[C]_0[S]_0}{K[S]_0 + [Y]}, \tag{6.30}$$

$$v = \frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[X][D] = \frac{k_2 K[C]_0[S]_0[D]}{K[S]_0 + [Y]}. \tag{6.31}$$

If $K[S]_0 \ll [Y]$, i.e., for low substrate concentrations, v changes linearly with $[S]_0$ resembling a first-order process with respect to the substrate, however, when $K[S]_0 \gg [Y]$, v is independent of $[S]_0$ resembling a zero-order reaction with respect to $[S]$ but changes linearly with $[C]_0$ (i.e., first-order with respect to the catalyst).

The second limiting condition (not common) applies when $[S] \ll [\mathrm{C}]$, so that $[C_{\mathrm{eq}}] \approx [C_{\mathrm{o}}]$, and hence the equilibrium constant simplifies to Eq. (6.32) and the rate of product formation is given by Eq. (6.33).

$$K = \frac{[X][Y]}{[C]_0([S]_0 - [X])} \Rightarrow [X] = \frac{K[C]_0[S]_0}{K[C]_0 + [Y]}, \tag{6.32}$$

$$v = \frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[X][D] = \frac{k_2 K[C]_0[S]_0[D]}{K[C]_0 + [Y]}. \tag{6.33}$$

If $K[C]_0 \ll [Y]$, v changes linearly with $[C]_0$; in this case the reaction is always first-order process with respect to substrate independently of its concentration. When $K[C]_0 \gg [Y]$, v is independent of $[S]_0$ resembling a zero-order reaction with respect to $[S]$ but changes linearly with $[C]_0$ (i.e., first-order with respect to the catalyst). If $K[C]_0 \gg [Y]$, v is independent of $[C]_0$ but changes linearly with $[S]_0$.

6.4.1.2 Van't Hoff Intermediates: Steady-State Approximation

This approximation applies when the concentration of the intermediate can be considered relatively low and constant throughout most of the reaction. The energy profile for the catalytic reactions is shown in Fig. 6.7.

When van't Hoff intermediates are formed,

$$k_2[X][D] \gg k_{-1}[X][Y]. \tag{6.34}$$

And hence, $[X]$ is small, and the steady-state approximation given by Eq. (6.35) can be applied to this mechanism.

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = k_1[C][S] - k_{-1}[X][Y] - k_2[X][D] = 0. \tag{6.35}$$

The corresponding mass balance makes one to be aware that part of C and S is transformed into X and P. As a common approach, initial rates are determined so that the amounts of products P formed can be considered negligible.

$$\begin{aligned} [S_{\mathrm{eq}}] &= [S]_0 - [X] - [P] \approx [S]_0 - [X], \\ [C_{\mathrm{eq}}] &= [C]_0 - [X]. \end{aligned} \tag{6.36}$$

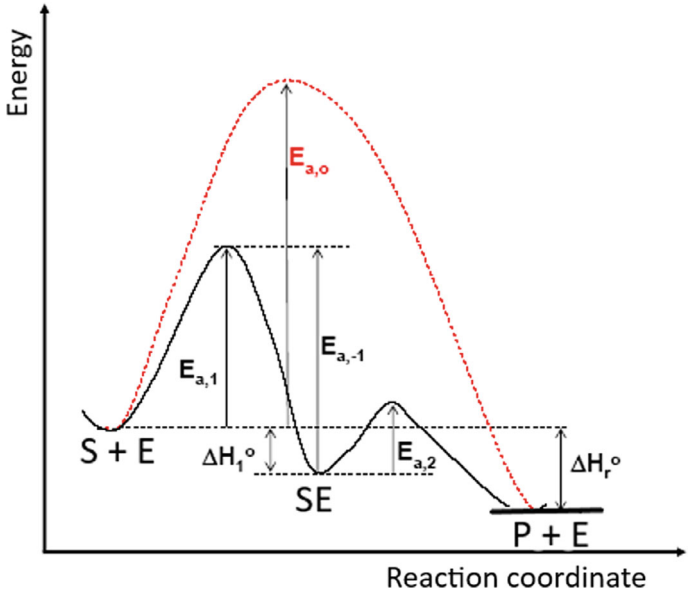


Fig. 6.7 Potential-energy profile for enzymatic catalyses, where the species Y and D does not exit (X = substrate–catalyst complex SE), and where the rate-determining step is the first step, which leads to the formation of van't Hoff intermediates. The activation energy of the overall process is given by $E = Ea_1$.

Taking into consideration the corresponding mass balances

$$k_1([C]_0[S]_0 - [C]_0[X] - [S]_0[X] + [X]^2) = (k_{-1}[Y] - k_2[D])[X]. \tag{6.37}$$

Reordering terms and bearing in mind that if the concentration of the intermediate is low, the quadratic term $[X]^2$ is very small and can be neglected. Hence

$$[X] = \frac{k_1([C]_0[S]_0}{k_1([C]_0 + [S]_0) + k_{-1}[Y] - k_2[D]}. \tag{6.38}$$

And the rate of the reaction is given by

$$v = \frac{\mathrm{d}[P]}{\mathrm{d}t} = k_2[X][D] = \frac{k_1k_2[C]_0[S]_0[D]}{k_1([C]_0 + [S]_0) + k_{-1}[Y] - k_2[D]}. \tag{6.39}$$

Several situations can be found depending on the relative values of the different terms in the denominator of the equation. A thorough analysis is not given here as it is very similar to that made for the Arrhenius intermediates.

6.4.2 Organometallic Catalysis

A large variety of effective organometallic catalysts have been described during the last 60 years. Due to their high efficiency, they have been utilized in numerous industrial and synthetic processes [21–23]. Among the organometallic catalysts, transition metal (Pt, Pd, Ru, Rh, etc.) complexes have achieved outstanding performance in several synthetic methods. However, the transition metals used are expensive, toxic, and scarce.

A very important problem in homogeneous organometallic catalysis is that most of the catalysts are deactivated by water and it is necessary to realize the reactions in organic solvents.

The kinetic model of these catalysts does not differ from that of any homogeneous catalyst, and the reaction model is through the formation of intermediates which can revert to reactants or undergo chemical changes that lead to products releasing the catalyst. The kinetic behavior of these catalytic processes is the same as those for a general homogeneous catalysis-shown in Sect. 6.4.1.

To resolve the stability and the loss of expensive and limited metals, two strategies have been adopted, but both imply to heterogenize the system.

(a) The possibility of anchoring the catalyst to a surface allowing the organometallic catalyst to be deposited on a surface [18].
(b) Another important advance in organometallic catalysis is the encapsulation of molecular guests into supramolecular hosts. The use of metal–ligand coordination to incorporate catalytically active molecules into supramolecular hosts has thus been an effective strategy for preparing catalysts with unusual selectivity and activity.

6.4.2.1 Acid–Base Catalysis

Many reactions are catalyzed by acids or bases, and this particular case of homogeneous catalysis can be achieved in different forms depending on the nature of the catalyst. Usually, chemists distinguish two types of acid–base catalyses:

Specific, where the catalyst is the acid or conjugated base of the solvent. Because water is the most universal solvent, in this type of catalysis H_3O^+ and/or OH^- are involved (coming from strong acids or bases), and the ions are transferred before the slow step of the reaction.

General, where the catalysts are Brönsted or Lewis acids or bases (usually weak) present in solution. In these cases, the proton is usually transferred in the slow step of the reaction. This is the case of reactions that are catalyzed by any basis other than OH^-.

The general catalytic mechanism can be described by Eq. (6.40). The overall rate of the reaction will be the sum of the rates for all alternative pathways, Eq. (6.41)

$$
\begin{aligned}
S &\xrightarrow{k_1} P,\\
S + [\mathrm{H}^+] &\xrightarrow{k_{\mathrm{H}^+}} P,\\
S + [\mathrm{OH}^-] &\xrightarrow{k_{\mathrm{OH}^-}} P,\\
S + [\mathrm{HA}] &\xrightarrow{k_{\mathrm{HA}}} P,\\
S + [\mathrm{A}^-] &\xrightarrow{k_{\mathrm{A}^-}} P,
\end{aligned}
\tag{6.40}
$$

$$
\begin{aligned}
v = -\frac{\mathrm{d}[S]}{\mathrm{d}t} = k[S] = k_1[S] + k_{\mathrm{H}^+}[\mathrm{H}^+][S]\\
+ k_{\mathrm{OH}^-}[OH^-][S] + k_{\mathrm{HA}}[\mathrm{HA}][S] + k_{\mathrm{A}^-}[\mathrm{A}^-][S].
\end{aligned}
\tag{6.41}
$$

In this expression, k_1 stands for the rate constant of the non-catalyzed reaction, and k_{H^+} and k_{OH^-} are the rate constants for catalysis by proton and hydroxide ion, and k_{HA} and k_{A^-} are the catalytic rate constants for the acid HA and the conjugated base A^-. The reader must be aware that in aqueous solutions, the protons and hydroxyl ions are strongly hydrated, and they are not present in the form of the free H^+ or OH^- ions. We will represent them indistinctly for simplicity.

Equation (6.41) shows the sum of the rates of the catalyzed and non-catalyzed reactions. In many catalytic systems, and particularly when working in aqueous solutions, only the general acid/base catalysis affects significantly the rate of the reaction. We also take into account that in aqueous systems, the ionic product of water is given by Eq. (6.42)

$$
K_W = [\mathrm{H}^+][\mathrm{OH}^-]. \tag{6.42}
$$

And hence the overall rate constant of an acid–base-catalyzed reaction is given by Eq. (6.43).

$$
k = k_1 + k_{\mathrm{H}^+}[H^+] + k_{\mathrm{OH}^-}[OH^-] + k_{\mathrm{HA}}[\mathrm{HA}] + k_{\mathrm{A}^-}[\mathrm{A}^-]. \tag{6.43}
$$

The reader must note that the rate of non-catalyzed reactions is usually much lower than that of the catalyzed ones. In the case of specific acid–base catalyses, the logarithmic expressions in Eq. (6.44) can be derived for the non-catalyzed, acid-catalyzed, and base-catalyzed reactions.

$$
\begin{aligned}
\log k &= \log k_1,\\
\log k_{\mathrm{cat}(\mathrm{H}^+)} &= \log k_{\mathrm{H}^+} - \mathrm{pH},\\
\log k_{\mathrm{cat}(\mathrm{OH}^-)} &= \log k_{\mathrm{OH}^-} + \log K_{\mathrm{W}} + \mathrm{pH}.
\end{aligned}
\tag{6.44}
$$

We note that for the non-catalyzed reaction, there is no dependence on the logarithm of the observed rate constant with pH; meanwhile a positive variation is

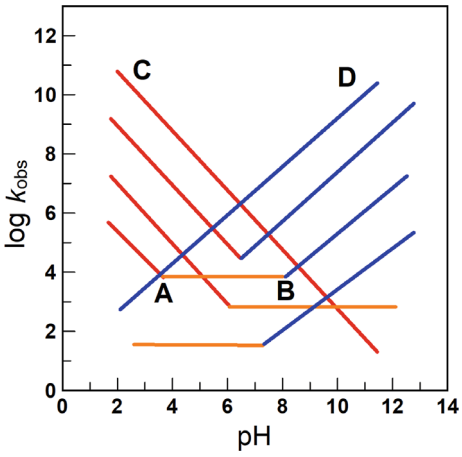


Fig. 6.8 Variations of the observed rate constant with pH for non-catalyzed reaction (slope null, A-B), acid catalyzed (negative slope as in C), and base catalyzed (positive slope, as in D). Note that there may be acid catalysis under basic conditions and base catalyze even under acidic conditions

predicted for specific base-catalyzed reaction and a positive variation for the specific acid–base catalysis as illustrated in Fig. 6.8.

6.4.3 General Acid Catalysis

The mechanisms for a general acid catalysis is shown in (6.45), from where Eqs. (6.46)–(6.48) can be easily derived.

$$\begin{aligned} &R + \mathrm{H}A \xrightarrow[\text{slow}]{k_1} R\mathrm{H}^+ + A^-, \\ &R\mathrm{H}^+ \xrightarrow[\text{fast}]{k_2} P\mathrm{H}^+, \\ &P\mathrm{H}^+ + A^- \xrightarrow[\text{fast}]{k_3} P + \mathrm{H}A, \end{aligned} \tag{6.45}$$

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_1[\mathrm{H}A][R], \tag{6.46}$$

$$\mathrm{H}A \overset{K_a}{\rightleftharpoons} A^- + \mathrm{H}^+, \tag{6.47}$$

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k_{\mathrm{obs}}[R] \Rightarrow k_{\mathrm{obs}} = k_1[\mathrm{H}A] = \frac{k_1[A^-][\mathrm{H}^+]}{K_a}. \tag{6.48}$$

And the typical variation in the observed rate constant with acidity for a general acid-catalyzed process is shown in Fig. 6.9

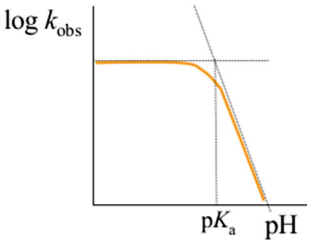


Fig. 6.9 Variation of log k_{obs} with pH for a general acid-catalyzed reaction

6.4.4 General Base Catalysis

The mechanisms for a general base catalysis are shown in (6.49), from where Eqs. (6.50)–(6.52) can be easily derived.

$$\begin{aligned} &RH + B + H_2O \xrightarrow[\text{slow}]{k_1} R^- + BH^+, \\ &R^- \xrightarrow[\text{fast}]{k_2} P^-, \\ &P^- + BH^+ \xrightarrow[\text{fast}]{k_3} PH + B, \end{aligned} \tag{6.49}$$

$$\frac{d[P]}{dt} = k_1[B][RH], \tag{6.50}$$

$$BH^+ \overset{K_B}{\rightleftharpoons} B + H_3O^+, \tag{6.51}$$

$$\frac{d[P]}{dt} = k_{obs}[RH] \Rightarrow k_{obs} = k_1[B] = \frac{k_1 K_B[BH^+]}{[H_3O^+]}. \tag{6.52}$$

And the typical variation in the observed rate constant with acidity for a general acid-catalyzed process is shown in Fig. 6.10

A summary of the mechanisms for acid–base catalysis is shown in Table 6.4. Each of these four mechanisms can be developed using the pre-equilibrium or the steady-state approximations to derive the corresponding rate laws.

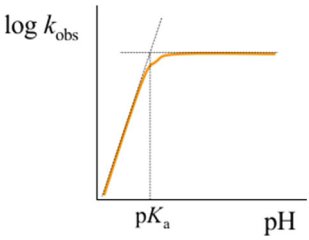


Fig. 6.10 Variation in the log k_{obs} with acidity for a general base-catalyzed process

Table 6.4 Summary of mechanisms for acid–base catalysis. Adapted from Arnaut et al. [20]

Acid catalysis	Arrhenius intermediate	Van't Hoff intermediate
H^+ transferred to solvent $BH^+ + S \rightleftarrows SH^+ + B$ $SH^+ + H_2O \rightleftarrows P + H_3O^+$	$k_2 \ll k_{-1}[B]$ Specific H catalysis With limiting rate	$k_2 \gg k_{-1}[B]$, $k_2 \gg k_1[BH^+]$ General catalysis No limiting rate
H^+ transferred to solute $BH^+ + S \rightleftarrows SH^+ + B$ $SH^+ + B \rightleftarrows P + BH^+$	$k_2 \ll k_{-1}$ General catalysis No limiting rate	$k_2 \gg k_{-1}$ General catalysis No limiting rate
H^+ transferred from solvent $B + SH \rightleftarrows S^- + BH^+$ $S^- + H_2O \rightleftarrows P + OH^-$	$k_2 \ll k_{-1}[BH^+]$ Specific OH^- catalysis With limiting rate	$k_2 \gg k_{-1}[BH^+]$, $k_2 \gg k_1[B]$ General catalysis No limiting rate
H^+ transferred from solute $B + SH \rightleftarrows S^- + BH^+$ $S^- + BH^+ \rightleftarrows P + B$	$k_2 \ll k_{-1}$ General catalysis No limiting rate	$k_2 \gg k_{-1}$ General catalysis No limiting rate

The above discussion shows that the dependence of the reaction rate upon the pH contains very important information on the reaction mechanism. Each rate must be measured at constant pH, which usually involves measuring it in a buffer solution. In addition, usually an inert salt is added to maintain ionic strength constant to avoid the salt effects. They also show that the dependence of the reaction rate on acidity contains very important and valuable information on the reaction mechanism. More developed treatments of reactions where protons or bases are involved and how their knowledge can be exploited to help proposing a reaction mechanism can be found in Chap. 8.

6.4.5 Proton Transfer Reaction. Brönsted Equation

Consider the acid–base reaction

$$HA + B^- \xrightarrow{k} A^- + BH.$$

Brönsted and Pedersen [24, 25] proposed that if the rate k at which proton is removed by a given base B^- were compared for various acids, then the base might remove the proton more rapidly from the stronger acid. This relationship has been in many cases and follows an equation known as Brönsted catalysis law

$$k = GK_a^{\alpha}, \tag{6.53}$$

where G and α are empirical constants, k the rate of the acid–base reaction, and K_a the acid dissociation constant. In the logarithmic form

$$\log k = \log G + \alpha \log K_a. \tag{6.54}$$

Equivalent procedures can also be carried out for base-catalyzed reactions. Taking into consideration the relationships between the rate constants and equilibrium constants with the Gibbs free energy (Eyring (6.55) and van't Hoff (6.56) equations),

$$k = \frac{k_B T}{h} \mathrm{e}^{-\left(\frac{\Delta G^{\pm}}{RT}\right)}, \tag{6.55}$$

$$K_a = \mathrm{e}^{-\left(\frac{\Delta G^{0}}{RT}\right)}. \tag{6.56}$$

From where the Brönsted law given by Eq. (6.57) can be obtained

$$\Delta G^{\pm} = \text{constant} + \alpha \Delta G^{0}. \tag{6.57}$$

Equation (6.57) shows that, at constant temperature, the Brönsted equation is a linear-free energy relationship (LFER). Chemists have tried for many years to use the Brönsted coefficient as a measure of the position of the transition state along the reaction coordinate, since $\alpha = 0$ for reactants and $\alpha = 1$ for products. That is, α corresponds to the transfer of the proton from the acid AH to the transition state of the reaction with B^-. Setting the coefficient β as that for the transfer of the proton from BH to the transition state of the reaction with A^-, $\alpha + \beta = 1$. The α and β values can be determined from the slopes of linear plots log k_1 versus pKa and log k_{-1} versus pKa, respectively, and one can get some valuable information about the charge development in the transition state of the reaction formation. If the α or β values are small (0.1–0.3), it means that very little proton transfer has occurred in the transition state and the transition state is like reactive; on the contrary, if the α or β values are large (0.8–1.0), the transition state is more like product. According to this, α or β values close to 0.5 suggests a concerted proton transfer [24, 25].

6.5 Microheterogeneous Catalysis

Microheterogeneous catalysis is a rather special case of catalysis lying somewhere between the homogeneous and heterogeneous catalyses. In most cases, both reactants and catalysts are in the same bulk phase, but at the molecular level, the catalysts are such a complex structure that creates "areas" or "regions" with properties different from those of the bulk solvent. Enzymatic catalysis, also called biocatalysis, and micellar catalysis are common examples of this type of catalysis [13, 26, 27].

6.5.1 Enzymatic Catalysis

Enzymes are the best catalysts known. They are mostly high molecular weight proteins mediating in a huge array of transformations in living organisms under mild conditions biological catalysts, and these compounds present in their structure some specific sites in which a substrate, and not others, can bind, which gives the quality of specific catalysts. Enzymes are extremely efficient catalysts, completing more than 1000 catalytic cycles per second.

Compared to this rates, conventional homogeneous and heterogeneous catalysts are "slow" and "inefficient" as they may provide as much as 100–10,000 cycles per hour. The achievable high rates of reactions are not the only advantage of enzymes. They are very specific in converting one specific reactant into another specific product and this specificity even includes chiral centers, so an enzyme will catalyze the reaction of one substrate enantiomer, but not of the other. Such enantiospecific reactions are extremely important in the pharmaceutical and food industries, because often only one enantiomer has the desired activity. This is the case for many drugs, food additives, flavorings, and fragrances. Further advantages of biocatalysis over common chemical catalysts include shorter synthesis routes and frequently much milder reaction conditions.

The velocities of reactions catalyzed by enzymes depend on the same factors as all other chemical reactions: nature of reactants, concentrations, temperature and the nature of the medium (solvent). The main reason for the enormous rate enhancements achieved by enzymes is the very special role played by the enzyme as a specific "medium" for the reaction. It is currently accepted that enzymatic catalysis involves sophisticated molecular recognition and that this accounts for a major part of the efficiency of enzymes. Molecular recognition is a consequence of the balance of complementarities of non-covalent forces. For one side, the active site of the enzyme stabilizes the transition state through steric effects, proximity, desolvation, entropic effects, electrostatic forces, and van der Waals forces. On the other hand, molecular recognition is enforced by partially covalent interactions such as hydrogen bonding and general acid–base catalysis. In extreme cases, the transient formation of an enzyme–substrate covalent bond also takes place. All these forces contribute to high enzyme–substrate association constants and to lower the energy of the transition state with respect to the isolated enzyme and substrate.

6.5.1.1 Michaelis–Menten Equation

The simplest mechanism to describe the action of an enzyme is that depicted in Eq. (6.58), where a substrate S reacts with an enzyme E to yield the enzyme–substrate complex ES which can either decompose to the starting material or to further react to yield the products P and regenerate the enzyme and Fig. 6.11 shows a typical energy-reaction coordinate diagram for enzymatic catalysis.

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} P + E. \tag{6.58}$$

In general, the real concentrations of the enzyme and the substrate are not known because, for example, part of the substrate can react being as "free" substrate, in form the substrate-enzyme complex or as a product. Under conditions where the time of reaction is sufficiently low to consider the production of product negligible, the stoichiometric amount of substrate is given by Eq. (6.59). In a similar way, the enzyme can be found either as free enzyme or as complex, Eq. (6.60).

$$[S]_T = [S] + [ES] + [P] \approx [S] + [ES], \tag{6.59}$$

$$[E_T] = [E] + [ES]. \tag{6.60}$$

The rate of the reaction is given by Eq. 6.61

$$v = k_2[ES]. \tag{6.61}$$

Assuming the steady-state approximation, Eq. (6.62), holds, Eq. (6.63) can be easily derived

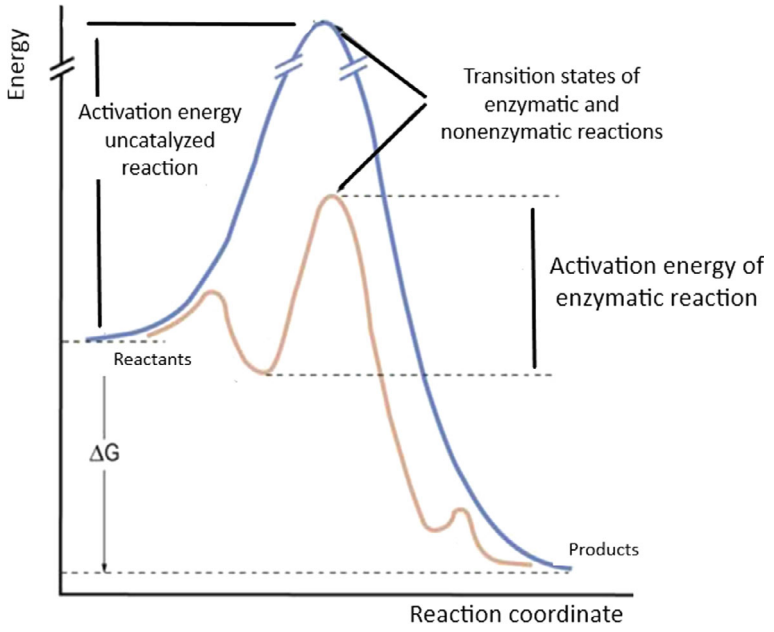


Fig. 6.11 Typical energy diagram of a reaction in the absence and presence of an enzyme

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] = 0. \tag{6.62}$$

Taking into account Eq. (6.60)

$$[ES](k_{-1} + k_2 + k_1[S]) = k_1[E_T][S]. \tag{6.63}$$

The concentration of the intermediate is given by Eq. (6.64)

$$[ES] = \frac{[E]_T[S]}{K_M + [S]}, \tag{6.64}$$

$$K_M = \frac{k_{-1} + k_2}{k_1}, \tag{6.65}$$

where K_M (Michaelis constant) represents the dissociation constant of the enzyme–substrate complex when $k_{-1} \gg k_2$. Low values of K_M thus correspond to strong enzyme–substrate complexes. The reaction rate is, therefore, given by Eq. (6.66)

$$v = k_2[ES] = k_2\frac{[E]_T[S]}{K_M + [S]} = \frac{v_{MAX}[S]}{K_M + [S]}. \tag{6.66}$$

When the concentration of the substrate is sufficiently small, $[S] << K_M$ and the reaction rate is first order with respect to the substrate. For high substrate concentrations, $[S] \gg K_M$ and the reaction rate is zero order with respect to the substrate, Fig. 6.12 left.

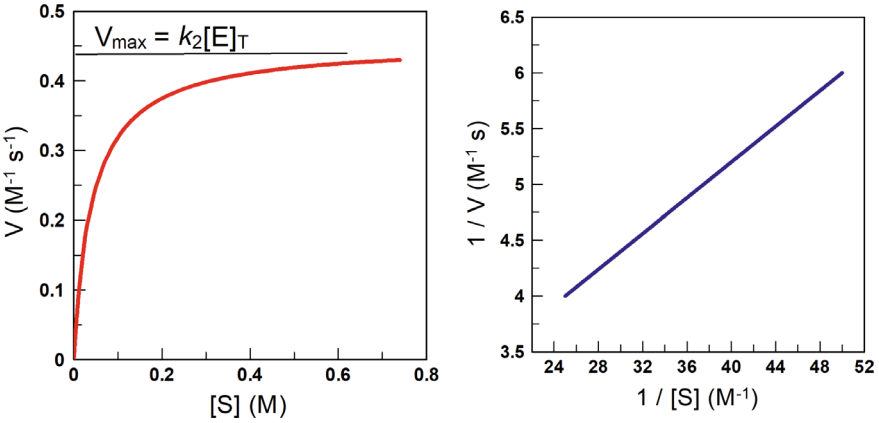


Fig. 6.12 Left. Simulated variation of the initial rate for the reaction of hydrolysis of sucrose catalyzed by the enzyme invertase. Right. Plot of the inverse of the initial rate with the inverse of substrate concentration (Lineaweaver–Burk plot). Simulation was carried out by employing the following values: $V_{max} = 0.45\ M^{-1}\ s^{-1}$, $K_M = 0.043\ M$

Equation (6.66) allows one to define the catalytic constant of an enzyme when $K_M = [S]$. This quantity is also known as the turnover number of an enzyme because it is the number of reaction processes (turnovers) that each active site catalyzes per unit time. It is obvious from Eq. (6.61) that, for the Michaelis–Menten mechanism, $k_{cat} = k_2$. However, for more complex mechanisms, k_{cat} may be a function of several rate constants.

$$k_{cat} = \frac{v_{max}}{[E_T]} \tag{6.67}$$

The double reciprocal plot of Eq. (6.66) predicts that, for the Michaelis–Menten mechanism, the dependence of $1/v$ with $1/[S]$ is linear, Fig. 6.12 right, with a slope $S = K_M/V_{max}$ and an extrapolated intercept $I = -\ 1/K_M$, from where the values of K_M and V_{max} can be determined. An alternative way of determining such parameters is through the Eadie–Hofstee plots, which result from a different mathematical rearrangement of Eq. (6.66) as indicated in Eq. (6.68)

$$\frac{v}{[S]} = \frac{v_{max}}{K_M} + \frac{1}{K_M} v. \tag{6.68}$$

Thus, plots of $v/[S]$ versus v should be linear with an intercept $I = v_{max}/K_M$ and a slope $S = 1/K_M$. The individual values of the rate constants k_1, k_{-1}, and k_2 cannot be determined from steady-state studies, and their determination requires the use of techniques to monitor fast reactions such as stopped-flow to monitor the initial moments of enzyme catalysis and measuring the length of the induction period.

6.5.1.2 Inhibition of Enzyme Catalysis

The efficiency of enzymes can be reduced or even become negligible in the presence of inhibitors I, which are chemicals with some structural resemblance with the enzyme and that competes to bind the substrate. The inhibition process can be reversible or irreversible depending on the degree of binding of the inhibitor to the enzyme. The mechanism for the enzymatic reaction in the presence of an inhibitor is that given by (6.69)

$$\begin{aligned} &E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} P + E, \\ &E + I \underset{k_{-3}}{\overset{k_3}{\rightleftharpoons}} EI. \end{aligned} \tag{6.69}$$

Assuming that the inhibitor binds reversibly to the enzyme in a rapid equilibrium and bearing in mind the definition of the Michaelis constant K_M, Eq. (6.70) can be derived.

$$K_I = \frac{[E][I]}{[EI]} \Rightarrow [E] = \frac{K_M[ES]}{[S]}. \tag{6.70}$$

Bering in mind the corresponding mass balance, Eq. (6.71), Eq. (6.72) can be derived

$$[E_0] = [E] + [ES] + [EI], \tag{6.71}$$

$$[E] = [ES]\left(\frac{K_M}{[S]}\left(1 + \frac{[I]}{K_I}\right) + 1\right) \Rightarrow [ES] = \frac{[E_0][S]}{K_M\left(1 + \frac{[I]}{K_I}\right) + [S]}. \tag{6.72}$$

And hence, the rate of the catalytic process is given by Eq. (6.73)

$$v = \frac{k_2[E_0][S]}{K_M\left(1 + \frac{[I]}{K_I}\right) + [S]}. \tag{6.73}$$

The double reciprocal of Eq. (6.73) yields the corresponding Lineweaver–Burk plot.

6.5.2 *Micellar Catalysis*

Chemists are continuously looking for biological transformations as models of handling essentially no waste, limited energy intake and as models of very efficient processes. From the chemical and environmental point of view, water is the perfect solvent to carry out organic reactions because of its intrinsic properties. It is the universal solvent; as it is available in large quantities, it is not toxic, not flammable, and can be stored indefinitely without any hazard occurring. Its physical properties such as its high thermal capacity and latent heat of evaporation provide some advantages in controlling exothermic reactions, mainly when upscaling preparations [14, 28].

However, the fact that most organics are hydrophobic limits the number of chemical reactions compatible with aqueous environments, and the common wisdom in attempting to dissolve reagents in a common phase to efficiently react is getting increasingly challenged. Nowadays, mainly because we are concerned about the environment, all chemical processes are finely tuned to reduce the amount of wastes, reduce energy consumption and, in some instances, resources depletion.

Some of these challenges have been achieved with the use of surfactants, Fig. 6.13. Surfactants are amphiphilic molecules that have a hydrophilic "head" and a hydrophobic hydrocarbon chain. They are capable of reducing the surface tension of water reducing the interfacial energy required to solubilize hydrophobic derivatives in aqueous environments because the hydrophilic domain of surfactants

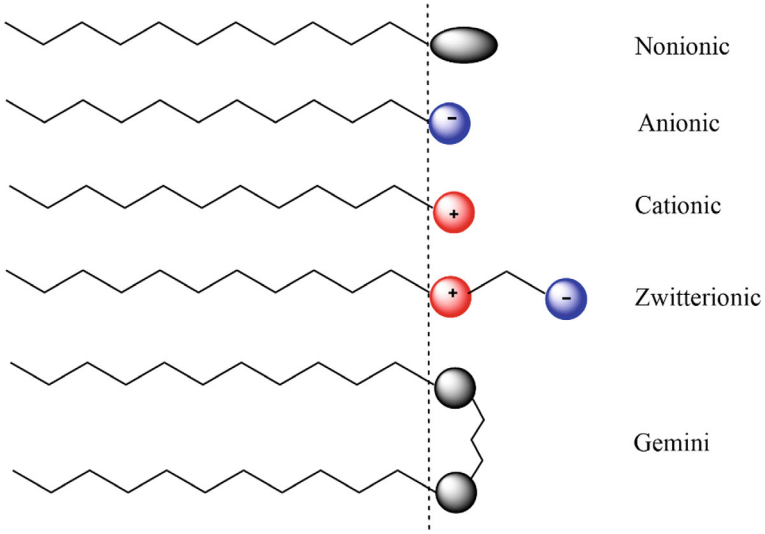


Fig. 6.13 Illustrative representation of different types of common surfactants showing their hydrophobic tail and the hydrophilic head

interacts with water, but the hydrophobic domain has a greater tendency to interact with oily substances.

Surfactants can be classified according to presence or not of a net charge, as in Fig. 6.13, according to their linear or branched structure and/or their molecular weight. They can also be classified in terms of their industrial availability. Industrial surfactants comprise all derivatives that are commercially available on an industrial scale and that have been optimized having in mind the purposes of formulation chemistry. Rather than being designed to perform chemistry, they are aimed at providing stable formulations of a given composition. Designer surfactants are not readily available and frequently used for a single specific purpose. Surfactants are generally classified and selected according to the corresponding hydrophilic lipophilic balance (HLB), a dimensionless number correlated with the mass ratio between the hydrophilic and lipophilic portions of a surfactant (see Chap. 8 for further information). The HLB scale enables to distinguish between neutral lipophilic (HLB $<$ 10) derivatives, prone to form water in oil emulsions, hydrophilic ones (HLB $>$ 10), suitable for oil in water emulsions, and balanced (HLB $=$ 10) surfactants. Charged surfactants have HLB values higher than 20. Formulations chemists select surfactants according to the affinity principle, so that the best derivative is the one matching the characteristics of the material to be dispersed. Further information about surfactants and different colloidal systems will be given in Chap. 8.

Above a certain concentration, most surfactants self-assemble in a variety of association colloids, and the most common is the spherical micelle, Fig. 6.14. The common characteristic of all colloidal systems is the presence of an interfacial region separating the bulk aqueous environment and the hydrophobic core where

hydrophobic molecules can be solubilized. These domains are usually denoted as "pseudophases" because their boundaries are not well-delimited as, for instance, in the phase-transfer or heterogeneous catalysis.

The idealized two-dimensional representation of the spherical micelle shows the hydrophobic core region formed by the surfactant tails and the interfacial region containing the polar (hydrated) head groups of the surfactant. Such an image is iconic as the surface of the micelles is fluid, the head groups do not define a smooth surface, and the tails are almost randomly distributed [27]. A key property of ionic micelles on water is their charge, which attracts counterions to the micellar surface. Their shape is a result of the balance between the hydrophobic forces, which tend to aggregate the unimers away from water, and the repulsive forces between the ionic head groups of the unimers. For a give ionic micelle, not all counterions are needed to form the micelle so that only a fraction of them is needed while the rest remains in the aqueous phase. This degree of ionization is usually denoted by β, which stands for the fraction of head groups that are neutralized by counterions on the Stern layer. Alternatively, one can design as α the fraction of counterions that do not neutralize head groups and are "lost" from a—hypothetical—neutral micelle, so that $\alpha + \beta = 1$. The micellar charge can be estimated by different techniques including electrophoresis, electrochemistry and light-scattering. Hence, the number of ions in bulk solution will be equal to the concentration of unimers (i.e., that of the critical micelle concentration) plus those generated by ionization of the micelle.

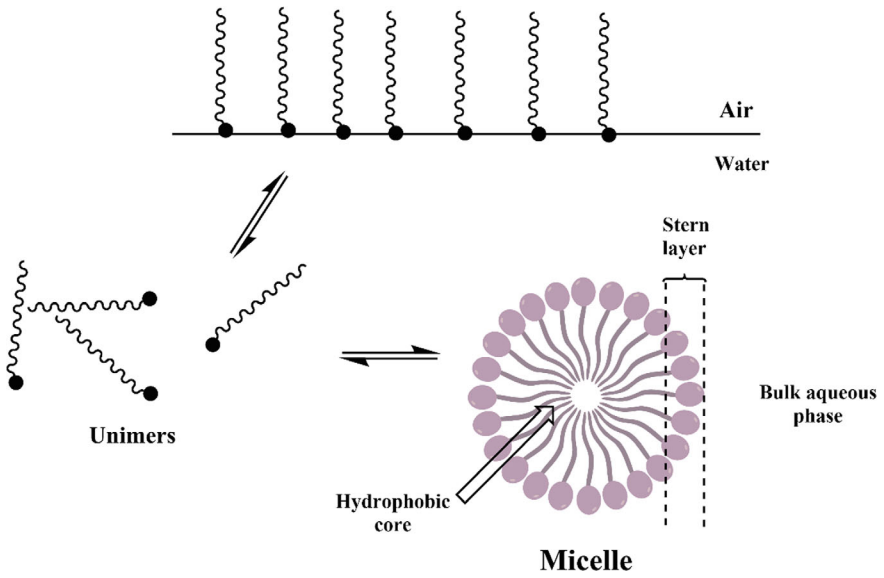


Fig. 6.14 Illustrative picture of the surfactant unimer distribution in aqueous solution and the spontaneous self-assembly of unimers into spherical micelles. The equilibrium arrows represent the (rapid) exchange of unimers between the air/water interface and micelles

Values of β are generally in the range 0.7–0.9, and they are relatively independent of the presence of added electrolyte in the solution. However, different experimental methods, and sometimes the same method in different hands, often give different values of β, and it may be that the value of this parameter does indeed depend on the methods used in its determination.

In summary, one can envisage micelles behave as if they have a lipid-like interior with a rough surface with the head groups in contact with water and counterions. The extent of water penetration is a matter of debate but in view of the rapid reforming and breakdown of micelles, one would expect that, on average, the alkyl groups of the surfactant will be exposed to water. Polar solutes may be located close to the micellar surface, dictating their local environment.

From the reactivity point of view, the most important feature of surfactants is probably their ability to, spontaneously, form self-assembly colloidal systems such as micelles. The self-assembly of micelles enables a steep increase in the solubility of lipophilic organics which can be in different regions of the micelle depending on the nature of the surfactant employed and on the polarity of the different reagents. Computational and experimental studies provide evidence that the lipophilic reagents tend to be in the interfacial region (Stern layer), where their local concentrations are much higher than the stoichiometric ones because of the much lower volume of the region. Hence, the presence of micelles causes significant changes in the reaction medium (compared to that of the bulk solvent) and because the micelles can change the local concentration of the substrates isolating species from the bulk solvent, or also due to the stabilization of substrates or/and intermediates, with the consequent change in the kinetics and, sometimes, on the mechanisms of chemical reactions.

6.5.2.1 Pseudophase Models

Pseudophase models consider the totality of surfactant aggregates (e.g., micelles) as a separate phase or pseudophase and do not include assumptions about aggregate size and shape [27]. The model assumes that surfactant unimers enter and exit micelles (or other association colloid) at near diffusion-controlled rates, so that the distribution of components is in dynamic equilibrium once the initial mixing is complete, and those rates are orders of magnitude faster than the typical rate constants for bimolecular reactions controlled by activation.

Aqueous micellar solutions contain two separate reaction media, micelles and water. And because components are in dynamic equilibrium, the value of the observed rate constant for a first-order reaction can be expressed as the sum of the rates in each region. These pseudophases have different solvent properties, and hence the rate constants values in each region are different. Such a reaction scheme can be illustrated, for the simplest unimolecular reactions, as illustrated in Fig. 6.15. Menger and Portnoy [29] developed the micellar pseudophase model considering micelles as a reaction medium different from pure water. This model assumes the reactants distribute thermodynamically between the bulk aqueous and micellar pseudophases.

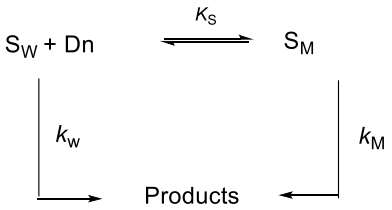


Fig. 6.15 Two-site model mechanism shows the incorporation of a substrate into the micellar environment and allowing the possibility of reaction in both the aqueous and micellar pseudophases

In Fig. 6.15, Dn represents the concentration of micellized surfactant, that the concentration of surfactant that forms micelles, and it is usually considered equal to the total or stoichiometric concentration of surfactant minus that necessary to form the micelles (the critical micelle concentration, CMC), Eq. (6.74), S_w the substrate in water, S_M the substrate associated with the micelles, and k_w and k_M the pseudo-first-order rate constants in water and in micelles, respectively.

$$D_n = [\text{Surf}_T] - \text{CMC}. \tag{6.74}$$

Assuming that the dynamic equilibrium condition holds, the rate of the reaction will be the sum of the rates in each pseudophase, Eq. (6.75), where $[S_T]$ stands for the stoichiometric concentration of the substrate S.

$$r = k_{\text{obs}}[S_T] = k_W\,[S_W] + k_M[S_M]. \tag{6.75}$$

The association constant of the substrate S to the micelle can be characterized by the equilibrium constant given in Eq. (6.76)

$$S_W + D_n \overset{K_S}{\leftrightarrows} S_M \Rightarrow K_S = \frac{[S_M]}{[S_W]D_n}. \tag{6.76}$$

The concentrations of the substrate in each pseudophase are given by Eq. (6.77)

$$\begin{aligned}[S_W] &= \frac{[S_T]}{1 + K_S D_n},\\ [S_M] &= \frac{K_S D_n [S_T]}{1 + K_S D_n}.\end{aligned} \tag{6.77}$$

After substitution in Eq. (6.75) and solving for k_{obs}, Eq. (6.78) can be easily derived

$$k_{\text{obs}} = \frac{k_W + k_M K_S D_n}{1 + K_S D_n}. \tag{6.78}$$

Equation (6.78) shows that the kinetic profile obtained for micellar-catalyzed unimolecular reactions is like those obtained from Michaelis–Menten enzymatic reactions, a linear increase in the rate constant upon increasing the surfactant concentration leveling off when the substrate is totally incorporated to the micellar aggregates (that is, high [Surf]).

If the substrate partitions between the micellar and aqueous pseudophase but only reacts in the aqueous region, then $k_M \approx 0$ and Eq. (6.79) holds, predicting that the observed rate constant decrease upon increasing the surfactant concentration showing hyperbolic-like kinetic patterns as shown in Fig. 6.16.

$$k_{\text{obs}} = \frac{k_W}{1 + K_S D_n}. \tag{6.79}$$

Pseudophase models also work for bimolecular reactions between two uncharged organic substrates and third-order reactions, reactions in vesicles and microemulsions, which may include partitioning into and reaction in the oil region, reactions of substrates with an ionizable (e.g., deprotonatable) second reactant, and the effect of association colloids on indicator equilibria. The observed or measured rate constant, k_{obs}, under pseudo-first-order conditions of a bimolecular reaction between a substrate S and a second neutral reactant B in micelles is the sum of the rates in the micellar and aqueous regions according to Fig. 6.17.

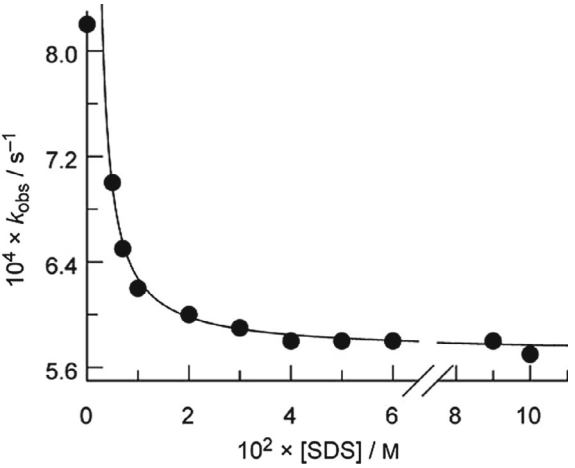


Fig. 6.16 Example of a micellar-induced inhibition reaction: Effects of sodium dodecyl sulfate on the spontaneous decomposition of 3-methybenzenediazonium ions [30]. Micelles are formed above the CMC

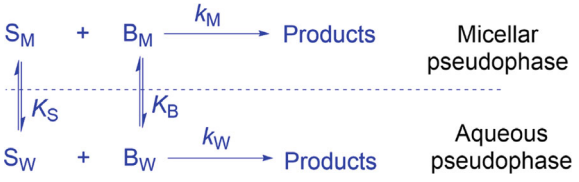


Fig. 6.17 Mechanism of a bimolecular reaction according to the pseudophase model

The rate of the reaction is given by Eq. (6.80), where the subscripts T, W, and M represent the stoichiometric, aqueous and interfacial concentrations. The second-order rate constants, k_2, ${k_2}^W$, and ${k_2}^M$, in units of M^{-1} s^{-1}, are for the observed, aqueous, and micellar reactions, respectively.

$$r = k_{obs}[S_T] = k_2[S_T][B_T] = k_2^W[S_W][B_W] + k_2^M[S_M][B_M]. \tag{6.80}$$

In general, reactions are run under first-order conditions with $[B_T] > [S_T]$ such that the concentration $[B_T]$ remains constant throughout the reaction, and $k_{obs} = k_2[N_T]$. The value of ${k_2}^W$ is set equal to the rate constant in water in the absence of micelles.

Bearing in mind the corresponding mass balance and the distribution of the reactants, Eq. (6.81), the relationship between the total (stoichiometric) concentrations of the reactants and the concentrations in each pseudophase is given by Eq. (6.82).

$$\begin{aligned} [S_T] &= [S_W] + [S_M] & [B_T] &= [B_W] + [B_M], \\ K_S &= \frac{[S_M]}{[S_W]D_n} & K_B &= \frac{[B_M]}{[B_W]D_n}, \end{aligned} \tag{6.81}$$

$$\begin{aligned} [S_W] &= \frac{[S_T]}{1 + K_S D_n} & [S_M] &= \frac{[S_T]K_S D_n}{1 + K_S D_n} \\ [B_W] &= \frac{[B_T]}{1 + K_B D_n} & [B_M] &= \frac{[B_T]K_B D_n}{1 + K_B D_n}. \end{aligned} \tag{6.82}$$

The concentration units of S and B in the aqueous and micellar pseudophases require special consideration [13, 27, 31]. The rate of the reaction in the micellar pseudophase depends on their concentrations in the interfacial region (Stern layer) and not on the whole solution volume. This point is crucial. The concentrations of S and B in the interfacial region are expressed in moles/liter of interfacial volume as defined by Eq. (6.83) showing the relationship between the interfacial concentration of B, B_M, and the stoichiometric concentrations of B and S_M (micellized surfactant) in micelles:

$$B_M = \frac{[B_M]}{[S_M]V_M}, \tag{6.83}$$

where V_M is the molar volume of reaction and is generally set equal to the estimated volume of the interfacial region in liter/mole and $[S_M]$ is the concentration of micellized surfactant ($[S_M] = [S_T] - CMC$). The value of B_M defines how concentrated reactant B is in the interfacial compared to its stoichiometric concentration, $[B_T]$. An equation of the same form is used to describe the interfacial concentration of S. The value of V_M is typically ~ 0.15 M^{-1} and can be estimated by assuming that micelles have a density of ~ 1 g cm^{-3}, and that the interfacial region occupies about 50% of the total micellar volume.

Equation (6.84) for the rate constant of the reaction can be derived by combining Eqs. (6.81)–(6.82) [13, 27, 31].

$$k_2 = \frac{k_2^W}{(1 + K_S D_n)(1 + K_B D_n)} + \frac{k_2^M K_S K_B (D_n)^2}{(1 + K_S D_n)(1 + K_B D_n)}. \tag{6.84}$$

Note that the value of k_2^{M}, which represents the second-order rate constant in the micellar pseudophase, is expressed in units of local molarity of B in its reaction volume in in the micellar pseudophase, V_M, and not its stoichiometric concentration in solution. Its value cannot be estimated independently but by fitting k_{obs} versus $[S_T]$ profiles using independent estimates of the other parameters.

6.5.2.2 Ionic Surfactants and Ion Exchange

If the surfactant is ionic, counterions may be exchanged because micelles and other aqueous ionic interfaces act as selective ion exchangers. For example, when SDS micelles are present, the Na^+ counterions surrounding the micelle may exchange with cations present in the bulk aqueous phase (e.g., H^+), or, in the case of cationic micelles such as CTAB, OH^-, or CN^- ions exchange with the Br^- ions surrounding the CTAB micelle.

The distribution of reactive counterions is described by an ion-exchange constant, Eq. (6.85).

$$N_M + X_W \overset{K_N^X}{\rightleftharpoons} N_W + X_M \Rightarrow K_N^X = \frac{[N_W][X_M]}{[N_M][X_W]}. \tag{6.85}$$

Typically, $K_N{}^X$ increases as the size of the X increases (and is more weakly hydrated) relative to N, what means that the order of effectiveness of X in displacing N from the micellar interface generally follows a Hofmeister series for both cations and anions [27]. Combining the rate expression with the equations for the association of S to micelles and ion exchange of N and X, the degree of counterion binding β, for X, with Eq. (6.86), and the corresponding mass balance, the observed second-order rate constant, k_2, is given by Eq. (6.87) [27, 31].

$$N_M = \frac{[N_M]}{[D_N]V_M} \quad X_M = \frac{[X_M]}{[D_N]V_M}, \tag{6.86}$$

$$k_2 = \frac{k_2^W}{(1 + K_S D_n)} + \frac{\frac{k_2^M K_S \beta D_n}{V_M}}{(1 + K_S D_n)([N_T] + [X_T]K_N^X)}. \tag{6.87}$$

The two terms of this equation are composed of independently measurable (or estimable) constants and experimentally fixed concentrations. The first term, the contribution in water decreases rapidly above the CMC and could be neglected, but makes an important contribution in those reactions that are inhibited. The second term of the equation, that reflects the contribution of the rate constant in the micellar system to the overall rate constant, describes a maximum in k_2 with increasing the

surfactant concentration [Surf$_T$], Fig. 6.18. Below the CMC, no micelles are present and $k_2 = k_2{}^W$. Above the CMC, increasing D_n (e.g., [Surf$_T$]) increases the micelle concentration and the fraction of reactants in the interfacial region of the micelles. The rate of increase in their bound fractions depends on the values of the association constant K_A and of the ion-exchange constant $K_N{}^X$. Concomitantly with the increase in the fraction of reactants in the interfacial region, their molarities in the interfacial region (moles per liter of interfacial volume) are decreasing continuously because the number of micelles is increasing and hence the volume of reactions increases. Thus, the rate maximum is a crossover point where the value of $k_{\text{rel}} = k_2/k_2{}^W$ changes from being determined by the transfer of reactants from the bulk aqueous solvent to the much smaller volume of the micellar pseudophase to dilution of the reactants within the micellar pseudophase.

At very high $[\text{Surf}]_T$, when the reactants are completely bound, Eq. (6.87) simplifies to Eq. (6.88).

$$k_2 = \frac{\frac{k_2^M \beta}{V_M}}{([N_T] + [X_T])K_N^X}. \tag{6.88}$$

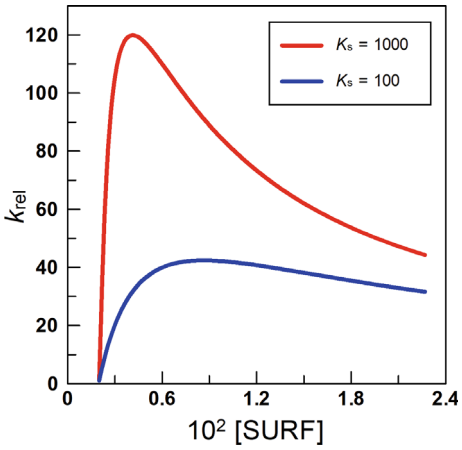


Fig. 6.18 Simulated effect of increasing the hydrophobicity of a substrate on $k_{\text{rel}} = k_2/k_2{}^W$ upon increasing the concentration of a monovalent surfactant ($[X_T] = [\text{Surf}]_T$. The value of $k_{\text{rel}} = 1$ at the CMC and below. The maxima in k_2 are a consequence of the incorporation of the substrates into the micelles and its further dilution once all substrates have been incorporated to the micelle. See the text for further information. Simulation was carried out by employing Eq. (6.87) and the following parameters: CMC = 0.002 M, $[S_T] = 0.01$ M, $K_N{}^X = 4, \beta = 0.7, V_M = 0.15\ \text{M}^{-1}, k_2{}^W = k_2{}^M$.

6.5.2.3 Reactive Counterion Surfactants

Micelles prepared with reactive counterion surfactants, where the counterion is a reactant, can be easily prepared. For example, H^+ in hydrogen dodecyl sulfate, HDS, or OH^- on cetyltrimethylammonium hydroxide, CTAOH, micelles. In these cases, the stoichiometric concentration of micellized reactant N (the reactive counterion) increases upon addition of surfactant, but the interfacial molarity, N_M, remains constant because it is assumed that the degree of ionization of the micelle, $\beta = [N_M]/[S_M]$ remains constant and independent of surfactant concentration and the concentration of any added salt. Hence, Eq. (6.87) reduces to Eq. (6.89), which has the same mathematical form as that for unimolecular reactions.

$$k_2 = \frac{k_2^W [N_W] + \frac{k_2^M K_S \beta D_n}{V_M}}{K_S D_n}. \tag{6.89}$$

6.6 Supramolecular Catalysis

Catalysis in nature often involves supramolecular interactions, for example, with enzymes to bring substrates into a configuration or position that is favorable for a certain reaction. Researchers tend to use biological systems in nature as models and try to mimic biological processes to gain a better understanding and subsequently transfer this knowledge to artificial systems. From the twentieth century to the modern day, several structures of macrocyclic compounds have been described including cyclodextrins, crown ethers, calix[n]arenes, carcerands, cyclobis(paraquat-p-phenylene), pillar[n]arenes, and cucurbit[n]urils [32, 33]. At the same time, the development of efficient enzyme-mimetic catalysts led researchers to design new macrocyclic host systems and to employ them to catalyze chemical reactions and to understand their mechanisms of action. General speaking, macrocycles are perfect compounds for these investigations. Because of their versatile geometries, preorganized cavities, and special binding affinities, they are able to coordinate and encapsulate a huge variety of different guests. Figure 6.19 shows the chemical structures of representative cyclodextrins, cucurbiturils, and calixarenes used in supramolecular catalysis.

The determining factors for binding properties of the supramolecular structures are their rigid, highly symmetrical structure and their hydrophobic cavity in contrast to the polarized outer rim. In general, an important binding property of these supramolecules is their applicability in aqueous media since potential guests can have high binding affinities. The water molecules inside the cavity are crucial for interactions in aqueous environments because they can compete with the guest by way of hydrogen bonds. Hosts can release high-energy water and in turn produce a driving force for guest complexation. Water molecules are bound in the cavities

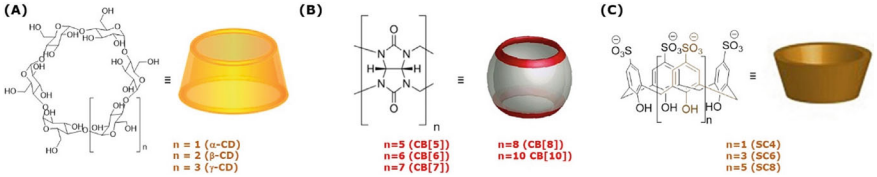


Fig. 6.19 Molecular representations of some illustrative cyclodextrins (**a**), cucurbiturils (**b**), and calixarenes (**c**)

of the supramolecules via hydrogen bonds, but the number of hydrogen bonds they can form is much smaller than in bulk water. These water molecules are called high-energy water. Hence the release of these water molecules by binding guest molecules lowers the energy of the system, reaching a state that is energetically preferred both from the enthalpic and entropic views. Cavitation energies play an important role as driving forces for the inclusion of guest molecules, especially for small CB[n]s. In general, solvent molecules must be removed from the cavity in order to offer space for the guest, Fig. 6.20. In the case of bulk solvents, the energy is quite high while macrocycles may have comparably low cavitation energy in water because the cavity water molecules are only weakly attached by hydrogen bonds.

All these host molecules may chelate guest molecules (with definite stability, selectivity, and kinetic features), react with them (with definite rate, selectivity, turnover), and, finally, release the products, thus regenerating the reagents for a new cycle. They are most frequently commercially available host molecules and have the added advantage of forming a vast number of inclusion complexes with a variety of binding moieties. The confinement imposed by supramolecular inclusion, e.g., by macrocyclic hosts, and the associated variations in substrate reactivity are especially important in these catalytic processes.

During the course of the reaction, these supramolecules can reversibly bind to a host–guest complex with the substrates via non-covalent bonding, thus forming a complex, which is responsible for altering product distribution during organic

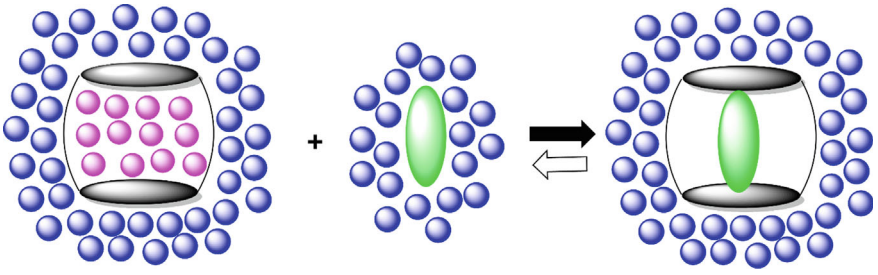


Fig. 6.20 Schematic illustration of host–guest complexation process where the high-energy water located in the cavity of the supramolecular is relief by the hosting of a hydrophobic molecule, displacing the equilibrium on the right side by energy gain from the release of the energetic water molecules

reactions. These characteristics are attributed to the development of geometrical constrains where hydrophobic forces are assumed to be responsible for driving a guest into the supramolecular cavity where 1:1 host–guest complexes usually form and where there is no interaction with other supramolecules. Supramolecular catalysis is, therefore, hallmarked by the ability of host molecules to stabilize encapsulated reactive species through a number of non-covalent interactions, thereby decreasing the free energy gap between transition states and reactants.

From the kinetic point of view, the reaction scheme is that shown in Fig. 6.21 for the hydrolysis of bis nitrophenyl carbonate (1) and thionocarbonate (2) catalyzed by cucurbituril, holds. The mechanism assumes that both the complexed and uncomplexed forms of the substrate are reactive.

This reaction mechanism is formally similar to those described for enzymatic or micellar catalyzes, i.e., a two-site reaction model as that shown in Fig. 6.15, and hence formally similar mathematical relationships will be obtained. The rate of the reaction will be the sum of the rates of the complexed and uncomplexed substrate, Eq. (6.90)

$$r = k_{\text{obs}}[S_T] = k_W\,[S_W] + k_{\text{CB7}}[S_{\text{CB7}}]. \tag{6.90}$$

The association constant of the substrate S to the micelle is given by Eq. (6.91)

$$S_W + \text{CB7} \overset{K_{\text{CB7}}}{\leftrightarrows} S_{\text{CB7}} \Rightarrow K_{\text{CB7}} = \frac{[S_{\text{CB7}}]}{[S_W]D_n} \tag{6.91}$$

Bearing in mind the mass balance for the substrate and the concentrations of the substrate in each pseudophase, the rate constant is given by Eq. (6.92).

$$k_{\text{obs}} = \frac{k_W + k_{\text{CB7}}K_{\text{CB7}}[\text{CB7}]}{1 + K_{CB7}[\text{CB7}]}. \tag{6.92}$$

Equation (6.92) shows that the kinetic profile obtained for supramolecular catalyzed unimolecular reactions are like those obtained from Michaelis–Menten

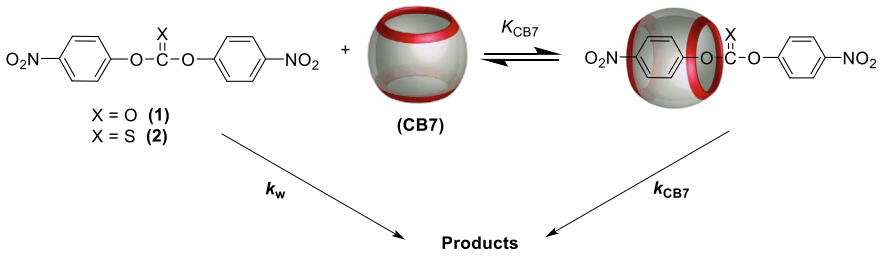


Fig. 6.21 Reaction model showing two competitive reaction pathways through the complexed and uncomplexed forms of the substrate. Adapted from [34] Copyright © 2020 American Chemical Society

enzymatic reactions, a linear increase in the rate constant upon increasing the surfactant concentration leveling off when the substrate is totally included into the cavities of the supramolecular system, that is, high [CB7].

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Chapter 7
Modeling Chemical Reactivity in Multiphasic Systems: Microemulsions and Emulsions



7.1 Introduction

The chemistry in the real world is much more complex than that in a test-tube, making frequently that bulk chemistry is hardly ever a good approximation to real-life chemistry. Most of the biologically relevant structures are multiphasic in the sense that they are composed of regions with different solvent properties. For the purpose of this chapter, the term "multiphasic" needs to be understood in a broad sense as it refers to environments where chemical reactions occur at the bulk and at interfaces. Dispersed systems with potential multiphasic implications include colloidal systems (micelles, microemulsions, liposomes, emulsions, etc.), atmospheric, environmental, biological environments, prebiotic, and synthetic organic chemistry [1–4].

Since interfaces are the boundary regions that separate bulk regions of matter having different solvent properties, intuition tells us that chemistry at liquid interfaces should be, somehow, different from that in bulk solution. The special physical, chemical, and/or biological properties of interfaces have captivated and drawn the attention of scientists of many fields. What makes interfaces unique regions is the two- or three-dimensional geometry of the interface in combination with the asymmetry of forces that experience molecules and atomic species located there. But in spite of their importance, it remains difficult to probe the chemistry taking place in such regions, and in particular in liquid interfaces, because it needs to be probed in the intact system to avoid equilibria disruption and because they are regions where properties such as density, polarity, hydrogen bonding, and viscosity are changing over molecular length scales [5]. Surface anisotropy can also cause interfacial solvent to organize and create local environments that differ from either adjacent bulk phases. In this chapter we will mostly focus on microemulsion and emulsion colloidal systems as the relevant concepts, assumptions and kinetic equations can be easily extended to other multiphasic systems.

In multiphasic systems, components and reactants are transferred between the various regions, where they can be concentrated, diluted, and even ordered. Changes

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(with respect to the bulk) in the chemical composition of interfaces, the geometrical arrangement of the species, the equilibrium constants, acidity, the motion of molecules, the thermodynamic are among the fundamental manifestations of the unique characteristics of interfaces modifying the thermodynamics and kinetics of ground-state and excited-state chemical changes and energy relaxations. Hence, under these circumstances, one anticipates that simple chemical transformations may be accelerated or suppressed, but confirming expectations is challenging [6–9].

For the sake of clarity and in attempting to keep focus on the main purpose of this book, we will limit ourselves to consider chemical reactivity in fluid liquid–liquid interfaces, where the motion of molecules is nor restricted by any component network. Interfaces of liquid water with solids and gas share some similarities with the liquid mentioned systems, but will not be considered here and the reader is referred to more specialized books or reviews.

The existence of these multiphasic regions makes that many chemical and photochemical reactions may be dramatically accelerated when they occur at aqueous interfaces, in comparison to when the same reactions occur in the bulk solvent. The reasons are various as interfaces, which are three-dimensional regions that bound two immiscible environments (usually denoted as water (polar) and oil (hydrophobic)), play a significant role as they are highly anisotropic regions with properties changing from those of the polar aqueous environment to those of the apolar hydrophobic environment. For example, the solvation of reactants may change dramatically on going from one environment to the other. Furthermore, reactions occurring at aqueous interfaces may have different mechanisms than those in bulk, which frequently complicates the implementation and interpretation of experimental results.

Colloidal systems provide suitable compartmentalized model systems where chemical reactions can be carried out and investigated [9–12]. Surfactant-based colloidal systems such as micelles, reverse micelles, liposomes, microemulsions, emulsions and, in particular, vesicles formed in bulk water have attracted considerable attention because their interface—the amphiphilic layer or bilayer separating the aqueous interior from the exterior media—exhibits some similarity to cell membranes. In these interfacial volumes, molecular crowding boosts the possibility of reactive encounters between the various chemical species and, simultaneously, sets some limits to the diffusion of products. Other parameters, including the interfacial acidity, the potential molecular alignment of reactive molecules, electrostatic repulsions, or attractions due to the presence of interfacial electric charges at aqueous interfaces also control the chemical reactivity inside the compartment. These parameters control the chemical activity inside the compartment and in many cases the reactivity is strongly favored in confined environments compared with similar reactions in bulk media [9, 13].

The information that provides these colloidal systems is indeed very valuable. For one side, the researcher can use it to modify chemical reactivity; on the contrary, we can select a suitable chemical reaction and learn how these colloidal systems modify chemical reactivity. The unique and extraordinary efficiency of these systems to selectively modify chemical reactivity and, eventually, catalyze chemical reactions in mild

aqueous environments is inspiring and it is continuously expanding in attempting to develop sustainable green chemistry and as bioinspired catalysts [3, 14].

7.2 A Brief Survey on Surfactant-Based Colloidal Systems

Before attempting to model chemical reactivity in colloidal systems, we will first provide a brief overview on the characteristics of the various colloidal with special emphasis on the interfacial region. Hence, in this section we will discuss briefly the interfacial phenomena dominated by the properties of soluble amphiphilic (also called surfactant) molecules. They are molecules, as those shown in Fig. 7.1, containing a polar group (water soluble) and a hydrophobic portion (oil soluble). The hydrophobic part is usually a long alkyl chain meanwhile the polar part can be anionic (e.g., sulfate group), cationic (e.g., thrimethylamonium halide), zwitterionic, or non-ionic (ethylene oxides).

Surfactants are present in everyday life and they are essential in many industrial processes [15, 16]. Examples of areas where they are widely used are in cleaning, detergency, as emulsifiers for foods, medicine (parenteral emulsions), cosmetics (pharmaceutics), and as flotation agents in the mining industry. Most of the surfactants used are anionic and non-ionic. The cationic ones often present environmental problems because they are not easily biodegradable, and the zwitterionic ones are usually very expensive and are only used for special applications.

One characteristic property of surfactants is that they spontaneously aggregate in water formed well-defined structures called association colloids. Among them, spherical micelles, cylinders, bilayers, and microemulsions can be formed. If there is some energy input, emulsions are formed. The simplest and best understood of these association colloids are probably the micelles, which will be sufficient to understand the behavior of surfactants in aqueous solution. Surfactants not only aggregate to spherical micelles but also form cylinders, bilayers, inverted micelles, microemulsions, emulsions, etc. Figure 7.2 shows a number of colloidal structures that can be formed by employing surfactants under different conditions.

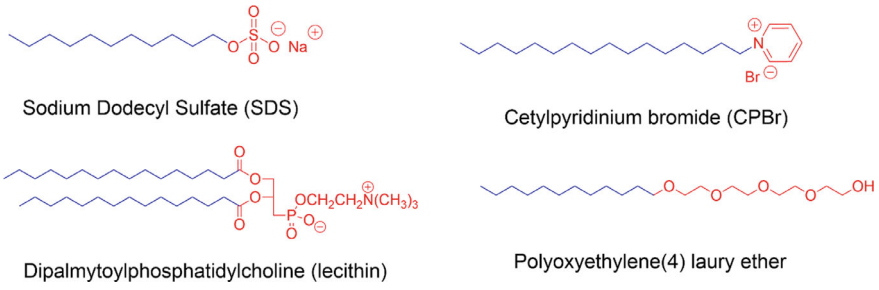


Fig. 7.1 Some common surfactants

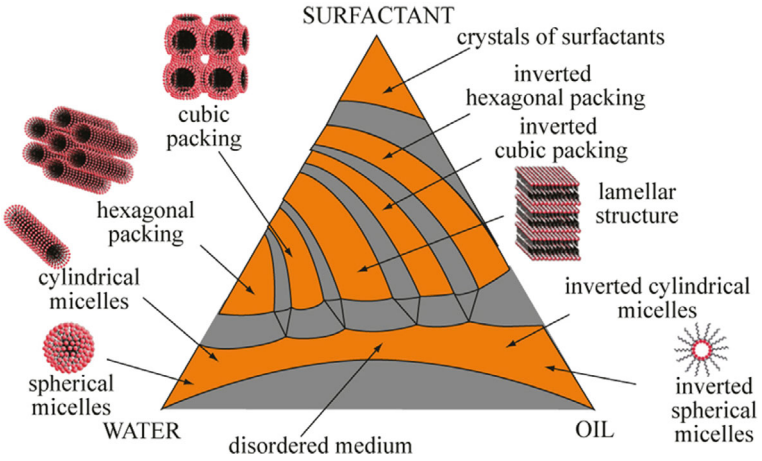


Fig. 7.2 Pseudoternary diagram showing different colloidal structures that can be formed depending on experimental conditions [9, 15]. The underlying commonality of all these mesophases is the interfacial region: a thin (1–3 nm) highly anisotropic layer that separates the bulk oil from the bulk aqueous regions

The type of aggregate structure formed depends on different factors but an important factor that controls their shape is the so-called packing **surfactant parameter**, also referred to as the packing ratio, defined by Eq. 7.1

$$N_s = \frac{V_C}{L_C \sigma_A}. \tag{7.1}$$

Here, V_C is the volume of the hydrophobic part of the surfactant, L_C is the length of the hydrocarbon chains, and σ_A is the effective area per head group. Figure 7.3 shows the spatial configuration of some of colloidal aggregates according to the packing parameter value.

When oil is present in the system, emulsions and microemulsions may be formed. Emulsions and microemulsions are very different in spite that they share some common functionalities. For instance, microemulsions are thermodynamically stable systems with droplet sizes ranging of 5–100 nm in diameter, too small to scatter the light effectively so that their solutions are optically transparent. Emulsions are metastable (or unstable) systems that tend to phase separate and contain droplets whose diameters are in of the order of the wavelength of visible light (0.5–10 μm), scattering light and their solutions are opaque, milky-like. Both microemulsions and emulsions are widely employed in many practical formulations and applications of science and technology, including oil recovery, production of creams for the pharmaceutical and cosmetic industry, in medicine (for example in the preparation of parenteral emulsions) and in the food industry (margarines, soups, sauces, chocolate drinks, etc.).

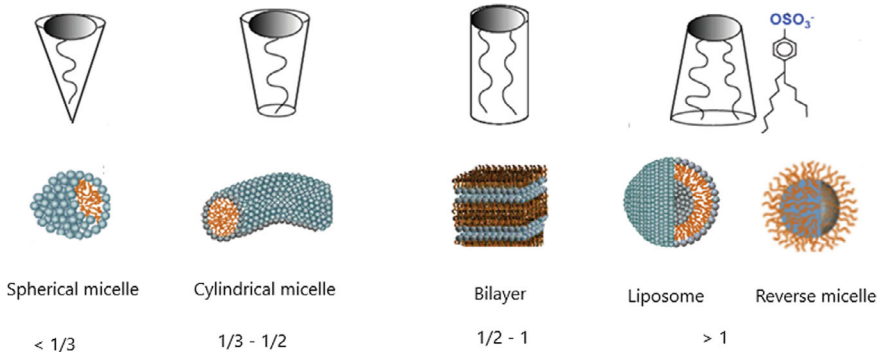


Fig. 7.3 Geometric shapes and type of aggregates formed depending on the values (indicated in the lowest row) of the packing parameter N_S (Eq. 7.1)

7.3 Importance of Dynamic Equilibrium in Modeling Chemical Reactivity in Emulsions: Diffusive Versus Reactive Systems

In fluid microemulsions and emulsions with no physical barriers between phases (for example, polymer networks) constraining the movement of molecular inert or reactive components, molecules and ions exchange rapidly because the various intermolecular forces (non-covalent coulombic, hydrogen bonding, acid/base, dipole, hydration, polarization, and dispersion interactions) are relatively weak compared to covalent bonds or complexes with a fixed stoichiometry.

Thus, after bulk mixing of the components of microemulsions or emulsions, reactants distribute according to their relative solubilities in each region, oil, interfacial, and aqueous so that the rate of thermal reactions is controlled by their activation energy values and temperature and is not limited by the diffusion of the reactants.

The number and size of the intermolecular and ionic interactions change in a three-component system when the bulk composition is changed, and these changes lead to changes in the number and shapes of the mesostructures as shown in Fig. 7.2. However, there is no reason to presume that small differences in composition on either side of a two-phase boundary should make a remarkable difference in the rates of diffusion of the molecules because the oil, surfactant, and aqueous components are chemically identical throughout the phase diagram.

Emulsion and microemulsion droplets are dynamic structures that are broken and reorganized constantly, being oil, surfactants, and other emulsion components (rapidly) transferred between droplets. Diffusion of reactive molecules within or between emulsion droplets depends not only on their molecular nature but also on the particular properties of the components of the emulsions (composition, polarity and viscosity of the oil, components of the water phase, nature of the emulsifier, etc.). This molecular diffusion in colloidal systems is important to control the kinetic processes of the various chemical reactions that occur at the liquid–liquid interfacial

boundaries because diffusion can be either faster or slower than the undergoing chemical reaction. If the observed rate of the chemical reaction k_{obs} depends on the rate of molecular diffusion, the reactions are said to be under *microscopic diffusion control*; meanwhile, if the observed rate depends on the rate at which solutions mix, they are said to be *macroscopic diffusion control* or under *activation control.*

Under activation control conditions, that is, when the rate of the reaction is slow in comparison with the emulsion dynamics, the reaction "sees" the emulsion droplets as static entities, and under these conditions the value of rate of the reaction can be computed as the sum of the rates in all regions of the emulsion. On the contrary, when the diffusion of reactants is rate-limiting, the dynamics of the droplets needs to be taken into account to explain the fate of the chemical reactions [17, 18].

Predicting if the dynamics of the droplets may, or may not, affect the chemical reaction can be, in a practical way, approached by employing the Smoluchowski theory for Brownian motion in combination with the the Acree–Curtin–Hammet principle [19, 20]. The Acree–Curtin–Hammett principle[21, 22] is described in many physical organic and kinetics textbooks commonly used to explain the selectivity ratios for stereoselective reactions as that depicted in Fig. 7.4a. The approach that we will develop considers a bimolecular reaction taking place in a multiphasic "ideal" systems where there are no physical barriers limiting the transport of molecules between the aqueous, interfacial, and oil regions. This is equivalent to assume a system with regions with different solvent properties and where there are no significant deviations from simple diffusion laws, molecular mobility, partitioning, and volume exclusion effects, etc.

Succinctly, the Acree–Curtin–Hammet principle asserts that: "if the values of the rate constants of the chemical reactions leading to the formation of products (k_A, k_B) are much lower than those of the equilibrium process between A and B (k_1, k_{-1}), Fig. 7.4a, then the system is under kinetic control and the final product ratio is given by Eq. (7.2), where K is the equilibrium constant ($K = k_1/k_{-1}$)". On the contrary, the

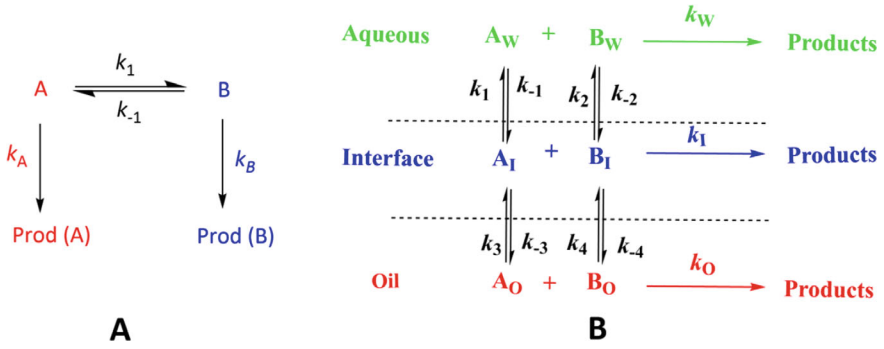


Fig. 7.4 **a** Descriptive representation of the formation of different products from two reactants in equilibrium with one another and the corresponding rate constants involved. **b** Illustrative representation of a bimolecular reaction taking place in a multiphasic system where the reactants A and B can distribute between the aqueous, interfacial and oil regions where they can react.

reaction is diffusion-controlled when the rates of the chemical reaction are higher that the diffusion rates and in this case the product distribution depends, in a complex manner, on both the relative size of the diffusion rate constants and the chemical rate constants[21].

$$\frac{[P_A]_\infty}{[P_B]_\infty} = \frac{k_A}{k_B}\frac{1}{K}. \tag{7.2}$$

The Acree–Curtin–Hammett outlines can be broadened to systems where the reactants partition between regions of distinctive solvent properties as that shown in Fig. 7.4b which represents a two-dimensional small part of an emulsion [18, 23]. Only when the rate constants (k_1, $k_{-1,}$ k_2, $k_{-2,}$ k_3, k_{-3} k_4, k_{-4}) of diffusion of molecules between regions are much higher than the rate constants of the chemical reaction in each region (k_W, k_I, k_O), the overall (measured) rate of the reaction is the sum of the rates of the reactions in each phase. When these requirements are fulfilled, the system is in "dynamic equilibrium" throughout the course of the chemical reaction, meaning that every time a molecule is consumed because of the chemical reaction, it is immediately replaced by another one in the same or different region. On the contrary, the dynamics of the droplets needs to be taken into account to explain the fate of the chemical reactions when the diffusion of reactants is rate-limiting [24–27].

Dynamic equilibrium warrants that, in colloidal systems (including kinetically stable or fluid emulsions), the reactants A and B are distributed throughout the oil, interfacial, and aqueous regions in microemulsions, and the distributions reflect their relative solubilities throughout the time course of the reaction (see Sect. 7.4 on the mathematical treatment.) Because the volumes of all three regions are large compared to the volumes of added reactants (A and B), the volume of each region determines, in part, the relative concentration of reactants in each region, i.e., their local molarities. Thus, the distributions of reactants and their reactivity are proportional to the volumes of each region and the reaction rates, that are proportional to local molarities, are independent of the size or shape of the oil, interfacial, and aqueous regions [6]. For example, the interfacial regions of non-ionic microemulsions and emulsions contain some oil, water, and polyoxyethylene chains and differences in shape of large aggregates should not significantly alter the medium properties of the interfacial regions. However, interfacial regions cannot be isolated as the emulsion would be broken and existing equilibria disrupted, and the rate constant within them cannot be determined independently, but only by fitting of kinetic models to measured rate constants[6, 19, 28].

A simple mathematical model, grounded on the transition state theory (TST) and diffusion laws, allows to analyze the conditions required to discriminate between kinetically controlled and diffusion-controlled reactive systems on the basis of simple molecular and solvent properties [23–25, 29]. The half-lives for the diffusions of the molecules are in the nano-time scale; meanwhile many thermal reactions (studied in surfactant systems) have half-lives in the minutes-to-hours time scale, that is, half-lives in the order of 10^2–10^4 s. This means that they are 10^6 to 10^9 times slower

than diffusion for most molecules and ions and of the second-order rate constants for entrance and exit of alkyl sulfate surfactant ions into micelles [22, 30].

For extremely rapid reactions, for example radical reactions where there is short-lived radicals, or photochemical reactions (where excited-state molecules may play a significant role) the reactions may be diffusion-controlled, as their activation energies are < 10 kJ mol^{-1}, and thus the rate of the reaction depends on the dynamics of fast processes (proton transfer, solvation of the probe, solute or counterion entry and exit from the aggregates, etc.) that take place in the nanoseconds to microsecond scales [17, 31, 32]. Some of these reactions, for instance the bimolecular quenching of the excited state of the probes by other molecules present in the medium, have been used to determine the mobilities of molecules, to estimate local microviscosities and encounter probabilities in the medium and to investigate ultrafast proton transfer reactions [17, 32]. The assumption of dynamic equilibrium, hence, does not apply to these systems and the distributions and rates of reactions are sensitive to specific interactions within the aggregates [17].

Both diffusion-controlled and activation-controlled are important and commonly observed in nature. As a rule of thumb, it can be assumed that diffusion-controlled reactions are those with low activation energies, so that the reaction time is negligible in comparison with that required for the diffusion of the molecules to collide. On the contrary, the reactions are controlled by activation if the reaction times are much higher those required for the diffusion of the reactants (high activation energies).

In summary, the rates of common thermally activated reactions are much lower than diffusion, and unless there are physical restrictions to movement (for example, if polymer networks are located in the interfacial region or if electrostatic forces are relevant), diffusion is not rate-limiting, and so the overall rate of the reaction is given by the sum of the rates in each compartment.

7.4 Modeling Chemical Reactivity in Emulsions

The transition state theory assumes that a bimolecular reaction $A + B \rightarrow P$ is initiated when the two reactive molecules diffuse to form the "encounter pair", which is followed by the formation of the activated complex that must progress to the formation of products for the reaction to take place; otherwise the reaction does not proceed. Thus, in bulk solution, the reactants need approach each other to form the so-called "encounter complex", so that the observed (or measured) rate of a bimolecular reaction depends first on the relative rates of these processes. The rate law is given by Eq. (7.3), where the square brackets indicate moles per liter of solution volume, and subscript T stands for total or stoichiometric concentration. When the second reactant is in large excess, i.e., $[B_T] \gg [A_T]$ (isolation method, see Chap. 3), the observed rate is expressed by a pseudo-first order rate constant, k_{obs} (s^{-1}) = $k_2[B_T]$, and where k_2 stands for the second-order rate constant (M^{-1} s^{-1}).

$$r = -\frac{\mathrm{d}[A_\mathrm{T}]}{\mathrm{d}t} = k_2[A_\mathrm{T}][B_\mathrm{T}] = k_\mathrm{obs}[A_\mathrm{T}]. \tag{7.3}$$

In multiphasic colloidal systems, reactants (and other molecules) are exchanged between the different regions (having different solvent properties), Fig. 7.4b, so that after that bulk mixing is complete the reactants distribute according to their relative solubilities between the oil, interfacial, and aqueous regions. Assuming that the dynamic equilibrium conditions holds (see Sect. 7.3), the overall rate of the reaction is given by the sum of the rates in each region, Eq. (7.4).

$$r = r_\mathrm{W} + r_\mathrm{I} + r_\mathrm{O} = k_\mathrm{W}(A_\mathrm{W})(B_\mathrm{W})\Phi_\mathrm{W} + k_\mathrm{I}(A_\mathrm{I})(B_\mathrm{I})\Phi_\mathrm{I} + k_\mathrm{O}(A_\mathrm{O})(B_\mathrm{O})\Phi_\mathrm{O}, \tag{7.4}$$

where subscripts O, I, and W indicate the oil, interfacial, and aqueous regions, respectively, k_W, k_I, and k_O and are the second-order rate constants, parenthesis () denote concentrations in moles per liter of the volume of a particular region, and Φ is the volume fraction of a region, defined as $\Phi = V_\mathrm{region}/V_\mathrm{Total}$.

Quantitative interpretation of chemical reactivity in colloidal systems requires a theoretical framework and different kinetic models. And this quantitative interpretation of experimental results, in terms of reactivity, can only be attained if the local reagent concentrations and the intrinsic rate constants in the various regions can be obtained from the overall, apparent, experimental kinetic data. Menger and Portnoy[33] first developed pseudophase models to interpret chemical reactivity in micellar systems. The model was later expanded by Romsted [34–36] to incorporate ion-exchange effects (see Chap. 6 for further details). García-Río et al. [37] developed a number of kinetic models to quantitatively explain the influence of microemulsions on chemical reactivity. Sanchez et al. [38] employed the Brönsted relationship to rationalize chemical reactivity in microemulsions and Romsted and Bravo [6, 18, 39] expanded the pseudophase model to emulsions to determine the distributions of reactants between the different regions as well as their effective concentrations.

However, working out reaction mechanisms such as that depicted in Fig. 7.1b, whose overall rate k_obs is given by Eq. (7.4), is extremely complex. Very frequently, extremely intricate kinetic equations are frequently obtained because of the huge amount of variables needed to fully describe the partitioning of reactants between regions. Moreover, the rate constants in each region are distinct because the solvent properties of each region are totally different (differences may be of several orders of magnitude).

To overcome the complex mathematical problem, judicious experimental simplifications are made by selecting suitable chemical reactions under appropriate conditions. For example, simplifications can be achieved through one or more of the following experimental approaches: (1) by making a careful choice of the chemical structure of the reactants, for example, by employing hydrophobic or hydrophilic reactants, to promote the chemical reaction in one region with respect to the other ones; (2) by using chemical probes with specific molecular or spectral characteristics so that they are located in a particular region or provide unique response to physical stimuli; (3) by choosing carefully appropriate experimental conditions, and (4) by

using ionic surfactants to promote the concentration of reactants or its electrostatic separation in the interfacial region [40–42].

For example, an intelligent choice of molecular structures of reactants undergoing nitrosation and denitrosation reactions has been exploited by Garcia-Río et al. [43, 44] and Iglesias et al. [45, 46] to investigate the use of microemulsions as reaction media. Aliaga et al. [47–49] employed a series of 4-alkanoyloxy-1,1,6,6-tetramethylpyperidineoxyl (TEMPO) radicals of different amphiphilic character as probes to investigate antioxidant activities in micelles and emulsions. Romsted et al. [6] employed a chemical probe that is both water and oil insoluble to probe the location and distribution of antioxidants in food-grade emulsions.

7.5 Modeling Chemical Reactivity in Microemulsions and Emulsions: Pseudophase Models

Menger et al. [33] introduced, more than 40 years ago, kinetic models for bimolecular reactions to interpret chemical reactivity in association colloid solutions. The model was later on expanded by Romsted [34] to include ion-exchange effects on bimolecular reactions, reactive counterions, and to introduce the ion-exchange concept grounded in thermodynamics. The model replaced attempts to interpret reactivity in terms of aggregate size and surface potentials by treating homogeneous association colloids as pseudophases and ionic aggregates as selective ion exchangers.

Pseudophase models are, in practice, also applied to any homogeneous mesophases and the emulsion regions in their vicinity as long as the mixtures are fluid enough to ensure good bulk mixing (see Chap. 8). In all pseudophase models, the colloidal solution is divided into two or three separate continuous reaction regions, aqueous (W), interfacial (I) and, eventually, oil (O), respectively [9, 18].

When applicable, the total volume of the colloidal droplet is assumed to be additive, Eq. (7.5), and equal to the total volume of the solution, V_T, the volume of oil within the droplets equals the volume of added oil, V_O, and the total water volume, V_W, determines the volume of the continuous aqueous pseudophase:

$$V_T = V_I + V_O + V_W. \tag{7.5}$$

Under dynamic equilibrium conditions (Sect. 7.3), the rapidly moving molecules are distributed between three separate continuous regions with distinct physical properties, oil, interface, and aqueous (described below). Both reactants, *A* and *B*, can, in principle, partition between the three regions, whose solvation properties are unique, so that the solubilities of *A* and *B* and the solvent effects on the ground and transition state stabilities of A and B in a reaction may be different in each region. The distributions of *A* and *B* can be described by employing the corresponding thermodynamic

partition constants between aqueous and interfacial regions, for A and B, $P'^{\mathrm{I}}_{\mathrm{W}}{}^{\mathrm{I}}$ and $P^{\mathrm{I}}_{\mathrm{W}}{}^{\mathrm{I}}$ and between oil and interfacial regions, $P'^{\mathrm{I}}_{\mathrm{O}}$ and $P^{\mathrm{I}}_{\mathrm{O}}$, respectively, Fig. 7.4b.

Hence, for the reaction $A + B \rightarrow P$, and assuming the order of the reaction is one for each reactant, the following Eqs. (7.6)–(7.7) hold:

$$v_{\mathrm{T}} = k_{\mathrm{obs}}(A_{\mathrm{T}})(B_{\mathrm{T}}) \tag{7.6}$$

$$v_{\mathrm{T}} = k_{\mathrm{obs}}[A_{\mathrm{T}}][B_{\mathrm{T}}] = k_{\mathrm{W}}(A_{\mathrm{W}})(B_{\mathrm{W}}) + k_{\mathrm{I}}(A_{\mathrm{I}})(B_{\mathrm{I}}) + k_{\mathrm{O}}(A_{\mathrm{O}})(B_{\mathrm{O}}). \tag{7.7}$$

Bearing in mind the distribution coefficients between the three pseudophases and the corresponding mass balance, a relationship between the total concentrations of the reactants and the concentrations in each of the pseudophases can be established. Simplifications can be done by, for instance, running experiments in which the stoichiometric concentration of one of the reactants is much higher than that of the other reactant, and by choosing appropriate probe molecules to simplify in as much as possible the complex kinetic scheme shown in Fig. 7.4b.

7.6 Pseudophase Model Applied to Microemulsions

Rather than presenting the mathematical relationships that can be derived for different situations, we will derive the rate equation for the general scheme in Fig. 7.4b so that the reader can adapt it according the particular experimental conditions.

The starting point will be a bimolecular reaction whose reactants can distribute thermodynamically between the different regions of the colloidal system, and where they can undergo chemical reaction (first order for each reactant), Fig. 7.4b. Assuming that the dynamic equilibrium condition holds (see Sect. 7.3), the observed or measured reaction rate is the sum of the rates in each region and Eq. (7.7) holds.

The mass balance on the three regions and equilibrium constants describing the partitioning of the reactants are given by Eqs. (7.8)–(7.9)

$$[A_{\mathrm{T}}] = [A_{\mathrm{W}}] + [A_{\mathrm{I}}] + [A_{\mathrm{O}}] \tag{7.8}$$

$$[B_{\mathrm{T}}] = [B_{\mathrm{W}}] + [B_{\mathrm{I}}] + [B_{\mathrm{O}}]. \tag{7.9}$$

To avoid having to define the volumes of the pseudophases, the partition coefficients are defined in terms of their mole ratios in each region, Eqs. (7.10)–(7.13), making the mathematical treatment simpler.

$$K^{\mathrm{I}}_{(A)\mathrm{O}} = \frac{[A_{\mathrm{I}}][\mathrm{oil}]}{[A_{\mathrm{O}}][\mathrm{Surf}_{\mathrm{T}}]} \tag{7.10}$$

$$K^{\mathrm{I}}_{(A)\mathrm{W}} = \frac{[A_{\mathrm{I}}][\mathrm{water}]}{[A_{\mathrm{W}}][\mathrm{Surf_T}]} \tag{7.11}$$

$$K^{\mathrm{I}}_{(B)\mathrm{O}} = \frac{[B_{\mathrm{I}}][\mathrm{oil}]}{[B_{\mathrm{O}}][\mathrm{Surf_T}]} \tag{7.12}$$

$$K^{\mathrm{I}}_{(B)\mathrm{W}} = \frac{[B_{\mathrm{I}}][\mathrm{water}]}{[B_{\mathrm{W}}][\mathrm{Surf_T}]}. \tag{7.13}$$

Denoting as w and z the molar ratios related with emulsion geometry.

$$w = \frac{[\mathrm{oil}]}{[\mathrm{Surf_T}]}, \tag{7.14}$$

$$z = \frac{[\mathrm{aqueous}]}{[\mathrm{Surf_T}]}. \tag{7.15}$$

The equilibrium constants are given by Eqs. (7.16)–(7.19)

$$K^{\mathrm{I}}_{(A)\mathrm{O}} = w\frac{[A_{\mathrm{I}}]}{[A_{\mathrm{O}}]} \tag{7.16}$$

$$K^{\mathrm{I}}_{(A)\mathrm{W}} = z\frac{[A_{\mathrm{I}}]}{[A_{\mathrm{W}}]} \tag{7.17}$$

$$K^{\mathrm{I}}_{(B)\mathrm{O}} = w\frac{[B_{\mathrm{I}}]}{[B_{\mathrm{O}}]} \tag{7.18}$$

$$K^{\mathrm{I}}_{(B)\mathrm{W}} = z\frac{[B_{\mathrm{I}}]}{[B_{\mathrm{W}}]}. \tag{7.19}$$

Equations (7.8) and (7.9) can be then rewritten as

$$[A_{\mathrm{T}}] = [A_{\mathrm{I}}] + \frac{[A_{\mathrm{I}}]w}{K^{\mathrm{I}}_{(A)\mathrm{O}}} + \frac{[A_{\mathrm{I}}]z}{K^{I}_{(A)\mathrm{W}}} \tag{7.20}$$

$$[B_{\mathrm{T}}] = [B_{\mathrm{I}}] + \frac{[B_{\mathrm{I}}]w}{K^{I}_{(B)\mathrm{O}}} + \frac{[B_{\mathrm{I}}]z}{K^{I}_{(\mathrm{B})\mathrm{W}}}. \tag{7.21}$$

And hence the concentrations of the reactants in each phase are given by Eqs. (7.22)–(7.27)

$$[A_{\mathrm{I}}] = \frac{[A_{\mathrm{T}}]}{1 + \frac{w}{K^{I}_{(A)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}}} \tag{7.22}$$

$$[B_{\mathrm{I}}] = \frac{[B_T]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)O}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}} \tag{7.23}$$

$$[A_{\mathrm{W}}] = \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}} \frac{[A_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(A)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}}} \tag{7.24}$$

$$[A_{\mathrm{O}}] = \frac{w}{K^{\mathrm{I}}_{(\mathrm{A})\mathrm{O}}} \frac{[A_T]}{1 + \frac{w}{K^{\mathrm{I}}_{(A)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}}} \tag{7.25}$$

$$[B_{\mathrm{W}}] = \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}} \frac{[B_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}} \tag{7.26}$$

$$[B_{\mathrm{O}}] = \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} \frac{[B_T]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}}. \tag{7.27}$$

The rate equation (7.7) is thus given by Eq. (7.28)

$$\begin{aligned} k_2[A_{\mathrm{T}}][B_{\mathrm{T}}] &= k_{\mathrm{w}}[A_{\mathrm{W}}][B_{\mathrm{w}}] + k_{\mathrm{I}}[A_{\mathrm{I}}][B_{\mathrm{I}}] + k_{\mathrm{O}}[A_{\mathrm{O}}][B_{\mathrm{O}}] \\ &= k_{\mathrm{w}} \left(\frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}} \frac{[A_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(A)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}}} \right) \left(\frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}} \frac{[B_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}} \right) \\ &\quad + k_{\mathrm{I}} \left(\frac{[A_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(A)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(A)\mathrm{W}}}} \right) \left(\frac{[B_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}} \right) \\ &\quad + k_{O} \left(\frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}} \frac{[A_T]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(\mathrm{B})\mathrm{W}}}} \right) \left(\frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} \frac{[B_{\mathrm{T}}]}{1 + \frac{w}{K^{\mathrm{I}}_{(B)\mathrm{O}}} + \frac{z}{K^{\mathrm{I}}_{(B)\mathrm{W}}}} \right). \end{aligned} \tag{7.28}$$

Operating, the following expression for the rate constant can be obtained, Eq. (7.29)

$$k_2 = \frac{k_{\mathrm{W}} z^2 K^{\mathrm{I}}_{(B)\mathrm{O}} K^{\mathrm{I}}_{(A)\mathrm{O}} + k_I K^{\mathrm{I}}_{(A)\mathrm{O}} K^{\mathrm{I}}_{(A)\mathrm{W}} K^{\mathrm{I}}_{(B)\mathrm{O}} K^{\mathrm{I}}_{(B)\mathrm{W}} + k_{\mathrm{O}} w^2 K^{\mathrm{I}}_{(A)\mathrm{W}} K^{\mathrm{I}}_{(A)\mathrm{W}}}{\left(K^{\mathrm{I}}_{(A)\mathrm{O}} K^{\mathrm{I}}_{(A)\mathrm{W}} + w K^{\mathrm{I}}_{(A)\mathrm{W}} + z K^{\mathrm{I}}_{(A)\mathrm{O}} \right) \left(K^{\mathrm{I}}_{(B)\mathrm{O}} K^{\mathrm{I}}_{(B)\mathrm{W}} + w K^{\mathrm{I}}_{(B)\mathrm{W}} + z K^{\mathrm{I}}_{(B)\mathrm{O}} \right)}. \tag{7.29}$$

This expression can be simplified considering pseudo-first order conditions and, of course, taking into consideration the partitioning of the reagents and the reaction site.

The reader must be aware that, to compare results, the rate constants values must be corrected taking into consideration the molar volumes of the oil, interfacial, and aqueous regions because pseudophase models consider that each region is evenly distributed in the total microemulsion volume. Further details and a complete derivatization of equations can be found elsewhere[50].

7.7 Key Equations Derived of the Application of the Pseudophase Model to Reactions in Emulsions

Romsted and coworkers[8, 18, 39, 51] exploited of the rich chemistry of arenediazonium ions to probe the interfacial composition of association colloids and to examine the distribution of polyphenolic antioxidants in intact emulsions under a variety of experimental conditions [8, 18, 39]. They synthetized the chemical probe 4-hexadecylbenzenediazonium tetrafluoroborate (16-ArN_2^+, reactant *A*) to determine the distribution and local concentrations of antioxidants (AOs, reactant *B*) in intact emulsions. The use of this chemical probe simplifies considerably the reaction scheme in Fig. 7.4b because 16-ArN_2^+ is both water and oil insoluble because of its long hydrophobic tail and because it is a cation. Therefore, the reactive group of 16-ArN_2^+ is located in the interfacial region of the emulsion where reacts with the phenolic antioxidants AOs. [39, 51–53] Thus, the mathematical treatment of Fig. 7.4b simplifies enormously because it is not necessary to consider the distribution of one of the reactants (reactant *A*). This implies that there is no reaction in the aqueous and oil regions, and the number of unknown parameters decreases significantly, so that the scheme in Fig. 7.5 now applies.

Equation (7.7) simplifies to Eq. (7.30) after bearing in mind that the oil and water concentrations of 16-ArN_2^+ are negligible because the reactive group of 16-ArN_2^+ is located only in the interfacial region, Fig. 7.5, i.e., [16-$ArN_2^+{}_O$] and [16-$ArN_2^+{}_W$] ≈ 0. Reactions are usually carried out under pseudo-first order conditions, so that $[B_T] >> [16\text{-}ArN_2^+{}_T]$. In Eq. (7.30), Φ_I is the surfactant volume fraction defined as $\Phi_I = V_{surf}/V_{total}$.

$$r = r_I = -\frac{d[16 - ArN_{2T}^+]}{dt} = k_I(16 - ArN_{2T}^+)(B_I)\Phi_I. \tag{7.30}$$

Figure 7.5 can be further simplified by, for instance, choosing carefully the chemical characteristics of the reactant B so that it can be located exclusively in the interfacial region or can only partition between two regions. Three limiting situations can be envisaged, (1) when *B* is very hydrophilic (low $P_W{}^O$ values), so that their concentration in the oil region is negligible and only partitions between the aqueous and interfacial regions, (2) *B* is very hydrophobic so that it is water insoluble and only partitions between the oil and interfacial regions, and (3) when *B* is both water and oil insoluble and hence is exclusively located in the interfacial region. We will consider these cases in the next subsections.

$P_W{}^O$ values cannot be extrapolated from the values determined in octanol–water binary systems (which is probably the most common system employed to determine the hydrophobicity of a molecule) because the solvents are totally different and need to be determined in the chosen oil [8, 18, 54, 55]. Predicting "a priori" the partitioning of a reactant between the oil, interfacial, and aqueous regions of emulsions is actually challenging because the values of the partition constant in binary oil/water systems $P_W{}^O$ are useless in emulsions because of the physical impossibility of isolating the

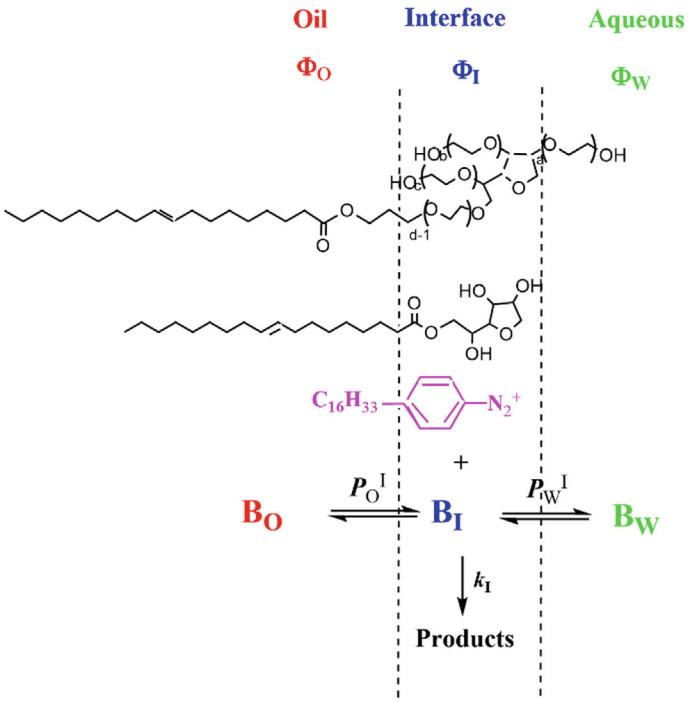


Fig. 7.5 Conceptual division of a non-ionic emulsion showing the oil (O), interfacial (I) and aqueous (W) regions at the molecular level. The dashed lines are not real and are drawn to help distinguish between regions. Because of the special molecular characteristics of the chemical probe, it does not partition. However, the second reactant *B* partitions between the three regions. The reaction between reactant *A* (16-ArN_2^+) and the reactant *B* (Fig. 7.4b) takes place exclusively in the interfacial region of the emulsion. Φ stands for the volume fraction of a region ($\Phi = V_{region}/V_{emulsion}$), *P* is the partition constant, and k_I is the rate constant for the reaction between the chemical probe 16-ArN_2^+ and the reactant *B* in the interfacial region. The chemical structures of some surfactants (Tween 20 and Span 80) employed to stabilize kinetically the emulsions are also shown

interfacial region from the oil and aqueous ones, which makes impossible to predict the values of the partition constants in emulsions ($P_O{}^I$ and $P_W{}^I$) from the value of the partition constant $P_W{}^O$ in binary systems. Only its ratio is known, since $P_W{}^O = P_W{}^I/P_O{}^I$ (see Sect. 7.7.1.1 below), but knowledge of $P_W{}^O$ values may, to a first approximation, help to envisage qualitatively how the distribution of the reactant B may be in emulsions. Very high $P_W{}^O$ values indicate that the reactant is water insoluble and thus should mainly distribute between the oil and interfacial regions; on the contrary, very low $P_W{}^O$ values indicate the reactant is oil insoluble and mainly distributes between the water and interfacial regions.

Mathematical relationships for the observed rate constant will be derived for the mentioned three cases in the next subsections. For the sake of simplicity, we will first consider non-ionic emulsions, where electrostatic effects are negligible since

the droplet surface is not charged, and will consider, in a different section, the case of ionic emulsions.

7.7.1 Reactions in Non-ionic Emulsions

We note that, according to Fig. 7.5, the reactive moiety of 16-ArN_2^+ is exclusively located in the interfacial region and (1) the reactant B is distributed between the three regions, (2) *B* is either very hydrophilic (only partitions between the aqueous and interfacial regions or very hydrophobic (only partitions between the oil and interfacial regions), and (3) *B* is exclusively located in the interfacial region.

7.7.1.1 Reactions Where the Second Reactant Partitions Between the Three Regions.

This is the case shown in Fig. 7.5. By combining Eq. (7.30) with the mass balance equations for AO and 16-ArN_2^+, and taking into consideration the definition of the relevant partition constants, Eqs. (7.31)–(7.32), Eq. (7.33) can be derived for k_{obs} [8].

$$P_W^I = \frac{(B_I)}{(B_W)} \tag{7.31}$$

$$P_O^I = \frac{(B_I)}{(B_O)} \tag{7.32}$$

$$k_{obs} = k_2[B_T] = \frac{[B_T]k_I P_W^I P_O^I}{\Phi_O P_W^I + \Phi_I P_W^I P_O^I + \Phi_W P_O^I}. \tag{7.33}$$

Equation (7.33) encloses three unknowns, k_I, $P_O{}^I$ and $P_W{}^I$, and their individual values cannot be obtained in a single experiment. A second independent mathematical relationship is thus needed. A first simple and straightforward option is to determine the partition constant in a binary oil–water mixture in the absence of surfactant, $P_W{}^O$, whose value equals to the ratio $P_W{}^I/P_O{}^I$, Eq. (7.34). This approach considers the binary oil–water system as a restraining condition of the emulsion when the surfactant concentration goes to zero, i.e., when no surfactant is present.

$$\frac{P_W^I}{P_O^I} = \frac{(B_O)}{(B_W)} = P_W^O. \tag{7.34}$$

Combining the value for $P_W{}^O$ (Eq. 7.34) with fits of k_{obs} versus Φ_I, Fig. 7.4, at a constant $[AO_T]$ and Φ_W:Φ_O (Eq. 7.33), the values for $P_O{}^I$ and $P_W{}^I$ are obtained by

solving two equations in two unknowns. Once $P_O{}^I$ and $P_W{}^I$ are known, Eq. 7.12 is used to determine the k_I value.

$P_O{}^I$ and $P_W{}^I$ values can also be obtained by employing a second set of kinetic experiments in emulsions with a different Φ_W/Φ_O ratio to get a second set of (k_{obs}, Φ_I) pairs of data [56–59]. Figure 7.6 illustrates the variations of k_{obs} with Φ_I at two Φ_W/Φ_O. Equation 7.33 is formally equivalent to Eq. 7.35, where the parameters $\boldsymbol{a}$ and $\boldsymbol{b}$ are defined by the Eqs. 7.36–7.37 [8, 18, 58]. The partition constants $P_O{}^I$ and $P_W{}^I$, defined by Eqs. (7.31)–(7.32), are then obtained by employing Eq. (7.37) from two sets of kinetic data at different Φ_W/Φ_O ratios and solving two equations in two unknowns. Once the partition constants are known, the k_I value is obtained from Eq. (7.36).

$$\frac{1}{k_{obs}} = \frac{b}{a}\Phi_I + \frac{1}{a} \tag{7.35}$$

$$a = \frac{[AO_T]k_I P_O^I P_W^I (1 + \Phi_W/\Phi_O)}{P_W^I + \Phi_W/\Phi_O P_O^I} \tag{7.36}$$

$$b = \frac{P_o^I P_w^I (1 + \frac{\Phi_w}{\Phi_o})}{P_w^I + \frac{\Phi_w}{\Phi_o} P_o^I} - 1. \tag{7.37}$$

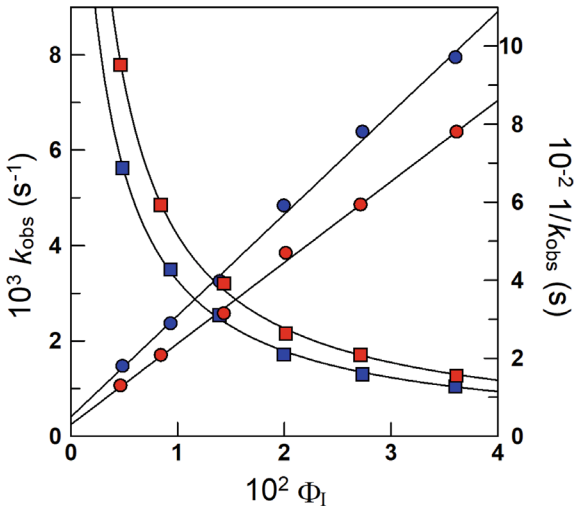


Fig. 7.6 Variations of k_{obs} with Φ_I for methyl caffeate in 4:6 (blue) and 1:9 (red) olive oil emulsions. Adapted from Ref. [60]

7.7.1.2 Reactions Where the Second Reactant *B* Partitions Only Between Two Regions

We will now consider those cases where the second reactant partitions only between two regions either because it is very hydrophobic (e.g., water insoluble) or because it is very hydrophilic (e.g., oil insoluble) as illustrated in the schemes in Fig. 7.7. In both cases, only one partition constant is needed to describe the distribution of the second reactant, that between the water interface, P_W^I, and that between the oil and interfacial region, P_O^I, (Eqs. 7.31–7.32).

Equations 7.38–7.39 can be derived after considering Eq. 7.33, the definition of the partition constants, Eqs. 7.31–7.32, and bearing in mind the corresponding mass balances. Equations 7.38–7.39 describe the dependence of k_{obs} on both the concentration of the antioxidant and the medium effects and can be used to calculate P_W^I, k_I and P_O^I, k_I values, respectively. Equations 7.40–7.41 are the reciprocal of Eqs. 7.31–7.32, respectively. They predict that plots of $1/k_{obs}$ versus Φ_I should be linear with a positive intercept. The values for k_I, P_W^I and P_O^I are obtained from the slopes and intercepts of Eqs. 7.38–7.39. Figure 7.8 is illustrative and shows the variations of k_{obs} versus Φ_I and $1/k_{obs}$ versus Φ_I for a hydrophilic and a hydrophobic reactant.

$$k_{obs} = \frac{k_I[B_T]P_W^I}{\Phi_I P_W^I + \Phi_W} \tag{7.38}$$

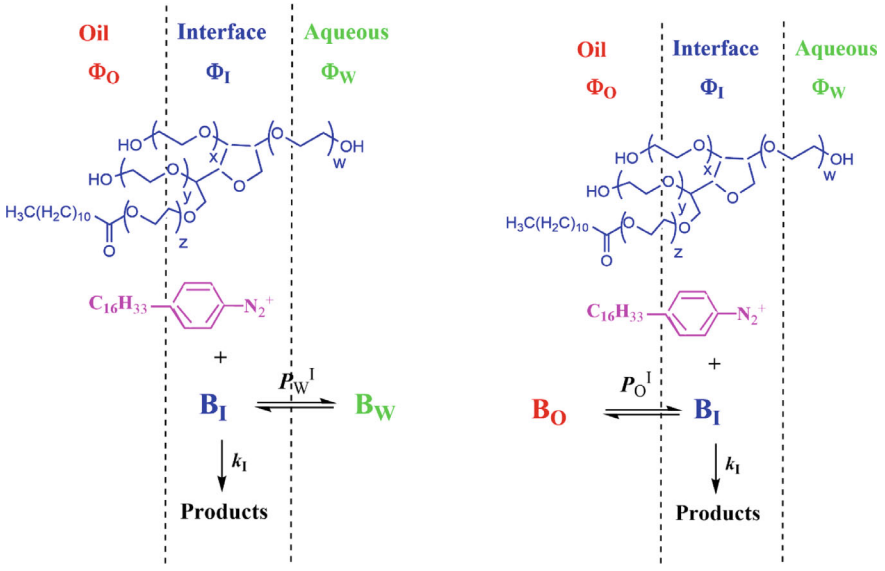


Fig. 7.7 Illustrative allocations of a hydrophilic (oil insoluble) reactant (left) and a hydrophobic (water insoluble) reactant (right) in emulsions. P_W^I and P_O^I stand for the partition constants and k_I is the interfacial rate constant

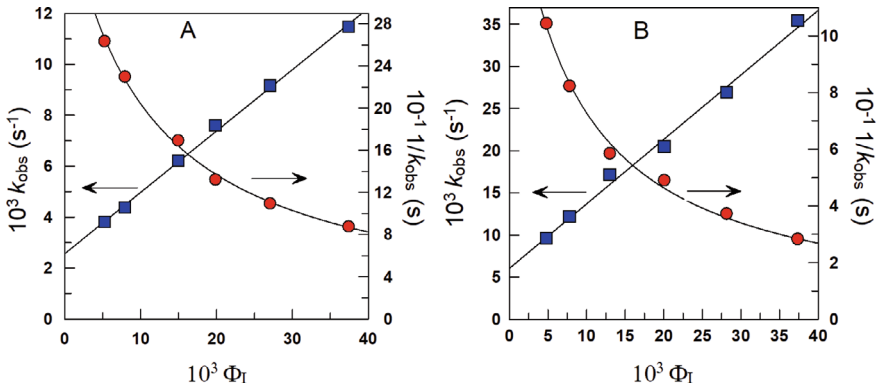


Fig. 7.8 Descriptive examples of the variations of k_{obs} with Φ_I for the reaction between 16-ArN_2^+ with gallic acid (**a**) and with lauryl gallate (**b**) in 1:9 fish oil-in-water emulsions at T = 25 °C. The solid lines are the theoretical curves obtained by fitting the experimental data to Eqs. 7.38 and 7.40 (**a**) and 7.39, and 7.41 (**b**). Adapted from Ref. [61]

$$k_{obs} = \frac{k_I[B_T]P_O^I}{\Phi_I P_O^I + \Phi_O} \tag{7.39}$$

$$\frac{1}{k_{obs}} = \frac{\Phi_W}{k_I[B_T]P_W^I} + \frac{1}{k_I[B_T]}\Phi_I \tag{7.40}$$

$$\frac{1}{k_{obs}} = \frac{\Phi_O}{k_I[B_T]P_O^I} + \frac{1}{k_I[B_T]}\Phi_I. \tag{7.41}$$

7.7.1.3 Reactions Taking Place Exclusively at the Interfacial Region

Reactions taking place exclusively at the interfacial region of emulsions represent a limiting case in which both reactants are both oil and water insoluble. They are entirely located in the interfacial region and hence do not partition as illustrated in Fig. 7.9. This case is formally equivalent to that of a bimolecular reaction taking place in a vessel of volume equal to that of the interfacial region, and the observed rate constant for the reaction is given by Eq. 7.42, that predicts that k_{obs} values should decrease upon increasing Φ_I, as illustrated in Fig. 7.10.

$$k_{obs} = k_2[B_T] = \frac{k_I[B_T]}{\Phi_I}. \tag{7.42}$$

The reciprocal of Eq. 7.42 predicts that plots of $1/k_{obs}$ versus Φ_I must be straight lines with intercepts at $1/k_{obs} = 0$ as illustrated in Fig. 7.10.

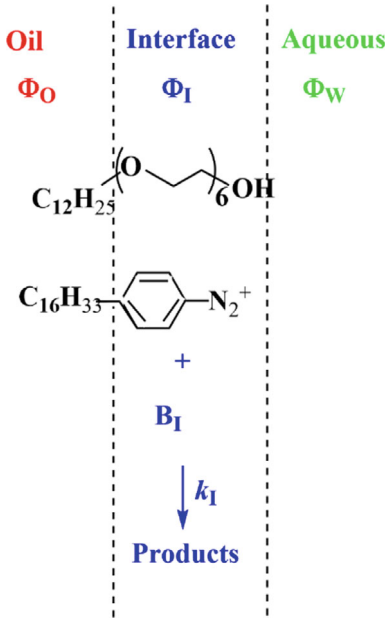


Fig. 7.9 Limiting case where both reactants *A* and *B* do not partition and are located in the interfacial region of a non-ionic emulsion

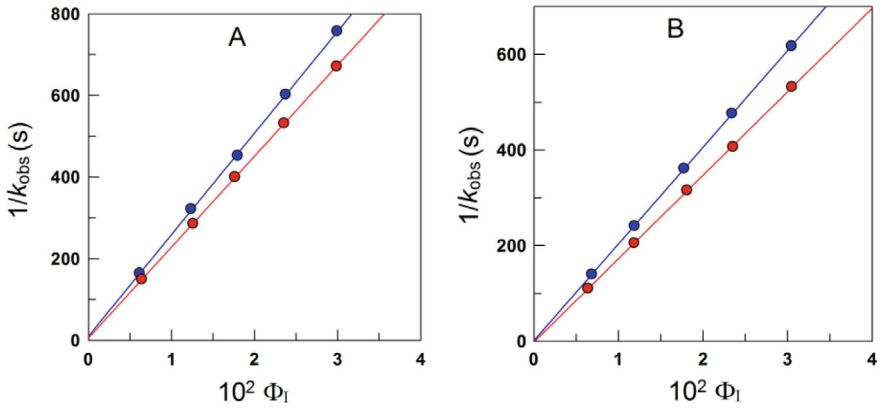


Fig. 7.10 Variations of $1/k$obs with Φ_I in 1:1 (blue) and 1:4 (red) v:v octane-in-water emulsions for the reaction of 16-ArN_2^+ with octyl (**a**) and lauryl (**b**) gallate. Extracted from Ref. [62]

7.7.2 Reactions in Ionic Emulsions

When ionic surfactants are employed to stabilize kinetically emulsions, the surfaces droplets become electrically charged (either negatively or positively), Fig. 7.11, and the presence of these charges has significant kinetic consequences because:

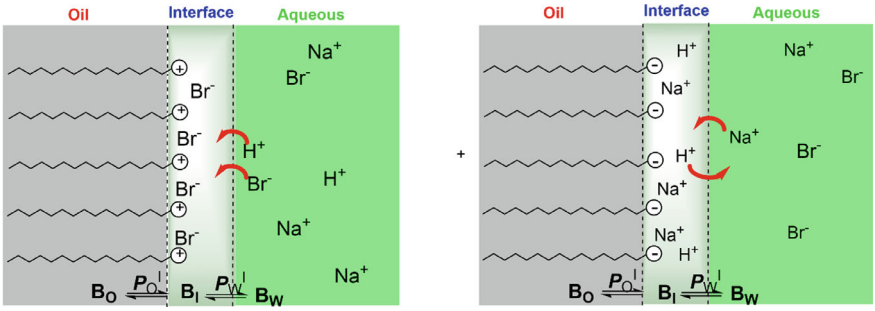


Fig. 7.11 Outlines of the application of the pseudophase model to ionic emulsions. Left. The Donnan equilibrium in cationic emulsions. Right ion exchange in anionic emulsions. *B* is a reactant that distributes between the oil, interfacial and aqueous regions. Adapted from Refs. [64, 65]

(1) Electrostatic forces become important. Droplet interfaces may repel or attract ions present in the aqueous phase, decreasing their concentration substantially in the interfacial region or producing high local concentrations.
(2) Charged interfaces may notably modify the interfacial molarities of ions such as H^+ or OH^-, thus having major effects on the ionization constant (pK_a) values of weak acids present in emulsions. Anionic interfaces have higher interfacial H^+ concentrations relative to those in the bulk but decrease the interfacial OH^- concentration with respect to that in the bulk aqueous phase. Cationic emulsions do the opposite.

The pseudophase ion-exchange (PIE) model (see Chap. 6) furnishes a suitable theoretical and operative framework to interpret chemical reactivity at charged emulsion interfaces[18]. The PIE model was first developed for modeling chemical reactivity in micellar solutions of ionic surfactants (Fig. 7.11) to describe specific ion effects and ion distributions [34]. It presumes counterions (ions of opposite charge to that of the surface) can exchange with H^+ or OH^-. i.e., the interfacial regions of ionic colloidal systems behave as selective ion exchangers. For example, when employing anionic sodium dodecyl sulfate micelles, Na^+ ions are counterions, as well as those from added NaCl. The dynamic equilibria of counterions between the aqueous and interfacial regions, Eq. 7.43, are defined by an empirical ion-exchange constant $K_H{}^{Na+}$, Eq. 7.44.

$$H_I^+ + Na_W^+ \overset{K_H^{Na}}{\rightleftarrows} H_W^+ + Na_I^+ \tag{7.43}$$

$$K_H^{Na} = \frac{[H_W^+][Na_I^+]}{[H_I^+][Na_W^+]} \tag{7.44}$$

$$H_I^+ + Br_I^- \overset{K_H^{Br}}{\rightleftarrows} H_W^+ + Br_W^- \tag{7.45}$$

$$K_H^{Br} = \frac{[H_W^+][Br_W^-]}{[H_I^+][Br_I^-]}. \quad (7.46)$$

H^+ is a co-ion in acidic solutions of cationic micelles, and to describe the effect of added Br^- on the proton distribution, a Donnan equilibrium such as that depicted in Eq. (7.45) is used [63]. To define the effects of added counterions, Br^-, on the H^+ co-ion distribution between the interfacial and aqueous regions, the Donnan equilibrium constant, K_H^{Br}, Eq. (7.46), is commonly employed. Equation (7.45) implies that, upon addition of NaBr, the Br^- concentration in the aqueous region increases, increasing the Br^- and H^+ concentration in the interfacial region. However, the ratio of the numerator and denominator, Eq. (7.46), remains constant.

The plots in Fig. 7.12 illustrate the kinetic profiles obtained for the reaction between a reactant B in cationic (hexadecyltrimethylammonium bromide (CTAB)) and anionic sodium dodecyl sulfate (SDS) emulsions [64, 65]. In cationic emulsions, Fig. 7.12 (left), in the absence of salt, the decrease in k_{obs} with increasing Φ_I is produced by dilution of B upon increasing the interfacial volume (i.e., increasing surfactant concentration) [18].

The observed effects on k_{obs} depend on the nature of the reaction considered. For example, for the reaction between ArN_2^+ ions (reactant *A*) with antioxidants (reactant *B*), ion exchange has significant consequences because most antioxidants are (poly)phenolic compounds and the reactive species is the deprotonated form of the phenol, AO^- [66]. Thus, at constant $[AO_T]$ and Φ_O:Φ_W, the main effect of increasing Φ_I (i.e., addition of surfactant CTAB) on k_{obs} is to dilute of both the anionic and neutral forms of B. The local (effective) H^+ concentration in the interfacial region (mole per liter of interfacial volume) should be reduced to approximately the same amount as the AO^- concentrations. Therefore, the decrease in k_{obs} with increasing

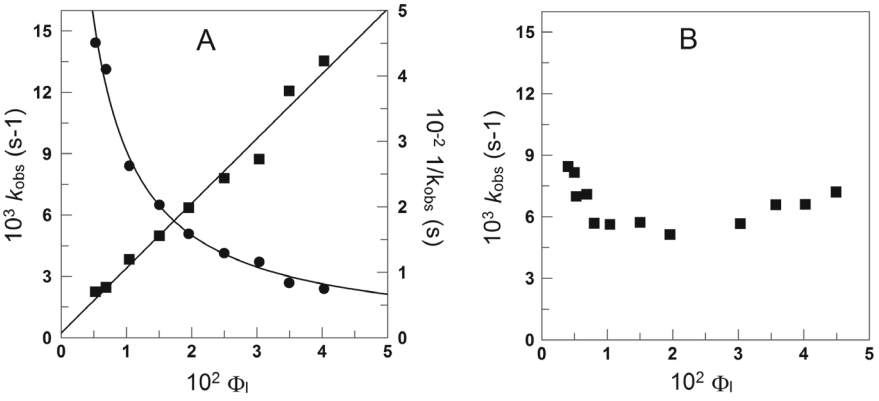


Fig. 7.12 Changes in k_{obs} upon increasing Φ_I for the reaction between 16-ArN_2^+ and tert-butylhydroquinone (TBHQ) in 1:1 (v:v) glyceryl trioctanoate-water emulsions stabilized with: **a** hexadecyltrymethylammonium bromide (CTAB), **b** sodium dodecyl sulfate (SDS). Figure extracted from Ref. [65]

Φ_I is caused by dilution of the anion within the increasing volume of the totality of the interfacial regions of the emulsions droplets as in anionic emulsions, Fig. 7.12a.

In anionic emulsions, Fig. 7.12b, the variation of k_{obs} with Φ_I decreases initially, passing through a shallow minimum and then increases. The shallow minimum was attributed to two competing effects, the simultaneous decrease in H^+ concentration in the interfacial region, and the dilution of B^-, which leads to an increase in k_{obs} because of the increase in the concentration of the reactive species B^-.

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Chapter 8
Analysis of Kinetic Data: Effects of Solution Composition



8.1 Introduction

The knowledge of the reaction rate equation frequently allows the researcher to draw deductive conclusions on the reaction mechanism. However, it does not usually provide practical information on the nature of the reactive forms of the species participating in the reaction. Very often, the knowledge of the reaction rate equation or the reaction order of individual species involved in the reaction may not be sufficient for drawing conclusions on the character and/or composition of the transition state and several possibilities remain, and making a choice to become a simple guess. The researcher must use deductive reasoning rather than guessing, and for the purpose, further information needs to be obtained to lay a solid scientific ground supporting the eventual hypothesis that can be put forward. This information can be obtained from the analysis of the effects of varying the composition of the reaction mixture on the reaction rates, and will be—certainly—crucial in the deduction—eliminating process to finally achieve the most probable reaction mechanism consistent with experimental observations. Particularly, the analyses of the effects of acidity, solvent, ionic strength and temperature on the values of the rate constants offer diagnostic tools providing a valuable and essential opportunity to discern between various potential reaction mechanisms. Researchers, therefore, can make deductions on the reaction mechanism from the knowledge of the nature and properties of the starting materials, products, and on the overall equation for the reaction rate. The aim of this chapter is, therefore, an attempt of showing the reader how to use deductive reasoning from variations in the observed rate constant upon changing the solution composition to finally propose a reasonable reaction mechanism.

Rate constants are determined by one or more of the methods described in Chap. 3, but the researcher must be aware that it is absolutely necessary to ensure that changes in the composition of the reaction mixture do not affect the kinetics and that the same reaction rate equation holds over the entire range in which the composition of the solution is changed. The effects of solvent on the rate constants have been described

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Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_8

briefly in Chap. 5. Here we will move further and will mainly focus on analyzing the effects of acidity, temperature, and ionic strength. The rates of numerous chemical reactions in solution are affected by changes acidity, and the analyses of the changes in temperature offer an opportunity to gain knowledge on the mechanism and on the energetics of the reactions.

8.2 Effects of Acidity or Basicity on Reaction Rates

Changes in the values of rate constants with acidity (or analogous functions when extended acidity scales need to be employed) can be observed when:

1. The rate-determining step (rds) is preceded by a rapidly stablished acid–base equilibrium causing changes in the actual concentration of the reactive form of one of the reactants.
2. The rds is the formation of an acid or a base.
3. The rds involves a proton transfer or the addition of OH^- as nucleophile. The conjugated acid or base of the solvent is consumed or transformed in the course of the reaction and is not regenerated.
4. The reaction is general acid–base catalyzed.

General catalyzed reactions usually involve types 1–3 as rate governing, meanwhile types 1 and 3 are sometimes described as specifically catalyzed reactions. In reactions where the slowest step is preceded by a rapidly established acid–base equilibria, either the reagent in excess R or the reactant B (employed to monitor the course of the reaction) can undergo changes in the degree of ionization by acting as acids or bases are described by formally equivalent (analogous) mathematical relationships. Case 3, where both R and B react as acids and/or bases will be discussed in a different section (see Sect. 8.3).

Only the simplest types of rds taking place subsequent to acid–base equilibria are considered in the subsequent discussion below. The mathematical treatment of more complex reactions can be done in a similar way as that shown here, only the rate equations needed for evaluation of the measured rate constant at each pH value are more complex.

For the sake of clarity, and in addition to the formal rate constant k which is a function of acidity (and possibly of other parameters too), the symbols in Table 8.1 will be employed:

Thus, in the discussion below, we will assume that series or reaction mixtures are prepared in such a way that their pH is varied, but in all cases, they have the same analytical (or stoichiometric) concentration of the reagent in excess (R) and of those of the other reactants A and B. Such a change in acidity can be carried out by either the addition of reaction components to appropriate buffers with sufficient buffer capacity or the reagent R is used as a buffer. In the latter case, reaction mixtures should be prepared from the reagent R and its conjugated base RH^+, varying the ratio $[RH^+]/[R]$ while keeping constant the total sum of concentrations, i.e., $[RH^+] + [R] = S_R$

Table 8.1 The table provides a list of generic parameters used in this chapter

K_1, K_2, K_3, K_4, …	Equilibrium constants indicating rapidly established acid–base reactions
k_1, k_2, k_3, k_4, …	Rate constants
R	Reagent present in excess, which, in some circumstance, may undergo acid–base reactions to form RH^+ and/or RH_2^{2+}
A	Reagent whose concentration changes in the course of the reaction but that remains in the same ionic form, i.e., its degree of ionization does not change with acidity within the studied acidity range
B, BH^+	Reactants whose concentration changes in the course of the reaction and whose ionization degree changes with acidity
$[S_R]$, $[S_A]$, $[S_B]$	Analytical or stoichiometric concentrations of the reagent R or other reactants A, B, …

= constant, in the series of experiments. The procedure limits the change in acidity to two pH units around the pK_1 of the reagent, $pH = pH_{RH+} \pm 2$, but this is not a problem at all because, as we will see, this pH region is usually the most relevant one. An alternative procedure is possible while keeping the concentration of one of the components (R or RH^+) constant and changing the pH value by changing the concentration of the other.

It is, therefore, necessary—and crucial—to ensure that all reaction mixtures in the compared series of reactions contain the same final concentrations of reactants. In addition, proper thermostatization of the series of reaction mixtures at the desired temperature is necessary to maintain the temperature constant and the ionic strength must also remain constant by (preferably) adding neutral salts to minimize electrostatic effects between reaction components.

For the purpose of detailing some of the various scenarios that the researcher can find upon investigating the solution composition effects on the reaction rates, we will show the different mechanisms that may operate and derive the corresponding acid-dependent rate equations so that changes in the rate constants with acidity can be easily visualized, allowing the researcher to quickly distinguish between different mechanistic alternatives.

8.2.1 Reactions Where the Acid Form of a Reagent Reacts

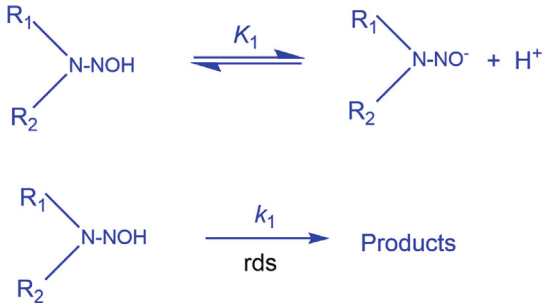
This case illustrates those reactions where only the acid form of the reagent RH^+ undergoes reaction at measurable rate (Eq. 8.1a), or that where only the acid form of a reactant BH^+ reacts with the unprotonated form of the reagent R (assuming

negligible the reaction with the B species, and where the ionization constant K_1' corresponds to the dissociation of the acid form of the reactant BH^+), Eq. 8.1b:

$$\begin{aligned} &RH^+ \overset{K_1}{\rightleftharpoons} R + H^+ \\ &RH^+ + A \xrightarrow{k_1} \text{Products} \end{aligned} \tag{8.1a}$$

$$\begin{aligned} &BH^+ \overset{K_1'}{\rightleftharpoons} B + H^+ \\ &BH^+ + R \xrightarrow{k_1} \text{Products}. \end{aligned} \tag{8.1b}$$

Examples of these reactions are those of the acid decompositions of dialkylthiocarbamates and of nitrosamines, Eq. 8.2.



(8.2)

Note that the above example is a particular case of the general Scheme 8.1 in which the reactive species A do not participate in the reaction. According to the general Scheme 8.1, the rate of reaction is thus given by Eq. 8.3

$$-\frac{d[A]}{dt} = k_1[RH^+][A]. \tag{8.3}$$

When the rate of the reaction is measured at various pH values and the ratio $[RH^+]/[R]$ is changed, we define the analytical concentration as the sum of the neutral and acid species, $[S_R] = [RH^+] + [R]$, and bearing in mind the ionization equilibrium in 8.1, Eqs. 8.4 and 8.5 can be derived

$$K_1 = \frac{[R][H^+]}{[RH^+]} \Rightarrow [R] = \frac{K_1[RH^+]}{[H^+]} \tag{8.4}$$

$$[S_R] = [RH^+]\left(1 + \frac{K_1}{[H^+]}\right). \tag{8.5}$$

And after substitution in Eqs. 8.3, 8.6 sets the rate equation of the reaction so that the observed or measured rate constant k_{obs} changes with acidity according to

Eq. 8.7

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1\left(\frac{[\mathrm{H}^+]}{[\mathrm{H}^+] + K_1}\right)[S_R][A] \tag{8.6}$$

$$k_{\mathrm{obs}} = k_1\left(\frac{[\mathrm{H}^+]}{[\mathrm{H}^+] + K_1}\right). \tag{8.7}$$

In the case that the mechanism 8.1b holds, Eqs. 8.8 and 8.9 can be derived in a similar way to that shown before

$$-\frac{\mathrm{d}[R]}{\mathrm{d}t} = k_1\left(\frac{[\mathrm{H}^+]}{[\mathrm{H}^+] + K_1'}\right)[R][\mathrm{BH}^+] \tag{8.8}$$

$$k'_{\mathrm{obs}} = k_1'\left(\frac{[\mathrm{H}^+]}{[\mathrm{H}^+] + K_1'}\right). \tag{8.9}$$

The plot of the variation of the measured rate constant k_{obs} with $[\mathrm{H}^+]$ or, even better, with pH has the form given in Fig. 8.1, and where the plot of the ratio k_{obs}/k_1 (or k'_{obs}/k_1) is employed to make them independent of the chosen scale.

It is apparent that the most suitable determinations of rate constants are within the pH region $\mathrm{pH} = pK_1 \pm 2$ because they produce the largest changes in the k_{obs}/k_1 ratio, meanwhile determining rate constants at pH values much higher or lower than the $\mathrm{p}K_1$ value is ill-suited for distinguishing the type of reaction mechanism involved as a consequence of the small changes in the in the k_{obs}/k_1 ratio.

It is worthwhile noting that it follows from Eqs. 8.7 and/or 8.9 that when $\mathrm{pH} << \mathrm{p}K_1$, the value of K_1 (or K_1' in Eq. 8.9) can be neglected against the value of $[\mathrm{H}^+]$ so that the ratio k_{obs}/k_1 approaches the unit, that is, the value of the rate constant determined in this pH range equals that of the true value of k_1. The same holds when

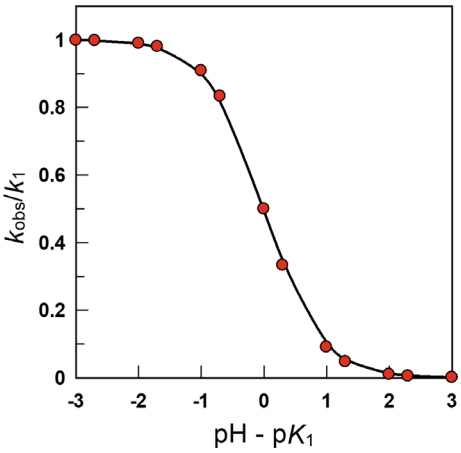


Fig. 8.1 Variation in the ratio k_{obs}/k_1 for pre-equilibrium protonation followed by reaction of the acid form (Eq. 8.1a) according to Eq. 8.7. An analogous plot is obtained for the acid dependence of reaction following the mechanism 8.1b (Eq. 8.9)

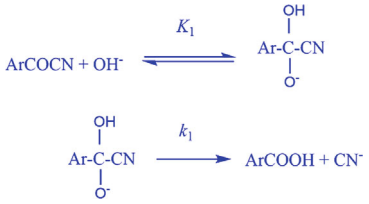
or pH >> pK_1, in this case, the value of [H^+] can be neglected against the K_1 (or K_1') value.

8.2.2 *Reactions Where the Conjugated Base Form of a Reagent RH^+ Reacts*

In these reactions, only the non-protonated (basic) form R of the reagent in excess reacts at a measurable rate, and they obey the following reaction mechanism:

$$\begin{aligned} &RH^+ \overset{K_1}{\rightleftharpoons} R + H^+ \\ &R + A \xrightarrow{k_1} \text{Products.} \end{aligned} \tag{8.10}$$

Examples of these reactions are the hydrolysis of acyl cianides 8.11, elimination reactions of Mannich bases and reactions of amines with carbon disulfide.



(8.11)

Note that the above example is a particular case of the general mechanism 8.10 in which the species A do not participate in the reaction. According to the general mechanism 8.10, the rate of the reaction is given by Eq. 8.12.

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1[RH^+][A]. \tag{8.12}$$

And the mathematical relationship for the analytical concentration of the reagent after introducing the equilibrium constant is given by Eq. 8.13.

$$[S_R] = [R]\left(1 + \frac{[H^+]}{K_1}\right). \tag{8.13}$$

And then, it follows that the mathematical relationship between the observed rate constant and the proton concentration is given by Eq. 8.14

$$k_{\text{obs}} = k_1\left(\frac{K_1}{[H^+] + K_1}\right). \tag{8.14}$$

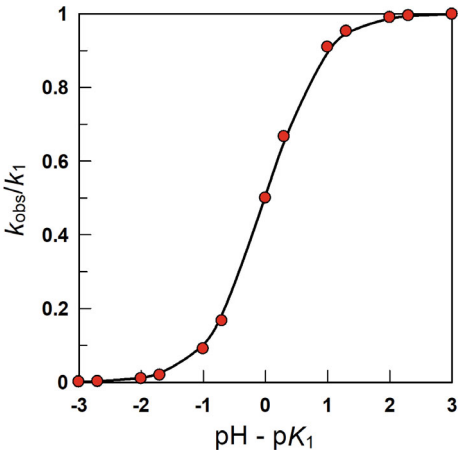


Fig. 8.2 Variation of the relative value of the ratio of rate constants k_{obs}/k_1 with pH for a reaction mechanism in which the reactant RH^+ undergoes a rapid acid–base dissociation process followed by reaction with the neutral (basic) form of the reagent (R)

Thus, the changes in $[A]$ with time follow simple kinetics at a given pH but the measured rate constant k_{obs} increases with pH according to Eq. 8.14 as illustrated in Fig. 8.2. At pH << pK_1, the value of K_1 can be neglected when compared to that of $[H^+]$ and Eq. 8.14 can be simplified showing an inverse relationship between k_{obs} and $[H^+]$, $k_{obs} = k_1 K_1/[H^+]$. When pH >> pK_1, the value of k_1 is identical to that of the value of the measured rate constant of k_{obs}. As illustrated in Fig. 8.2, the most suitable acidity range for determining changes in k_{obs} is, therefore, pH = p$K_1 \pm 2$ as in case 1.

8.2.3 *Both the Acid RH^+ and Conjugated Base R Forms of the Reagent React*

When both the acid and basic forms of the reagent react with reactant A at similar rates, i.e., when the values of the rate constants are comparable, Eqs. 8.15 hold. If, for whatever reason, one of the rate constants is much smaller than the other, the present case leads to one of the above discussed cases 1 (Eq. 8.1) or case 2 (Eq. 8.10).

$$\begin{aligned} &RH^+ \overset{K_1}{\rightleftharpoons} R + H^+ \\ &RH^+ + A \xrightarrow{k_1} \text{Products} \\ &R + A \xrightarrow{k_2} \text{Products}. \end{aligned} \tag{8.15}$$

The rate of the reaction is then given by Eq. 8.16

$$-\frac{d[A]}{dt} = k_1[RH^+][A] + k_2[R][A]. \tag{8.16}$$

Taking into account the preceding equilibria, Eq. 8.17 can be easily derived, from where the dependence of the observed rate constant with acidity can be obtained, Eq. 8.18

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1\left(\frac{[\mathrm{H}^+]}{[\mathrm{H}^+]+K_1}\right)[S_R][A] + k_2\left(\frac{K_1}{[\mathrm{H}^+]+K_1}\right)[S_R][A] \tag{8.17}$$

$$k_{\mathrm{obs}} = \frac{k_1[\mathrm{H}^+]+k_2K_1}{[\mathrm{H}^+]+K_1}. \tag{8.18}$$

A thorough analysis of Eq. 8.18 leads us to several potential cases depending on the relative values of $[\mathrm{H}^+]$, K_1, k_1 and k_2.

a. When the reaction is carried out at acidities such that pH >> pK_1, $[\mathrm{H}^+]$ can be neglected against K_1, and the ratio $k_1[\mathrm{H}^+]/K_1$ can be neglected against k_2, so that $k_{\mathrm{obs}} \approx k_2$.
b. When the reaction is carried out at acidities so that pH << pK_1, K_1 can be neglected against $[\mathrm{H}^+]$, and the ratio $k_2[\mathrm{H}^+]/K_1$ can be neglected against k_1, so that $k_{\mathrm{obs}} \approx k_1$.
c. If the experimental conditions are such, the pH value equals that of the pK_1, i.e., pH = pK_1, $k_{\mathrm{obs}} = (k_1 + k_2)/2$.
d. In the acidity region pH = p$K_1 \pm 3$, the variation of the measured rate constant k_{obs} follows a sigmoidal-like curve, the exact profile depending on whether $k_1 > k_2$ or $k_1 < k_2$ as illustrated in Fig. 8.3a, b, respectively.

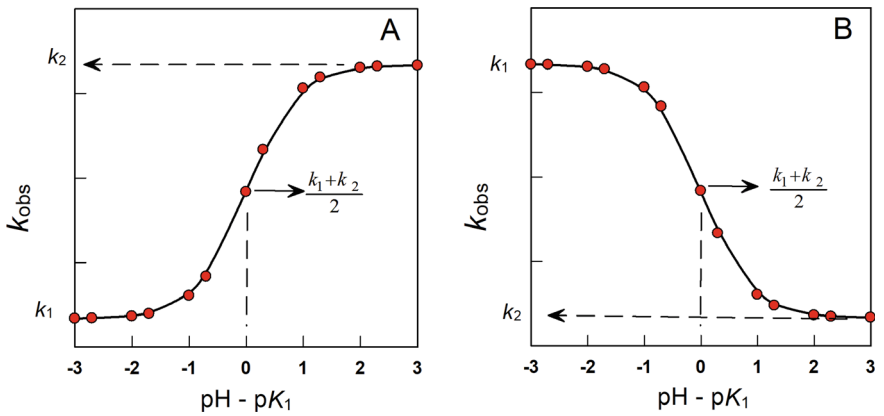


Fig. 8.3 Variation of the measured rate constant k_{obs} for reactions preceded by a rapidly established acid–base equilibrium where both the acid and base forms of the reagent react according to Eq. 8.15 **a** $k_1 < k_2$, **b** $k_1 > k_2$

8.2.4 The Reacting Species is the Bivalent Acid Form, RH_2^{2+} of the Reagent

This case relates to those reactions where the reagent is a bivalent acid that reacts with A only when it is fully protonated, so that Eq. 8.19 holds

$$\begin{aligned} &\mathrm{RH_2^{2+}} \overset{K_1}{\rightleftharpoons} \mathrm{RH^+} + \mathrm{H^+} \\ &\mathrm{RH^+} \overset{K_2}{\rightleftharpoons} R + \mathrm{H^+} \\ &\mathrm{RH_2^{2+}} + A \xrightarrow{k_1} \text{Products.} \end{aligned} \tag{8.19}$$

The rate equation is given by Eq. 8.20, and the actual concentration of the reagent is given by Eq. 8.21

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1[\mathrm{RH_2^{2+}}][A] \tag{8.20}$$

$$[S_R] = \left[\mathrm{RH_2^{2+}}\right] + \left[\mathrm{RH^+}\right] + [R]. \tag{8.21}$$

Considering the mathematical equations for the dissociation process in 8.19, the actual concentration of the reagent is given by Eq. 8.22

$$[S_R] = [\mathrm{RH_2^{2+}}]\left(1 + \frac{K_1}{[\mathrm{H^+}]} + \frac{K_1K_2}{[\mathrm{H^+}]^2}\right). \tag{8.22}$$

And, therefore, the rate of the reaction and the measured rate constant k_{obs} are given by Eqs. 8.23 and 8.24, respectively

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = \frac{k_1}{1 + \frac{K_1}{[\mathrm{H^+}]} + \frac{K_1K_2}{[\mathrm{H^+}]^2}}[S_R][A] \tag{8.23}$$

$$k_{\mathrm{obs}} = \frac{k_1}{1 + \frac{K_1}{[\mathrm{H^+}]} + \frac{K_1K_2}{[\mathrm{H^+}]^2}}. \tag{8.24}$$

As we can see, the dependence of k_{obs} with acidity is quite complex. But some limiting situations can be analyzed depending on the relative values of the dissociation constants K_1 and K_2. For instance, when $\mathrm{p}K_1 < \mathrm{p}K_2 - 2$, the value of the ratio $K_1K_2/[\mathrm{H^+}]^2$ can be neglected against $1 + K_1/[\mathrm{H^+}]$, and the equation for the measured rate constant degenerates to Eq. 8.7, showing a dependence with the acidity as that in Fig. 8.1.

On the contrary, when $\mathrm{p}K_1 > \mathrm{p}K_2 - 2$, the shape of the k_{obs} dependence with acidity corresponds to steeper curves as shown in Fig. 8.4. In this case, the effect of

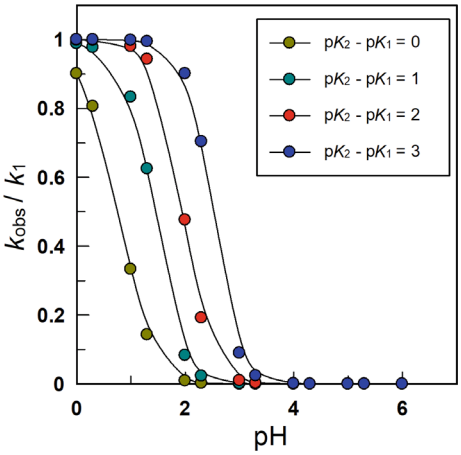


Fig. 8.4 Variation of the ratio k_{obs}/k_1 with pH for a reaction of the acid form of a bivalent acid following Eq. 8.19

the second dissociation step can be detected only for such a ratio of the dissociation constants.

8.2.5 *The Reacting Species is the Monobasic Form, RH^+, of a Bivalent Acid, RH_2^{2+}*

This case holds when the reagent is a bivalent acid RH_2^{2+} that reacts with the reactant A only in the protonated form RH^+, according to Eqs. 8.25

$$\begin{aligned} &RH_2^{2+} \overset{K_1}{\rightleftharpoons} RH^+ + H^+ \\ &RH^+ \overset{K_2}{\rightleftharpoons} R + H^+ \\ &RH^+ + A \xrightarrow{k_1} \text{Products.} \end{aligned} \tag{8.25}$$

Examples of this reaction are that of ethyleneglycol with periodic acid (which, in fact, is a bit more complicated because of the acid–base equilibria of an intermediate formed in the course of the reaction) or the hydrolysis reaction of pyridoxal-5-phosphate.

The rate equation is thus given by Eq. 8.26. Bearing in mind the mathematical expression for the actual concentration of the reagent S_R and the mathematical equations defining the dissociation constants K_1 and K_2, the actual concentration of the reagent is given by Eq. 8.26

$$-\frac{d[A]}{dt} = k_1[RH^+][A] = \frac{k_1}{1 + \frac{[H^+]}{K_1} + \frac{K_2}{[H^+]}}[S_R][A]. \tag{8.26}$$

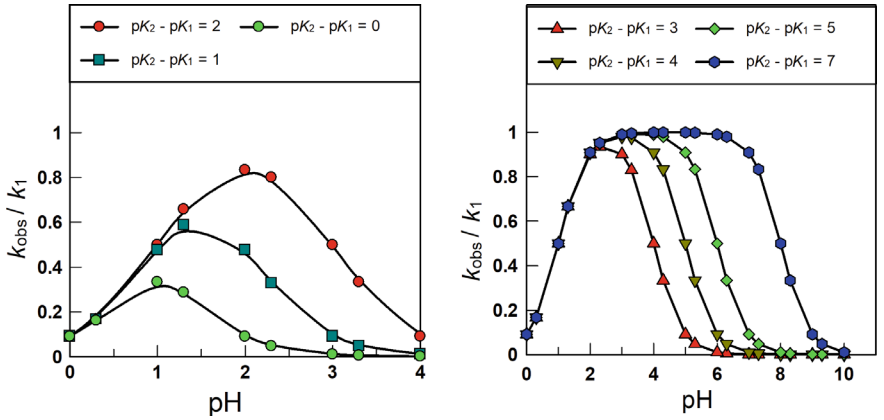


Fig. 8.5 Dependence of the ratio k_{obs}/k_1 with pH for a reaction where the reactive species is the conjugated monobasic form of a bivalent acid according to Eq. 8.25. The different plots were obtained by assuming a value of $pK_1 = 1$ for the first dissociation constant.

The measured rate constant k_{obs} is thus given by Eq. 8.27

$$k_{obs} = \frac{k_1}{1 + \frac{[H^+]}{K_1} + \frac{K_2}{[H^+]}}. \tag{8.27}$$

The dependence of the measured rate constant depends, as expected, of the relative values of the acidity and those of the dissociation constants. The pH-dependence of k_{obs} is displayed in Fig. 8.5 for $pK_2 - pK_1$ values ranging 0–8.

8.2.6 *The Reacting Species is the Dibasic Form, R, of a Bivalent Acid, RH_2^{2+}*

When the regent is a divalent acid RH_2^{2+} that reacts with reactant A only when completely dissociated, the mechanism given by Eq. 8.28 holds

$$\begin{aligned} RH_2^{2+} &\overset{K_1}{\rightleftharpoons} RH^+ + H^+ \\ RH^+ &\overset{K_2}{\rightleftharpoons} R + H^+ \\ R + A &\xrightarrow{k_1} \text{Products}. \end{aligned} \tag{8.28}$$

The rate equation is given by Eq. 8.29 and the actual concentration of the reagent is given by Eq. 8.30

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = k_1[R][A] \tag{8.29}$$

$$[S_R] = \left[\mathrm{RH}_2^{2+}\right] + \left[\mathrm{RH}^{+}\right] + [R]. \tag{8.30}$$

Considering the mathematical equations for the dissociation process in 8.28, the rate of the reaction and the measured rate constant k_{obs} are given by Eqs. 8.31 and 8.32, respectively

$$-\frac{\mathrm{d}[A]}{\mathrm{d}t} = \frac{k_1}{1 + \frac{[\mathrm{H}^+]}{K_2} + \frac{[\mathrm{H}^+]^2}{K_1 K_2}}[S_R][A] \tag{8.31}$$

$$k_{\mathrm{obs}} = \frac{k_1}{1 + \frac{[\mathrm{H}^+]}{K_2} + \frac{[\mathrm{H}^+]^2}{K_1 K_2}}. \tag{8.32}$$

An analysis of the Eq. 8.32 shows that when $\mathrm{p}K_1 << \mathrm{p}K_2$, the equation simplifies to Eq. 8.33

$$k_{\mathrm{obs}} = \frac{k_1 K_2}{K_2 + [\mathrm{H}^+]}. \tag{8.33}$$

When the dissociation constants are similar, asymmetric variations in the k_{obs}/k_1 ratio as those shown in Fig. 8.6 are obtained.

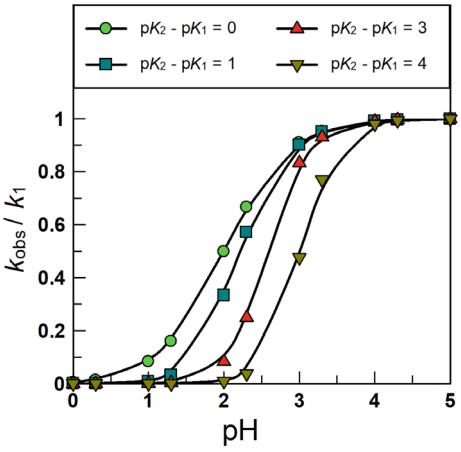


Fig. 8.6 Variation of the ratio k_{obs}/k_1 with pH for a reaction where the reactive species is the conjugated dibasic form of a bivalent acid according to the reaction scheme in Eq. 8.28. The different plots were obtained by assuming a value of $\mathrm{p}K_1 = 2$ for the first dissociation constant

8.2.7 All Three Forms of the Bivalent Acid RH_2^{2+} (RH_2^{2+}, RH^+, R) Are Reactive

When reactant *A* reacts, at comparable rates, with all three possible forms of a bivalent acid RH_2^{2+}, the reaction mechanism given by Eq. 8.34 holds.

$$\begin{aligned} &RH_2^+ \overset{K_1}{\rightleftharpoons} RH^+ + H^+ \\ &RH^+ \overset{K_2}{\rightleftharpoons} R + H^+ \\ &RH_2^+ + A \xrightarrow{k_1} P_1 \\ &RH^+ + A \xrightarrow{k_2} P_2 \\ &R + A \xrightarrow{k_3} P_3. \end{aligned} \tag{8.34}$$

The rate of the reaction is then given by Eq. 8.35

$$\begin{aligned} -\frac{d[A]}{dt} = {} & k_1[RH_2^+][A] + k_2[RH^+][A] \\ & + k_3[R][A]. \end{aligned} \tag{8.35}$$

The analytical concentration of the reagent is given by Eq. 8.36

$$[S_R] = \left[RH_2^{2+}\right] + \left[RH^+\right] + [R]. \tag{8.36}$$

And the concentration of the different ionized species is thus given by Eqs. 8.37–8.39

$$[RH_2^{2+}] = \frac{[S_R]}{1 + \frac{K_1}{[H^+]} + \frac{K_1K_2}{[H^+]^2}} \tag{8.37}$$

$$[RH^+] = \frac{[S_R]}{1 + \frac{[H^+]}{K_1} + \frac{K_2}{[H^+]}} \tag{8.38}$$

$$[R] = \frac{[S_R]}{1 + \frac{[H^+]}{K_2} + \frac{[H^+]^2}{K_1K_2}}. \tag{8.39}$$

And the rate equation is given by Eq. 8.40

$$-\frac{d[A]}{dt} = \left(\frac{k_1}{1 + \frac{K_1}{[H^+]} + \frac{K_1K_2}{[H^+]^2}} + \frac{k_2}{1 + \frac{[H^+]}{K_1} + \frac{K_2}{[H^+]}} \right.$$

$$+\frac{k_3}{1+\frac{[H^+]}{K_2}+\frac{[H^+]^2K_2}{K_1K_2}}\Bigg)[S_R][A]. \tag{8.40}$$

And the observed or measured rate constant is given by Eq. 8.41

$$k_{obs}=\frac{k_1[H^+]^2+k_3K_1K_2+k_2K_1[H^+]}{[H^+]^2+K_1[H^+]+K_1K_2}. \tag{8.41}$$

The cases when only two or one of the three species react at measurable rates and the others are too slow to be considered negligible can be derived from the above mathematical treatment for all three possible combinations by simple setting the value of the appropriate rate constants equal to zero.

Simplifications to Eq. 8.40 can be made depending on the relative values of the dissociation constants. We will first consider those cases where $K_1 >> K_2$. Depending on the proton concentration, several limiting cases can be found:

When $[H^+] >> K_1$, then $K_1\ K_2 << K_1[H^+] << [H^+]^2$, and therefore $k_{obs} \approx k_1$.

When $K_1 >> [H^+] >> K_2$, then $K_1\ K_2 << [H^+]^2 << K_1[H^+]$, and therefore $k_{obs} \approx k_2$.

When $[H^+] << K_2$, then $[H^+]^2 << K_1[H^+] << K_1\ K_2$, so that K_1.

When $[H^+] = K_1$, $k_{obs} = (k_1 + k_2)/2$.

When $[H^+] = K_2$, $k_{obs} = (k_2 + k_3)/2$.

The variation of the observed rate constant with acidity is given in Fig. 8.7 and, as illustrated, the particular profile depends on the relative values of the rate constants.

When the values of the dissociation constants K_1 and K_2 are comparable (i.e., $pK_2 - pK_1 \leq 1$), the dependence of the observed rate constant with acidity becomes more complicated as it depends on the relative values of the rate constants, Fig. 8.8.

So far, the cases discussed above assume that the acid–base equilibria preceding the rate-determining step is established rapidly for only one reactant and mathematical relationships for the pH-dependence of the observed or measured rate constant were derived for different cases, all illustrated with the theoretical dependence of k_{obs} with the acidity for different relative values of the rate constants and of the dissociation constants involved. In the next paragraphs, we will discuss reactions where two reactants undergo acid–base processes within the investigated acidity range.

8.2.8 *The Base Form R of the Reagent RH^+ Reacts with the Acid Form BH^+ of the Reactant B*

The reaction scheme that holds when the acid form of the reagent, RH^+, reacts with the non-ionized reactant *B* is that given in Eq. 8.42a. Alternatively, a similar reaction mechanism, Eq. 8.42b, holds when the base form *R* of the reagent reacts with the acid form of the reactant BH^+.

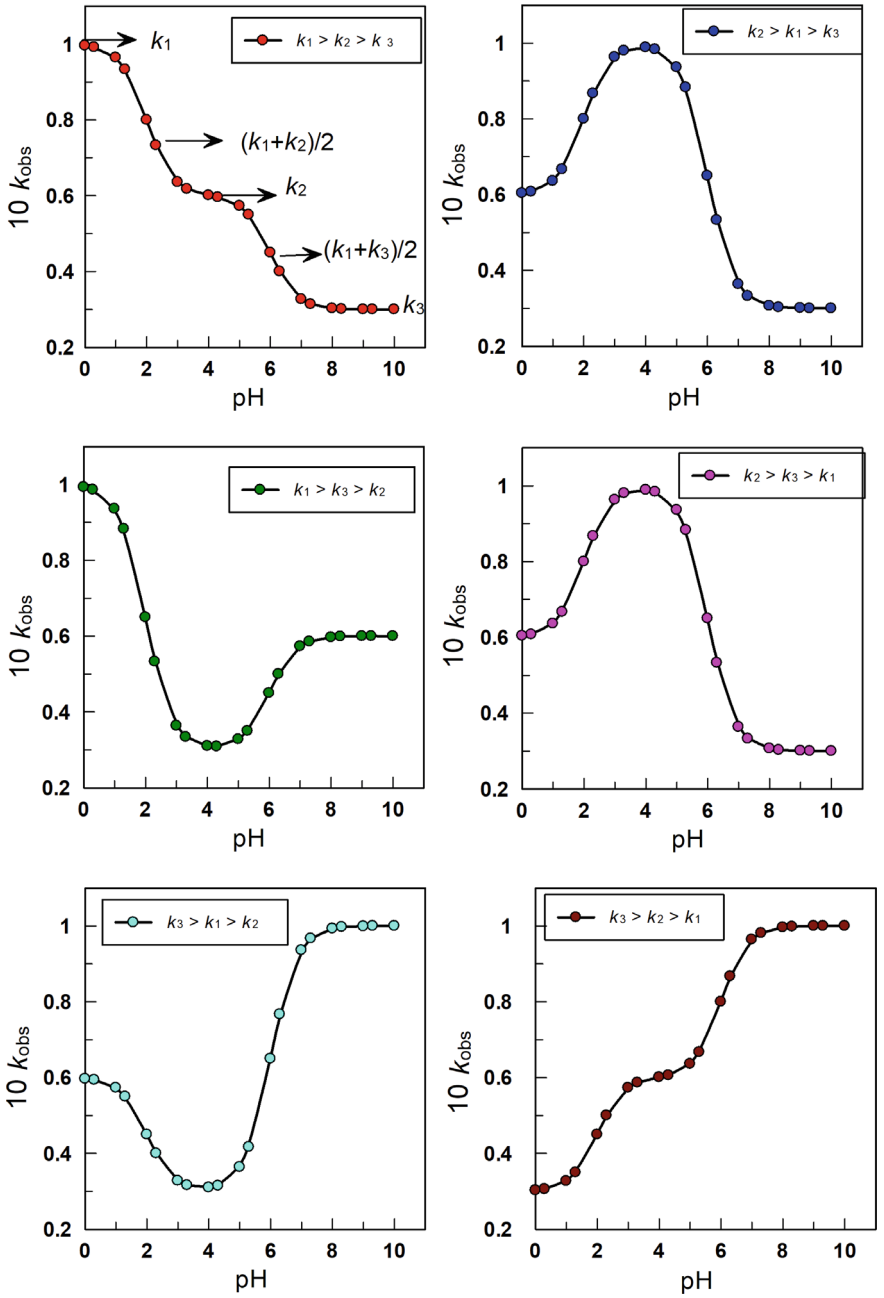


Fig. 8.7 Variation of the observed rate constant with pH for a reaction in which all three forms of a bivalent acid are reactive according to Eq. 8.1. k_{obs} values in plots were obtained by employing Eq. 8.40 and the values of 0.1, 0.03, 0.06 (arbitrary units) for the rate constants k_1, k_2, k_3 and $pK_1 = 2$ and $pK_2 = 6$ for the dissociation constants

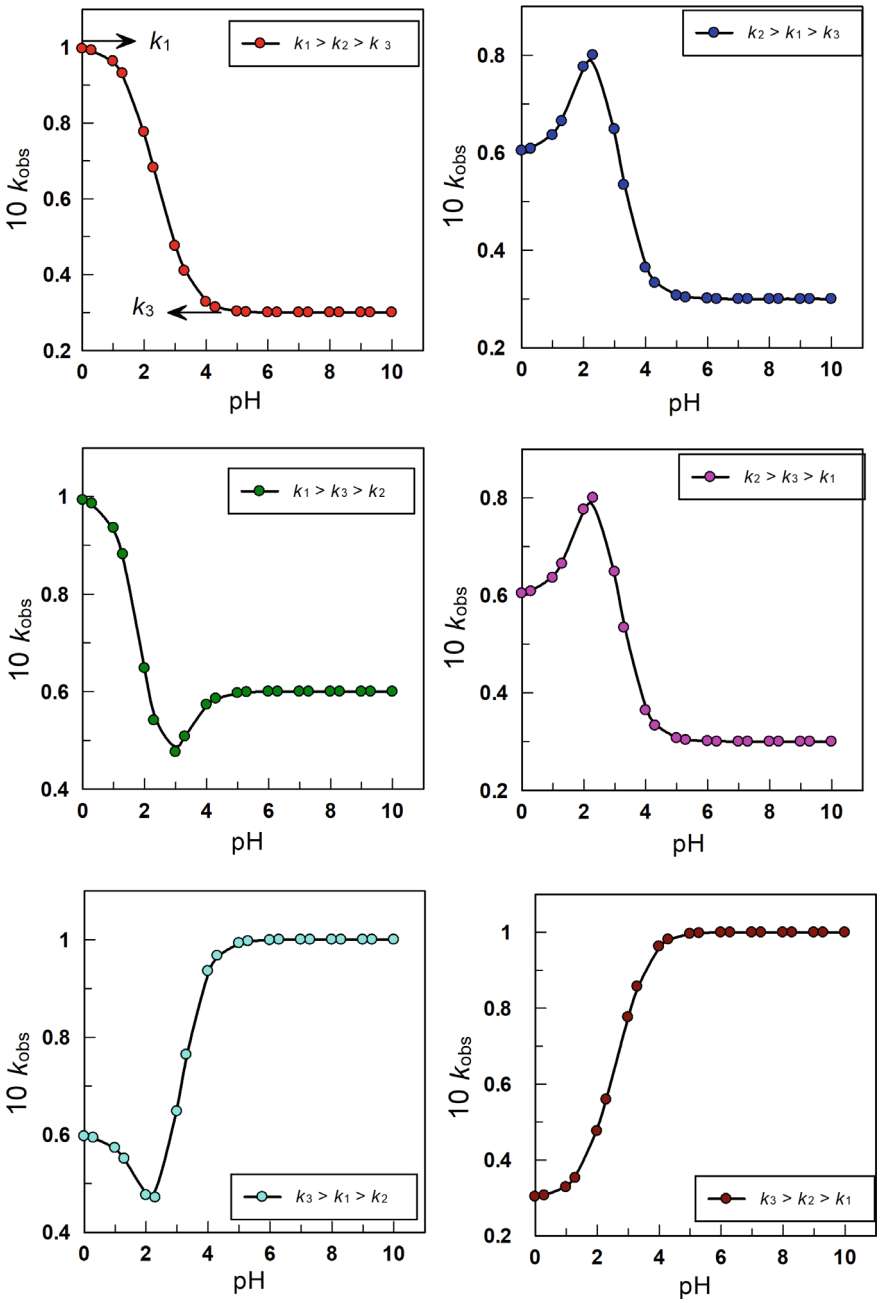


Fig. 8.8 Variation of the observed rate constant with pH for a reaction in which all three forms of a bivalent acid are reactive according to Eq. 8.34 and where the values of the dissociation constants are comparable ($pK_2 - pK_1 \leq 1$). Data in plots were obtained by employing Eq. 8.40 and the same values for the rate constants k_1, k_2, k_3 as in Fig. 8.7 and the values of $pK_1 = 2$ and $pK_2 = 3$

$$\begin{aligned} &\mathrm{RH}^+ \overset{K_1}{\rightleftharpoons} R + \mathrm{H}^+ \\ &\mathrm{BH}^+ \overset{K_B}{\rightleftharpoons} B + \mathrm{H}^+ \\ &\mathrm{RH}^+ + B \xrightarrow{k_1} P_1 \end{aligned} \tag{8.42a}$$

$$\begin{aligned} &\mathrm{RH}^+ \overset{K_1}{\rightleftharpoons} R + \mathrm{H}^+ \\ &\mathrm{BH}^+ \overset{K_B}{\rightleftharpoons} B + \mathrm{H}^+ \\ &\mathrm{BH}^+ + R \xrightarrow{k_2} P_2. \end{aligned} \tag{8.42b}$$

According to the mechanism 8.42a, the rate of the reaction is given by Eq. 8.43

$$-\frac{\mathrm{d}[P_1]}{\mathrm{d}t} = k_1[\mathrm{RH}^+][B] \tag{8.43}$$

and the actual concentration of the reagent is given by Eq. 8.44

$$[S_R] = \left[\mathrm{RH}^+\right] + [R]. \tag{8.44}$$

Considering the mathematical equations for the dissociation process in 8.42, the actual concentrations of the reagent S_R and of the reactant S_B are given by Eqs. 8.45 and 8.46, respectively

$$[S_R] = [\mathrm{RH}^+]\left(1 + \frac{K_1}{[\mathrm{H}^+]}\right) \tag{8.45}$$

$$[S_B] = [B]\left(1 + \frac{[\mathrm{H}^+]}{K_B}\right). \tag{8.46}$$

And, therefore, the rate of the reaction and the measured rate constant k_{obs} are given by Eqs. 8.47 and 8.48, respectively

$$-\frac{\mathrm{d}[P_1]}{\mathrm{d}t} = \frac{k_1[S_R][S_B]}{\left(1 + \frac{K_1}{[\mathrm{H}^+]}\right)\left(1 + \frac{[\mathrm{H}^+]}{K_B}\right)} \tag{8.47}$$

$$\frac{k_{\mathrm{obs}}}{k_1} = \frac{K_2[\mathrm{H}^+]}{\left(K_1 + [\mathrm{H}^+]\right)\left(K_B + [\mathrm{H}^+]\right)}. \tag{8.48}$$

When considering the mechanism given by 8.42a, Eq. 8.49 can be derived

$$\frac{k_{\mathrm{obs}}}{k_2} = \frac{K_1[\mathrm{H}^+]}{\left(K_1 + [\mathrm{H}^+]\right)\left(K_B + [\mathrm{H}^+]\right)}. \tag{8.49}$$

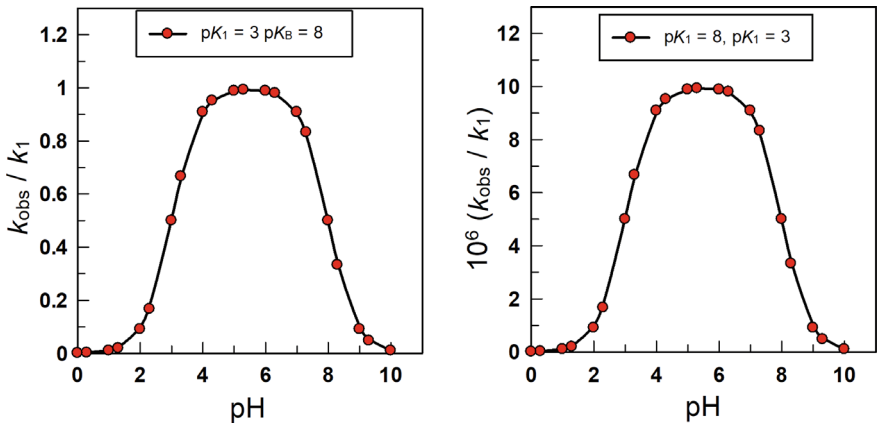


Fig. 8.9 Variation of the ratio k_{obs}/k_1 with acidity in which the base form of the reagent reacts with the acid form of the reactant according to the reaction mechanism shown in Eq. 8.42a

We will assume that the values of the dissociation constants can be determined by an independent experiment and not as an output of the fitting of experimental data so that we can decide which of the mechanisms 8.42a or 8.42b holds. However, since they are formally equivalent, we will analyze only the mechanism given by 8.42a, and for which Eq. 8.48 applies. The dependence of the k_{obs}/k_1 ratio with acidity will depend on the relative values of the dissociation constants and so we will consider the three cases in which the values are equal to each other, or one is much higher than the other.

a. $K_B >>> K_1$. Under these conditions, the pH-dependence of the ratio k_{obs}/k_1 is bell-shaped as illustrated in Fig. 8.9.
b. $K_B <<< K_1$. Under these conditions, the variation of the ratio k_{obs}/k_1 with acidity follows a similar bell-shaped profile as in the previous case, Fig. 8.9. However, the ratio k_{obs}/k_1 is very small, which means that the observed rate constant k_{obs} is only a small fraction of k_1, and therefore, unless k_1 has a very high value, the rate of the reaction may be too small (and even it may not be possible to measure it) and the reaction can be considered as not taking place.
c. K_B similar to K_1.

When the values of K_B and K_1 are comparable (i.e., $pK_B - 2 < pK_1 < pK_B + 2$), the shape of the k_{obs} dependence with pH does not depend on whether $K_B < K_1$ or $K_B > K_1$ so that from such plots, one can obtain estimates of the values of both dissociation constants. The maximum value of the observed rate constant k_{obs} is always a fraction of that of k_1, Fig. 8.10.

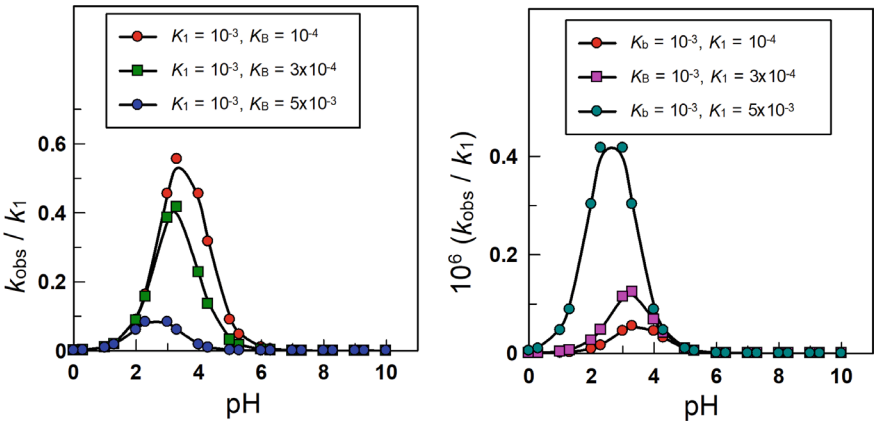


Fig. 8.10 Variations in the relative values of the rate constants k_{obs}/k_1 for a reaction following the mechanism depicted in Eq. 8.42a and where the values of the dissociation constants are comparable

8.2.9 *The Two Acid or the Two Base Forms of the Reagent and of the Reactant React*

These two cases can be described by the mechanism involving the reactions in Eqs. 8.50a and 8.50b, respectively.

$$\begin{aligned} &\mathrm{RH}^+ \overset{K_1}{\rightleftharpoons} R + \mathrm{H}^+ \\ &\mathrm{BH}^+ \overset{K_B}{\rightleftharpoons} B + \mathrm{H}^+ \\ &\mathrm{RH}^+ + \mathrm{BH}^+ \xrightarrow{k_1} P_1 \end{aligned} \tag{8.50a}$$

$$\begin{aligned} &\mathrm{RH}^+ \overset{K_1}{\rightleftharpoons} R + \mathrm{H}^+ \\ &\mathrm{BH}^+ \overset{K_B}{\rightleftharpoons} B + \mathrm{H}^+ \\ &R + B \xrightarrow{k_2} P_1. \end{aligned} \tag{8.50b}$$

For the reaction involving the two acid forms (8.50a), the rate of the reaction is given by Eq. 8.51

$$-\frac{\mathrm{d}[B]}{\mathrm{d}t} = k_1[\mathrm{RH}^+][\mathrm{BH}^+]. \tag{8.51}$$

Bearing in mind that the actual concentrations of the reagent and of the reactant are given by Eqs. 8.52 and 8.53, respectively. Taking into consideration the mathematical equations for the dissociation processes in 8.50a, the analytical concentrations of the

reagent S_R and of the reactant S_B are given by Eqs. 8.54 and 8.55, respectively.

$$[S_R] = \left[\mathrm{RH}^+\right] + [R] \tag{8.52}$$

$$[S_B] = \left[\mathrm{BH}^+\right] + [B] \tag{8.53}$$

$$[S_R] = [\mathrm{RH}^+]\left(1 + \frac{K_1}{[\mathrm{H}^+]}\right) \tag{8.54}$$

$$[S_B] = [B]\left(1 + \frac{K_B}{[\mathrm{H}^+]}\right). \tag{8.55}$$

And, therefore, the rate of the reaction and the measured rate constant k_{obs} are given by Eqs. 8.56 and 8.57, respectively

$$-\frac{\mathrm{d}[B]}{\mathrm{d}t} = \frac{k_1[S_R][S_B]}{\left(1 + \frac{K_1}{[\mathrm{H}^+]}\right)\left(1 + \frac{K_B}{[\mathrm{H}^+]}\right)} \tag{8.56}$$

$$\frac{k_{\mathrm{obs}}}{k_1} = \frac{[\mathrm{H}^+]^2}{\left(K_1 + [\mathrm{H}^+]\right)\left(K_B + [\mathrm{H}^+]\right)}. \tag{8.57}$$

As in previous examples, the variation of the ratio k_{obs}/k_1 with acidity shows the form of a normal dissociation profile, depending on the relative values of the dissociation constants, Fig. 8.11. Note that in those cases where the K_B and K_1 values are comparable, the shape of the curves may be unsymmetrical.

When the basic forms are the reacting species, a mathematical equation for the observed rate constant can be derived in a similar way as before, Eq. 8.58

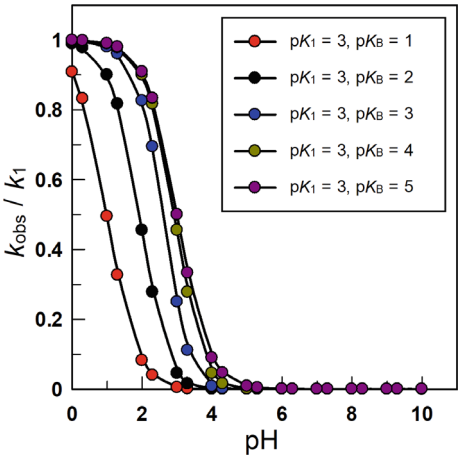


Fig. 8.11 Dependence of the ratio $k_{\mathrm{obs}}/\mathrm{k}_1$ with acidity for a reaction following the mechanism depicted in Eq. 8.50a (reaction of the acid form of the reagent with the acid form of the reactant)

$$\frac{k_{\text{obs}}}{k_1} = \frac{K_1 K_B}{\left(K_1 + [\text{H}^+]\right)\left(K_B + [\text{H}^+]\right)}. \tag{8.58}$$

8.3 Reactions in Which H^+ or OH^- Participate in the Rate Determining Step

A word of caution must be indicated for reactions where the rate-determining step can be the dissociation of a proton or addition of a proton, or a nucleophilic attach by OH^- ions, or a cleavage resulting in the formation of OH^- ions.

In such cases, the variation of the measured rate constant k_{obs} with acidity is a linear function of $[H^+]$ or $[OH^-]$ with an intercept approaching zero (assuming that the reaction with the solvent either as acid or base can be neglected). It is not easy to distinguish reactions of this type from those where a rapidly established acid–base equilibrium precedes the slow step. At acidities where $pK_a - pH \geq 2$ or $pK_a - pH < 2$. In both cases, the measured rate constant is a linear function of $[H^+]$ or $[OH^-]$. In attempting to distinguish between both alternatives, it is recommended to monitor the variation of k_{obs} with acidity in the widest range possible detect potential deviations from linearity, which could indicate that observed dependence is a part of an exponential curve rather than a linear function.

8.4 Effects of Temperature

As shown in Chap. 2, reactions are rather sensitive to temperature changes. For most reactions, the temperature dependence of the rate constant is described by the Arrhenius equation, where two empirical constants, the preexponential factor A and the activation energy E_a, are usually obtained from the intercept and slope, respectively, of the fitting of the experimental data to the logarithmic form of the Arrhenius equation, which should be a straight line (see Chap. 2). Deviations from linearity can be caused by the fact that a complex reaction is involved, which at the originally chosen temperature corresponds to a simple reaction rate equation. It may also be indicative that one of the reactants is sensitive to temperature so that it may be degrading or otherwise transforming into a non-reactive species. In these cases, a careful re-consideration of the reaction mechanism may be necessary so that a more complex reaction mechanism than that originally assumed can be proposed. On the other hand, a strictly linear relationship is not by itself sufficient evidence that the reaction rate equation used fully describes the observed changes in concentration with time, and hence, that the proposed mechanism is correct.

Although the Arrhenius equation was derived empirically, two alternative formulations were offered based on two different theoretical approaches: collision theory

and the thermodynamic-based transition state theory (TST, also called theory of absolute rates). While the collision theory, grounded on the kinetic theory of gases, is rather limited and mainly applies to gas reactions where neutral species are involved, the TST is more general and much widely used. It assumes that an "activated complex" is, in statistical equilibrium, with the reactants. Applying the vant´Hoff equation from the classical thermodynamics to the TST, it follows those thermodynamic parameters such as the entropy, $\Delta S^{\ddagger}$, and enthalpy, $\Delta H^{\ddagger}$, of activation can be determined.

Usually, determining $\Delta S^{\ddagger}$, and $\Delta H^{\ddagger}$ for a single reaction offers limited useful mechanistic information. Only changes of $\Delta H^{\ddagger}$ for properly chosen reaction series (for example, by choosing a group of structurally related compounds) can offer information on the origin of such changes and hence can be related to structure and geometry of the presumed transition state. Enthalpic changes are usually related to changes in energy along the reaction coordinate and are frequently interpreted in terms of bond changes. It must be noted, however, that similar information can be obtained from changes in the Gibbs free energy $\Delta G^{\ddagger}$ without the need of evaluating separately the activation enthalpy, $\Delta H^{\ddagger}$.

Entropic changes are more often given to mechanistic interpretation since $\Delta S^{\ddagger}$ values indicate changes in the order/disorder of the structures of the activated complex and the reactants, including the solvent molecules. Negative $\Delta S^{\ddagger}$ values indicate that the transition state is more ordered than the ground state, and the increase in order can result from several sources including increasing solvation, steric effects, or a decrease in the number of separate molecules.

8.5 Effects of Ionic Strength

The rates of chemical reactions are affected by the nature of the solvent used as discussed in Chap. 5 and by the nature and concentration of ionic species present in the reaction mixture. Changes in the solvent and nature and concentration of neutral salts added to the reaction mixture results in complex variations of the solute–solvent and solute–solute interactions. The variability of the individual types of interactions makes a general—rigorous—treatment of such effects intractable. However, for some types of reactions, it is possible get some insights and conclusions on the charge of the species participating in the reaction. Sometimes, counterion effects can offer some information about the composition of the transition state and in some instances; it is possible to obtain some information about solvation of the starting materials and the transition state.

The effect of charged species on reaction kinetics is known as the salt effect. If the concentration of an electrolyte affects the activity of the reacting species, it is known as the primary salt effect; on the other hand, if the addition of an electrolyte affects the concentration of one of the reacting species, the kinetic effect is called the secondary salt effect.

8.5.1 *Effect of the Ionic Strength in the Rate Constant of Reactions Between Ions*

In the reaction between ions, coulombic forces predominate over those of other types of neutral interactions, and a better quantitative interpretation of the salt effects is probably prevented by the lack of detailed information on the behavior of ions, their precursors, and the behavior of polar molecules in the strong electrical fields adjacent to ions.

Let´s consider the following reaction 8.59 taking place in solution

$$A + B \rightleftharpoons X \rightarrow P. \tag{8.59}$$

The ionic strength (I) of the solution corresponds to the concentration of *all* charged species in a solution represented by Eq. 8.60

$$I = \frac{1}{2}\sum_i c_i z_i^2, \tag{8.60}$$

where c_i is the concentration of the ion i and z_i its charge.

The equilibrium constant of the reaction 8.59, K_e, is given by 8.60

$$K_e = \frac{a_x}{a_A a_B}, \tag{8.61}$$

where a_X, a_A, and a_B correspond to the activities of X, A, and B, respectively. The activity represents a measure of deviation from ideality and is defined in terms of the activity coefficient γ and the concentration of the species, $a = \gamma$ []. The equilibrium constant can be expressed by Eq. 8.62, and [X] is given by Eq. 8.63

$$K_e = \frac{\gamma_x[X]}{\gamma_A[A]\gamma_B[B]} \tag{8.62}$$

$$[X] = \frac{K_e[A][B]\gamma_A\gamma_B}{\gamma_x}. \tag{8.63}$$

The rate of the reaction is given by Eq. 8.64

$$\frac{\mathrm{d}[P]}{\mathrm{d}t} = k[X] = k\frac{K_e[A][B]\gamma_A\gamma_B}{\gamma_x}. \tag{8.64}$$

When $\gamma_A . \gamma_B/\gamma_x = 1$ (dilute solutions), the rate constant is

$$k_0 = kK_e. \tag{8.65}$$

Therefore,

$$k = k_0 \frac{\gamma_A \gamma_B}{\gamma_x}. \tag{8.66}$$

Taking logaritms and bearing in mind the Debye–Hückel limiting law for the species A, B and X, Eqs. 8.67 and 8.68 can be derived

$$\log k = \log k_0 + \log \gamma_A + \log \gamma_B - \log \gamma_x \tag{8.67}$$

$$\begin{aligned} \log \gamma_A &= -AZ_A^2 I^{1/2} \\ \log \gamma_B &= -AZ_B^2 I^{1/2} \\ \log \gamma_x &= -AZ_X^2 I^{1/2} \end{aligned} \tag{8.68}$$

where Z_A *y* Z_B are the total charges of A and B, respectively, and A is a constant ($A = 0.51$ mol$^{1/2}$ dm$^{3/2}$ for reactions in water solution at 25 °C). After substituing Eqs. 8.68 in 8.67, Eqs. 8.69 and 8.70 can be easily derived

$$\log k = \log k_0 - A[Z_A^2 + Z_B^2 - (Z_A + Z_B)^2] I^{1/2} \tag{8.69}$$

$$\log\left(\frac{k}{k_0}\right) = 2AZ_A Z_B I^{1/2}. \tag{8.70}$$

Equation 8.70 is the well-known **Bronsted–Bjerrum** equation and allows us to predict the effect of the ionic strength on reactions where the reactants are ionic and non-ionic species. Three cases are presented in this equation:

i. If the charges of A and B are of the same sign (positive or negative), an increase in the ionic strength will cause an increase in the rate constant
ii. If the charges of A and B are of different signs, the constant of velocity will undergo a decrease with the increase of the ionic force and
iii. If any of the reactants have no charge, then the Z_A Z_B product will be zero and the rate constant will be independent of the ionic strength.

According to Eq. 8.70, the rate of an ionic reaction is not exclusively determined by the concentration of the reactants. Any ion present in the solution affects the rate through the ionic strength dependent rate coefficient, even if those ions are inert with respect to the reaction.

Chapter 9
Nonlinear Regression Analyses: Principles and Comparisons Between Experimental and Fitting Data



9.1 Background

Dynamic chemical processes such as the description of chemical kinetics are typically modeled with a system of ordinary differential equations (ODEs). These models usually contain many parameters representing different physicochemical quantities, such as rates of chemical reactions, formation and degradation rates of diverse species, etc. [1]. These parameters values cannot always be experimentally measured and are often inferred using computational algorithms, which can be slow because there are infinitely many possible parameter values to explore.

An ODE model can enhance optimization, while accounting for limits, equilibria, competitive reactions, and many other effects, which are otherwise easily overlooked. An accurate system of ODEs would represent a reaction model, creating a direct link between data and proposed reaction pathways, which could be crucial for extracting valuable insights from chemical kinetic data [2]. Recently, there has been a growing interest in the application of machine learning and AI methods for the analysis of chemical kinetics.

Use of nonlinear regression to analyze kinetic or binding data involves a nonlinear dependence of the observed parameter upon that varied, for example, the rate of a reaction on substrate concentration or the extent of reaction with time. Early different reports have mentioned that it is important to note that all rearrangement of the data to give a linear plot, which is then fitted by linear regression to give rate or equilibrium constants, perturb the error distribution of the data [3]. For example, Leatherbarrow [3] reported this issue associated with enzyme kinetic data, as shown in Fig. 9.1. A reason why the data may have a complex error distribution is if they have been manipulated in some way, for example by taking reciprocals (Fig. 9.1b) or logarithms to obtain a linear plot.

An approach, which is more generally applicable than the derivation and application of complex weighting functions for linear regression of rearranged data, is

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Physical Chemistry in Action, https://doi.org/10.1007/978-3-031-90702-9_9

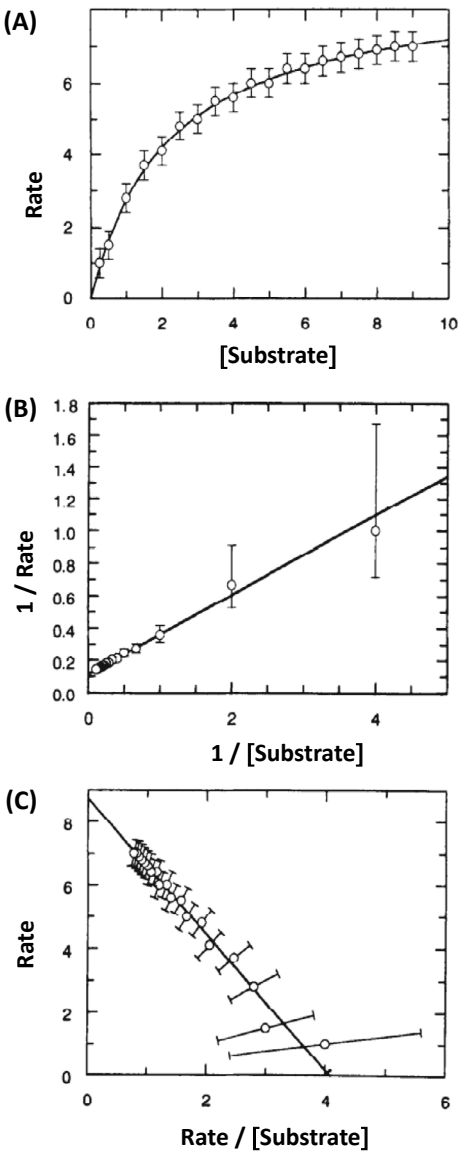


Fig. 9.1 Transformations effect to give linear plot on the error structure of enzyme kinetic data [3]. Reprinted from [3], Copyright (1990), with permission from Elsevier

nonlinear regression on the original data. Nonlinear regression minimizes the sums-of-squares differences between experimental and calculated data for any equation of the form $y = f(x)$, which covers essentially any equation describing an experimental process:

$$y = f(x, p_1, p_2, \ldots). \tag{9.1}$$

Equation (9.1) shows that y is a unique function of x and one or more unknown parameters, p_1, p_2, etc. The best-fit values for these parameters are determined by the calculations. The method is completely general, in that any equation can be used.

Regarding the statistical aspects of the fitting, nonlinear regression offers considerable practical advantages over analysis involving extensive rearrangements followed by linear regression. However, the statistics generated during the fitting procedures allow further insight into the significance of the data. In both examples described by there is the addition of a background parameter to a more standard equation. It is therefore pertinent to question the significance of adding this extra variable, as it is always possible to fit a dataset better by adding extra parameters to an equation.

In quantitative biological assays, for their analysis, although each experiment involves unique considerations, the use of regression analysis is an essential component of the meaningful interpretation of biological data [4]. In simplest terms, regression analysis is a statistical method used to evaluate the relationship between at least two continuous variables. Typically, biological assays define a dependent, or response, variable (y) and one or more independent (predictor) variables (x) to which the dependent variable responds. Such experiments seek to measure the response in y when x is defined and systematically altered. In defining the relationship between or among variables, regression provides a computational model against which the data generated are compared. The scientist then uses a variety of statistical parameters to assess how well that model fits the data. Regression can provide the ability to predict values for one variable based on the values of the other. Regression quantifies the observed behavior of these molecules, allowing for statistically validated comparisons of the desired biological activity.

In the case of biological assays, the most frequent types of regression analyses applied are linear and nonlinear regression. As we shall see, the comparative analysis of multiple molecules through binding or activity profiles in a biological system most frequently involves nonlinear analysis, whereas linear methods are typically applied for the determination of unknowns [5, 6]. In this chapter, we focus on the principles of curve fitting from the perspective of a research scientist designing and conducting experiments without an emphasis on the underlying mathematical formulae.

9.2 Residual Plot Analysis

Residual plots: Residual plots are a measure of how much a regression line vertically misses a data point. Therefore, it can be used to assess the regression quality, and it is one of the most important skills of a data analyst. Examining the underlying statistical assumptions about residuals, such as constant variance, variables' independence, and the distribution's normality is possible. A satisfactory residuals plot should give no trend or pattern [7]; $\Sigma r_i = 0$. For more details, see Draper and Smith [7]. In addition, as previously mentioned by El-Seoud et al. [8], checking the residuals is a reliable approach to judging kinetic data quality. Mathematically, residual (r_i) has been defined as Eq. (9.2) [9]:

$$r_i = y - \hat{y}$$
$$= \text{observed value of } y \text{ - predicted(fitted) value of } y. \tag{9.2}$$

9.3 Sum-of-Squares

Sum-of-squares (SS) from least squares nonlinear regression is the sum of the squares of the vertical distances of the points from the curve. It is expressed in the units used for the y values squared, and it is defined as Eq. (9.3):

$$\sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2, \tag{9.3}$$

where n is the sample size, $\hat{y}_i$ is the predicted value of y for observation i, and $\overline{y}$ is the mean of y value. Standard (least squares) nonlinear regression works by varying the values of the model parameters to minimize SS. If you choose to weigh the values and minimize the relative distance squared (or some other weighting function), goodness of fit is quantified with the weighted sum of squares. You'll only find the SS value useful if you do additional calculations to compare models [9].

9.4 Coefficient of Determination

This coefficient R^2 is a fraction between 0.0 and 1.0, which has no units and provides statistical accounts for the agreement of the fit with the values. Higher values indicate that the curve comes closer to the data. The coefficient determination is defined as Eq. (9.4):

$$R^2 = \frac{\sum_{i=1}^{n} (\hat{y}_i - \overline{y})^2}{\sum_{i=1}^{n} (y_i - \overline{y})^2}. \tag{9.4}$$

Its value can be interpreted from nonlinear regression very much like you interpret R^2 from linear regression [10]. When R^2 equals 0.0, the best-fit curve fits the data no better than a horizontal line going through the meaning of all y values. In this case, knowing x does not help you predict Y. When R^2 equals 1.0, all points lie exactly on the curve with no scatter. If X is known, it is possible to calculate Y exactly.

9.5 Size of Dataset

As much in statistics as in chemistry, is very important to get confident values to be analyzed. In this sense, the size of the dataset plays an important role [11]. In practice, when the presence of systematic errors in the measurements is avoided, the mean value of the samples provides a value that is close to the true value. However, various other factors affect the measured values, moving them away from the true value. For this reason, giving a range of values is more useful and is likely to include the true value. The width of this range depends on two factors: the precision of the individual measurements, which in turn depends on the population's standard deviation and the number of measurements in the sample. The very fact that we repeat measurements implies that we have more confidence in the meaning of several values than in a single value. In other words, the higher the dataset you get, the more confident the values will be. Therefore, to get an adequate size dataset, it could be obtained as the following Eq. (9.5) [12]:

$$n = \frac{Z^2 P(1-P)}{d^2}, \tag{9.5}$$

where n is the sample size, Z is the statistic corresponding to the level of confidence, P is expected prevalence, and d is precision (corresponding to the effect of size). Equation 9.5 allows get to the size dataset necessary to obtain a confident value. This will be useful for kinetic profile measures. For instance, El-Seoud et al. [8] compared the kinetics profile with different datasets. The authors concluded that the reduction of the number of data points by a factor of 6 does not compromise the value of the rate constants for either the disappearance of a reagent or formation of products, however, the rate constants related to the intermediate species have been appreciably affected.

9.6 Noise

In chemistry, there are other factors that affect the measures, such as noise. It results from aleatoric or random interaction between the sample and the medium, specifically, the molecules with the environment [13]. However, the noise is not always a problem. In the previous example, the number of points in the measures doesn't affect the kinetic parameters. Moreover, Yamasaki et al. [14] showed the influence of noise on getting rate constants and concluded that the method they proposed was never affected by noise. Although a strong method can improve the noise, data something before fitting may influence the value of the rate constant.

9.7 Fitting Function

With respect to fitting function, there are criteria to consider. It is well-known that R^2 and common chi-squared (χ^2) tests for goodness can give a reason to think that the fit is good, but the fit does not guarantee that a given model fits the data better than another. Therefore, the fitting model should be chosen together with the kinetic model must be based on a reasonable reaction mechanism. In fact, tests that can solve what model could be better are the F-test and the Akaike Information Criterion (AIC). El-Seoud et al. explain that the F-test assumes that the models are nested (one model is the simplified version of the other). If the models are not nested, the model with the lower AIC best fits a given dataset [8].

9.8 Limits of the Fitting Algorithm

Considering that the second and higher derivatives are not zero for nonlinear regression, an iterative process is required to calculate the optimal parameter values. Several different algorithms can be used in nonlinear regression, including the Gauss–Newton, the Marquardt–Levenberg Algorithm (LMA), the Nelder–Mead, and the steepest descent methods [2]. In the case of SOLVER, however, it uses another iteration protocol, which is based on the robust and reliable generalized reduced gradient (GRG) method. A detailed description of the evolution and implementation of this code can be found elsewhere [3, 4]. All these algorithms have similar properties. They all require the user to input initial parameter values and use them to better estimate the parameters using an iterative process. Some limitations have been reported; with the same set of data, all these methods should yield the same parameter values. Different methods are used to obtain kinetic parameters from nonlinear regressions, for instance, the LMA. This type of algorithm is useful for finding local minima, but not for finding global minima. However, these limitations would be decreased by applying a correct parameterization of the results and a correct and orderly work plan to arrive at a good model with a real result.

Recently, Zielinski et al. [15] depicted the parameterization workflow for an example of enzymatic kinetic, which is divided into six steps: (1) Gather available enzyme kinetic data and associated experimental conditions, (2) Construct a mass action enzyme mechanism, consisting of known individual reaction steps involved in catalysis and regulation of the enzyme. (3) Based upon the enzyme mechanism, define kinetic equations corresponding to the macroscopic kinetic data types (e.g., K_M, k_{cat}) to be fit. (4) Preprocess the kinetic data to allow direct comparison to the equations defined in Step 3. (5) Fit the equations defined in step 3 to the processed data in step 4 using a nonlinear least squares optimization approach to obtain a locally optimal set of microscopic rate constants that reproduces the experimental data. Finally, (6) Cluster the resulting rate constant sets to identify a reduced number of characteristic rate constant sets for the enzyme, each of which reproduces the experimentally

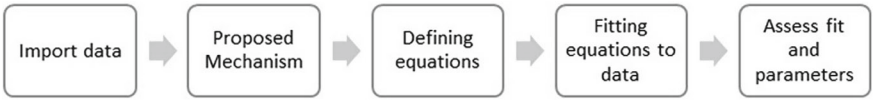


Fig. 9.2 General workflow for parameterization of rate constants using kinetic data. **a** Gathering of kinetic data, **b** Defining a reaction mechanism, **c** Defining equations that relate model behavior to each data type, **d** Fitting the available kinetic data, and **e** Assessing the fit for goodness of fit and parameter uncertainty. Further details are given elsewhere [15]

determined kinetic behavior of the enzyme. This workflow is resumed in Fig. 9.2 and each step could be analyzed in more detail in the study by Zielinski et al. [15].

9.9 Commercial Software for Performing Nonlinear Regression

Firstly, it is important to note that an accessible method is to use Microsoft Excel to fit nonlinear regressions. An advantage of this method is that Excel is probably included in the computer package as part of Microsoft Office, and thus no additional expense is required. Additionally, Excel offers a friendly user-interface, flexible data manipulation, built-in mathematical functions, and instantaneous graphing of data [16]. As above mentioned, Excel contains the **SOLVER function**, which is ideally suited to fitting data with nonlinear functions via an iterative algorithm, which minimizes the sum of the squared difference between data points and the function describing the data.

In the case of numerical methods useful to find numerical approximations to the solutions of ordinary/partial differential highlights the classical Runge–Kutta methods, which are able to solve the initial-value problems of differential equations. These methods can be used to construct high-order accurate numerical method by functions' self without needing the high-order derivatives of functions [17].

These methods coordinate with the solution of Taylor's series up to the term in h_r, where r varies from method to method, representing the order of that method. One of the most significant advantages of Runge–Kutta formulae is that it requires the function's values at some specified points. In this context a **Runge–Kutta fourth-order (RK4)** is the most commonly used method to find the solution of a differential equation. The Runge–Kutta method provides the approximate value of y for a given point x. Only the first-order ODEs can be solved using the RK4 method.

Consider an ordinary differential equation of the form $\mathrm{d}y/\mathrm{d}x = f(x, y)$ with initial condition $y(x_0) = y_0$.

For example, the formula for the fourth-order Runge–Kutta method is given by:

$$y_1 = y_0 + (1/6)(k_1 + 2k_2 + 2k_3 + k_4).$$

Here,

$$k_1 = hf(x_0, y_0)$$
$$k_2 = hf\left[x_0 + (1/2)h, y_0 + (1/2)k_1\right]$$
$$k_3 = hf\left[x_0 + (1/2)h, y_0 + (1/2)k_2\right]$$
$$k_4 = hf(x_0 + h, y_0 + k_3),$$

where h represents the step size, x_n and y_n are the values of the independent and dependent variables at the current step, $f(x_n, y_n)$ is the derivative function, and k_1, k_2, and k_3 are intermediate calculations based on the derivative function. Thus, these methods can be easily applicable to a large class of kinetic applications with high accuracy.

With the aim of fitting kinetic models to experimental data using nonlinear regression, different methods and commercial programs are available which allow the user simply to type in a new equation, to which the data are then fitted. All that is required is to translate the equation into a form slightly more akin to a computer language.

Moreover, the choice of freely available and commercial software for performing nonlinear regression, numerical integration, and other tasks is a matter of personal taste, but any well-written package is expected to generate the same results with a given set of data. There are different packages available, for example:

- KAPattern based on the schematic method of King and Altman (KA), it uses linear graph theory for generating directed graphs, allowing the derivation of rate equations for enzymes and transporters, Qi et al. [18]. The underlying algorithm allows for the assumption steady-state, rapid equilibrium-binding, and/or irreversibility for individual steps in enzyme-catalyzed reaction mechanism. The program can automatically generate MathML and MATLAB output files that users can easily incorporate into simulation programs. A computer program, called KAPattern, for generating rate equations for enzymatic systems is freely available and can be accessed at http://www.biocoda.org [18].
- Maple can be used in kinetics for almost all purposes, provided the user is skilled in writing the algorithms. Very useful for the verification of algebraic expressions, limits calculation, equations solutions, three-dimensional graphics, among other (MathWorks—Maker of MATLAB and Simulink—MATLAB and Simulink). In fact, Korobov and Ochkov [19] have explained the principles of chemical kinetics and approaches to computerized calculations in a software suite such as Maple.
- MATLAB is useful for producing simulated datasets with/without added random scatter. A practical study was reported by Fisher [20] developing a tutorial of MATLAB on the simulation of a pseudo first-order reaction.
- KinTek Explorer provides the foundation for teaching kinetics and fitting data based on numerical integration of rate equations. Various experiments differ only by the starting concentrations of reactants and the nature of the measured signal. This information is entered into KinTek Explorer software to model kinetics and fit data, allowing multiple experiments to be fit simultaneously using a single, unified model, and thereby transcending the need for a different equation to fit each experiment.

- BestCurvFit performs nonlinear regression curve fitting of data to kinetic models either defined by the user or selected from in internal library of specific models. This contains several models that embrace a broad range of kinetic applications, including all basic enzyme-modifier mechanisms. The user can also write in a simple way her/his own functions. The search for the best fit is achieved through *"multiple nonlinear regression methods in tandem"* and, in model discrimination, goodness of fit is ascertained by the smallest normalized Akaike information criterion [21].
- Origin, a complete graphing and data analysis software package provides a suite of features catering to the needs of scientists and engineers. Origin is one of several software packages designed specifically for plotting and analyzing quantitative data. It offers a beginner-friendly interface and advanced customization capabilities. Origin can be expanded with free applications and connected to MATLAB™, LabVIEW™, or Microsoft© Excel.
- DynaFit (BioKin Ltd.) is a software package to fit kinetic models to experimental data. The main purpose of the program DynaFit is to perform nonlinear least-squares regression of chemical kinetic, enzyme kinetic, or ligand-receptor binding data. The experimental data can be either initial reaction velocities in dependence on the concentration of varied species (e.g., inhibitor concentration versus velocity), or the reaction progress curves (e.g., time versus absorbance). Fully functional DynaFit [22] continues to be freely available to all academic researchers from http://www.biokin.com. It uses a combination of nonlinear regression and numerical integration [23].

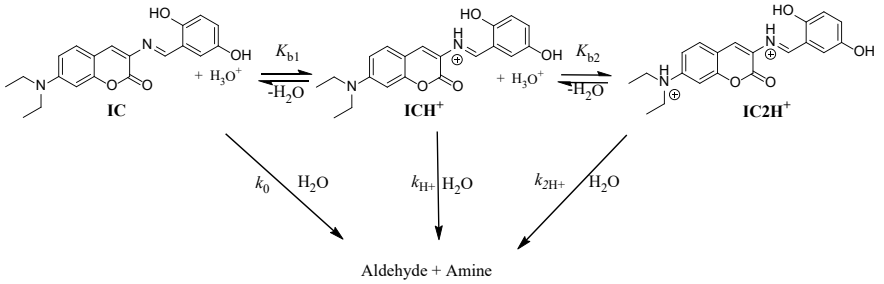
As mentioned above, there are different softwares capable of being useful in finding a suitable kinetic model with accurate values, all of them have the same objectives and differ in some minor aspects. For instance, Baici [21] has detailed supporting software and simulation for kinetic of enzyme data. Other studies have carried out a comparison of relevant software tools for kinetic evaluation of chemical degradation data [24].

9.10 Example: Comparisons Between Fitting Data and Experimental Data for the Hydrolysis of a Schiff Base

The kinetics and mechanisms of the hydrolysis of Schiff bases in different media have been well-reported. In particular, a kinetic study has been carried out, using UV–vis spectroscopy as technique [25], with the aim to determine the constants of the hydrolysis of Schiff base (imine coumarin; IC) and the rate constants of the reaction of the protonated imines (ICH^+ and $IC2H^+$) with water (kH^+ and $k2H^+$, respectively) according to Scheme 9.1. To determine the rate constant values, the reaction was studied at different pH values according to the experimental data, shown in Table 9.1.

Table 9.1 Kinetic data for the hydrolysis reaction of imine coumarin **IC** at different pH values

pH	2.0	2.5	3.0	3.5	4.0	4.5	5.0
$10^4\ k_{obs}/s^{-1}$	65.0	21.0	5.60	1.41	0.256	0.730	0.394



Scheme 9.1 Proposed mechanism for the hydrolysis reaction of the substituted aromatic Schiff bases. The term **IC** represents the studied 7-diethylaminocoumarin-Schiff base. Reproduced from Ref. [25]

Considering the general workflow for parameterization of rate constants, firstly, kinetic data from Table 9.1 were assessed, then the following proposed Scheme 9.1 was used to define the Eqs. (9.6–9.9), to fit and to determine the constants values for each step using Solver from Excel and other fitting program. Rate and equilibrium constants for the hydrolysis of Schiff bases (Scheme 9.1) were obtained and are presented in (Table 9.2).

According to the mechanism described in Scheme 9.1, k_{obs} for the hydrolysis reaction of **IC** can be expressed as the following Eqs. (9.6–9.9):

$$k_{obs} = \frac{k_0 + k_{H^+}\left[H_3O^+\right]10^{pKa1} + k_{2H^+}10^{(pKa1+pKa2)}\left[H_3O^+\right]^2}{10^{(pKa1+pKa2)}\left[H_3O^+\right]^2 + 10^{pKa1}\left[H_3O^+\right] + 1}, \tag{9.6}$$

where [H$_3$O+] is determined by the equation:

$$a\left[H_3O^+\right]^3 + b\left[H_3O^+\right]^2 + c\left[H_3O^+\right] - \left[H_3O^+\right]_0 = 0 \tag{9.7}$$

with $a = 10^{(pKa1+pKa2)}$;

Table 9.2 Fitting kinetic parameters for the hydrolysis reaction of **IC** using Eq. 9.6

Schiff base	pK_{a1}	pK_{a2}	$10^4\ k_{H+}$/ $s^{-1}\ M^{-1}$ (a)	$k_{2H+}/s^{-1}\ M^{-1}$ (a)	$10^3\ k_0$/ s^{-1} (b)
IC	3.26 ± 0.07	1.17 ± 0.18	1.3 ± 0.4	0.052 ± 0.001	< 0.03

$$b = 10^{(pKa1+pKa2)}\left(\frac{1}{10^{pKa2}} + 2[\mathrm{IC}]_0 - \left[\mathrm{H_3O^+}\right]_0\right); \tag{9.8}$$

and

$$c = 10^{pKa1}\left(\frac{1}{10^{pKa1}} + [\mathrm{IC}]_0 - \left[\mathrm{H_3O^+}\right]_0\right). \tag{9.9}$$

This example was carried out according to the general workflow for parameterization of rate constants using kinetic experimental data: (1) Gather available kinetic experimental data (Table 9.1), (2) Construct a reaction mechanism consisting of known individual reaction steps involved in the hydrolysis reaction (Scheme 9.1). (3) Based upon the mechanism, define kinetic equations corresponding to the macroscopic kinetic data (Eqs. 9.6–9.9) to be fit. (4) Fit the equations defined in step 3 using a nonlinear least squares optimization approach (OriginPro) to obtain the rate constant and pK_a values (Table 9.2). Finally, (5) as shown in Fig. 9.3, the comparison of the experimental with the calculated data shows that the non linear regression reproduces the experimental kinetic behavior of the IC hydrolysis reaction.

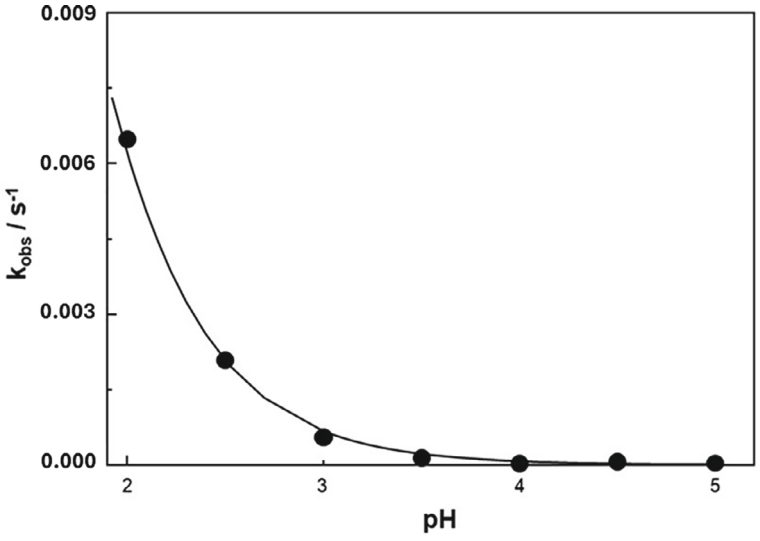


Fig. 9.3 pH profile for the hydrolysis reactions of IC at 25.0 °C. Experimental data (●) presented in Table 9.1 and curve fitted by Eq. 9.6 with parameters presented in Table 9.2. Reproduced from Ref. [25]

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